

Total captured: 229

Question 2 of 230

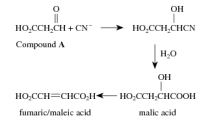
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PASSAGE

Passage 1 (Questions 1-5)

In order to understand the chemical evolution of life, scientists have studied the production of amino acids under conditions that simulate those of primitive Earth.

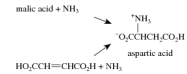
Malic acid is thought to have formed via hydrolysis of a cyanohydrin intermediate (Scheme I).



Scheme I

Subsequent dehydration of malic acid results in the formation of fumaric acid (mp 286°C) and its cis isomer, maleic acid (mp 130°C).

Ammonia was thought to be a constituent of the early atmosphere, and it was proposed that it reacted with fumaric, maleic, and malic acids to afford a prebiological synthesis of aspartic acid, an amino acid (Scheme II).



Scheme II

In an attempt to verify this hypothesis, chemists tried to reproduce the synthesis of aspartic acid under simulated primitive Earth conditions.

An open flask containing malic acid was treated with a continuous ammonia feed, and the mixture was heated at 150°C for 1.5 h. The experiment was repeated with a mixture of fumaric and maleic acids. The results were analyzed by paper chromatography; the *R_f* values of the reaction mixture were compared to the *R_f* value of a standard sample of aspartic acid. (Note: *R_f* = distance moved by compound/distance moved by solvent front.)

QUESTION + ANSWER CHOICES

What assumption is being made if scientists conclude that aspartic acid was formed by the prebiological synthesis in the passage?

- A. Aspartic acid is unstable at temperatures below 150°C.
- B. ~~All of the malic acid underwent the dehydration reaction to form fumaric/maleic acid.~~
- C. Compound A and cyanide were available on primitive Earth.
- D. The reaction between ammonia and fumaric acid was catalyzed by the presence of water.

SOLUTION

Solution: The correct answer is C.

- Because the prebiotic experiment was performed at 150°C in order to form aspartic acid, it is contradictory that the researchers would expect this temperature to degrade at lower temperatures. In addition, aspartic acid is found in living organisms at 37°C.
- There is no requirement given that all of the malic acid dehydrate to fumaric/maleic acid, only that some does.
- Without Compound A and cyanide, no aspartic acid can form.**
- There is no assumption in the passage that water acts as a catalyst in the reaction between ammonia and fumaric acid.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Separations and purifications](#).

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
X
Time Spent:
3 mins 17 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS3
CC5C
FC5
OCH

Question 3 of 230

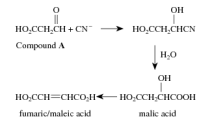
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PASSAGE

Passage 1 (Questions 1-5)

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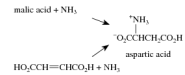
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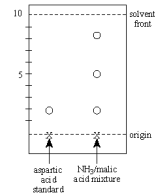
Scheme II

In an attempt to verify this hypothesis, chemists tried to reproduce the synthesis of aspartic acid under simulated primitive Earth conditions.

An open flask containing malic acid was treated with a continuous ammonia feed, and the mixture was heated at 150°C for 1.5 h. The experiment was repeated with a mixture of fumaric and maleic acids. The results were analyzed by paper chromatography; the R_f values of the reaction mixture were compared to the R_f value of a standard sample of aspartic acid. (Note: R_f = distance moved by compound/distance moved by solvent front.)

QUESTION + ANSWER CHOICES

According to the developed chromatography plate shown below, what is the approximate R_f value of aspartic acid?



- A. 0.20
- B. 0.50
- C. 5
- D. 10

SOLUTION

Solution: The correct answer is A.

- The R_f of aspartic acid = (distance traveled by aspartic acid)/(distance to the solvent front) = 2/10 = 0.20
- This value is the R_f of the middle spot of the reaction mixture, not aspartic acid.
- This value is the distance traveled by the middle spot of the reaction mixture. It is not an R_f value.
- This value is the distance to the solvent front.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
42 secs
Difficulty Level:
Easy
Content & Skills:
SIRS4
CC5C
FC5
OCH

Question 4 of 230

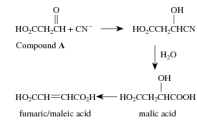
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PASSAGE

Passage 1 (Questions 1-5)

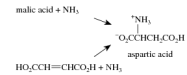
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Time Spent:
1 min 44 secs
Difficulty Level:
Expert
Content & Skills:
SIRS4
CC5C
FC5
OCH

QUESTION + ANSWER CHOICES

Which of the following statements does NOT correctly describe the dehydration of malic acid to fumaric acid and maleic acid?

- A.
The reaction occurs most readily with tertiary alcohols.
- B.
The reaction involves the loss of a water molecule.
- C.
The reaction has a carbocation intermediate.
- D.
The reaction is stereospecific.

SOLUTION

Solution: The correct answer is D.

- Tertiary alcohols readily dehydrate.
- Dehydration is loss of a water molecule by definition.
- During dehydration, a carbocation forms.
- Because of the carbocation intermediate in dehydration, the subsequent loss of a proton will lead to a mixture of cis- and trans-alkenes. Thus, dehydration is NOT stereospecific.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer

A
B
C
D
✓

Question 5 of 230

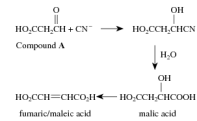
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PASSAGE

Passage 1 (Questions 1-5)

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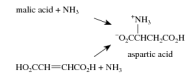
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45 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC5C
FC5
OCH

QUESTION + ANSWER CHOICES

What type of functional group is formed when aspartic acid reacts with another amino acid to form a peptide bond?

- A.
An amine group
- B.
An aldehyde group
- C.
An amide group
- D.
A carboxyl group

SOLUTION

Solution: The correct answer is C.

1. An amine is one of the functional groups that reacts to form a peptide bond, not the peptide bond itself.
2. An aldehyde is RCHO, whereas the product is an amide, RCONHR'.
3. The functional group that represents a peptide bond is called an amide group.
4. A carboxyl group reacts with an amine to form a peptide bond.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:

Question 6 of 230

Data ID: 529570
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PASSAGE

Passage 2 (Questions 6-9)

The nucleophilic alkyl substitution reactions are among the most thoroughly studied transformations in organic chemistry. A substitution by an S_N2 reaction mechanism is shown in Figure 1 and a substitution by an S_N1 mechanism is shown in Figure 2.



Figure 1 S_N2 Reaction mechanism

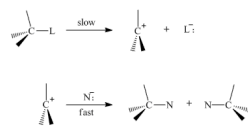


Figure 2 S_N1 Reaction mechanism

The S_N2 mechanism involves a direct displacement of a leaving group (L) by a nucleophile (N). The reaction results in an inversion of configuration at the carbon bearing the leaving group, because the incoming nucleophile attacks it from the side opposite the leaving group. The S_N1 reaction mechanism involves an intermediate carbocation that is formed by loss of the leaving group. A nucleophile can react with the planar carbocation from either side, leading to the inversion or retention of the configuration at the site of the substitution.

The product distribution of two reactions is shown in figures 3 and 4.

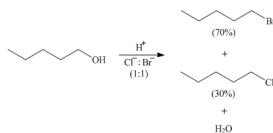


Figure 3 S_N2 Reaction of 1-pentanol

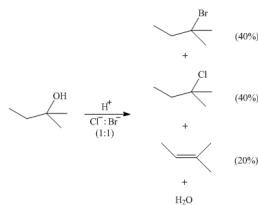


Figure 4 Reaction of 2-methyl-2-butanol

The rate of the reaction shown in Figure 3 increases as the concentration of alcohol, halide, or H^+ increases. The rate of the reaction shown in Figure 4 increases as the concentration of alcohol or H^+ increases. This reaction is independent of the halide concentration.

QUESTION + ANSWER CHOICES

If 2-pentanol replaces 1-pentanol in the reaction shown in Figure 3, the rate of substitution is less because:

- A.

the C–O bond in 2-pentanol is stronger than the C–O bond in 1-pentanol.

- B. there is a competing elimination reaction that slows the rate of substitution.
- C. there is more steric hindrance at the oxygen atom in 2-pentanol than in 1-pentanol, making protonation less likely.
- D. there is more steric hindrance at the 2-position of 2-pentanol than at the 1-position of 1-pentanol.

SOLUTION

Solution: The correct answer is D.

- This statement is false. For both heterolytic and homolytic bond breaking, the more substituted C–O bond is weaker. Relative C–O bond strength is not the relevant parameter for this question.
- Competing elimination does not slow the rate of substitution. Their rates are independent of each other.
- Protonation at OH is a fast, pre-equilibrium step, thus it has no effect on the reaction rate.
- At a secondary carbon, there is more crowding from the adjacent carbon chains, which makes nucleophilic substitution slower.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Amino Acids](#), [Dinitiles](#), [Dinitiles](#), [Lipids](#), [Perfluorinated](#), [Aldehydes](#), and

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
2 mins 15 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC5D
FC5
OCH

Question 7 of 230

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PASSAGE

Passage 2 (Questions 6-9)

The nucleophilic alkyl substitution reactions are among the most thoroughly studied transformations in organic chemistry. A substitution by an S_N2 reaction mechanism is shown in Figure 1 and a substitution by an S_N1 mechanism is shown in Figure 2.



Figure 1 S_N2 Reaction mechanism

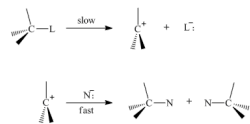


Figure 2 S_N1 Reaction mechanism

The S_N2 mechanism involves a direct displacement of a leaving group (L) by a nucleophile (N). The reaction results in an inversion of configuration at the carbon bearing the leaving group, because the incoming nucleophile attacks it from the side opposite the leaving group. The S_N1 reaction mechanism involves an intermediate carbocation that is formed by loss of the leaving group. A nucleophile can react with the planar carbocation from either side, leading to the inversion or retention of the configuration at the site of the substitution.

The product distribution of two reactions is shown in figures 3 and 4.

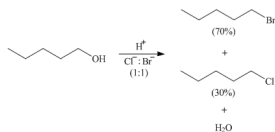


Figure 3 S_N2 Reaction of 1-pentanol

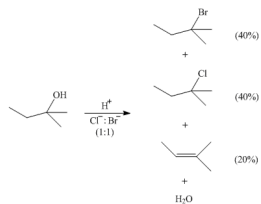
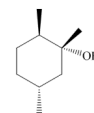


Figure 4 Reaction of 2-methyl-2-butanol

The rate of the reaction shown in Figure 3 increases as the concentration of alcohol, halide, or H^+ increases. The rate of the reaction shown in Figure 4 increases as the concentration of alcohol or H^+ increases. This reaction is independent of the halide concentration.

QUESTION + ANSWER CHOICES

If Reaction 2 (Figure 4) is repeated with HCl and the compound shown below, which of the following compounds is NOT a direct product (without rearrangement)?



- A.
- B.
- C.
- D.

SOLUTION

Solution: The correct answer is A.

- The only way that this product can form is by rearrangement, a 1,2-hydride shift following the initial formation of a tertiary carbocation.
- This product forms directly from deprotonation after the formation of the tertiary carbocation.
- This product forms directly via chloride nucleophile covalently binding to the tertiary carbocation intermediate from below the plane of the carbocation.
- This product forms directly via chloride nucleophile covalently binding to the tertiary carbocation intermediate from above the plane of the carbocation.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Amino Acids](#), [Peptides](#), [Proteins](#), [Lipids](#), [Carbohydrates](#), [Aldehydes](#), and [Ketones](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A ✓
B
C
D
Time Spent:
2 mins 18 secs
Difficulty Level:
Easy
Content & Skills:
SIRS2
CC5D
FC5
OCH

Question 8 of 230

Data ID: 629656
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PASSAGE

Passage 2 (Questions 6-9)

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Figure 1 S_N2 Reaction mechanism

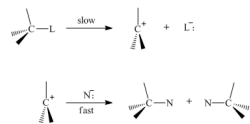


Figure 2 S_N1 Reaction mechanism

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The product distribution of two reactions is shown in figures 3 and 4.

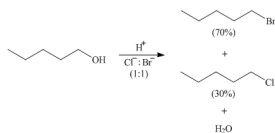


Figure 3 S_N2 Reaction of 1-pentanol

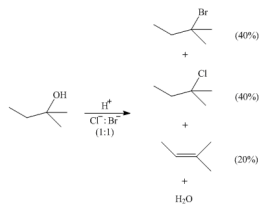


Figure 4 Reaction of 2-methyl-2-butanol

The rate of the reaction shown in Figure 3 increases as the concentration of alcohol, halide, or H^+ increases. The rate of the reaction shown in Figure 4 increases as the concentration of alcohol or H^+ increases. This reaction is independent of the halide concentration.

QUESTION + ANSWER CHOICES

If a solution containing the compounds shown in Figure 4, is injected into a gas-liquid chromatograph, the first peak observed in the gc trace is attributable to which compound?

- A.

2-Methyl-2-butanol

- B. 2-Methyl-2-butene
- C. 2-Chloro-2-methylbutane
- D. 2-Bromo-2-methylbutane

SOLUTION

Solution: The correct answer is B.

1. 2-Methyl-2-butanol is highly polar and makes hydrogen bonds, so it is the least volatile of the options.
2. In gas-liquid chromatography, the first peak to emerge will be from the least polar, most volatile compound. 2-Methyl-2-butene is nonpolar and the most volatile because it only has London forces for intermolecular attraction.
3. Unlike 2-methyl-2-butene, 2-chloro-2-methylbutane possesses dipole-dipole forces in addition to greater London forces, so it will not be the most volatile.
4. Unlike 2-methyl-2-butene, 2-bromo-2-methylbutane possesses dipole-dipole forces in addition to greater London forces, so it will not be the most volatile.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect

Correct Answer Your Answer

A

B

C

X

D

Time Spent:

1 min 20 secs

Difficulty Level:

Difficult

Content & Skills:

SIRS2

CC5C

FC5

OCH

Question 9 of 230

Data ID: 529567
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PASSAGE

Passage 2 (Questions 6-9)

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Figure 1 S_N2 Reaction mechanism

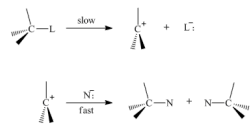


Figure 2 S_N1 Reaction mechanism

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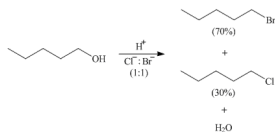


Figure 3 S_N2 Reaction of 1-pentanol

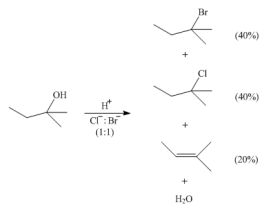
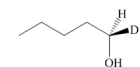


Figure 4 Reaction of 2-methyl-2-butanol

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QUESTION + ANSWER CHOICES

What is the product of the reaction of Compound 1 (shown below) with HBr by the pathway shown in Figure 3?



Compound 1

- A. (R) -1-bromo-1-deuteriohexane
- B. (S) -1-bromo-1-deuteriohexane
- C. (S) -1-bromo-1-deuteriopentane
- D. (R) -1-bromo-1-deuteriopentane

SOLUTION

Solution: The correct answer is C.

1. This product has both the wrong stereochemistry and the wrong number of carbon atoms.
2. Although this product has correct stereochemistry, it has the wrong number of carbon atoms.
3. **Because this is an S_N2 reaction, there will be inversion of configuration when bromide ion displaces water as a leaving group, changing the stereochemistry from R to S.**
4. Although this product has the correct number of carbons, the stereochemistry should be S, not R.

Khan Academy Lessons

RESULT / TIMING / TAGS

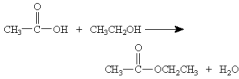
Incorrect
Correct Answer Your Answer
A
B
C
D
X
Time Spent:
1 min 35 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC5B
FC5
OCH

Question 10 of 230

Data ID: 529699
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QUESTION • ANSWER CHOICES

Acetic acid and ethanol react to form an ester product as shown below.



In determining which reactant loses the –OH group, which of the following isotopic substitutions would be most useful?

- A. Replace the acidic H of acetic acid with D.
- B. Replace the alcoholic H of ethanol with D.
- **C.** Replace the carbonyl oxygen of acetic acid with O-18.
- D. Replace the hydroxyl oxygen of ethanol with O-18.

SOLUTION

Solution: The correct answer is D.

1. In order to determine which reactant loses the –OH group, an oxygen atom on ethanol must be radiolabeled.
2. In order to determine which reactant loses the –OH group, an oxygen atom on ethanol must be radiolabeled. Replacement the alcoholic H by D will not provide this information because of rapid H/D exchange between acetic acid and ethanol.
3. Replacing the carbonyl oxygen in acetic acid with O-18 will yield a mixture of O-18-labeled water and ester, further confusing the situation. There may be a mechanism of O-18 exchange with ethanol via O-18 labeled water that would obscure which reactant actually loses the –OH during ester formation.
4. **Labeling the oxygen in ethanol with O-18 definitely determines which –OH is lost because 100% of the O-18 in radiolabeled ethanol will be retained in the ester, not the water. Thus, only acetic acid supplies –OH that is lost to form water.**

Khan Academy Lessons

For more help on the skills, concepts, or content associated by this question, see the Khan Academy lessons on [Amino Acids](#), [Polptides](#), [Proteins](#), [Lipids](#), [Carbohydrates](#), [Alcohols](#), and

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer

- A
B
C
x
D

Time Spent:
2 mins 12 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS3
CC5D
FC5
OCH

Question 11 of 230

Data ID: 529675
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QUESTION • ANSWER CHOICES

A person, whose eye has a lens-to-retina distance of 2.0 cm, can only clearly see objects that are closer than 1.0 m away. What is the strength S of the person's eye lens? (Note: Use the thin lens formula $1/O + 1/I = S$.)

- A. –50 D
- B. –10 D
- **C.** 51 D
- D. 55 D

SOLUTION

Solution: The correct answer is C.

1. The negative sign of the lens strength is indicative of a diverging lens, whereas the eye lens is a converging lens.
2. The negative sign of the lens strength is indicative of a diverging lens, whereas the eye lens is a converging lens.
3. **According to the thin lens formula, the strength of the eye lens is equal to the inverse of the focal length of the eye lens. Its numerical value is given by $(1\text{ m})^{-1} + (0.02\text{ m})^{-1} = 1\text{ D} + 50\text{ D} = 51$**
4. This answer implies the person can only clearly see objects that are closer than 20 cm away.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer

- A
B
C
✓

D
Time Spent:
2 mins 52 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC4D
FC4
PHY

Question 12 of 230

Data ID: 529579
Captured: 6/3/2026, 4:39:57 PM

QUESTION • ANSWER CHOICES

Which statement correctly describes how enzymes affect chemical reactions? Stabilization of:

- A.
the substrate changes the free energy of the reaction.
- B.
the transition state changes the free energy of the reaction.
- C.
the substrate changes the activation energy of the reaction.
- D.
the transition state changes the activation energy of the reaction.

SOLUTION

Solution: The correct answer is **D**.

1. Although an enzyme binds to substrate, it does not affect the free energy of a reaction.
2. Although an enzyme does bind the transition state, it does not affect the free energy of a reaction.
3. An enzyme does not stabilize a substrate to change the activation energy of a reaction. Substrate stabilization would increase the activation energy and slow down the reaction.
4. **An enzyme stabilizes the transition state, thus lowering the activation energy of a reaction.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
39 secs
Difficulty Level:
Easy
Content & Skills:
SIRS1
CC5E
FC5
BCH

Question 13 of 230

Data ID: 529586
Captured: 6/3/2026, 4:39:57 PM

QUESTION • ANSWER CHOICES

Which of the following will *decrease* the percentage ionization of 1.0 M acetic acid, $\text{CH}_3\text{CO}_2\text{H}(aq)$?

- A.
Chlorinating the CH_3 group
- B.
~~Diluting the solution~~
- C.
Adding concentrated $\text{HCl}(aq)$
- D.
~~Adding a drop of basic indicator~~

SOLUTION

Solution: The correct answer is **C**.

1. Replacing H with Cl will stabilize the formation of negative charge on the nearby carboxylic acid group. This is because chlorine is an electronegative atom. This change will therefore increase the percentage ionization of the acid.
2. Diluting the solution will increase the percentage ionization by Le Châtelier's Principle.
3. **HCl is a strong acid that will increase the amount of H^+ in solution and thus decrease the percentage of $\text{CH}_3\text{CO}_2\text{H}$ that ionizes.**
4. Adding a drop of basic indicator will slightly increase the amount of dissociated (ionized) acid due to the acid–base reaction that occurs between the two.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
1 min 57 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC5A
FC5
GCH

Question 14 of 230

Data ID: 525942
Captured: 6/3/2026, 4:39:58 PM

PASSAGE

Passage 3 (Questions 14–17)

Researchers are designing a micro-robot equipped with an oxygen sensor for intraocular procedures. The micro-robot will be injected into the patient’s eye to determine if a blood clot is present. The micro-robot is powered by electromagnetic radiations that induce a current in its interior circuit. The current powers a motor that propels the micro-robot through the fluid of the vitreous humor. Inside the eye, the micro-robot experiences a viscous drag force F dependent on the vitreous humor viscosity η , and the micro-robot’s diameter d and velocity v :

$$F \approx -3\pi\eta dv$$

Equation 1

The micro-robot uses a luminescent sensor film to determine the oxygen concentration, and potential blood clots, in the retina. In this method, a blue LED is shined into the eye and strikes the surface of the sensor. The sensor coating absorbs the photons from the LED and a short time later emits photons that are subsequently detected by a photodiode (Figure 1). Experiments in which oxygen parameters were controlled showed that oxygen, if present in the retina, affects both the time it takes the sensor to emit the photons and the intensity of the emitted radiation.

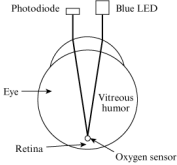


Figure 1 Micro-robot equipped with oxygen sensor used to detect blood clots

If the results of the test indicate that the patient has a clot that is obstructing blood flow, a second micro-robot, coated with tissue plasminogen activator (t-PA), is steered to the thrombus site where it punctures and docks with a retinal vein. The enzyme t-PA is a serine protease that converts plasminogen into plasmin, which subsequently degrades the clot by fibrinolysis.

QUESTION + ANSWER CHOICES

When the current in the micro-robot’s circuit increases and the resistance of the circuit remains constant, the voltage of the circuit:

- A. decreases.
- B. increases.
- C. is constant.
- D. is zero.

SOLUTION

Solution: The correct answer is B.

1. This option contradicts Ohm’s law which states that current is equal to voltage divided by resistance. If current increases and resistance is constant, then voltage cannot decrease.
2. This option is correct because according to Ohm’s law, current is equal to voltage divided by resistance. If current increases and resistance is constant, then voltage increases as well.
3. This option contradicts Ohm’s law which states that current is equal to voltage divided by resistance. If current increases and resistance is constant, then voltage cannot remain constant.
4. This option contradicts Ohm’s law that states that current is equal to voltage divided by resistance. If current is not zero, then voltage cannot be zero, too.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B ✓
C
D
Time Spent:
49 secs

Difficulty Level:
Easy
Content & Skills:
SIRS2
CC4C
FC4
PHY

Question 15 of 230

Data ID: 529954
Captured: 6/3/2026, 4:39:59 PM

PASSAGE

Passage 3 (Questions 14–17)

Researchers are designing a micro-robot equipped with an oxygen sensor for intraocular procedures. The micro-robot will be injected into the patient’s eye to determine if a blood clot is present. The micro-robot is powered by electromagnetic radiations that induce a current in its interior circuit. The current powers a motor that propels the micro-robot through the fluid of the vitreous humor. Inside the eye, the micro-robot experiences a viscous drag force **F** dependent on the vitreous humor viscosity η , and the micro-robot’s diameter d and velocity v :

$$F = -3\pi d\eta v$$

Equation 1

The micro-robot uses a luminescent sensor film to determine the oxygen concentration, and potential blood clots, in the retina. In this method, a blue LED is shined into the eye and strikes the surface of the sensor. The sensor coating absorbs the photons from the LED and a short time later emits photons that are subsequently detected by a photodiode (Figure 1). Experiments in which oxygen parameters were controlled showed that oxygen, if present in the retina, affects both the time it takes the sensor to emit the photons and the intensity of the emitted radiation.

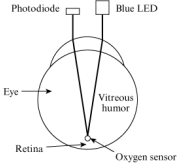


Figure 1 Micro-robot equipped with oxygen sensor used to detect blood clots

If the results of the test indicate that the patient has a clot that is obstructing blood flow, a second micro-robot, coated with tissue plasminogen activator (t-PA), is steered to the thrombus site where it punctures and docks with a retinal vein. The enzyme t-PA is a serine protease that converts plasminogen into plasmin, which subsequently degrades the clot by fibrinolysis.

QUESTION + ANSWER CHOICES

Knowing that the speed of light in the vitreous humor is 2.1×10^8 m/s, what is the index of refraction of the vitreous humor? (Note: The speed of light in a vacuum is 3.0×10^8 m/s.)

- A. 0.7
- B. 1.4
- C. 2.1
- D. 3.0

SOLUTION

Solution: The correct answer is **B**.

1. This option implies light travels faster through the vitreous humor than through vacuum, which is a scientific impossibility.

2. The index of refraction of a medium is equal to the ratio of the speed of light in vacuum to the speed of light in the medium, thus in this case is equal to $(3.0 \times 10^8 \text{ m/s}) / (2.1 \times 10^8 \text{ m/s}) = 1.4$.

3. This is 50% larger than the index of refraction of the vitreous humor, which is equal to $(3.0 \times 10^8 \text{ m/s}) / (2.1 \times 10^8 \text{ m/s}) = 1.4$.

4. The index of refraction of the vitreous humor is equal to $(3.0 \times 10^8 \text{ m/s}) / (2.1 \times 10^8 \text{ m/s}) = 1.4$; however, this (3.0) number is 114% larger.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
X
B
C
D
Time Spent:

1 min 23 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC4D
FC4
PHY

Question 16 of 230

Data ID: 525962
Captured: 6/3/2026, 4:39:59 PM

PASSAGE

Passage 3 (Questions 14–17)

Researchers are designing a micro-robot equipped with an oxygen sensor for intraocular procedures. The micro-robot will be injected into the patient’s eye to determine if a blood clot is present. The micro-robot is powered by electromagnetic radiations that induce a current in its interior circuit. The current powers a motor that propels the micro-robot through the fluid of the vitreous humor. Inside the eye, the micro-robot experiences a viscous drag force **F** dependent on the vitreous humor viscosity η , and the micro-robot’s diameter d and velocity v :

$$F \approx -3\pi\eta dv$$

Equation 1

The micro-robot uses a luminescent sensor film to determine the oxygen concentration, and potential blood clots, in the retina. In this method, a blue LED is shined into the eye and strikes the surface of the sensor. The sensor coating absorbs the photons from the LED and a short time later emits photons that are subsequently detected by a photodiode (Figure 1). Experiments in which oxygen parameters were controlled showed that oxygen, if present in the retina, affects both the time it takes the sensor to emit the photons and the intensity of the emitted radiation.

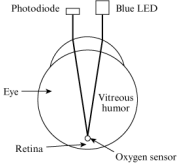


Figure 1 Micro-robot equipped with oxygen sensor used to detect blood clots

If the results of the test indicate that the patient has a clot that is obstructing blood flow, a second micro-robot, coated with tissue plasminogen activator (t-PA), is steered to the thrombus site where it punctures and docks with a retinal vein. The enzyme t-PA is a serine protease that converts plasminogen into plasmin, which subsequently degrades the clot by fibrinolysis.

QUESTION + ANSWER CHOICES

The intensity of the radiation emitted by the oxygen sensor is directly proportional to the:

- A. propagation speed of the radiation.
- B. wavelength of the radiation.
- C. polarization of photons emitted.
- D. number of photons emitted.

SOLUTION

Solution: The correct answer is D.

1. The intensity of radiation is not related to its propagation speed, but to the number of photons emitted per unit time and their respective frequency.
2. The intensity of radiation is not related to the wavelength of radiation, but to the number of photons emitted per unit time and their respective frequency.
3. The intensity of radiation is not related to the polarization state, but to the number of photons emitted per unit time and their respective frequency.
4. The energy of electromagnetic radiation is directly proportional to the number of photons, and the intensity of electromagnetic radiation is defined as energy emitted per unit time. Thus, intensity is directly proportional to the number of photons emitted.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
X
B
C
D
Time Spent:
1 min 26 secs

Difficulty Level:
Difficult
Content & Skills:
SIRS1
CC4D
FC4
PHY

Question 17 of 230

Data ID: 525965
Captured: 6/3/2026, 4:40:00 PM

PASSAGE

Passage 3 (Questions 14–17)

Researchers are designing a micro-robot equipped with an oxygen sensor for intraocular procedures. The micro-robot will be injected into the patient’s eye to determine if a blood clot is present. The micro-robot is powered by electromagnetic radiations that induce a current in its interior circuit. The current powers a motor that propels the micro-robot through the fluid of the vitreous humor. Inside the eye, the micro-robot experiences a viscous drag force F dependent on the vitreous humor viscosity η , and the micro-robot’s diameter d and velocity v :

$$F = -3\pi d\eta v$$

Equation 1

The micro-robot uses a luminescent sensor film to determine the oxygen concentration, and potential blood clots, in the retina. In this method, a blue LED is shined into the eye and strikes the surface of the sensor. The sensor coating absorbs the photons from the LED and a short time later emits photons that are subsequently detected by a photodiode (Figure 1). Experiments in which oxygen parameters were controlled showed that oxygen, if present in the retina, affects both the time it takes the sensor to emit the photons and the intensity of the emitted radiation.

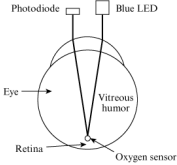


Figure 1 Micro-robot equipped with oxygen sensor used to detect blood clots

If the results of the test indicate that the patient has a clot that is obstructing blood flow, a second micro-robot, coated with tissue plasminogen activator (t-PA), is steered to the thrombus site where it punctures and docks with a retinal vein. The enzyme t-PA is a serine protease that converts plasminogen into plasmin, which subsequently degrades the clot by fibrinolysis.

QUESTION + ANSWER CHOICES

What is the energy of the photons emitted by the LED at a frequency of 610 THz? (Note: $h = 6.6 \times 10^{-34}$ J·s)

- A. 9.2×10^{-12} J
- B. 1.6×10^{-16} J
- C. 1.1×10^{-18} J
- D. 4.0×10^{-19} J

SOLUTION

Solution: The correct answer is D.

- This is about seven orders of magnitude larger than the energy of a photon of frequency 610 THz which is 4×10^{-19} J.
- This is about three orders of magnitude larger than the energy of a photon of frequency 610 THz which is 4×10^{-19} J.
- This is about three times larger than the energy of a photon of frequency 610 THz which is 4×10^{-19} J.
- The energy of a photon of frequency 610 THz is equal to 6.6×10^{-34} J·s $\times$ 610×10^{12} Hz = 4×10^{-19} J.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
x
D
Time Spent:
3 mins 51 secs

Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC4D
FC4
PHY

Question 18 of 230

Data ID: 529579
Captured: 6/3/2026, 4:40:00 PM

PASSAGE

Passage 4 (Questions 18-21)

Blood pressure can be measured in the brachial artery, which has a cross-sectional area of 25 mm² in a healthy adult. The blood flow rate is related to the power the person generates to move their body, as shown Figure 1.

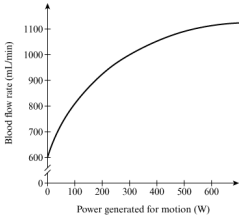


Figure 1 Relationship between the brachial artery blood flow rate of a healthy adult and the power generated for body motion

An athlete running on a horizontal treadmill maintained a brachial artery blood flow rate of 1.1 L/min for 30 s. The treadmill had an electric circuit used to charge a parallel-plate capacitor of capacitance 1 mF.

The maximum and minimum blood pressures of a healthy adult (the systolic P_s and the diastolic P_d pressures, respectively) are about $P_s = 120$ mmHg and $P_d = 75$ mm Hg. The mean arterial pressure P_{MPA} is given by

$$P_{MPA} = P_d + (P_s - P_d)/3$$

Equation 1

and the pulse pressure P_p is defined as

$$P_p = P_s - P_d$$

Equation 2

Ca^{2+} ions are normally found in the blood at a concentration of 100 mg/L. The location of internal organs where changes in the calcium content occur can be found by placing positron-emitting isotopes such as ^{18}O (with a half-life $T_{1/2} = 2$ min) into the patient's bloodstream. When positrons annihilate with electrons, they produce pairs of gamma photons, each with an energy of 500 keV. The location where gamma photons originate gives the location of the organ of interest.

QUESTION + ANSWER CHOICES

What is the relationship between P_p , P_{MPA} , and P_d ?

- A. $P_p = P_{MPA} + 3P_d$
- B. $P_p = 3P_{MPA} + P_d$
- C. $P_p = 3P_{MPA} - 3P_d$
- D. $P_p = P_{MPA} - 3P_d$

SOLUTION

Solution: The correct answer is C.

- This option incorrectly implies that Equation 1 is $P_{MPA} = 3P_d$.
- This option incorrectly implies that $P_d = 0$.
- Substituting Equation 2 in Equation 1 yields $P_{MPA} = P_d + \frac{P_p}{3}$ that can be arranged as $P_p = 3P_{MPA} - 3P_d$.
- This option incorrectly implies that $P_{MPA} = 0$.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
1 min 25 secs
Difficulty Level:
Easy
Content & Skills:
SIRS1
CC4B
FC4
PHY

Question 19 of 230

Data ID: 529994
Captured: 6/3/2026, 4:40:01 PM

PASSAGE

Passage 4 (Questions 18-21)

Blood pressure can be measured in the brachial artery, which has a cross-sectional area of 25 mm² in a healthy adult. The blood flow rate is related to the power the person generates to move their body, as shown Figure 1.

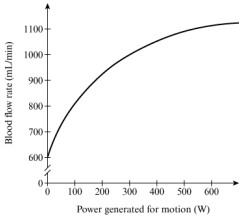


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$$P_{MPA} = P_d + (P_s - P_d)/3$$

Equation 1

and the pulse pressure P_p is defined as

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Equation 2

Ca^{2+} ions are normally found in the blood at a concentration of 100 mg/L. The location of internal organs where changes in the calcium content occur can be found by placing positron-emitting isotopes such as ^{18}O (with a half-life $T_{1/2} = 2$ min) into the patient's bloodstream. When positrons annihilate with electrons, they produce pairs of gamma photons, each with an energy of 500 keV. The location where gamma photons originate gives the location of the organ of interest.

QUESTION + ANSWER CHOICES

What percentage of standard atmospheric pressure is the pulse pressure of a healthy adult?

- A. 10%
- B. 6%
- C. 2%
- D. 1%

SOLUTION

Solution: The correct answer is B.

- Atmospheric pressure is 760 mmHg. If pulse pressure was 10% of this value, it would be 76 mmHg.
- The pulse pressure in a healthy adult is (120 - 75) mmHg = 45 mmHg, and so the percentage is 45 mmHg/760 mmHg = 6%.
- A pulse pressure that is 2% of atmospheric pressure is 0.02 × 760 mmHg = 15.2 mmHg.
- A pulse pressure that is 1% of atmospheric pressure would be 0.01 × 760 mmHg = 7.6 mmHg.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer

- A
B
C
X
D

Time Spent:
4 mins 29 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC4B
FC4
GCH

Question 20 of 230

Data ID: 529997
Captured: 6/3/2026, 4:40:02 PM

PASSAGE

Passage 4 (Questions 18–21)

Blood pressure can be measured in the brachial artery, which has a cross-sectional area of 25 mm² in a healthy adult. The blood flow rate is related to the power the person generates to move their body, as shown Figure 1.

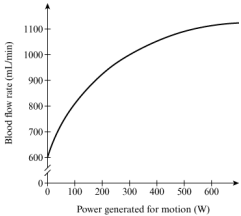


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$$P_{MPA} = P_d + (P_s - P_d)/3$$

Equation 1

and the pulse pressure P_p is defined as

$$P_p = P_s - P_d$$

Equation 2

Ca^{2+} ions are normally found in the blood at a concentration of 100 mg/L. The location of internal organs where changes in the calcium content occur can be found by placing positron-emitting isotopes such as ^{15}O (with a half-life $T_{1/2} = 2$ min) into the patient's bloodstream. When positrons annihilate with electrons, they produce pairs of gamma photons, each with an energy of 500 keV. The location where gamma photons originate gives the location of the organ of interest.

QUESTION + ANSWER CHOICES

What fraction of a ^{15}O sample decays in 10 min?

- A. $\frac{1}{8}$
- B. $\frac{9}{16}$
- C. $\frac{3}{4}$
- D. $\frac{31}{32}$

SOLUTION

Solution: The correct answer is D.

- Based on the radioactive decay law $N(t) = N (1/2)^{t/T}$ where $T = 2$ min, this fraction of ^{15}O decays in $t = 2$ min $\times \ln(7/8) \div \ln(1/2) = 22$ seconds.
- Based on the radioactive decay law $N(t) = N (1/2)^{t/T}$ where $T = 2$ min, this fraction of ^{15}O decays in $t = 2$ min $\times \ln(7/16) \div \ln(1/2) = 2.4$ minutes.
- Based on the radioactive decay law $N(t) = N (1/2)^{t/T}$ where $T = 2$ min, this fraction of ^{15}O decays in $t = 2$ min $\times \ln(1/4) \div \ln(1/2) = 4$ minutes.
- This answer is correct because based on the radioactive decay law $N(t) = N (1/2)^{t/T}$ where $T = 2$ min, this fraction of ^{15}O decays in $t = 2$ min $\times \ln(1/32) \div \ln(1/2) = 10$ minutes.

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
X
D
Time Spent:
2 mins 3 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC4E
FC4
PHY

Question 21 of 230

Data ID: 529999
Captured: 6/3/2026, 4:40:02 PM

PASSAGE

Passage 4 (Questions 18–21)

Blood pressure can be measured in the brachial artery, which has a cross-sectional area of 25 mm^2 in a healthy adult. The blood flow rate is related to the power the person generates to move their body, as shown Figure 1.

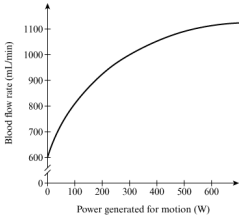


Figure 1 Relationship between the brachial artery blood flow rate of a healthy adult and the power generated for body motion

An athlete running on a horizontal treadmill maintained a brachial artery blood flow rate of 1.1 L/min for 30 s . The treadmill had an electric circuit used to charge a parallel-plate capacitor of capacitance 1 mF .

The maximum and minimum blood pressures of a healthy adult (the systolic P_s and the diastolic P_d pressures, respectively) are about $P_s = 120\text{ mmHg}$ and $P_d = 75\text{ mm Hg}$. The mean arterial pressure P_{MPA} is given by

$$P_{\text{MPA}} = P_d + (P_s - P_d)/3$$

Equation 1

and the pulse pressure P_p is defined as

$$P_p = P_s - P_d$$

Equation 2

Ca^{2+} ions are normally found in the blood at a concentration of 100 mg/L . The location of internal organs where changes in the calcium content occur can be found by placing positron-emitting isotopes such as ^{15}O (with a half-life $T_{1/2} = 2\text{ min}$) into the patient's bloodstream. When positrons annihilate with electrons, they produce pairs of gamma photons, each with an energy of 500 keV . The location where gamma photons originate gives the location of the organ of interest.

QUESTION + ANSWER CHOICES

What is the work generated by a healthy adult who circulates 9 L of blood through the brachial artery in 10 min ?

- A. 2 kJ
- B. 12 kJ
- C. 20 kJ
- D. 120 kJ

SOLUTION

Solution: The correct answer is D.

1. This option is 60 times smaller than the correct work, because a volume flow rate of 9 liters in 10 minutes means 900 mL/min , and according to the figure, this flow rate correlates to a power of 200 W . The correct work is then $200\text{ W} \times 600\text{ s} = 120\text{ kJ}$.
2. This option is 10 times smaller than the correct work, because a volume flow rate of 9 liters in 10 minutes means 900 mL/min and according to the figure, this flow rate correlates to a power of 200 W . The correct work is then $200\text{ W} \times 600\text{ s} = 120\text{ kJ}$.
3. This option is one order of magnitude smaller than the correct work, because a volume flow rate of 9 liters in 10 minutes means 900 mL/min , and according to the figure, this flow rate correlates to a power of 200 W . The correct work is then $200\text{ W} \times 600\text{ s} = 120\text{ kJ}$.
4. A volume flow rate of 9 liters in 10 minutes means 900 mL/min . According to the figure, this flow rate correlates to a power of 200 W . The work is then $200\text{ W} \times 600\text{ s} = 120\text{ kJ}$.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
x
B
C
D
Time Spent:
3 mins 10 secs
Difficulty Level:
Expert
Content & Skills:
SIRS4
CC4A
FC4
PHY

Data ID: 830000
Captured: 6/3/2026, 4:40:03 PM

Passage 5 (Questions 22-25)

Researchers in tissue engineering have developed a peptide amphiphile (Compound 1, Figure 1) that can be used to make a nanostructured fibrous scaffold (composite) similar to an extracellular matrix found in bone.

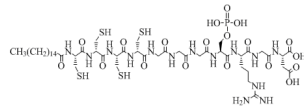
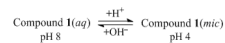
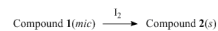


Figure 1	Structure of peptide amphiphile, Compound 1, used to build scaffolds for synthetic bone formation
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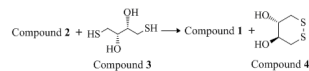
Molecules of Compound **1** spontaneously self-assemble into cylindrical micelles in water at pH 4 but freely redissolve at pH 8 (Equation 1). Oxidation of self-assembled, micellar Compound **1** with iodine resulted in a solid material (Compound **2**) that was stable to alkaline solutions (Equation 2). Reaction of Compound **2** with dithiol (Compound **3**) resulted in reformation of Compound **1** (Equation 3).



Equation 1



Equation 2



Equation 3

Adapted from J. D. Hartgerink, E. Beniash, and S. I. Stupp, *Science*. ©2001 American Association for the Advancement of Science.

- A. Intermolecular hydrogen bonding
- ~~B. Intermolecular covalent bonding~~
- ~~C. Intramolecular hydrogen bonding~~
- ~~D. Intramolecular covalent bonding~~

Solution: The correct answer is **B**.

- Khan Academy Lessons

[illegible]

Incorrect
Correct Answer Your Answer

Incorrect
Correct Answer
A
B
C
X
D

Question 23 of 230

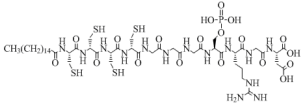
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PASSAGE

Passage 5 (Questions 22-25)

The natural development and repair of bone tissue provides a good example for the design of mineralized tissue fabrication schemes. Besides cells, bone tissue consists of both inorganic and organic components. The inorganic component is a carbonated, calcium-deficient, nano-crystalline hydroxyapatite [HA = Ca₅(PO₄)₃(OH)]. The organic component is a matrix composed of type I collagen (95%) and non-collageneous proteins.

Researchers in tissue engineering have developed a peptide amphiphile (Compound 1, Figure 1) that can be used to make a nanostructured fibrous scaffold (composite) similar to an extracellular matrix found in bone.

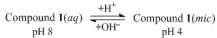


Compound 1

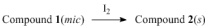
Figure 1 Structure of peptide amphiphile, Compound 1, used to build scaffolds for synthetic bone formation

Compound 1 contained five regions: (1) a hydrophobic region, (2) residues that could be cross-linked for additional structural integrity, (3) spacer residues, (4) a residue that was phosphorylated for future attachment of nanocrystals of HA, and (5) residues with affinity for cells.

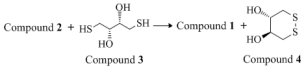
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Equation 1



Equation 2



Equation 3

The researchers wanted to mimic the alignment that occurs between crystals of HA and collagen fibrils in bone tissue. They created a peptide amphiphile that assembles into cylindrical micelles as shown in Figure 2. Subsequent capture of nanocrystals of HA followed by mineralization was used to form synthetic bone.

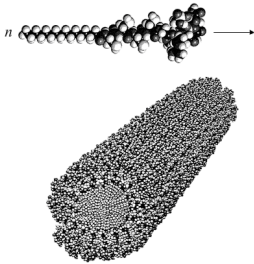


Figure 2 Space-filling molecular model illustrating cylindrical micelle formation by the peptide amphiphile, Compound 1

Adapted from J. D. Hartgerink, E. Beniash, and S. I. Stupp, Science, ©2001 American Association for the Advancement of Science.

QUESTION + ANSWER CHOICES

Which amino acid residues were incorporated into Compound 1 to promote the adhesion of cells on the scaffold surfaces?

- A. Arg and Gly
- B. Cys and Gly
- C. Cys and Asp
- D. Asp and Arg

SOLUTION

Solution: The correct answer is D.

- Arg has a side chain that can promote strong adhesion via electrostatic interaction, but Gly does not.
- Cys is part of the cross-linking region of Compound 1, not the cell adhesion region, and Gly has no side chain that can form strong attractive forces of any type.
- Although the side of Asp is in the cell adhesion region of Compound 1 and can form electrostatic interactions, Cys is part of the cross-linking region of Compound 1, not the cell adhesion region.
- Asp and Arg are part of the cell adhesion region of Compound 1. Both have side chains that can participate in electrostatic interactions.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Lipids](#), [Amino Acids](#), [Peptides](#), [Proteins](#), [The](#)

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
X
C
D
Time Spent:
1 min 26 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC5D
FC5
BCH

Question 24 of 230

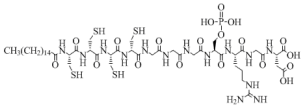
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PASSAGE

Passage 5 (Questions 22-25)

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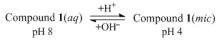


Compound 1

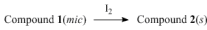
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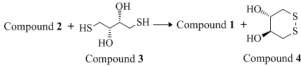
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Equation 1



Equation 2



Equation 3

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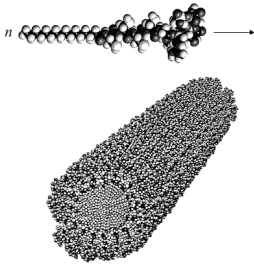


Figure 2 Space-filling molecular model illustrating cylindrical micelle formation by the peptide amphiphile, Compound 1

Adapted from J. D. Hartgerink, E. Beniash, and S. I. Stupp, Science, ©2001 American Association for the Advancement of Science.

QUESTION + ANSWER CHOICES

Which amino acid was incorporated into Compound 1 as a future site of covalent attachment to HA prior to mineralization?

- A. Ser
- B. Ala
- C. Tyr
- D. Thr

SOLUTION

Solution: The correct answer is A.

1. Ser is the residue that was phosphorylated at its primary alcohol side chain for future attachment of nanocrystals of HA
2. Ala has a methyl side chain, not a primary alcohol.
3. The side chain of Tyr is a phenol, not a primary alcohol.
4. The side chain of Thr is a secondary alcohol, not a primary alcohol.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Lipids](#), [Amino Acids](#), [Peptides](#), [Proteins](#), [The](#)

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
X
C
D
Time Spent:
2 mins 5 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC5D
FC5
BCH

Question 25 of 230

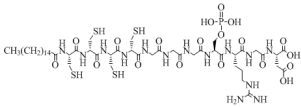
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PASSAGE

Passage 5 (Questions 22-25)

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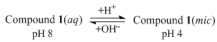


Compound 1

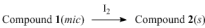
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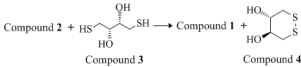
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Equation 1



Equation 2



Equation 3

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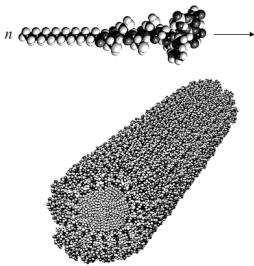


Figure 2 Space-filling molecular model illustrating cylindrical micelle formation by the peptide amphiphile, Compound 1

Adapted from J. D. Hartgerink, E. Beniash, and S. I. Stupp, Science, ©2001 American Association for the Advancement of Science.

QUESTION + ANSWER CHOICES

Compound 1 was designed to exhibit pH-dependent self-assembly. What feature(s) of the molecule is(are) responsible for the pH dependence of aggregation (Equation 1)?

- A. Thiol side chains that can hydrogen bond
- B. Long alkyl tail that exhibits predominantly London forces
- C. Side chains whose net charge responds to pH
- D. Covalent linkages that reversibly hydrolyze

SOLUTION

Solution: The correct answer is C.

- Charge development is the key to pH dependence of aggregation, not hydrogen bonding.
- Long alkyl chains are not affected by pH.
- In order for pH to affect self-assembly of Compound 1, there must be side chains whose net charge respond to pH. At pH 4, there is a smaller net negative charge, self-assembly; at pH 8, the net negative charge is much higher, causing repulsion of Compound 1 molecules and blocking self-assembly.
- There are no pH-dependent, reversibly hydrolyzable covalent bonds formed when Compound 1 self-assembles.

Khan Academy Lessons

For more help on the skills, concepts, or content associated with this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Imino Acids](#), [Amino Acids](#), [Dinitrogen](#), [Dinitrogen](#), [The](#)

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
X
C
D
Time Spent:
57 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS3
CCSD
FCS
BCH

Question 26 of 230

Data ID: 530005
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QUESTION • ANSWER CHOICES

Why is the velocity of blood flow slower in capillaries than in arteries?

- A. Capillary walls are more elastic than arterial walls.
- B. Capillaries have less resistance to blood flow than arteries.
- C. **The total cross-sectional area of capillaries exceeds that of arteries.**
- D. Blood pressure is higher in the capillaries than in arteries.

SOLUTION

Solution: The correct answer is C.

- The velocity of blood flow is primarily determined by the total cross-sectional area, not by elasticity of that level. The velocity of blood flow is inversely proportional to the total cross-sectional area.
- The velocity of blood flow is primarily determined by the total cross-sectional area, not by resistance to blood flow of that level.
- The high number of capillaries in the body means that the total cross-sectional area of these vessels is larger than any other vessel type in the circulatory system. This causes the velocity of the blood to decrease.**
- There is an increased ratio of surface area to blood volume in capillaries compared to arteries. This means that the blood is in contact with more blood vessel surface area. This slows down the velocity of the blood within capillaries and thus drops the blood pressure.

Khan Academy Lessons

RESULT / TIMING / TAGS

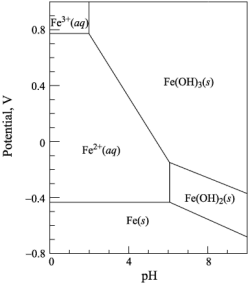
Incorrect
Correct Answer Your Answer
A
B
X
C
D
Time Spent:
2 mins 6 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC4B
FC4
BIO

Question 27 of 230

Data ID: 530010
Captured: 6/3/2026, 4:40:06 PM

QUESTION • ANSWER CHOICES

The graph below shows the relationship between the predominant form of iron as a function of solution pH and applied potential. Based on the graph, which of the following statements is true?



- A. At a potential of -0.4 V , as pH increases, Fe^{2+} is reduced and precipitates as $\text{Fe}(\text{OH})_2$.
- B. **At a potential of -0.44 V , the equilibrium between Fe and Fe^{2+} is independent of solution pH below pH 6.**
- C. At pH = 1, as the potential is changed from -0.2 to $+0.8$, Fe^{3+} is reduced to Fe^{2+} .
- D. At pH = 8 and $V = -0.1\text{ V}$, $\text{Fe}(\text{OH})_2$ is the predominant form of iron.

SOLUTION

Solution: The correct answer is B.

- The conversion of Fe^{2+} to $\text{Fe}(\text{OH})_2(\text{s})$ represents an oxidation. From the graph, at a potential of -0.4 V , no oxidation or reduction occurs. $\text{Fe}(\text{OH})_2(\text{s})$ precipitates.
- The Pourbaix diagram for iron is given, but only one statement correctly conveys information presented by the diagram. At an applied potential of -0.44 V and pH below 6, there is an equilibrium between $\text{Fe}(\text{s})$ and $\text{Fe}^{2+}(\text{aq})$, as shown by the line of demarcation. The fact that this line moves horizontally with increasing pH up until pH = 6 means that this equilibrium is unaffected by changing the pH.**
- The opposite is true. At this pH as the potential is raised, Fe^{2+} is oxidized to Fe^{3+} .
- At pH = 8 and $V = -0.1\text{ V}$, $\text{Fe}(\text{OH})_2(\text{s})$ is the predominant form of iron.

Khan Academy Lessons

For more help on the skills, concepts, or content covered by this question, see the Khan Academy lessons on [Atomic Structure](#), [Electrolytic Reaction](#), [The Periodic Table](#), [Classification of](#)

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
2 mins 24 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS4
CC4E
FC4
GCH

Question 28 of 230

Data ID: 530013
Captured: 6/3/2026, 4:40:06 PM

QUESTION • ANSWER CHOICES

Which of the following types of orbitals of the central atom are involved in bonding in octahedral compounds?

- A. sp
- B. sp^3
- C. p
- D. $d^2 sp^3$

SOLUTION

Solution: The correct answer is D.

1. sp hybrid orbitals can be used to make linear AX_2E_n compounds.
2. Tetrahedral AX_4 compounds and other compounds based on a tetrahedral arrangement of electron pairs utilize sp^3 hybrid orbitals.
3. A pure p type orbital will typically be involved in π bonding, or will represent a lone pair of electrons.
4. **Octahedral compounds have six σ bonds and no stereochemically active lone pairs. According to valence bond theory, the central atom requires the hybridization of six atomic orbitals, $d^2 sp^3$.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
☒ D
Time Spent:
2 mins 18 secs
Difficulty Level:
Easy
Content & Skills:
SIRS1
CC5B
FCS
GCH

Question 29 of 230

Data ID: 530016
Captured: 6/3/2026, 4:40:07 PM

QUESTION • ANSWER CHOICES

The term "ideal gas" refers to a gas for which certain assumptions have been made. Which of the following is such an assumption?

- A. The law $PV = nRT^2$ is strictly obeyed.
- B. Intermolecular molecular forces are infinitely large.
- C. Individual molecular volume and intermolecular forces are negligible.
- D. One gram-mole occupies a volume of 22.4 L at 25°C and one atmosphere pressure.

SOLUTION

Solution: The correct answer is C.

1. The Ideal Gas Law is $PV = nRT$.
2. Large intermolecular forces are associated with condensed phases (solids and liquids), not gases.
3. **A property of an "ideal" gas is that it is composed of particles that have negligible volume and do not exert intermolecular forces.**
4. The gram-mole is an old way to refer to the molar mass of a substance. The molar volume of an ideal gas at 25°C is 24.4 L, not 22.4 L.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
☒ C
☒ D
Time Spent:
43 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC4B
FC4
GCH

Question 30 of 230

Data ID: 530090
Captured: 6/3/2026, 4:40:08 PM

PASSAGE

Passage 6 (Questions 30-33)

Human rhodopsin is a G protein-coupled receptor (GPCR) found in rod cells of the retina. Rhodopsin is composed of opsin, a family of seven helix transmembrane proteins, and a covalently attached coenzyme, retinal. Retinal is derived from all-trans-retinol, also known as vitamin A (Figure 1). When rhodopsin is exposed to light, the 11-cis-retinal (Figure 1) converts to all-trans-retinal. Subsequent structural changes in rhodopsin allow it to interact with and activate the intracellular G protein transducin. The signal is turned off via phosphorylation of rhodopsin by rhodopsin kinase and subsequent binding of the protein arrestin to rhodopsin. Rhodopsin has a maximum absorption at 500 nm. Additional opsin based receptors are expressed in specialized cone cells and have maximum absorptions at 420 nm, 530 nm, and 560 nm when reconstituted with 11-cis-retinal.

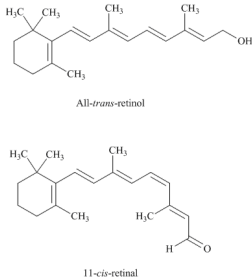


Figure 1 Structural representations of all-trans-retinol and 11-cis-retinal

Previous studies indicated that various intracellular loops of rhodopsin are important for interacting with arrestin. In order to determine which amino acids within the loops have the greatest impact on binding to arrestin, variants of rhodopsin containing one or more amino acid changes were prepared, and their binding to arrestin measured. Researchers focused on two loops, Loop I (Figure 2) and Loop II (Figure 3). Wild-type or variant rhodopsin was reconstituted with 11-cis-retinal and phosphorylated by rhodopsin kinase in the presence of ATP before examining arrestin binding. In the absence of phosphorylation, arrestin binding to wild-type rhodopsin decreased by 90%.

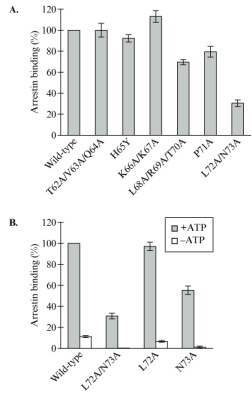


Figure 2 Arrestin binding to Loop I variants of rhodopsin: (A) Reconstituted with 11-cis-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-cis-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (-ATP)

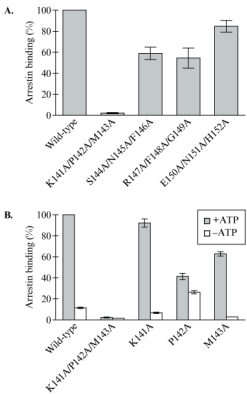


Figure 3 Arrestin binding to Loop II variants of rhodopsin: (A) Reconstituted with 11-cis-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-cis-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (-ATP)

Adapted from Ramani, D., Osawa, S.I. and Weiss, E. ©1999 by American Chemical Society

QUESTION + ANSWER CHOICES

The production of a variety of opsins functions to:

- A. increase sensitivity to low light.
- B. enable the detection of different colors.
- C. ensure fast recovery of 11-cis-retinal after exposure.
- D. increase refractive index of the eye lens.

SOLUTION

Solution: The correct answer is B.

- There is no information in the passage indicating that some opsins are more sensitive to light than others.
- Because the wavelengths of maximum light absorption are significantly different for the specialized opsins in the cone cells, these opsins enable the detection of different colors of light.
- There is no information in the passage indicating that some opsins have faster recovery of 11-cis-retinal after light exposure than others.
- There is no information in the passage indicating that opsins have any effect on the refractive index of the eye lens.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
x
B
C
D
Time Spent:
1 min 45 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC4D
FC4

Question 31 of 230

Data ID: 530062
Captured: 6/3/2026, 4:40:08 PM

PASSAGE

Passage 6 (Questions 30-33)

Human rhodopsin is a G protein-coupled receptor (GPCR) found in rod cells of the retina. Rhodopsin is composed of opsin, a family of seven helix transmembrane proteins, and a covalently attached coenzyme, retinal. Retinal is derived from all-trans-retinol, also known as vitamin A (Figure 1). When rhodopsin is exposed to light, the 11-cis-retinal (Figure 1) converts to all-trans-retinal. Subsequent structural changes in rhodopsin allow it to interact with and activate the intracellular G protein transducin. The signal is turned off via phosphorylation of rhodopsin by rhodopsin kinase and subsequent binding of the protein arrestin to rhodopsin. Rhodopsin has a maximum absorption at 500 nm. Additional opsin based receptors are expressed in specialized cone cells and have maximum absorptions at 420 nm, 530 nm, and 560 nm when reconstituted with 11-cis-retinal.

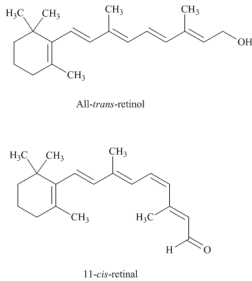


Figure 1 Structural representations of all-trans-retinol and 11-cis-retinal

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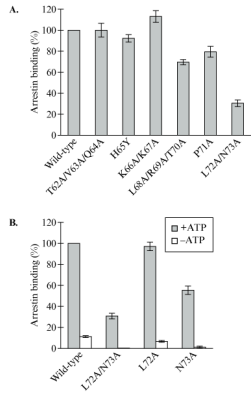


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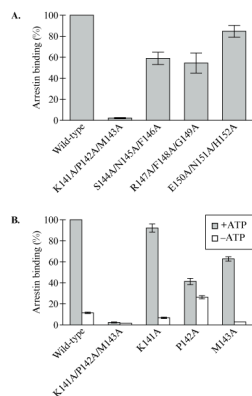


Figure 3 Arrestin binding to Loop II variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (-ATP)

Adapted from Raman, D., Osawa, S., and Weiss, E. ©1999 by American Chemical Society

QUESTION + ANSWER CHOICES

The environment of the retinal binding site is most likely:

- A. hydrophilic.
- B. positively charged.
- C. negatively charged.
- D. hydrophobic.

SOLUTION

Solution: The correct answer is **D**.

1. Retinal consists almost entirely of C-H bonds and is non-polar. Hydrophilic molecules have polar bonds and are especially capable of hydrogen-bonding.
2. A positively charged binding domain is well-suited for positioning substrates that have areas of negative charge. Retinal is non-polar.
3. A negatively charged binding domain is well-suited for positioning substrates that feature positive charge. Retinal is non-polar.
4. **Retinal is composed of mainly carbon and hydrogen, making it largely hydrophobic.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer

- A
B
C
D
✓

Time Spent:

2 mins 1 sec

Difficulty Level:

Difficult

Content & Skills:

SIRS2

CC5B

FC5

GCH

Question 32 of 230

Data ID: 530054

Captured: 6/3/2026, 4:40:09 PM

PASSAGE

Passage 6 (Questions 30-33)

Human rhodopsin is a G protein-coupled receptor (GPCR) found in rod cells of the retina. Rhodopsin is composed of opsin, a family of seven helix transmembrane proteins, and a covalently attached coenzyme, retinal. Retinal is derived from all-*trans*-retinol, also known as vitamin A (Figure 1). When rhodopsin is exposed to light, the 11-*cis*-retinal (Figure 1) converts to all-*trans*-retinal. Subsequent structural changes in rhodopsin allow it to interact with and activate the intracellular G protein transducin. The signal is turned off via phosphorylation of rhodopsin by rhodopsin kinase and subsequent binding of the protein arrestin to rhodopsin. Rhodopsin has a maximum absorption at 500 nm. Additional opsin based receptors are expressed in specialized cone cells and have maximum absorptions at 420 nm, 530 nm, and 560 nm when reconstituted with 11-*cis*-retinal.

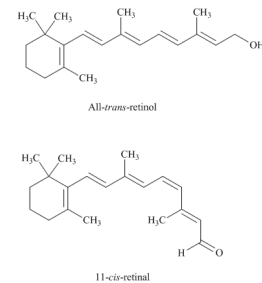


Figure 1 Structural representations of all-*trans*-retinol and 11-*cis*-retinal

Previous studies indicated that various intracellular loops of rhodopsin are important for interacting with arrestin. In order to determine which amino acids within the loops have the greatest impact on binding to arrestin, variants of rhodopsin containing one or more amino acid changes were prepared, and their binding to arrestin measured. Researchers focused on two loops, Loop I (Figure 2) and Loop II (Figure 3). Wild-type or variant rhodopsin was reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase in the presence of ATP before examining arrestin binding. In the absence of phosphorylation, arrestin binding to wild-type rhodopsin decreased by 90%.

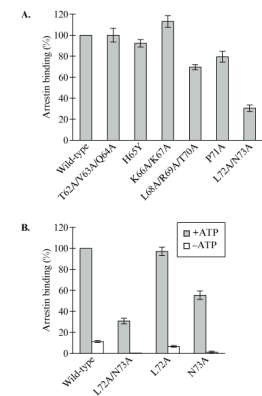


Figure 2 Arrestin binding to Loop I variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (-ATP)

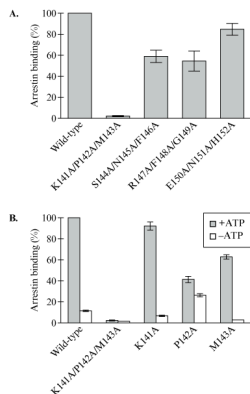


Figure 3 Arrestin binding to Loop II variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (-ATP)

Adapted from Haman, D., Osawa, S., and Weiss, E. ©1999 by American Chemical Society

QUESTION + ANSWER CHOICES

The source of the phosphate groups that are added to rhodopsin is:

- A. arrestin.
- B. rhodopsin kinase.
- C. ATP.
- D. all-*trans*-retinol.

SOLUTION

Solution: The correct answer is **C**.

1. Arrestin binds to rhodopsin after rhodopsin has been phosphorylated. Arrestin is not the source of the phosphate groups.
2. Rhodopsin kinase catalyzes the reaction between ATP and rhodopsin that causes rhodopsin to be phosphorylated, but ATP is the source of the phosphate.
3. **The substrate for rhodopsin kinase that the source of phosphate is ATP.**
4. Retinol contains no phosphate groups.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer

- A
- B
- C ✓
- D

Time Spent:
52 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC5E
FC5
BCH

Question 33 of 230

Data ID: 530065
Captured: 6/3/2026, 4:40:09 PM

PASSAGE

Passage 6 (Questions 30-33)

Human rhodopsin is a G protein-coupled receptor (GPCR) found in rod cells of the retina. Rhodopsin is composed of opsin, a family of seven helix transmembrane proteins, and a covalently attached coenzyme, retinal. Retinal is derived from all-*trans*-retinol, also known as vitamin A (Figure 1). When rhodopsin is exposed to light, the 11-*cis*-retinal (Figure 1) converts to all-*trans*-retinal. Subsequent structural changes in rhodopsin allow it to interact with and activate the intracellular G protein transducin. The signal is turned off via phosphorylation of rhodopsin by rhodopsin kinase and subsequent binding of the protein arrestin to rhodopsin. Rhodopsin has a maximum absorption at 500 nm. Additional opsin based receptors are expressed in specialized cone cells and have maximum absorptions at 420 nm, 530 nm, and 560 nm when reconstituted with 11-*cis*-retinal.

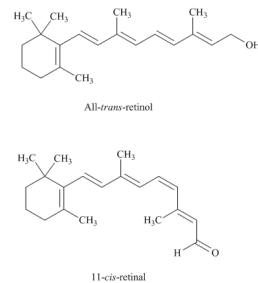


Figure 1 Structural representations of all-*trans*-retinol and 11-*cis*-retinal

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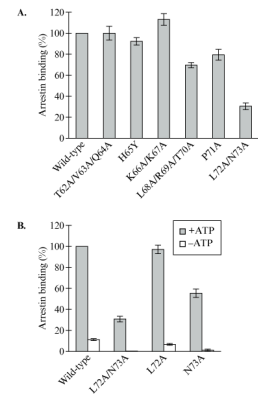


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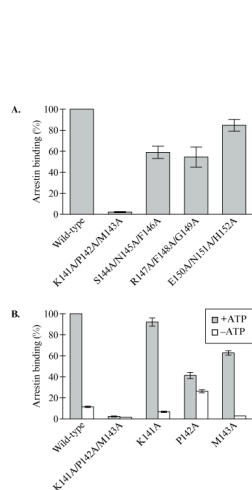


Figure 3 Arrestin binding to Loop II variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (-ATP)

Adapted from Raman, D., Osawa, S., and Weiss, E. ©1999 by American Chemical Society

QUESTION + ANSWER CHOICES

What are the hybridization states of the carbon atoms involved in the conversion of *trans* to *cis* retinal?

- A. sp
- B. sp^2
- C. sp^3
- D. sp^3d

SOLUTION

Solution: The correct answer is B.

1. A carbon atom that is sp hybridized is part of an AX_2 system. AX_2 systems are linear according to VSEPR.
2. The reacting carbon atoms are both central to AX_3 systems with 3 bonded atoms and no lone pairs. The preferred geometry for such a system is trigonal planar and the hybridization scheme that facilitates this geometry is sp^2 .
3. Tetrahedral AX_4 compounds and other compounds based on a tetrahedral arrangement of electron pairs utilize sp^3 hybrid orbitals. What is depicted is a geometric rearrangement at AX_3 systems, which are trigonal planar.
4. Transition metals are capable of using d orbitals in bonding. An sp^3d system has 5 electron pair domains and is based on a trigonal bipyramidal or other 5-coordinate geometry.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
1 min 29 secs
Difficulty Level:
Easy
Content & Skills:
SIRS1
CC5B
FC5

GCH

Question 34 of 230

Data ID: 630071
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PASSAGE

Passage 7 (Questions 34-37)

The crystal structure of the protein lysosomal lipase (LipA) was determined. Crystal structure B factors, which are an indication of structural rigidity, were used to identify regions of high flexibility. Sites within these regions were varied in an attempt to create versions of LipA with higher thermal stability. All variants were then screened for retention of enzymatic activity. The three most thermally stable variants that retained enzymatic activity are shown in Table 1.

Table 1 Variants of LipA

Variant	Amino acid substitutions
IX	K112D, M134D, Y139C, I157M
X	R33Q, D34N, K35D, K112D, M134D, Y139C, I157M
XI	R33G, K112D, M134D, Y139C, I157M

All proteins were expressed in *Escherichia coli* and purified chromatographically. After protein purity was verified via SDS-PAGE, wild-type and variant LipA proteins were characterized. The enzymatic activity of the proteins was measured at 37°C and then again after the protein was briefly exposed to higher temperatures and cooled down to 37°C (Figure 1). The thermal unfolding of the proteins was followed by loss of secondary structure (Table 2). Structural integrity was verified using NMR spectroscopy.

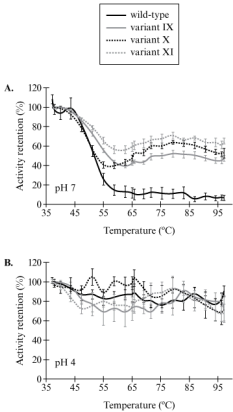


Figure 1 Enzymatic activity retention as a function of exposure temperature for wild-type and variant LipA proteins (Note: The solution contained either a pH 7 buffer (A) or a pH 4 buffer (B). The x-axis represents the temperature that the protein was heated to prior to the activity assay.)

Table 2 Parameters for Thermal Unfolding as Monitored by Loss of Secondary Structure

Parameter		Reference	2 M Urea	4 M Urea
T_f (°C)	Wild-type	51.8	39.6	26.3
	Variant IX	45.6	32.0	24.9
	Variant X	48.5	33.1	24.0
	Variant XI	49.3	34.3	25.4
Slope, deg cm ² /(mmol K)	Wild-type	8.2	3.9	3.5
	Variant IX	5.7	3.3	2.6
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Adapted from W. Augustyniak, A. Brzezinska, R. Pilning, H. Wienk, R. Boelens, et al., *Protein Science*. ©2012 The Protein Society.

QUESTION + ANSWER CHOICES

Which reaction is catalyzed by LipA?

- A. ATP hydrolysis
- B. Peptide bond cleavage
- C. Hydrolysis of triacylglycerides
- D. Transfer of carboxyl groups

SOLUTION

Solution: The correct answer is C.

- LipA is a lipase, which does not hydrolyze ATP.
- LipA is a lipase, not a protease.
- LipA is a lipase, which hydrolyzes triacylglycerides.
- LipA is a lipase, not a transferase.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
44 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CCSE
FCS
BCH

Question 35 of 230

Data ID: 530072
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PASSAGE

Passage 7 (Questions 34-37)

The crystal structure of the protein lysosomal lipase (LipA) was determined. Crystal structure B factors, which are an indication of structural rigidity, were used to identify regions of high flexibility. Sites within these regions were varied in an attempt to create versions of LipA with higher thermal stability. All variants were then screened for retention of enzymatic activity. The three most thermally stable variants that retained enzymatic activity are shown in Table 1.

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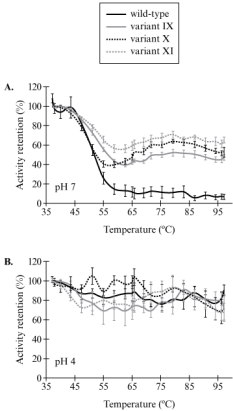


Figure 1 Enzymatic activity retention as a function of exposure temperature for wild-type and variant LipA proteins (Note: The solution contained either a pH 7 buffer (A) or a pH 4 buffer (B). The x-axis represents the temperature that the protein was heated to prior to the activity assay.)

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Note: T_f is the temperature at the start of unfolding and the slope is measured for the unfolding transition. The experiments were performed in a pH 7 buffer (reference) and in the presence of 2 M and 4 M urea, which is a chemical denaturant.

Adapted from W. Augustyniak, A. Brzezinska, R. Pilning, H. Wienk, R. Boelens, et al., *Protein Science*. ©2012 The Protein Society.

QUESTION + ANSWER CHOICES

Compared to the wild-type LipA, what is the change in net charge in variant XI at pH 7?

- A. +4
- B. +3
- C. -3
- D. -4

SOLUTION

Solution: The correct answer is D.

1. This change in net charge has the wrong sign but the correct magnitude.
2. Variant XI loses positive charge and gains negative charge.
3. This is an insufficient change in net negative charge.
4. Variant XI has a change of -1 charge with R33G ($+1$ to 0), -2 charge with K112D ($+1$ to -1), and -1 charge with M134D (0 to -1) for a net change of -4 .

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Lipids](#), [Amino Acids](#), [Peptides](#), [Proteins](#); [The](#)

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
x
D
Time Spent:
1 min 31 secs
Difficulty Level:
Expert
Content & Skills:
SIRS2
CC5D
FC5
BCH

Data ID: 830074
Captured: 6/3/2026, 4:40:11 PM

Passage 7 (Questions 34-37)

Table 1	Variants of LipA
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Variant	Amino acid substitutions
IX	K112D, M134D, Y139C, I157M
X	R33Q, D34N, K35D, K112D, M134D, Y139C, I157M
XI	R33G, K112D, M134D, Y139C, I157M

A.

Activity retention (%)

pH 7

Temperature (°C)

— wild-type
- - variant IX
... variant X
- . variant XI

B.

Activity retention (%)

pH 4

Temperature (°C)

— wild-type
- - variant IX
... variant X
- . variant XI

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Adapted from W. Augustyniak, A. Brzezinska, R. Pijning, H. Wienk, R. Boelens, et al., *Protein Science*. ©2012 The Protein Society.

The data in Figure 1 show that:

The data in Figure 1 show that:

- SOLUTION**

1. At pH 4, the enzymatic activity of LipA proteins does not change even after heating and cooling.
2. In Figure 1, the highest temperatures used still show enzymatic activity for variants.
3. **Figure 1 shows that temperature has less effect on the enzymatic activity of the variants than wild-type LipA.**
4. At pH 7, wild-type LipA loses some but not all activity when first heated to 50°C.

Enzyme has an effect on the value of constant as seen by this question see the Khan Academy lecture on [Michaelis-Menten and Michaelis-Menton Equations](#). I include Amino Acids, Dantrolene, Dantrolene, Thapsigargin, and Thapsigargin.

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
1 min 27 secs
Difficulty Level:
Easy
Content & Skills:
SIRS4
CC5D
FC5
BCH

Question 37 of 230

Data ID: 630076
Captured: 6/3/2026, 4:40:12 PM

PASSAGE

Passage 7 (Questions 34-37)

The crystal structure of the protein lysosomal lipase (LipA) was determined. Crystal structure B factors, which are an indication of structural rigidity, were used to identify regions of high flexibility. Sites within these regions were varied in an attempt to create versions of LipA with higher thermal stability. All variants were then screened for retention of enzymatic activity. The three most thermally stable variants that retained enzymatic activity are shown in Table 1.

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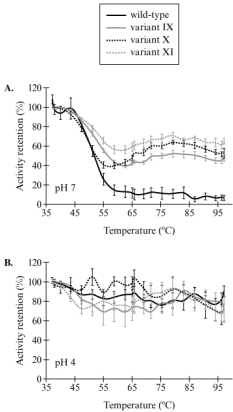


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Adapted from W. Augustyniak, A. Brzezinska, R. Pijning, H. Wienk, R. Boelens, et al., *Protein Science*. ©2012 The Protein Society.

QUESTION + ANSWER CHOICES

By what factor is the proton concentration increased in the experiments shown in Figure 1B from those shown in Figure 1A?

- A. 2
- B. 10
- C. 200
- D. 1000

SOLUTION

Solution: The correct answer is D.

- Doubling the concentration of H_3O^+ involves decreasing the pH by only 0.3 units.
- If the solution pH was lower by 1 pH unit, this would correspond to an increase in $[H_3O^+]$ by a factor of 10.
- An increase of $[H_3O^+]$ by a factor of 200 corresponds to a decrease in 2.3 pH units.
- The pH scale is logarithmic ($pH = -\log([H^+])$). A difference of 3 pH units corresponds to a $10^3 = 1000$ -fold difference in proton concentrations.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
2 mins 20 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC5A
FCS
GCH

Question 38 of 230

Data ID: 530084
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PASSAGE

Passage 8 (Questions 38-43)

Glycogen synthase catalyzes the lengthening of glycogen polymers through the addition of glucose. The reaction consists of transferring a glucosyl moiety from uridine diphosphate glucose (UDP-glucose, Figure 1) to the nonreducing end of the glycogen molecule along with the release of UDP. UDP-glucose is formed from the enzymatically-catalyzed reaction between glucose 1-phosphate and UTP. It was discovered that approximately one out of every 600–1500 glucose molecules in muscle glycogen contains phosphate groups covalently linked to the C2 and C3 positions as monoesters. It is hypothesized that glycogen synthase is responsible for the phosphorylation.

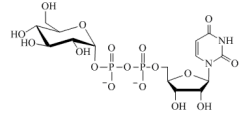


Figure 1 The structure of UDP-glucose

Lafora disease is a fatal progressive epilepsy that is characterized by the formation of deposits of sparsely branched, insoluble glycogen-like polymers called Lafora bodies (LBs). LBs contain a much larger amount of covalently-linked phosphate compared to normal glycogen. The abundance of phosphate has been linked to a mutation in the gene for laforin, a phosphatase specific to carbohydrates.

In order to test if glycogen synthase is responsible for glycogen phosphorylation, both UDP-[U-¹⁴C]glucose and [β-³²P]UDP-glucose were incubated with muscle extracts from wild-type (WT) mice and mice lacking muscle glycogen synthase (MGSKO). The reaction products were separated by SDS-PAGE and visualized by autoradiography that detects the products of β⁺ decay (Figure 2). The data were normalized to account for the different decay rates. The samples were also treated with glucosidase, an enzyme that breaks down glycogen, and assayed in the same manner.

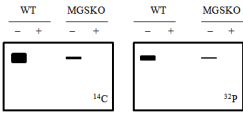


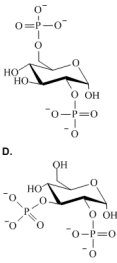
Figure 2 SDS-PAGE analysis from muscle extracts visualized with either ¹⁴C or ³²P radioactive decay (Note: The + and – lanes represent the presence or absence of glucosidase treatment, respectively.)

Adapted from Tagliafrazzi, V. et al. Cell Metabolism, © 2011 Elsevier Inc.

QUESTION + ANSWER CHOICES

Which structure represents a phosphorylated subunit of glycogen as described in the passage?

- A.
- B.
- C.



SOLUTION

Solution: The correct answer is D.

- This glucose molecule is phosphorylated at C1 instead of at C2 and C3.
- This glucose molecule is phosphorylated at C1 and C6 instead of at C2 and C3.
- This glucose molecule is phosphorylated at C2 and C6. The phosphorylation at C6 should be at C3.
- According to the passage, glycogen has phosphate monoesters at C2 and C3 in one out of every 600–1500 glucose molecules. This glucose molecule is phosphorylated at C2 and C3.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Lipids](#), [Amino Acids](#), [Polymers](#), [Proteins](#), [The](#)

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
Time Spent:
1 min 53 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS1
CC5D
FC5
BCH

Question 39 of 230

Data ID: 530086
Captured: 6/3/2026, 4:40:13 PM

PASSAGE

Passage 8 (Questions 38-43)

Glycogen synthase catalyzes the lengthening of glycogen polymers through the addition of glucose. The reaction consists of transferring a glucosyl moiety from uridine diphosphate glucose (UDP-glucose, Figure 1) to the nonreducing end of the glycogen molecule along with the release of UDP. UDP-glucose is formed from the enzymatically-catalyzed reaction between glucose 1-phosphate and UTP. It was discovered that approximately one out of every 600–1500 glucose molecules in muscle glycogen contains phosphate groups covalently linked to the C2 and C3 positions as monoesters. It is hypothesized that glycogen synthase is responsible for the phosphorylation.

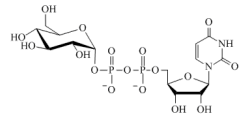


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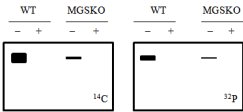


Figure 2 SDS-PAGE analysis from muscle extracts visualized with either ¹⁴C or ³²P radioactive decay (Note: The + and – lanes represent the presence or absence of glucosidase treatment, respectively.)

Adapted from Tagliafracci, V. et al. Cell Metabolism, © 2011 Elsevier Inc.

QUESTION + ANSWER CHOICES

Which type of bond is formed by glycogen synthase upon release of UDP?

- A. α-1,4-Glycosidic bond
- B. α-1,6-Glycosidic bond
- C. β-1,4-Glycosidic bond
- D. β-1,6-Glycosidic bond

SOLUTION

Solution: The correct answer is A.

- In glycogen, each glucose subunit is connected to an adjacent glucose subunit by an α-1,4-glycosidic bond.
- Glycogen contains glucose subunits connected by α-1,4-glycosidic bonds, not α-1,6-glycosidic bonds.
- Glycogen contains glucose subunits connected by 1,4-glycosidic bonds that are α, not β.
- Glycogen contains glucose subunits connected by α-1,4-glycosidic bonds, not β-1,6-glycosidic bonds.

Khan Academy Lessons

For more help on this skill, or content associated to this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Lipids](#), [Amino Acids](#), [Polysaccharides](#), [Proteins](#), [The](#)

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
X
D
Time Spent:
1 min 6 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS1
CC5D
FC5
BCH

Question 40 of 230

Data ID: 530088
Captured: 6/3/2026, 4:40:14 PM

PASSAGE

Passage 8 (Questions 38-43)

Glycogen synthase catalyzes the lengthening of glycogen polymers through the addition of glucose. The reaction consists of transferring a glucosyl moiety from uridine diphosphate glucose (UDP-glucose, Figure 1) to the nonreducing end of the glycogen molecule along with the release of UDP. UDP-glucose is formed from the enzymatically-catalyzed reaction between glucose 1-phosphate and UTP. It was discovered that approximately one out of every 600–1500 glucose molecules in muscle glycogen contains phosphate groups covalently linked to the C2 and C3 positions as monoesters. It is hypothesized that glycogen synthase is responsible for the phosphorylation.

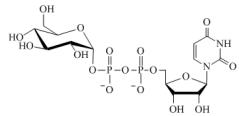


Figure 1 The structure of UDP-glucose

Lafora disease is a fatal progressive epilepsy that is characterized by the formation of deposits of sparsely branched, insoluble glycogen-like polymers called Lafora bodies (LBs). LBs contain a much larger amount of covalently-linked phosphate compared to normal glycogen. The abundance of phosphate has been linked to a mutation in the gene for laforin, a phosphatase specific to carbohydrates.

In order to test if glycogen synthase is responsible for glycogen phosphorylation, both UDP-[U-¹⁴C]glucose and [β-³²P]UDP-glucose were incubated with muscle extracts from wild-type (WT) mice and mice lacking muscle glycogen synthase (MGSKO). The reaction products were separated by SDS-PAGE and visualized by autoradiography that detects the products of β⁺ decay (Figure 2). The data were normalized to account for the different decay rates. The samples were also treated with glucosidase, an enzyme that breaks down glycogen, and assayed in the same manner.

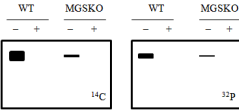


Figure 2 SDS-PAGE analysis from muscle extracts visualized with either ¹⁴C or ³²P radioactive decay (Note: The + and – lanes represent the presence or absence of glucosidase treatment, respectively.)

Adapted from Tagliafracci, V. et al. Cell Metabolism, © 2011 Elsevier Inc.

QUESTION + ANSWER CHOICES

In addition to glucose, what other monosaccharide is part of the UDP-glucose structure?

- A. Xylose
- B. Fructose
- C. Ribose
- D. Arabinose

SOLUTION

Solution: The correct answer is C.

- Xylose is pentofuranose in which the C2 and C3 hydroxyl groups are *trans*, and the C3 hydroxyl group and the C4 hydroxymethyl group are *cis*. Although the other monosaccharide in UDP glucose is a pentofuranose, the C2 and C3 hydroxyl groups are *cis*, and the C3 hydroxyl group and the C4 hydroxymethyl group are *trans*.
- Fructose is a hexose, not a pentose.
- The other monosaccharide in UDP-glucose is ribose, a pentofuranose in which the C2 and C3 hydroxyl groups are *cis*, and the C3 hydroxyl group and the C4 hydroxymethyl group are *trans*.
- Although arabinose can be a pentofuranose, the C2 and C3 hydroxyl groups would be *trans*, not *cis*.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Lipids](#), [Amino Acids](#), [Polysaccharides](#), [Proteins](#), [The](#)

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
X
C
D
Time Spent:
20 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC5D
FC5
BCH

Question 41 of 230

Data ID: 530092
Captured: 6/3/2026, 4:40:14 PM

PASSAGE

Passage 8 (Questions 38-43)

Glycogen synthase catalyzes the lengthening of glycogen polymers through the addition of glucose. The reaction consists of transferring a glucosyl moiety from uridine diphosphate glucose (UDP-glucose, Figure 1) to the nonreducing end of the glycogen molecule along with the release of UDP. UDP-glucose is formed from the enzymatically-catalyzed reaction between glucose 1-phosphate and UTP. It was discovered that approximately one out of every 600–1500 glucose molecules in muscle glycogen contains phosphate groups covalently linked to the C2 and C3 positions as monoesters. It is hypothesized that glycogen synthase is responsible for the phosphorylation.

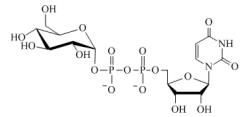


Figure 1 The structure of UDP-glucose

Lafora disease is a fatal progressive epilepsy that is characterized by the formation of deposits of sparsely branched, insoluble glycogen-like polymers called Lafora bodies (LBs). LBs contain a much larger amount of covalently-linked phosphate compared to normal glycogen. The abundance of phosphate has been linked to a mutation in the gene for laforin, a phosphatase specific to carbohydrates.

In order to test if glycogen synthase is responsible for glycogen phosphorylation, both UDP-[U-¹⁴C]glucose and [β-³²P]UDP-glucose were incubated with muscle extracts from wild-type (WT) mice and mice lacking muscle glycogen synthase (MGSKO). The reaction products were separated by SDS-PAGE and visualized by autoradiography that detects the products of β⁺ decay (Figure 2). The data were normalized to account for the different decay rates. The samples were also treated with glucosidase, an enzyme that breaks down glycogen, and assayed in the same manner.

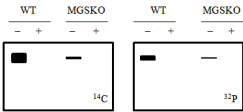


Figure 2 SDS-PAGE analysis from muscle extracts visualized with either ¹⁴C or ³²P radioactive decay (Note: The + and – lanes represent the presence or absence of glucosidase treatment, respectively.)

Adapted from Tagliafrazzi, V. et al. Cell Metabolism, © 2011 Elsevier Inc.

QUESTION + ANSWER CHOICES

Addition of laforin to the reaction mixtures used to generate the data in Figure 2 will result in:

- A. a decrease in the intensity of the band representing WT without glucosidase when visualized with ¹⁴C decay.
- B. an increase in the intensity of the band representing WT without glucosidase when visualized with ¹⁴C decay.
- C. a decrease in the intensity of the band representing WT without glucosidase when visualized with ³²P decay.
- D. an increase in the intensity of the band representing WT without glucosidase when visualized with ³²P decay.

SOLUTION

Solution: The correct answer is C.

1. A decrease in the intensity of this band will not be observed because laforin will not cleave off the ¹⁴C-glucose residues from glycogen.
2. An increase in the intensity of this band will not be observed because laforin will not cleave off the ¹⁴C-glucose residues from glycogen.
3. Laforin will cleave off phosphate groups from glycogen, thus removing radioactive phosphate from the WT band and decreasing the intensity of the band.
4. Laforin is a phosphatase, not a kinase. It will remove radioactive phosphate from glucose subunits in glycogen, not add them.

Khan Academy Lessons

Incorrect
Correct Answer Your Answer
A
B
X
C
D
Time Spent:
2 mins 55 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS3
CC5E
FC5
BCH

Question 42 of 230

Data ID: 530094
Captured: 6/3/2026, 4:40:15 PM

PASSAGE

Passage 8 (Questions 38-43)

Glycogen synthase catalyzes the lengthening of glycogen polymers through the addition of glucose. The reaction consists of transferring a glucosyl moiety from uridine diphosphate glucose (UDP-glucose, Figure 1) to the nonreducing end of the glycogen molecule along with the release of UDP. UDP-glucose is formed from the enzymatically-catalyzed reaction between glucose 1-phosphate and UTP. It was discovered that approximately one out of every 600–1500 glucose molecules in muscle glycogen contains phosphate groups covalently linked to the C2 and C3 positions as monoesters. It is hypothesized that glycogen synthase is responsible for the phosphorylation.

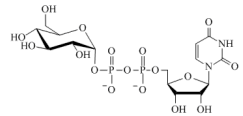


Figure 1 The structure of UDP-glucose

Lafora disease is a fatal progressive epilepsy that is characterized by the formation of deposits of sparsely branched, insoluble glycogen-like polymers called Lafora bodies (LBs). LBs contain a much larger amount of covalently-linked phosphate compared to normal glycogen. The abundance of phosphate has been linked to a mutation in the gene for laforin, a phosphatase specific to carbohydrates.

In order to test if glycogen synthase is responsible for glycogen phosphorylation, both UDP-[U-¹⁴C]glucose and [β-³²P]UDP-glucose were incubated with muscle extracts from wild-type (WT) mice and mice lacking muscle glycogen synthase (MGSKO). The reaction products were separated by SDS-PAGE and visualized by autoradiography that detects the products of β[−] decay (Figure 2). The data were normalized to account for the different decay rates. The samples were also treated with glucosidase, an enzyme that breaks down glycogen, and assayed in the same manner.

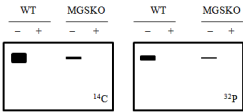


Figure 2 SDS-PAGE analysis from muscle extracts visualized with either ¹⁴C or ³²P radioactive decay (Note: The + and − lanes represent the presence or absence of glucosidase treatment, respectively.)

Adapted from Taghizadeh, V. et al. Cell Metabolism, © 2011 Elsevier Inc.

QUESTION + ANSWER CHOICES

The radioactive decay described in the passage results in the formation of which two atoms?

- A. N and S
- B. B and Si
- C. B and S
- D. N and Si

SOLUTION

Solution: The correct answer is A.

- The β[−] radioactive decay of ¹⁴C occurs according to the reaction ¹⁴C → ¹⁴N + e[−] + antineutrino, and that of ³²P according to ³²P → ³²S + e[−] + antineutrino, so the resulting atoms are N and S.
- The resulting atoms are neither B nor Si because the β[−] radioactive decay of ¹⁴C occurs according to the reaction ¹⁴C → ¹⁴N + e[−] + antineutrino, and that of ³²P according to ³²P → ³²S + e[−] + antineutrino.
- B is not formed because the β[−] radioactive decay of ¹⁴C occurs according to the reaction ¹⁴C → ¹⁴N + e[−] + antineutrino, and that of ³²P according to ³²P → ³²S + e[−] + antineutrino.
- Si is not formed because the β[−] radioactive decay of ¹⁴C occurs according to the reaction ¹⁴C → ¹⁴N + e[−] + antineutrino, and that of ³²P according to ³²P → ³²S + e[−] + antineutrino.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
X
Time Spent:
2 mins 30 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC4E
FC4
PHY

Question 43 of 230

Data ID: 530096
Captured: 6/3/2026, 4:40:16 PM

PASSAGE

Passage 8 (Questions 38-43)

Glycogen synthase catalyzes the lengthening of glycogen polymers through the addition of glucose. The reaction consists of transferring a glucosyl moiety from uridine diphosphate glucose (UDP-glucose, Figure 1) to the nonreducing end of the glycogen molecule along with the release of UDP. UDP-glucose is formed from the enzymatically-catalyzed reaction between glucose 1-phosphate and UTP. It was discovered that approximately one out of every 600–1500 glucose molecules in muscle glycogen contains phosphate groups covalently linked to the C2 and C3 positions as monoesters. It is hypothesized that glycogen synthase is responsible for the phosphorylation.

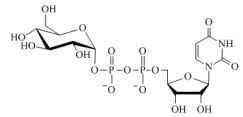


Figure 1 The structure of UDP-glucose

Lafora disease is a fatal progressive epilepsy that is characterized by the formation of deposits of sparsely branched, insoluble glycogen-like polymers called Lafora bodies (LBs). LBs contain a much larger amount of covalently-linked phosphate compared to normal glycogen. The abundance of phosphate has been linked to a mutation in the gene for laforin, a phosphatase specific to carbohydrates.

In order to test if glycogen synthase is responsible for glycogen phosphorylation, both UDP-[U-¹⁴C]glucose and [β-³²P]UDP-glucose were incubated with muscle extracts from wild-type (WT) mice and mice lacking muscle glycogen synthase (MGSKO). The reaction products were separated by SDS-PAGE and visualized by autoradiography that detects the products of β⁺ decay (Figure 2). The data were normalized to account for the different decay rates. The samples were also treated with glucosidase, an enzyme that breaks down glycogen, and assayed in the same manner.

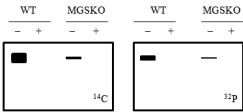


Figure 2 SDS-PAGE analysis from muscle extracts visualized with either ¹⁴C or ³²P radioactive decay (Note: The + and – lanes represent the presence or absence of glucosidase treatment, respectively.)

Adapted from Tagliabracci, V. et al. Cell Metabolism, © 2011 Elsevier Inc.

QUESTION + ANSWER CHOICES

Compared to normal glycogen, the amount of what type of bond is decreased in Lafora bodies?

- A. Phosphomonoester
- B. Phosphodiester
- C. α-1,4-Glycosidic bond
- D. α-1,6-Glycosidic bond

SOLUTION

Solution: The correct answer is D.

- According to the passage, Lafora bodies should have more phosphomonoester bonds than normal glycogen.
- There is no evidence from the passage that either Lafora bodies or normal glycogen have phosphodiester bonds.
- The passage did not indicate that the polymerization of glucose subunits was less in Lafora bodies than normal glycogen.
- The passage stated that Lafora bodies have sparse branching compared to normal glycogen. In glycogen, branching is due to the formation of α-1, 6-glycosidic bonds. Thus, these bonds are decreased in Lafora bodies.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Monosaccharides and Disaccharides](#), [Lipids](#), [Amino Acids](#), [Nucleotides](#), [Proteins](#), [The](#)

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
1 min 42 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS1
CC5D
FC5
BCH

Question 44 of 230

Data ID: 530101
Captured: 6/3/2026, 4:40:16 PM

QUESTION • ANSWER CHOICES

Each of the following equations shows the dissociation of an acid in water. Which of the reactions occurs to the LEAST extent?

- A. $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$
- B. $\text{HPO}_4^{2-} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{PO}_4^{3-}$
- C. $\text{H}_2\text{SO}_4 + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{HSO}_4^-$
- D. $\text{H}_3\text{PO}_4 + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{H}_2\text{PO}_4^-$

SOLUTION

Solution: The correct answer is **B**.

1. Hydrochloric acid is a strong acid. This reaction will occur to completion.
2. **HPO_4^{2-} has a high negative charge and so dissociation of it will occur to the least extent.**
3. Sulfuric acid is a strong acid. This means that the dissociation shown will go to completion as water is a stronger base than the hydrogen sulfate anion.
4. Phosphoric acid is a much stronger Brønsted acid than the hydrogen phosphate anion. This is readily seen due to the increased charge on the anion which will make it much harder to lose another proton.

Khan Academy Lessons

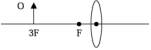
RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
1 min 0 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC5A
FC5
GCH

Question 45 of 230

Data ID: 530126
Captured: 6/3/2026, 4:40:17 PM

QUESTION • ANSWER CHOICES



In the above figure, an object O is at a distance of three focal lengths from the center of a convex lens. What is the ratio of the height of the image to the height of the object?

- A. 1/3
- B. 1/2**
- C. 2/3
- D. 3/2

SOLUTION

Solution: The correct answer is **B**.

1. The ratio of the image height to the object height is equal to the ratio of the lens-image distance to the object-lens distance. According to the thin lens equation, the distance between the lens and the image is $\frac{3F \times 1F}{(3F - 1F)} = \left(\frac{3}{2}\right)F$. The ratio sought is then equal to $\frac{(3/2)F}{3F} = \frac{1}{2}$, and not $\frac{1}{3}$.
2. **This is correct because the ratio of the image height to the object height is equal to the ratio of the lens-image distance to the object-lens distance. According to the thin lens equation, the distance between the lens and the image is $\frac{3F \times 1F}{(3F - 1F)} = \left(\frac{3}{2}\right)F$. The ratio sought is then equal to $\frac{(3/2)F}{3F} = \frac{1}{2}$.**
3. The ratio of the image height to the object height is equal to the ratio of the lens-image distance to the object-lens distance. According to the thin lens equation, the distance between the lens and the image is $\frac{3F \times 1F}{(3F - 1F)} = \left(\frac{3}{2}\right)F$. The ratio sought is then equal to $\frac{(3/2)F}{3F} = \frac{1}{2}$, and not $\frac{2}{3}$.
4. The ratio of the image height to the object height is equal to the ratio of the lens-image distance to the object-lens distance. According to the thin lens equation, the distance between the lens and the image is $\frac{3F \times 1F}{(3F - 1F)} = \left(\frac{3}{2}\right)F$. The ratio sought is then equal to $\frac{(3/2)F}{3F} = \frac{1}{2}$, and not $\frac{3}{2}$.

Khan Academy Lessons

For more help on this difficulty, concepts or content associated with this question, see the Khan Academy lessons on [Ray and Light](#), [Electromagnetic Radiation](#), [Geometrical Optics](#), [Spherical mirrors](#).

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
X
Time Spent:
1 min 39 secs
Difficulty Level:
Expert
Content & Skills:
SIRS2
CC4D
FC4
PHY

Question 46 of 230

Data ID: 530127
Captured: 6/3/2026, 4:40:17 PM

QUESTION • ANSWER CHOICES

Gamma decay occurs when a nucleus emits:

- A. a photon.
- B. a positron.
- C. a neutron.
- D. an electron.

SOLUTION

Solution: The correct answer is A.

1. By definition, gamma decay means the emission of photons by the nucleus.
2. The emission of positrons by the nucleus is called beta-plus decay.
3. The emission of neutrons by the nucleus occurs only during stimulated nuclear fission.
4. The emission of an electron by the nucleus is called beta-minus decay.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
X
D
Time Spent:
31 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC4E
FC4
PHY

Question 47 of 230

Data ID: 530128
Captured: 6/3/2026, 4:40:18 PM

QUESTION • ANSWER CHOICES

When a 0.1 M H₂SO₄ solution is added to pulverized blackboard chalk, the following reaction takes place.



At 25°C, the reaction is spontaneous and has:

- A. negative ΔG° and positive ΔS°.
- B. negative ΔG° and negative ΔS°.
- C. positive ΔG° and negative ΔS°.
- D. positive ΔG° and positive ΔS°.

SOLUTION

Solution: The correct answer is A.

1. Spontaneous reactions must have a negative ΔG°, and reactions that create gaseous products from solids and liquids must have a positive ΔS°.
2. The reaction does not have negative ΔS because it produces a gas.
3. The reaction must have negative ΔG because it is spontaneous. It is also apparent that the reaction has positive ΔS, because the reaction produces a gas.
4. Spontaneous reactions have negative ΔG.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
31 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC5E
FC5
GCH

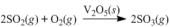
Question 48 of 230

Data ID: 530129
Captured: 6/3/2026, 4:40:19 PM

PASSAGE

Passage 9 (Questions 48-51)

A catalyst is a substance that speeds up a reaction without being consumed. For example, V_2O_5 is used as a catalyst to oxidize SO_2 to SO_3 in the first step of the production of sulfuric acid (reactions 1 and 2).



Reaction 1



Reaction 2

Furthermore, H_2O_2 can be decomposed with the aid of a catalyst, as shown in reactions 3 and 4.



Reaction 3



Reaction 4

Catalysts are widely utilized in automobile catalytic converters, where the exhaust gases are passed through beads of a solid catalyst. The catalyst aids in the conversions of carbon monoxide to carbon dioxide, hydrocarbons to carbon dioxide and water, and nitric oxide to nitrogen gas. Unfortunately, catalytic converters also can catalyze the oxidation of SO_2 to SO_3 , which subsequently reacts with moisture to form unwanted sulfuric acid.

Catalysts are classified as either *homogeneous* or *heterogeneous*. A homogeneous catalyst exists in the same phase as the reactant molecules, whereas a heterogeneous catalyst exists in a different phase than the reactants.

QUESTION + ANSWER CHOICES

Which of the following is the overall reaction for the decomposition of H_2O_2 that is shown in reactions 3 and 4?

- A.
 $2H_2O_2(aq) + IO^-(aq) \longrightarrow 2H_2O(l) + I^-(aq) + O_2(g)$
- B.
 $2H_2O_2(aq) + 2I^-(aq) \longrightarrow 2H_2O(l) + I_2(g) + O_2(g)$
- C.
 $H_2O_2(aq) \longrightarrow H_2O(l) + O_2(g)$
- D.
 $2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$

SOLUTION

Solution: The correct answer is D.

- Both of the anions shown in this equation are intermediates of the net of reactions 3 and 4. This reaction is also not balanced for O atoms.
- Iodide is an intermediate in reactions 3 and 4. Furthermore, iodine does not appear in either reaction. The reaction shown in this response is not balanced for charge.
- This reaction is not balanced for O atoms.
- This response depicts the net sum of reactions 3 and 4: $2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Atomic Structure](#), [Electrolytic Structure](#), [The Periodic Table](#), [Classification of](#)

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
☒ D
Time Spent:
1 min 1 sec
Difficulty Level:

Moderate
Content & Skills:
SIRS1
CC4E
FC4
GCH

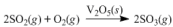
Question 49 of 230

Data ID: 530130
Captured: 6/3/2026, 4:40:20 PM

PASSAGE

Passage 9 (Questions 48-51)

A catalyst is a substance that speeds up a reaction without being consumed. For example, V_2O_5 is used as a catalyst to oxidize SO_2 to SO_3 in the first step of the production of sulfuric acid (reactions 1 and 2).



Reaction 1



Reaction 2

Furthermore, H_2O_2 can be decomposed with the aid of a catalyst, as shown in reactions 3 and 4.



Reaction 3



Reaction 4

Catalysts are widely utilized in automobile catalytic converters, where the exhaust gases are passed through beads of a solid catalyst. The catalyst aids in the conversions of carbon monoxide to carbon dioxide, hydrocarbons to carbon dioxide and water, and nitric oxide to nitrogen gas. Unfortunately, catalytic converters also can catalyze the oxidation of SO_2 to SO_3 , which subsequently reacts with moisture to form unwanted sulfuric acid.

Catalysts are classified as either *homogeneous* or *heterogeneous*. A homogeneous catalyst exists in the same phase as the reactant molecules, whereas a heterogeneous catalyst exists in a different phase than the reactants.

QUESTION + ANSWER CHOICES

Which of the following diagrams does NOT illustrate a function of an automobile catalytic converter?

- A.
 $CH_4 \rightarrow \text{converter} \rightarrow CO_2 + 2H_2O$
- B.
 $CO \rightarrow \text{converter} \rightarrow CO_2$
- C.
 $SO_3 \rightarrow \text{converter} \rightarrow SO_2 + H_2O$
- D.
 $NO \rightarrow \text{converter} \rightarrow N_2$

SOLUTION

Solution: The correct answer is C.

- Catalytic converters were stated to aid in the conversion of hydrocarbons to carbon dioxide and water.
- Catalytic converters were stated to aid in the conversion of carbon monoxide to carbon dioxide.
- This response depicts a chemical transformation ($SO_3 \rightarrow SO_2 + H_2O$) that is NOT catalyzed by an automobile's catalytic converter.
- Catalytic converters were stated to aid in the conversion of nitric oxide to nitrogen gas.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B

C
✓
D
Time Spent:
56 secs
Difficulty Level:
Easy
Content & Skills:
SIRS2
CCSE
FCS
GCH

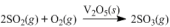
Question 50 of 230

Data ID: 530131
Captured: 6/3/2026, 4:40:20 PM

PASSAGE

Passage 9 (Questions 48-51)

A catalyst is a substance that speeds up a reaction without being consumed. For example, V_2O_5 is used as a catalyst to oxidize SO_2 to SO_3 in the first step of the production of sulfuric acid (reactions 1 and 2).



Reaction 1

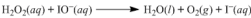


Reaction 2

Furthermore, H_2O_2 can be decomposed with the aid of a catalyst, as shown in reactions 3 and 4.



Reaction 3



Reaction 4

Catalysts are widely utilized in automobile catalytic converters, where the exhaust gases are passed through beads of a solid catalyst. The catalyst aids in the conversions of carbon monoxide to carbon dioxide, hydrocarbons to carbon dioxide and water, and nitric oxide to nitrogen gas. Unfortunately, catalytic converters also can catalyze the oxidation of SO_2 to SO_3 , which subsequently reacts with moisture to form unwanted sulfuric acid.

Catalysts are classified as either *homogeneous* or *heterogeneous*. A homogeneous catalyst exists in the same phase as the reactant molecules, whereas a heterogeneous catalyst exists in a different phase than the reactants.

QUESTION + ANSWER CHOICES

Which of the following most likely will occur if a homogeneous catalyst CANNOT be separated from the products at the end of a reaction?

- A. The catalyst will become heterogeneous.
- B. The products will be contaminated.**
- C. The reaction will not occur.
- D. The reaction rate will speed up.

SOLUTION

Solution: The correct answer is **B**.

- Of the many separation techniques available, only fractional distillation might cause the catalyst to become heterogeneous and that would facilitate separation by filtration.
- If a suitable separation technique is not found, the product will be contaminated by the catalyst.**
- Separation of products from unreacted starting materials or catalysts does not affect the outcome of a reaction, which has already taken place.
- The reaction rate is unaffected. The separation process will occur after the reaction is completed.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
X
Time Spent:
59 secs
Difficulty Level:
Moderate
Content & Skills:

SIRS3
CC5E
FC5
GCH

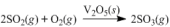
Question 51 of 230

Data ID: 530132
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PASSAGE

Passage 9 (Questions 48-51)

A catalyst is a substance that speeds up a reaction without being consumed. For example, V_2O_5 is used as a catalyst to oxidize SO_2 to SO_3 in the first step of the production of sulfuric acid (reactions 1 and 2).



Reaction 1

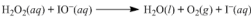


Reaction 2

Furthermore, H_2O_2 can be decomposed with the aid of a catalyst, as shown in reactions 3 and 4.



Reaction 3



Reaction 4

Catalysts are widely utilized in automobile catalytic converters, where the exhaust gases are passed through beads of a solid catalyst. The catalyst aids in the conversions of carbon monoxide to carbon dioxide, hydrocarbons to carbon dioxide and water, and nitric oxide to nitrogen gas. Unfortunately, catalytic converters also can catalyze the oxidation of SO_2 to SO_3 , which subsequently reacts with moisture to form unwanted sulfuric acid.

Catalysts are classified as either *homogeneous* or *heterogeneous*. A homogeneous catalyst exists in the same phase as the reactant molecules, whereas a heterogeneous catalyst exists in a different phase than the reactants.

QUESTION + ANSWER CHOICES

How will the rate of a catalyzed reaction be affected if the solid catalyst is finely ground before it is added to the reaction mixture?

- A. The rate will be faster because a greater mass of catalyst will be present.
- B. The rate will be faster because a greater surface area of catalyst will be exposed.**
- C. The rate will be slower because the fine catalyst particles will interfere with product formation.
- D. The rate will remain the same because the mass of catalyst will be the same.

SOLUTION

Solution: The correct answer is **B**.

- Grinding a solid does not change its mass.
- Grinding a heterogeneous catalyst increases the amount of catalyst available to the reaction and therefore increases its rate.**
- Product formation occurs on a scale so small that the presence of more suspended solid cannot interfere with the course of reaction.
- While it is true that the mass is the same, the rate will increase due to the greater amount of catalyst that is interacting with the liquid solution.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
1 min 59 secs
Difficulty Level:
Moderate
Content & Skills:

SIRS3
CC5E
FC5
GCH

Question 52 of 230

Data ID: 531050
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PASSAGE

Passage 10 (Questions 52-56)

A group of students investigated the reactions of several metals with solutions containing metallic ions.

Experiment 1

The students placed strips of a metal, X(s), into solutions containing different metallic ions, Y^{m+}(aq), as shown in Figure 1.

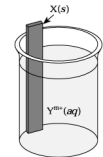


Figure 1 Metal strip in solution

After several hours, they observed whether or not Y(s) formed on the surface of the metal strip (see Table 1).

Table 1 Results of Metal Strip Experiment

Identity of metal ions	Composition of metal strip					
	Al(s)	Cu(s)	Zn(s)	Ag(s)	Au(s)	Pb(s)
Al ³⁺ (aq)	*	no change	no change	no change	no change	no change
Cu ²⁺ (aq)	new solid	*	new solid	no change	no change	new solid
Zn ²⁺ (aq)	new solid	no change	*	no change	no change	no change
Ag ⁺ (aq)	new solid	new solid	new solid	*	no change	new solid
Au ³⁺ (aq)	*	*	*	new solid	*	*
Pb ²⁺ (aq)	new solid	no change	new solid	no change	no change	*

* = not studied

Experiment 2

The students placed a solid strip of each metal into an aqueous solution containing ions of the same metal. Complete circuits were established (Figure 2).

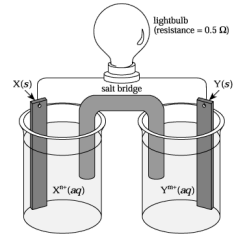
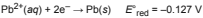
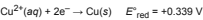


Figure 2 Complete circuit

Finally, the students checked a general chemistry text to determine the standard reduction potentials for the following reactions.



Reaction 1



Reaction 2

QUESTION + ANSWER CHOICES

A student can most effectively increase the current passing through the circuit in Figure 2 by doing which of the following?

- A. Using electrical wire with a smaller diameter
- B. Increasing the temperature of the electrical wire
- C. Decreasing the concentrations of X^{m+}(aq) and Y^{m+}(aq)
- D. Replacing the lightbulb with one that has a resistance of 0.2 Ω

SOLUTION

Solution: The correct answer is D.

- Using an electrical wire with a smaller diameter will increase the resistance of the wire. According to Ohm's law, this will decrease the current in the circuit.
- Increasing the temperature of the electrical wire will increase the resistance of the wire. According to Ohm's law, this will decrease the current in the circuit.
- Decreasing the concentrations will decrease the amount of charges that flow through the circuit that effectively decreases the current.
- This option is correct because according to Ohm's law, current is equal to voltage divided by resistance. In order to increase the current when the voltage is fixed, the resistance must be decreased, such as by replacing the lightbulb with one that has a resistance of 0.2 Ω instead of 0.5 Ω.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
42 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS3
CC4C
FC4
PHY

Question 53 of 230

Data ID: 531123
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PASSAGE

Passage 10 (Questions 52-56)

A group of students investigated the reactions of several metals with solutions containing metallic ions.

Experiment 1

The students placed strips of a metal, X(s), into solutions containing different metallic ions, Y^{m+}(aq), as shown in Figure 1.

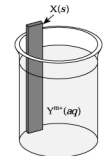


Figure 1 Metal strip in solution

After several hours, they observed whether or not Y(s) formed on the surface of the metal strip (see Table 1).

Table 1 Results of Metal Strip Experiment

Identity of metal ions	Composition of metal strip					
	Al(s)	Cu(s)	Zn(s)	Ag(s)	Au(s)	Pb(s)
Al ³⁺ (aq)	*	no change	no change	no change	no change	no change
Cu ²⁺ (aq)	new solid	*	new solid	no change	no change	new solid
Zn ²⁺ (aq)	new solid	no change	*	no change	no change	no change
Ag ⁺ (aq)	new solid	new solid	new solid	*	no change	new solid
Au ³⁺ (aq)	*	*	*	new solid	*	*
Pb ²⁺ (aq)	new solid	no change	new solid	no change	no change	*

* = not studied

Experiment 2

The students placed a solid strip of each metal into an aqueous solution containing ions of the same metal. Complete circuits were established (Figure 2).

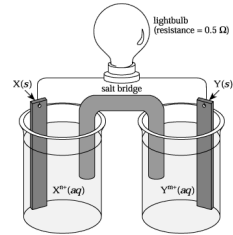
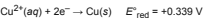


Figure 2 Complete circuit

Finally, the students checked a general chemistry text to determine the standard reduction potentials for the following reactions.



Reaction 1



Reaction 2

QUESTION + ANSWER CHOICES

When a strip of Zn is placed in a beaker containing 0.1 M HCl, H₂(g) evolves. If a strip of Al is placed in a beaker containing 0.1 M HCl, does H₂(g) evolve?

- A. Yes; Al is reduced and H⁺(aq) is oxidized.
- B. Yes; Al is oxidized and H⁺(aq) is reduced.
- C. No; Al is reduced and Cl⁻(aq) is oxidized.
- D. No; Al is oxidized and H₂O(l) is produced.

SOLUTION

Solution: The correct answer is B.

- Al is an electropositive metal. It will not form anions. Formation of H₂ from H⁺ is a reduction.
- Since a new solid forms when Al(s) is mixed with Zn²⁺(aq), it is reasonable to assume that Al(s) is more susceptible to oxidation than Zn(s).
- Electropositive metals are good reducing agents. Only exceptionally good oxidizing agents can oxidize chloride.
- If Al is oxidized, something has to be reduced. Of the possible choices, reduction of H⁺ occurs at a potential of 0.0 V and is most likely.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer

A

B

✓

C

D

Time Spent:

1 min 5 secs

Difficulty Level:

Difficult

Content & Skills:

SIRS2

CC4C

FC4

GCH

Question 54 of 230

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PASSAGE

Passage 10 (Questions 52-56)

A group of students investigated the reactions of several metals with solutions containing metallic ions.

Experiment 1

The students placed strips of a metal, X(s), into solutions containing different metallic ions, Y^{m+}(aq), as shown in Figure 1.

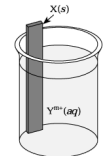


Figure 1 Metal strip in solution

After several hours, they observed whether or not Y(s) formed on the surface of the metal strip (see Table 1).

Table 1 Results of Metal Strip Experiment

Identity of metal ions	Composition of metal strip					
	Al(s)	Cu(s)	Zn(s)	Ag(s)	Au(s)	Pb(s)
Al ³⁺ (aq)	*	no change	no change	no change	no change	no change
Cu ²⁺ (aq)	new solid	*	new solid	no change	no change	new solid
Zn ²⁺ (aq)	new solid	no change	*	no change	no change	no change
Ag ⁺ (aq)	new solid	new solid	new solid	*	no change	new solid
Au ³⁺ (aq)	*	*	*	new solid	*	*
Pb ²⁺ (aq)	new solid	no change	new solid	no change	no change	*

* = not studied

Experiment 2

The students placed a solid strip of each metal into an aqueous solution containing ions of the same metal. Complete circuits were established (Figure 2).

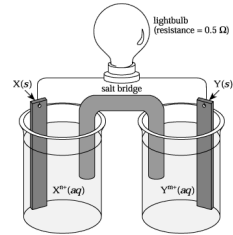
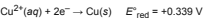


Figure 2 Complete circuit

Finally, the students checked a general chemistry text to determine the standard reduction potentials for the following reactions.



Reaction 1



Reaction 2

QUESTION + ANSWER CHOICES

Assume that a circuit similar to that in Figure 2 is set up in which X = Al, X^{m+} = Al³⁺, Y = Cu, and Y^{m+} = Cu²⁺. Which of the following reactions will occur?

- A:
 $2\text{Al}^{3+}(\text{aq}) + 3\text{Cu}(\text{s}) \rightarrow 2\text{Al}(\text{s}) + 3\text{Cu}^{2+}(\text{aq})$
- B:
 $3\text{Al}^{3+}(\text{aq}) + 2\text{Cu}(\text{s}) \rightarrow 3\text{Al}(\text{s}) + 2\text{Cu}^{2+}(\text{aq})$
- C:
 $2\text{Al}(\text{s}) + 3\text{Cu}^{2+}(\text{aq}) \rightarrow 2\text{Al}^{3+}(\text{aq}) + 3\text{Cu}(\text{s})$
- D:
 $3\text{Al}(\text{s}) + 2\text{Cu}^{2+}(\text{aq}) \rightarrow 3\text{Al}^{3+}(\text{aq}) + 2\text{Cu}(\text{s})$

SOLUTION

Solution: The correct answer is C.

- While this reaction is balanced, it does not fit with the data in the table. Al³⁺(aq) does not react with Cu(s).
- This reaction is not balanced for charge. Furthermore, it does not fit with the data in the table. Al³⁺(aq) does not react with Cu(s).
- This response depicts the balanced chemical reaction that occurs when the four substances [Al(s), Al³⁺(aq), Cu(s), and Cu²⁺(aq)] are mixed.**
- This reaction is not balanced for charge. There is a net +9 on the product side, but only net +4 on the reactant side.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Atomic Nucleus](#), [Electronic Structure](#), [The Periodic Table - Classification of](#)

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
1 min 10 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC4E
FC4
GCH

Question 55 of 230

Data ID: 531125
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PASSAGE

Passage 10 (Questions 52-56)

A group of students investigated the reactions of several metals with solutions containing metallic ions.

Experiment 1

The students placed strips of a metal, X(s), into solutions containing different metallic ions, Y^{m+}(aq), as shown in Figure 1.

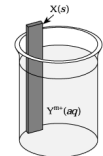


Figure 1 Metal strip in solution

After several hours, they observed whether or not Y(s) formed on the surface of the metal strip (see Table 1).

Table 1 Results of Metal Strip Experiment

Identity of metal ions	Composition of metal strip					
	Al(s)	Cu(s)	Zn(s)	Ag(s)	Au(s)	Pb(s)
Al ³⁺ (aq)	*	no change	no change	no change	no change	no change
Cu ²⁺ (aq)	new solid	*	new solid	no change	no change	new solid
Zn ²⁺ (aq)	new solid	no change	*	no change	no change	no change
Ag ⁺ (aq)	new solid	new solid	new solid	*	no change	new solid
Au ³⁺ (aq)	*	*	*	new solid	*	*
Pb ²⁺ (aq)	new solid	no change	new solid	no change	no change	*

* = not studied

Experiment 2

The students placed a solid strip of each metal into an aqueous solution containing ions of the same metal. Complete circuits were established (Figure 2).

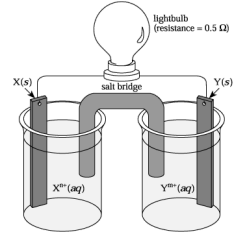
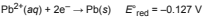
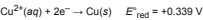


Figure 2 Complete circuit

Finally, the students checked a general chemistry text to determine the standard reduction potentials for the following reactions.



Reaction 1



Reaction 2

QUESTION + ANSWER CHOICES

A circuit similar to that in Figure 2 is set up in which X = Pb, X^{m+} = Pb²⁺, Y = Cu, and Y^{m+} = Cu²⁺. Will an oxidation–reduction reaction occur spontaneously?

- A. No; E°_{cell} = −0.212.
- B. No; E°_{cell} = +0.212.
- C. Yes; E°_{cell} = −0.466.
- D. Yes; E°_{cell} = +0.466.

SOLUTION

Solution: The correct answer is D.

1. This is an incorrect calculation of E°_{cell}. In order to use the potentials provided, one of the half-reactions must be reversed. This changes the sign of the reduction potential. The calculated value is the result of reversing both half-reactions and adding the resulting potentials.
2. This is not the real value of E°_{cell}. It results from adding both reduction potentials listed. One of the reductions needs to be reversed in order to have a complete oxidation–reduction reaction.
3. This value of E°_{cell} is obtained by combining the oxidation of Cu(s) to Cu²⁺(aq) with the reduction of Pb²⁺(aq) to Pb(s). The fact that the value is negative indicates that this reaction will not occur spontaneously.
4. **A spontaneous reaction occurs when E° is greater than zero, and this occurs if the oxidation of Pb(s) is combined with the reduction of Cu²⁺(aq), resulting in a net E° of +0.466 V.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
10 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC4C
FC4
GCH

Question 56 of 230

Data ID: 531126
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PASSAGE

Passage 10 (Questions 52-56)

A group of students investigated the reactions of several metals with solutions containing metallic ions.

Experiment 1

The students placed strips of a metal, X(s), into solutions containing different metallic ions, Y^{m+}(aq), as shown in Figure 1.

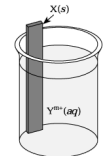


Figure 1 Metal strip in solution

After several hours, they observed whether or not Y(s) formed on the surface of the metal strip (see Table 1).

Table 1 Results of Metal Strip Experiment

Identity of metal ions	Composition of metal strip					
	Al(s)	Cu(s)	Zn(s)	Ag(s)	Au(s)	Pb(s)
Al ³⁺ (aq)	*	no change	no change	no change	no change	no change
Cu ²⁺ (aq)	new solid	*	new solid	no change	no change	new solid
Zn ²⁺ (aq)	new solid	no change	*	no change	no change	no change
Ag ⁺ (aq)	new solid	new solid	new solid	*	no change	new solid
Au ³⁺ (aq)	*	*	*	new solid	*	*
Pb ²⁺ (aq)	new solid	no change	new solid	no change	no change	*

* = not studied

Experiment 2

The students placed a solid strip of each metal into an aqueous solution containing ions of the same metal. Complete circuits were established (Figure 2).

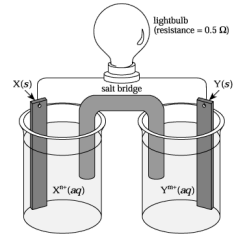
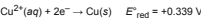


Figure 2 Complete circuit

Finally, the students checked a general chemistry text to determine the standard reduction potentials for the following reactions.



Reaction 1



Reaction 2

QUESTION + ANSWER CHOICES

A circuit similar to that in Figure 2 is set up. It has a potential of 2.0 V. Assume that the resistance in the circuit is negligible compared to that of the lightbulb. Approximately how much current passes through the lightbulb?

- A. 0.5 A
- B. 1.0 A
- C. 2.0 A
- D. 4.0 A

SOLUTION

Solution: The correct answer is D.

- The current is equal to 2 V divided by 0.5 Ω equals 4.0 A, not 1.0 A.
- The current is equal to 2 V divided by 0.5 Ω equals 4.0 A, not 1.0 A.
- The current is equal to 2 V divided by 0.5 Ω equals 4.0 A, not 2.0 A.
- The current is equal to 2 V divided by 0.5 Ω equals 4.0 A.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
X
D
Time Spent:
3 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC4C
FC4
PHY

Question 57 of 230

Data ID: 531127
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QUESTION + ANSWER CHOICES

The equilibrium

$$\text{BaCrO}_4(s) \rightleftharpoons \text{Ba}^{2+}(aq) + \text{CrO}_4^{2-}(aq)$$

exists in a saturated aqueous solution of BaCrO₄. Dissolution of Na₂CrO₄ in a saturated aqueous BaCrO₄ solution would:

- A. have no effect on the position of this equilibrium.
- B. shift this equilibrium left.
- C. shift this equilibrium right.
- D. shift this equilibrium first right and then left.

SOLUTION

Solution: The correct answer is **B**.

1. Adding the chromate anion to solution will drive the indicated equilibrium to the left by Le Châtelier's Principle.

2. **Dissolution of Na₂CrO₄ would introduce the common ion, CrO₄²⁻, which would reduce the solubility of BaCrO₄ due to the common ion effect.**

3. The reverse is true. Adding a product anion stresses the existing equilibrium. Upon addition of chromate, the reaction quotient Q will exceed K_{sp} and the equilibrium shifts left until Q and K_{sp} are equal.

4. This response does not follow the principles of chemical dynamics. While both reactions are constantly occurring (forward and reverse), the net reaction will only go to the left to compensate for the added anion.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
1 min 15 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC5A
FCS
GCH

Question 58 of 230

Data ID: 531128
Captured: 6/3/2026, 4:40:25 PM

QUESTION + ANSWER CHOICES

What volume of a 0.120 M CaI₂ solution would contain 0.078 mol of the solute?

- A. 35.0 mL
- B. 65.0 mL
- C. 350 mL
- D. 650 mL

SOLUTION

Solution: The correct answer is **D**.

1. A volume of 35 mL corresponds to only 0.035 L × 0.120 mol/L = 0.0042 mol of solute.

2. A volume of 65 mL corresponds to only 0.065 L × 0.120 mol/L = 0.0078 mol of solute.

3. A volume of 350 mL corresponds to only 0.35 L × 0.120 mol/L = 0.042 mol of solute.

4. **In order to obtain the volume of solution necessary to provide a given amount of solute in moles, one needs to take the number of moles and divide by the solution concentration in molarity: 0.078 mol ÷ 1 L/0.120 mol = 0.65 L = 650 mL.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
Time Spent:
1 min 29 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC5A
FCS
GCH

Question 59 of 230

Data ID: 531129
Captured: 6/3/2026, 4:40:26 PM

QUESTION + ANSWER CHOICES

Which measurement unit CANNOT be used to express power?

- A. $\text{kg}\cdot\text{m}^2\cdot\text{s}^2$
- B. $\text{J}\cdot\text{s}^{-1}$
- C. $\text{ft}\cdot\text{lb}\cdot\text{s}^{-1}$
- D. W

SOLUTION

Solution: The correct answer is A.

- This answer is correct because the measurement unit of power is watt, defined as $J/s = ft \bullet lb/s = kg \bullet m^2/s^3$.
- This unit is consistent with the definition of the watt, which is used to express power, because by definition $1W = J \bullet s^{-1}$.
- This unit is consistent with the definition of the watt, which is used to express power, because by definition $1W = J \bullet s^{-1} = 1 ft \bullet lb \bullet s^{-1}$.
- This unit is the watt, which is by definition is used to express power.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
0 secs
Difficulty Level:
Expert
Content & Skills:
SIRS1
CC4A
FC4
PHY

Question 60 of 230

Data ID: 531139
Captured: 6/3/2026, 4:40:26 PM

PASSAGE

Passage 1 (Questions 1-5)

The practical ideal of democracy consists in the self-government of the masses in conformity with the decisions of popular assemblies. But while this system limits the extension of the principle of delegation, it fails to provide any guarantee against the formation of an oligarchical ruling clique. Undoubtedly it deprives the natural leaders of their quality as functionaries, for this quality is transferred to the people themselves. The crowd, however, is always subject to suggestion, being readily influenced by the eloquence of popular orators; moreover, direct government by the people, admitting of no serious discussions or thoughtful deliberations, greatly facilitates sudden insurrections of all kinds by individuals who are exceptionally bold, energetic, and adroit.

It is easier to dominate a large crowd than a small audience. The adhesion of the crowd is tumultuous, summary, and unconditional. Once the suggestions have taken effect, the crowd does not readily tolerate contradiction from a small minority, and still less from isolated individuals. A great multitude assembled within a small area is unquestionably more accessible to panic alarms, to unreflective enthusiasm, and the like than is a small meeting, whose members can quietly discuss matters among themselves.

It is a fact of everyday experience that enormous public meetings commonly carry resolutions by acclamation or by general assent, while these same assemblies, if divided into small sections, say of fifty persons each, would be much more guarded in their assent. Great party congresses, in which the elite of the membership are present, usually act in this way. Words and actions are far less deliberately weighed by the crowd than by the individuals of the little groups of which this crowd is composed. The fact is incontestable—a manifestation of the pathology of the crowd. The individual disappears in the multitude and therewith disappears also personality and sense of responsibility.

The most formidable argument against the sovereignty of the masses is, however, derived from the mechanical and technical impossibility of its realization. The sovereign masses are incapable of undertaking the most necessary resolutions. The impotence of direct democracy, like the power of indirect democracy, is a direct outcome of the influence of number.

It is obvious that a gigantic number of persons belonging to a unitary organization cannot do any practical work using a system of direct discussion. The regular holding of deliberative assemblies of a thousand members encounters the gravest difficulties in respect to room and distance; from the topographical point of view, such an assembly would become altogether impossible if the members numbered ten thousand. Even if we imagined the means of communication to become much better than those that now exist, how would it be possible to assemble such a multitude in a given place, at a stated time, and with the frequency demanded by the exigencies of political life? In addition must be considered the physiological impossibility even for the most powerful orators of making themselves heard by a crowd of ten thousand persons.

There are, however, other reasons of a technical and administrative character that render impossible the direct self-government of large groups. If Peter wrongs Paul, it is out of the question that all the other citizens should hasten to the spot to undertake a personal examination of the matter in dispute and to take the part of Paul against Peter. By parity of reasoning, in the modern democratic state, it is impossible for the collectivity to undertake the direct settlement of all the controversies that may arise.

Adapted from R. Michels, *Political Parties: A Sociological Study of the Oligarchical Tendencies of Modern Democracy*. In public domain. Published 1998 by The Free Press.

QUESTION + ANSWER CHOICES

Which of the following statements from the passage most strongly indicates that the passage was written prior to the twenty-first century?

- A. "The regular holding of deliberative assemblies of a thousand members encounters the gravest difficulties in respect of room and distance"
- B. "The crowd . . . is . . . readily influenced by the eloquence of popular orators."
- C. "[T]hese same assemblies, if divided into small sections, . . . would be much more guarded in their assent."
- D. "The sovereign masses are incapable of undertaking the most necessary resolutions."

SOLUTION

Solution: The correct answer is A.

- This is a Social Science passage that falls under the content category "Government." This Comprehension question tests your understanding of passage statements as they appear in the content of the passage. The correct answer is A, because that option clearly reflects an earlier set of technological possibilities. In the twenty-first century, the Internet and other forms of electronic communication would certainly ameliorate "the gravest difficulties" in "holding . . . deliberative assemblies of a thousand members." The sentence in paragraph five that follows the sentence in option A ("Even if we imagined the means of communication . . .") clearly signals as well that the passage was written before the twenty-first century.
- There is no reason to think that eloquent orators do not retain the power to influence a crowd in the twenty-first century.
- There is no reason to think that the influence of large groups of people, as opposed to smaller groups, is less powerful in the twenty-first century than it was in previous eras.
- Nothing about this claim signals that it was made before the twenty-first century, or that it could not be made today.

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
1 min 23 secs

Difficulty Level:
Difficult
Content & Skills:
Skill: CMP
GOV
SSC

Question 61 of 230

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PASSAGE

Passage 1 (Questions 1-5)

The practical ideal of democracy consists in the self-government of the masses in conformity with the decisions of popular assemblies. But while this system limits the extension of the principle of delegation, it fails to provide any guarantee against the formation of an oligarchical ruling clique. Undoubtedly it deprives the natural leaders of their quality as functionaries, for this quality is transferred to the people themselves. The crowd, however, is always subject to suggestion, being readily influenced by the eloquence of popular orators; moreover, direct government by the people, admitting of no serious discussions or thoughtful deliberations, greatly facilitates sudden insurrections of all kinds by individuals who are exceptionally bold, energetic, and adroit.

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The most formidable argument against the sovereignty of the masses is, however, derived from the mechanical and technical impossibility of its realization. The sovereign masses are incapable of undertaking the most necessary resolutions. The impotence of direct democracy, like the power of indirect democracy, is a direct outcome of the influence of number.

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There are, however, other reasons of a technical and administrative character that render impossible the direct self-government of large groups. If Peter wrongs Paul, it is out of the question that all the other citizens should hasten to the spot to undertake a personal examination of the matter in dispute and to take the part of Paul against Peter. By parity of reasoning, in the modern democratic state, it is impossible for the collectivity to undertake the direct settlement of all the controversies that may arise.

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QUESTION + ANSWER CHOICES

Which of the following assertions in the passage does the author support with a practical example?

- A: ~~Direct government by the people . . . greatly facilitates sudden insurrection of all kinds by individuals who are exceptionally bold, energetic, and adroit.~~
- B: ~~"[A]nd therewith disappears also personality and sense of responsibility."~~
- C: ~~"The adhesion of the crowd is tumultuous, summary, and unconditional."~~
- D: ~~It is impossible for the collectivity to undertake the direct settlement of all the controversies that may arise."~~

SOLUTION

Solution: The correct answer is D.

- The quotation in this option is not supported by any kind of practical, real-life example.
- The quotation in this option is not supported by any kind of practical, real-life example.
- The quotation in this option is not supported by any kind of practical, real-life example.
- This is a Reasoning Within the Text question because it asks you to assess the nature of the support that the author provides for various claims. In this case, the question asks which passage assertion is supported with a "practical example." The answer is D because the claim about the impossibility of a "collectivity" settling all of the "controversies that may arise" in a society, is supported by the example (in the final paragraph) of what happens when "Peter wrongs Paul." That example demonstrates why it is absurd to imagine that the entire "collectivity" would be able to examine the circumstances of the controversy and adjudicate it.

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RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
X
D
Time Spent:
2 mins 33 secs
Difficulty Level:

Difficult
Content & Skills:
Skill: RWT
GOV
SSC

Question 62 of 230

Data ID: 531148
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PASSAGE

Passage 1 (Questions 1-5)

The practical ideal of democracy consists in the self-government of the masses in conformity with the decisions of popular assemblies. But while this system limits the extension of the principle of delegation, it fails to provide any guarantee against the formation of an oligarchical ruling clique. Undoubtedly it deprives the natural leaders of their quality as functionaries, for this quality is transferred to the people themselves. The crowd, however, is always subject to suggestion, being readily influenced by the eloquence of popular orators; moreover, direct government by the people, admitting of no serious discussions or thoughtful deliberations, greatly facilitates sudden insurrections of all kinds by individuals who are exceptionally bold, energetic, and adroit.

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QUESTION + ANSWER CHOICES

Suppose the author were asked to advise a newly independent nation about the form of government it should establish. Based on the passage, the author would probably recommend:

- A.
a monarchy, in which decisions are made by one supreme leader.
- B.
a representative democracy, in which decisions are made by a limited number of elected representatives.
- C.
~~a direct democracy, in which decisions are made by all the adults in the country.~~
- D.
~~an oligarchy, in which decisions are made by a small group of capable individuals.~~

SOLUTION

Solution: The correct answer is B.

1. The author wishes to avoid "oligarchy" and the sway of a powerful "orator" (first paragraph), so it is clear that he prefers democratic values over the reign of a supreme leader.
2. This is a Reasoning Beyond the Text question, because it asks you to apply reasoning from the passage to a scenario that is not addressed in the passage. The answer is B because the passage focuses on the problems inherent in a direct democracy, in which "the masses" would govern themselves entirely. The author worries about the susceptibility of crowds to passionate enthusiasms and the tendency of very large groups to avoid "serious discussion or thoughtful deliberations" (first paragraph). Based on the author's preference for smaller assemblies rather than larger ones (paragraph 3) and his concern for the impracticality of huge assemblies—coupled with a dislike of "oligarchy" or domination in any form, the author would certainly recommend option B.
3. Much of the passage is devoted to discussions of the intellectual and practical disadvantages of the political efforts of very large groups, so the author would not recommend a direct democracy.
4. In the first paragraph, the author mentions the problem of failing to "guarantee against the formation of" an oligarchy, so this is not the form of government that he seeks.

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:

53 secs
Difficulty Level:
Moderate
Content & Skills:
Skill: RBT
GOV
SSC

Question 63 of 230

Data ID: 531149
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PASSAGE

Passage 1 (Questions 1-5)

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QUESTION + ANSWER CHOICES

Consider that some groups are defined as “cults”; members of these extremist sects generally live communally and obey an authoritarian, charismatic leader. Assume that all cults have fewer than 100 members. This information would tend to *weaken* the passage’s assertions about the:

- A. ability of great orators to sway large groups.
- B. loss of personality and sense of responsibility.
- C. difficulties involved in convening large assemblies.
- D. reasonableness of small groups compared to large groups.

SOLUTION

Solution: The correct answer is D.

1. The information in the question only applies to the ability of orators to sway smaller groups; it has no impact on claims about the effect of orators on larger groups.
2. The reference to “loss of personality and sense of responsibility” refers to the behavior of the individual in a large group, so the information about people’s behavior in a small group is irrelevant.
3. The information in the question is irrelevant to passage claims about the difficulty involved in convening large assemblies.
4. **This is a Reasoning Beyond the Text question because it presents information that is beyond the scope of the passage and then asks you to consider how this information would affect passage claims. The answer is D, because the question describes a small group in which people “obey an authoritarian, charismatic leader.” In the passage, however, the author claims that a “great multitude” is more likely to succumb to “unreflective enthusiasm . . . than is a small meeting” (paragraph 2). So if the cults that the question describes are small groups of fewer than 100 members, this would weaken passage claims about the more considered behavior of small groups when compared to larger ones.**

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓

Time Spent:
4 mins 44 secs
Difficulty Level:
Difficult
Content & Skills:
Skill: RBT
GOV
SSC

Question 64 of 230

Data ID: 631151
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PASSAGE

Passage 1 (Questions 1-5)

The practical ideal of democracy consists in the self-government of the masses in conformity with the decisions of popular assemblies. But while this system limits the extension of the principle of delegation, it fails to provide any guarantee against the formation of an oligarchical ruling clique. Undoubtedly it deprives the natural leaders of their quality as functionaries, for this quality is transferred to the people themselves. The crowd, however, is always subject to suggestion, being readily influenced by the eloquence of popular orators; moreover, direct government by the people, admitting of no serious discussions or thoughtful deliberations, greatly facilitates sudden insurrections of all kinds by individuals who are exceptionally bold, energetic, and adroit.

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QUESTION + ANSWER CHOICES

When using the term “the sovereignty of the masses” (paragraph 4), the author seems to mean the ability of the masses to:

- A. vote in elections that decide who their leaders will be.
- B. make every decision concerning their governance.
- C. discuss all the potential consequences of specific actions.
- D. ~~understand the difference between popular orators and natural leaders.~~

SOLUTION

Solution: The correct answer is B.

1. Option A refers to what the passage calls “indirect” democracy, and “the sovereignty of the masses” is another way of describing “direct” democracy.
2. This is a Comprehension question, as it asks you to determine the meaning of a phrase as it is used in the context of the passage. The answer is B, because the author first refers to the “mechanical and technical impossibility of [the] realization” of the “sovereignty of the masses” (paragraph 4). He later states that some “technical and administrative” factors “render impossible the direct self-government of large groups” (final paragraph), and it is clear that this is another way of stating “the sovereignty of the masses.”
3. The author claims that very large groups do a poor job of “serious discussions or thoughtful deliberations,” and there is no indication that “sovereignty of the masses” suggests that large groups are indeed able to engage in this kind of deliberation.
4. The author suggests that large groups tend to be susceptible to popular orators, so it is unlikely that he implies that “the sovereignty of the masses” suggests that the masses can see popular orators for what they are.

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:

2 mins 4 secs
Difficulty Level:
Moderate
Content & Skills:
Skill: CMP
GOV
SSC

Question 65 of 230

Data ID: 631155
Captured: 6/3/2026, 4:40:29 PM

PASSAGE

Passage 2 (Questions 6-12)

In many ways, readers must be a severe disappointment to writers. Why don't they just get on with it—buy the books and read the words? That's what the business is about, after all. But instead, they want autographs, cigar ends, bits of hair, bits of your time, bits of your life. Fandom, obsequious yet demanding, is not confined to the music business.

Reverent souvenir hunting is often deeply installed in those who love art. It seems like proof of proper intent: If you preserve the trivial, then you must truly value the serious. Take the case of Robert Louis Stevenson's hair. When the novelist died at forty-four, he was survived by his old Scottish nurse, who would regularly be sought out by literary enthusiasts. Asked if she had any mementos of the young genius, she would admit to possessing a single lock of hair she had cut from Louis's head some forty years earlier. She was naturally unwilling to part with it, but, well, perhaps by the time the canny nanny had finished trading, there was enough Stevenson hair around to open a wig shop.

Many writers learn to handle the idolatry of their fans with a measure of grace, but if you are a writer who has the misfortune to be dead as well as famous, matters are less under your control. "He became his admirers," wrote Auden on the death of Yeats. The dead poet has been ingested by his readers; from now on he will be read, admired, and loved in a different way—owned without opposition:

Now he is scattered among a hundred cities
And wholly given over to unfamiliar affections,
To find his happiness in another kind of wood
And be punished under a foreign code of conscience.

Writers, as they grow older, anticipate and increasingly fear those unfamiliar affections, that foreign code of conscience. In the present century more than ever before, their punishment is likely to include betrayal by biography. And the more famous you are, the more likely that betrayal. After all, you belong to the world, not to yourself—don't you? You can't expect loyalty now.

The wider public assumes that writers don't want their biographies written because they don't want the gossip to come out—that mad spouse, that embarrassing disease, the booze, the laziness. Doubtless this element exists (although no more than with other categories of biography), but the main motive is usually larger: It's a defiant attempt to concentrate the reader's mind on the work. The writer's life usually isn't (as it seems to the writer) an especially interesting or instructive one. It is full of frailty and defeat, like any other life. What counts is the work. Yet the work can quite easily be buried, or half-buried, by the life. Think of Lord Byron, Oscar Wilde, Robert Lowell, Samuel Johnson. And even the biography that glorifies the life, that styles itself as homage, may be just another, less obvious form of betrayal. What novelist, given the choice, wouldn't prefer you to reread one of his or her novels rather than read his or her biography?

You may feel "close" to a writer when you walk around the writer's house and examine a lock of his or her hair, but the only time you are truly close is when you are reading words on the page. This is the only pure act: the rest is dilution, marginality, betrayal—the higher sentimentality. Biography is only sophisticated hair collecting.

We persuade ourselves that a writer's life helps illuminate the writer's work, but I wonder if we really believe this. The life offers false and easy explanations—T. S. Eliot's first marriage went wrong, so he took a glum view of things in *The Waste Land*. Only the work can really explain the work. So perhaps we want to approach and touch the writer's life for a more basic, a more magical reason. Parallels between religion and art have generally been much overworked, but perhaps literary biography has come to occupy the place once owned by hagiography—except that in these disenchanted times, the instructive lives of saints have been replaced by the instructive lives of sinners: The reader now thrills to the bad life rather than the good life. And just as the saint's relics were kept in a gold casket beneath the altar, so now we preserve the writer's relics, although we expect no miracles from them.

Adapted from J. Barnes, *The follies of writer worship*. ©1995 J. Barnes.

QUESTION + ANSWER CHOICES

The statement that "if you preserve the trivial, then you must truly value the serious" (paragraph 2) functions in the passage to:

- A. demonstrate that the motives of souvenir hunters are superior to those of biographers.
- B. convey the author's belief about the motives of souvenir hunters.
- C. explain the way souvenir hunters justify their enterprise to themselves.
- D. suggest that souvenir hunting is more respectable than it might seem.

SOLUTION

Solution: The correct answer is C.

- The author actually considers all preoccupation with the artist herself—with her biography, and with souvenirs of the author's body or life—to be equivalent. She does not suggest that souvenir hunters are superior to biographers (or vice versa).
- This statement does not provide evidence of the author's views of the souvenir hunters' motives; the author gives perhaps the clearest hint of her theories about their motives when she says, "You may feel 'close' to a writer when you . . . examine locks of his or her hair" (paragraph 6).
- This is a Humanities passage that falls under the "Literature" content category. It is a Reasoning Within the Text question because it asks you to evaluate the function of a statement within the passage. The answer is C. The statement quoted in the question is introduced this way: "It seems like proof of proper intent: If you preserve the trivial, then you must truly value the serious." The author argues throughout that readers should value an author's words over tangible objects or "souvenirs," and clearly does not endorse the statement quoted in the question. Instead, she says that valuing souvenirs "seems like proof of proper intent." This implies that those "who love art" believe or want to believe that their preoccupation with tangible remnants of artists is evidence of their own seriousness of purpose.
- The author does not use this statement to suggest anything about the "respectability" of the souvenir hunters' motives.

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Incorrect
Correct Answer Your Answer

- A
B
X
C
D

Time Spent:
2 mins 12 secs
Difficulty Level:
Difficult
Content & Skills:
Skill: RWT
HUM
LIT

Question 66 of 230

Data ID: 631157
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PASSAGE

Passage 2 (Questions 6-12)

In many ways, readers must be a severe disappointment to writers. Why don't they just get on with it—buy the books and read the words? That's what the business is about, after all. But instead, they want autographs, cigar ends, bits of hair, bits of your time, bits of your life. Fandom, obsequious yet demanding, is not confined to the music business.

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We persuade ourselves that a writer's life helps illuminate the writer's work, but I wonder if we really believe this. The life offers false and easy explanations—T. S. Eliot's first marriage went wrong, so he took a glum view of things in *The Waste Land*. Only the work can really explain the work. So perhaps we want to approach and touch the writer's life for a more basic, a more magical reason. Parallels between religion and art have generally been much overworked, but perhaps literary biography has come to occupy the place once owned by hagiography—except that in these disenchanted times, the instructive lives of saints have been replaced by the instructive lives of sinners: The reader now thrills to the bad life rather than the good life. And just as the saint's relics were kept in a gold casket beneath the altar, so now we preserve the writer's relics, although we expect no miracles from them.

Adapted from J. Barnes, *The follies of writer worship*. ©1995 J. Barnes.

QUESTION • ANSWER CHOICES

The story of Stevenson's Scottish nanny (paragraph 2) is probably included in order to:

- A. ~~demonstrate that those who preserve the trivial also value the serious.~~
- B. expose the nanny as a charlatan.
- C. establish the pervasiveness of the desire to collect an author's mementos.
- D. demonstrate the popularity of Stevenson's books.

SOLUTION

Solution: The correct answer is C.

1. This anecdote does not testify to the "seriousness" of those who value the trivial; as is clear in item -02, the author sees that contention as a rationalization on the part of the collectors.
2. While the anecdote does humorously expose the nanny as a fraud, this story is not included in order to demonstrate something about the nanny; rather, it is intended to demonstrate something about the souvenir hunters.
3. **This is a Reasoning Within the Text question because it asks you to establish the likely purpose or function of an anecdote that the author includes in the passage. The answer is C because the story of Stevenson's nanny, who purported to have a "single" lock of his hair, but actually sold countless locks of dubious origin, shows how common is the desire to own something that apparently belonged to an author. The author writes archly that "by the time" the nanny was finished with her sales, "there was enough Stevenson hair around to open a wig shop."**
4. The story really does not say anything about Stevenson's books, which is part of the reason that the author dislikes souvenir hunting.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer

A
B
C
✓
D

Time Spent:
4 mins 54 secs
Difficulty Level:
Easy
Content & Skills:
Skill: RWT
HUM
LIT

Question 67 of 230

Data ID: 631158
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PASSAGE

Passage 2 (Questions 6-12)

In many ways, readers must be a severe disappointment to writers. Why don't they just get on with it—buy the books and read the words? That's what the business is about, after all. But instead, they want autographs, cigar ends, bits of hair, bits of your time, bits of your life. Fandom, obsequious yet demanding, is not confined to the music business.

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Adapted from J. Barnes, *The follies of writer worship*. ©1995 J. Barnes.

QUESTION + ANSWER CHOICES

The passage assertion that a dead poet is "owned without opposition" (paragraph 3) probably means that:

- A. the poet is no longer able to challenge the public's response to the poet's life and work.
- B. other writers will no longer oppose the poet's views.
- C. readers will no longer intrude on the poet's private life.
- D. the poet's admirers will buy up all the poet's books and relics.

SOLUTION

Solution: The correct answer is A.

- This is a Comprehension question, because it requires you to understand the meaning of a phrase in the context of the passage. The answer is A, because, before claiming that a dead poet is "owned without opposition," the author writes, "if you [have] the misfortune to be dead as well as famous, matters are less under your control . . . The dead poet has been ingested by his readers; from now on he will be read, admired, and loved in a different way." "Ingested by his readers" clearly implies that readers have taken over the subjectivity of the dead author. The dead author is unable to determine how he is perceived and represented by fans.
- This is a misreading of the quotation, which does not address an author's treatment by other writers, and there is no implication that those other writers "own" him.
- The quotation implies that readers have more, not less, capacity to intrude on a writer when he is dead.
- The quotation implies that readers may gain a figurative kind of "ownership" over the dead writer, who can become (in the readers' minds) what the readers imagine or desire him to be—the dead author can no longer respond to or contest this version of himself. This option involves a literal ownership of objects, which is not what the quotation implies.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer

A

B

C

D

X

Time Spent:

1 min 27 secs

Difficulty Level:

Easy

Content & Skills:

Skill: CMP

HUM

LIT

Question 68 of 230

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PASSAGE

Passage 2 (Questions 6-12)

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Adapted from J. Barnes, *The follies of writer worship*. ©1995 J. Barnes.

QUESTION + ANSWER CHOICES

Implicit in the use of the phrase "betrayal by biography" (paragraph 4) is the idea that biography:

- A. often explores the writer's personal life against the writer's wishes.
- B. reveals the true intentions writers had for their works.
- C. offers no rewards to the interested reader.
- D. is only worthwhile when it glorifies its subject.

SOLUTION

Solution: The correct answer is A.

- This is a Comprehension question that asks you to understand the meaning of a phrase in the passage by considering passage context. The answer is A. When the author describes the possibility that authors will suffer "betrayal by biography," she elaborates on the plight of the famous author: "After all, you belong to the world, not to yourself." This quotation, together with the passage claim that aging authors increasingly fear being judged by a "foreign code of conscience," supports the idea that "betrayal by biography" implies that biography is something inflicted on the author against his wishes.
- The passage author explicitly argues that readers should study an author's works, not his life (especially in paragraphs 5-7), so this option is at odds with the passage.
- This option is not implied by "betrayal by biography," so it is not supported by the passage.
- This option is never suggested by the author, who is not interested in "glorification," so it is not supported by the passage.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A

✓
B
C
D

Time Spent:
1 min 27 secs
Difficulty Level:
Easy
Content & Skills:
Skill: CMP
HUM
LIT

Question 69 of 230

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PASSAGE

Passage 2 (Questions 6-12)

In many ways, readers must be a severe disappointment to writers. Why don't they just get on with it—buy the books and read the words? That's what the business is about, after all. But instead, they want autographs, cigar ends, bits of hair, bits of your time, bits of your life. Fandom, obsequious yet demanding, is not confined to the music business.

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Many writers learn to handle the idolatry of their fans with a measure of grace, but if you are a writer who has the misfortune to be dead as well as famous, matters are less under your control. "He became his admirers," wrote Auden on the death of Yeats. The dead poet has been ingested by his readers; from now on he will be read, admired, and loved in a different way—owned without opposition:

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QUESTION + ANSWER CHOICES

Herbert Gorman's 1939 biography of the writer James Joyce was written with Joyce's cooperation and published in Joyce's lifetime. These facts would appear to *challenge* which of the following assertions made in the passage?

- A. Writers prefer that readers concentrate on the writer's work rather than read a biography of the writer.
- B. As they grow older, writers fear what will become of their reputation after death.
- C. Biographies cannot explain the meaning of a writer's work.
- D. Most readers are interested in reading about the lives of their favorite writers.

SOLUTION

Solution: The correct answer is A.

- This is a Reasoning Beyond the Text question, because it introduces information that is not in the passage and asks you to consider how that information would affect passage claims. The answer is A, because the quotation describes a biography of James Joyce that Joyce agreed to and saw published in his own lifetime. The passage author argues that writers are usually uncomfortable with the prospect of biography because they want "to concentrate the reader's mind on the work" (paragraph 5). So if Joyce cooperated with the biography, this would challenge the author's claim that writers prefer to avoid biographers and to emphasize their works instead.
- This claim is not challenged by Joyce's participation in the biography. In fact, one could argue that he might have participated in an effort to exert control over what the biography would contain.
- Joyce's participation in the project does not challenge (or support) this claim.
- Joyce's participation in the biography does not challenge this assertion of reader interest.

Khan Academy Lessons

Correct
Correct Answer Your Answer

A

✓

B

C

D

Time Spent:

55 secs

Difficulty Level:

Easy

Content & Skills:

Skill: RBT

HUM

LIT

Question 70 of 230

Data ID: 531166

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PASSAGE

Passage 2 (Questions 6-12)

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QUESTION • ANSWER CHOICES

According to the passage, if a reader who thinks highly of Virginia Woolf's fiction came upon a biography of Woolf, that reader should:

- A.
use the biography as a way of ascertaining the meaning of Woolf's fiction.
- B.
read the biography as a replacement for reading Woolf's fiction.
- C.
approach the biography as faithfully as people in earlier times revered biographies of the saints.
- D.
hesitate to read the biography, because to do so would be to betray Woolf.

SOLUTION

Solution: The correct answer is D.

1. The author explicitly challenges the idea that biography offers insight into a writer's work.
2. This option is precisely the opposite of what the passage would recommend.
3. The author suggests that in contemporary times, literary biography has perhaps come to replace the study of the lives of the saints, but she certainly does not advocate this approach to literary biography.
4. **This is a Reasoning Beyond the Text question because it asks you to apply passage arguments to a hypothetical scenario that is not discussed in the passage. The answer is D because the passage author argues explicitly that "the only time [a reader is] truly close [to a writer] is when [the reader is] reading words on the page . . . the rest is dilution, marginality, betrayal." In the next sentence, the author makes clear that biography is included among other forms of souvenir collecting. Thus, based on the passage, a reader should not read the biography of Woolf she has found; in the passage author's mind, doing so would constitute a betrayal of Woolf.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct

Correct Answer Your Answer

A
B
C
D
✓

Time Spent:
53 secs
Difficulty Level:
Easy
Content & Skills:
Skill: RBT
HUM
LIT

Question 71 of 230

Data ID: 631170
Captured: 6/9/2026, 4:40:33 PM

PASSAGE

Passage 2 (Questions 6-12)

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QUESTION + ANSWER CHOICES

According to the passage, a fan of William Faulkner who buys a pipe that allegedly belonged to Faulkner should:

- A. accept the authenticity of the pipe without question.
- B. realize that the purchase has nothing to do with appreciating Faulkner's work.
- C. investigate the authenticity of the pipe thoroughly before agreeing to purchase it.
- D. preserve the pipe as carefully as a saint's relics.

SOLUTION

Solution: The correct answer is B.

- The authenticity of the pipe is not really relevant here, but as the question suggests that the pipe "allegedly belonged to Faulkner," and the passage contains the story of the fraudulent locks of Stevenson's, the passage certainly does not support this option.
- This is another Reasoning Beyond the Text question, as it asks you to apply passage logic to a hypothetical scenario not discussed in the passage. The answer is B, because the passage author argues explicitly that relics of an author's life have nothing to do with appreciating or understanding that author's work ("only the work can really explain the work," she says in the final paragraph). So the fan of Faulkner who has purchased a pipe that is supposed to have once belonged to Faulkner should recognize that this souvenir will offer no insight into Faulkner's work.
- Again, the authenticity of the pipe is not the author's concern; she argues that souvenirs (such as the pipe) offer no help in appreciating an artist's work. That comes only from immersing oneself in the work, according to the author.
- The author rejects the equation of artists and saints, so this option is not supported.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct

Correct Answer Your Answer

A
B

✓

C

D

Time Spent:

21 secs

Difficulty Level:

Easy

Content & Skills:

Skill: R&T

HUM

LIT

Question 72 of 230

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PASSAGE

Passage 3 (Questions 13-19)

News of the French Revolution came to a Portugal that was struggling to return to the quiet order of the Middle Ages after the violent and scandalous attempt of the Marquis de Pombal to bring it abreast, in culture and law, with the France of Louis XV and the Spain of Charles III. The Pyrenees obstructed the flow of ideas between France and the Peninsula; the movement of ideas from Spain was hindered by Spain's recurrent eagerness to swallow her sister state; and in both countries the agents of the Inquisition loomed like lions at a palace gate to repel any word or thought that might question the ancient creed.

At the bottom of the social scale stood other guardians of the past: the simple, mostly unlettered commoners—peasants, craftspeople, tradespeople, soldiers—who were fondly habituated to their transmitted faith, comforted by its legends, awed by its miracles, thrilled by its ritual. At the top were the feudal barons, models of manners and owners of the soil, a timid, feeble-minded Queen Maria Francisca and her son John, regent (1799) and then (1816–1826) king—all dependably protective of the Church as the indispensable support of private morals, social order, and absolute, divine-right monarchy.

Amid these diverse sentinels lurked a small minority—students, Freemasons, scientists, poets, businesspeople, a few officials, even a noble or two—who were irked by the despotism of the past, furtively flirted with philosophy, and dreamed of representative government, free trade, free assembly, free press, free thought, and a stimulating participation in the International of the Mind.

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But when Louis XVI was deposed by a Paris uprising (1792), Queen Maria felt her throne tremble and surrendered the government to her son. The future John VI turned with fury on the liberals of Portugal and encouraged his intendant of police to arrest, or expel, or keep under unremitting surveillance, every Freemason, every important alien, every writer who advocated political reform. Francisco da Silva, leader of the liberals, was imprisoned; liberal nobles were banished from the court; Manuel du Bocage, leading Portuguese poet of the age, who had written a powerful sonnet against despotism, was jailed in 1797 and supported himself in prison by translating Ovid and Virgil. In 1793, infuriated by the execution of Louis XVI, the Portuguese government followed Spain in a holy war against France and sent a squadron to join the British fleet in the Mediterranean. Soon Spain negotiated a separate peace (1795); Portugal asked for a like accommodation, but France refused, alleging that Portugal was in effect a colony and ally of England. The quarrel simmered till Napoleon, after conquering half of Europe, reached out for the little state that was refusing to join in his Continental blockade of Britain.

Adapted from W. Durant and A. Durant, *The Age of Napoleon*. ©1975 by Simon and Schuster.

QUESTION + ANSWER CHOICES

Which of the following groups played a role in conserving the older social, religious, and political order in Portugal?

1. Peasants
2. Freemasons
3. The royal family

- A.
II only
- B.
III only
- C.
I and III only
- D.
II and III only

SOLUTION

Solution: The correct answer is C.

1. In paragraph 3, the author characterizes the “small minority” of the population “irked by the despotism of the past” as a group of students and Freemasons. Thus, proposition II is incorrect.
2. Proposition III is correct, but so is Proposition I, so this option is incorrect.
3. **This is a Social Science passage, which falls under the content category, “History.” This is a Comprehension question because it tests your understanding of passage language and argument. C is the answer because Commoners and peasants are referred to as “other guardians of the past” (paragraph 2); therefore, proposition I is correct. The opening paragraph of the passage makes clear that “Queen Maria Francisca and her son [were] dependably protective of the Church as the indispensable support of private morals, social order.” Proposition III is, therefore, also correct. Option C is correct.**
4. Proposition II is incorrect, so this option is incorrect.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C

✓

D

Time Spent:

5 mins 9 secs

Difficulty Level:

Easy

Content & Skills:

Skill: CMP

HIS

SSC

Question 73 of 230

Data ID: 5311179
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PASSAGE

Passage 3 (Questions 13-19)

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QUESTION + ANSWER CHOICES

It may be inferred from the passage that Portugal might have avoided the unfavorable attention of Napoleon if it had:

- A: ~~refused to have commerce with Britain.~~
- B: supported the French Revolution more enthusiastically.
- C: ~~not expelled French aliens.~~
- D: not followed Spain into war in the Mediterranean.

SOLUTION

Solution: The correct answer is A.

- This is a Reasoning Beyond the Text question because it asks you to use passage information to assess the effects of a hypothetical scenario not discussed in the passage. The last sentence of the passage contains the statement that “Napoleon, after conquering half of Europe, reached out for the little state that was refusing to join in his continental blockade of Britain.” The implication, then, is that if Portugal had joined the blockade, it might have avoided Napoleon’s action. Option A is correct.
- Some constituents of Portugal did support the French Revolution enthusiastically, according to paragraph 4.
- The passage does mention that Portugal arrested, expelled, or kept under surveillance “every important alien” (paragraph 5), but this mention is not made in connection with Napoleon’s action.
- Again, the passage does say that “the Portuguese government followed Spain in a holy war against France” (paragraph 5), but Spain negotiated a peace with France. When “Portugal asked for a like accommodation. . . . France refused” (paragraph 5). However, nothing in the passage suggests that this refusal was based on Portugal’s entry into war.

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RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
X
Time Spent:
1 min 38 secs

Difficulty Level:
Expert
Content & Skills:
Skill: RBT
HIS
SSC

Question 74 of 230

Data ID: 531182
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PASSAGE

Passage 3 (Questions 13-19)

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QUESTION • ANSWER CHOICES

Which of the following consequences would have been most likely had the Church ceased to support divine-right monarchy?

- A. The common people would have fallen away from the Church.
- B. Freemasonry would have picked up the former political role of the Church.
- C. ~~Revolution could have been prevented without John's stringent measures.~~
- D. John would have included priests in his list of political suspects.

SOLUTION

Solution: The correct answer is D.

- The common people “were fondly habituated to their transmitted faith, comforted by its legends, awed by its miracles, thrilled by its ritual” (paragraph 2), and there is no indication that this allegiance is in any way tied specifically to Church support of divine-right monarchy. There is nothing to suggest that a change in the Church’s support would weaken its appeal to the common people.
- Freemasons “dreamed of representative government” (paragraph 3), it is unlikely they would have taken over the role of supporting divine-right monarchy had the Church ceased to do so.
- It is unclear, given the information provided in the passage, whether the driving purpose of revolution in Portugal would have been, or would have been only, to do away with divine-right monarchy. The attempt of the Marquis de Pombal was to bring Portugal “abreast, in culture and law,” (paragraph 1) with France and Spain. Culture and law at the time seems to have been defined as much as (if not more so) by the Church as it was by the monarchy, including the queen’s son.
- This is a Reasoning Beyond the Text question because it asks you to use passage information to assess the potential effects of a scenario not described in the passage. The answer is D; paragraph 5 supports this choice with its mention of John’s “fury” against reformers and his instructions to police to “to arrest, or expel, or keep under unremitting surveillance” every dissident writer including nobles.

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D



Time Spent:
1 min 27 secs
Difficulty Level:
Difficult
Content & Skills:
Skill: RBT
HIS
SSC

Question 75 of 230

Data ID: 531185
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PASSAGE

Passage 3 (Questions 13-19)

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QUESTION + ANSWER CHOICES

Which of the following facts cited in the passage gives the strongest support for the claim that Portuguese liberals were in the minority politically?

- A:
~~The queen and her son were opposed to them.~~
- B:
Several important liberals were jailed by John.
- C:
~~The Inquisition was still active in Portugal.~~
- D:
The commoners were politically conservative.

SOLUTION

Solution: The correct answer is D.

1. This would be correct only if the passage indicated that the queen and her son were opposed to all political minorities, but the passage doesn't do so.
2. This does not address whether or not these liberals were in the minority.
3. The Inquisition is a conservative counter-force in opposition to liberalism, but it by no means logically follows that its existence necessarily indicates that liberals were in the minority.
4. This is a Reasoning Within the Text question, because it asks you to assess the strength of evidence presented in the text. The correct answer is D, because paragraph 2 defines the components of the commoners as a highly inclusive segment of the populace: "peasants, craftspeople, tradespeople, soldiers." The aggregate number of their ranks would, therefore, vastly outnumber the relatively few groups of individuals (including scientists and poets) that comprise the liberal faction alluded to in the passage (paragraph 3).

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
2 mins 21 secs
Difficulty Level:

Difficult
Content & Skills:
Skill: RWT
HiS
SSC

Question 76 of 230

Data ID: 531188
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PASSAGE

Passage 3 (Questions 13-19)

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QUESTION + ANSWER CHOICES

After the Napoleonic Wars, the Portuguese people demanded democratic reforms, including major restrictions on the power of monarchs. Together with information in the passage, this fact most clearly indicates that:

- A. John had good reason to fear that new ideas from abroad would undermine monarchical power.
- B. the Catholic Church had lost its stabilizing social influence.
- C. the policies of Napoleon had become increasingly popular in Portugal.
- D. Freemasonry was coming into the social and political mainstream.

SOLUTION

Solution: The correct answer is A.

- This is a Reasoning Beyond the Text question, because it provides information that is not included in the passage, and asks you to integrate it with passage information to draw a conclusion. The answer is A. The passage describes John's repressive reaction after the French king Louis XVI was deposed by a popular uprising in 1792: "the future John VI turned with fury on the liberals of Portugal" (final paragraph). So, if the Portuguese people "demanded . . . major restrictions" on monarchical power after the Napoleonic Wars, this certainly suggests that John's earlier fears that the power of the throne could be undermined by "new ideas from abroad" had some rational basis.
- The passage says that the monarchs depended on the Church for its support of "social order and absolute, divine-right monarchy," but the demands of the Portuguese people for democratic reforms cannot necessarily be seen as an indication that the Catholic Church "had lost its stabilizing social influence" in Portugal. It may have been that other influences just became stronger at that moment. Further, the Church may have retained some aspects of its influence, without maintaining its sway on the question of divine right.
- The passage provides no conclusive link between the policies of Napoleon and support for "democratic reforms." Because these demands for reform came after the Napoleonic Wars does not mean that they were inspired by support for Napoleon's positions.
- Although the Freemasons were part of the "minority" who were "irked by the despotism of the past," a move away from such despotism does not necessarily imply support for the Freemasons' ideas specifically.

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B

C
D
Time Spent:
1 min 31 secs
Difficulty Level:
Difficult
Content & Skills:
Skill: RBT
HIS
SSC

Question 77 of 230

Data ID: 531190
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PASSAGE

Passage 3 (Questions 13-19)

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QUESTION + ANSWER CHOICES

Which of the following facts most strongly supports the authors' image of John VI as resistant to social change in his realm?

- ~~A.~~
~~When Napoleon invaded Portugal, he fled to exile in Brazil.~~
- B.**
On returning from exile, he agreed to reforms but quickly revoked them.
- ~~C.~~
~~As soon as he returned to power, Brazil declared its independence from Portugal.~~
- ~~D.~~
~~After his return, he reigned as monarch of Portugal for only five years.~~

SOLUTION

Solution: The correct answer is B.

- To assert that this option was correct, we would need to know more about conditions in Brazil – for example, whether it was like France and Spain, in culture and law, or like Portugal.
- This is a Reasoning Within the Text question, as it asks you to evaluate the strength of evidence for passage claims. The answer is B. “Reforms,” especially in the context of this passage, represent clear efforts at making progressive social changes. Retraction of such measures by a highly repressive King (possibly initially consented to as the price for being allowed to return from exile) would strongly illustrate this king's resistance to change.
- Without a link being established between John and Portugal, on the one hand, and Brazil on the other, this option would be irrelevant.
- To assert this option, we would need to know why he stopped reigning as monarch after five years. This information is not included in the passage.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
X
B
C
D
Time Spent:
1 min 36 secs
Difficulty Level:
Moderate

Content & Skills:
Skill: RWT
HiS
SSC

Question 78 of 230

Data ID: 531191
Captured: 6/3/2026, 4:40:37 PM

PASSAGE

Passage 3 (Questions 13-19)

News of the French Revolution came to a Portugal that was struggling to return to the quiet order of the Middle Ages after the violent and scandalous attempt of the Marquis de Pombal to bring it abreast, in culture and law, with the France of Louis XV and the Spain of Charles III. The Pyrenees obstructed the flow of ideas between France and the Peninsula; the movement of ideas from Spain was hindered by Spain's recurrent eagerness to swallow her sister state; and in both countries the agents of the Inquisition loomed like lions at a palace gate to repel any word or thought that might question the ancient creed.

At the bottom of the social scale stood other guardians of the past: the simple, mostly unlettered commoners—peasants, craftspeople, tradespeople, soldiers—who were fondly habituated to their transmitted faith, comforted by its legends, awed by its miracles, thrilled by its ritual. At the top were the feudal barons, models of manners and owners of the soil, a timid, feeble-minded Queen Maria Francisca and her son John, regent (1799) and then (1816–1826) king—all dependably protective of the Church as the indispensable support of private morals, social order, and absolute, divine-right monarchy.

Amid these diverse sentinels lurked a small minority—students, Freemasons, scientists, poets, businesspeople, a few officials, even a noble or two—who were irked by the despotism of the past, furtively flirted with philosophy, and dreamed of representative government, free trade, free assembly, free press, free thought, and a stimulating participation in the International of the Mind.

Upon that timid minority, those shocked commoners, those startled dignitaries and inquisitors, the news of the French Revolution, however dulled by delay, came as an exhilarating or terrifying revelation. Some reckless spirits openly rejoiced; Masonic lodges in Portugal celebrated the event; the Portuguese ambassador in Paris, who may have read Rousseau or heard Mirabeau, applauded the French National Assembly; the Portuguese Minister for Foreign Affairs allowed the official gazette to salute the fall of the Bastille; copies of the Revolutionary Constitution of 1791 were sold by French booksellers in Portugal.

But when Louis XVI was deposed by a Paris uprising (1792), Queen Maria felt her throne tremble and surrendered the government to her son. The future John VI turned with fury on the liberals of Portugal and encouraged his intendant of police to arrest, or expel, or keep under unremitting surveillance, every Freemason, every important alien, every writer who advocated political reform. Francisco da Silva, leader of the liberals, was imprisoned; liberal nobles were banished from the court; Manuel du Bocage, leading Portuguese poet of the age, who had written a powerful sonnet against despotism, was jailed in 1797 and supported himself in prison by translating Ovid and Virgil. In 1793, infuriated by the execution of Louis XVI, the Portuguese government followed Spain in a holy war against France and sent a squadron to join the British fleet in the Mediterranean. Soon Spain negotiated a separate peace (1795); Portugal asked for a like accommodation, but France refused, alleging that Portugal was in effect a colony and ally of England. The quarrel simmered till Napoleon, after conquering half of Europe, reached out for the little state that was refusing to join in his Continental blockade of Britain.

Adapted from W. Durant and A. Durant, *The Age of Napoleon*. ©1975 by Simon and Schuster.

QUESTION + ANSWER CHOICES

The authors imply that the Portuguese ambassador applauded the French National Assembly because of genuine sympathy for their ideals. Alternatively, the ambassador's action could be explained as indicating that:

- **A.**
living in France, he was in fear of the new revolutionary French government.
- **B.**
his life outside of Portugal had broadened his political sympathies.
- **C.**
he was at odds with the medieval policies of Queen Maria.
- **D.**
he had no strong political commitments one way or the other.

SOLUTION

Solution: The correct answer is **A**.

1. This is a Reasoning Beyond the Text question as it asks you to identify the novel situation most consistent with passage information. This is the most plausible: pretending support for those with power over you can offer protection from them. Option A is correct.
2. Since this is noted in paragraph 4, it cannot be an alternative explanation, which is what the question requires.
3. Although being at odds with the medieval policies of Queen Maria might imply having genuine sympathy for the ideals of the French National Assembly (an opposition between the two is implied in the passage), "imply" doesn't mean "explain"—and the question asks what can explain the ambassador's action.
4. If he had no strong political commitments one way or the other, one would think he would be neutral, abstaining from applause one way or the other (unless he feared a power).

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
X
C
D
Time Spent:
1 min 24 secs
Difficulty Level:

Expert
Content & Skills:
Skill: RBT
HiS
SSC

Question 79 of 230

Data ID: 531198
Captured: 6/3/2026, 4:40:38 PM

PASSAGE

Passage 4 (Questions 20-26)

Photography is a medium with endless possibilities: Anything can be said and any purpose served by it. Objects are discrete, but photographic images join them. By means of a photograph, an exploding bomb becomes an advertisement for a safe. Two premises underlie the presumption in the United States that everything is material for the camera: There is nothing that should not be seen; there is nothing that should not be recorded.

Seen through the acute eye of the camera, any object acquires beauty or appears interesting—the most mundane subject constitutes art. The camera empowers everyone to make artistic judgments about importance, interest, or beauty, to assert “that would make a good picture.” Photographs can also be useful, facilitating estimates, decisions, and predictions. Cameras implement the instrumental view of reality by condensing information that enables quick, informed responses for a variety of purposes. Video equipment can be used to identify criminal offenders; X-rays can help to save lives.

The aesthetic approach and the instrumental approach evoke incompatible feelings about human beings and situations—a contradiction inevitable in any society that divorces the sphere of individual contemplation and expression from the sphere of surveillance and social utility. Picture-taking lends itself brilliantly to both realms. Cameras capture and “fix” impressions from that mythical space known as private perception. Cameras also arm vision in the service of power—of the state, of industry, of science.

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The logic of consumption provides the ultimate reason for photographing everything in modern, capitalist society. To consume means to *burn*, to use up—and, therefore, to *require replacement*. As images are made and consumed, the consumers need more and more of them. Images are manufactured at an ever-faster rate. Cameras are both the antidote and the disease. Photographic images add to the natural world the manufactured images that help to bolster a depleted sense of reality. But by so doing, these images further deplete the real world, making it appear drab and obsolete by comparison.

The increasing supply of photographic images, never exhausted by consumerist waste, can overwhelm and threaten to obliterate reality. Because photographic images have come to resemble real things to a greater degree than anyone could have predicted, a distinction between images and real objects, between copies and originals, is less and less plausible. The remedy lies in a new form of conservation—a conservation of manufactured images, as well as natural resources, that would provide ecological balance between the real world and the re-created one.

Adapted from S. Sontag, *A Sivan Sontag Reader*. ©1983 Susan Sontag.

QUESTION + ANSWER CHOICES

The author’s main purpose in exploring the aesthetic and the instrumental approaches to photography is to suggest that a new form of conservation must:

- A. allow ordinary people to make judgments about beauty.
- B. increase constantly the production of entertaining images.
- C. solve the problem of the continual proliferation of images.
- D. develop a coherent ideology to facilitate the task of governing.

SOLUTION

Solution: The correct answer is C.

1. The author asserts that “The camera empowers everyone to make artistic judgments about importance, interest, or beauty” (paragraph 2). This happens, according to the passage, each time someone decides “that would make a good picture.” In other words, the author asserts that this is already happening; it is not the goal of a “new form of conservation.”
2. The problem, for which “a new form of conservation” is the solution, is due to “the increasing supply of photographic images” (paragraph 6). Therefore, it wouldn’t make sense, for a new form of conservation to increase the production of images; instead a solution would involve a decrease in the production of images.
3. **This is a Social Science passage, belonging to the content category “Sociology.” It is a Comprehension question, as it asks you to identify an aspect of the author’s argument. The answer is C. The author postulates that the increasing supply of photographic images is a problem because it “can overwhelm and threaten to obliterate reality” (final paragraph). It threatens reality because these images do not actually depict “reality”; they are framed and positioned versions made by the picture-taker and reproduced by some form of media. According to the passage author, these second-hand, mediated images confront individuals in modern technological society, outnumbering and drowning out direct perceptions. “The remedy lies in a new form of conservation” (paragraph 6) that would reduce the number of artificial images and restore balance between what is actually seen with the naked eye and what is staged or manufactured by means of technology. Option C is the best answer.**
4. The only references to, or examples of, the use of images as an instrument for governing relate to surveillance, verification, or documentation of identity; for example, the photographic image is referred to as “an object of scrutiny to assist officials responsible for governing.” In terms of the “instrumental approach” the images are part of a “sphere of surveillance and social utility.” Conserving or limiting the number of these images would therefore result in less rather than more control by the government. It would not, moreover, further any “coherent ideology.”

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct

Correct Answer Your Answer

A
B
C
✓
D

Time Spent:
4 mins 21 secs
Difficulty Level:
Moderate
Content & Skills:
Skill: CMP
SOC
SSC

Question 80 of 230

Data ID: 631201
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PASSAGE

Passage 4 (Questions 20–26)

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Seen through the acute eye of the camera, any object acquires beauty or appears interesting—the most mundane subject constitutes art. The camera empowers everyone to make artistic judgments about importance, interest, or beauty, to assert “that would make a good picture.” Photographs can also be useful, facilitating estimates, decisions, and predictions. Cameras implement the instrumental view of reality by condensing information that enables quick, informed responses for a variety of purposes: Video equipment can be used to identify criminal offenders; X-rays can help to save lives.

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Adapted from S. Sontag, *A Sontag Reader*. ©1983 Susan Sontag.

QUESTION + ANSWER CHOICES

The author’s analysis of image-making rests on the assumption that a capitalist society separates:

- A.
advertising from entertainment.
- B.
private perception from social order.
- C.
~~diseases that afflict society from their cause.~~
- D.
~~free political choice from free economic consumption.~~

SOLUTION

Solution: The correct answer is B.

1. The fourth paragraph says a capitalist society “needs to generate images of new commodities” [which refers to advertising] and “forms of entertainment in order to stimulate buying.” Both are used in order to stimulate buying; it’s not that the first, advertising, is used to stimulate buying, and the second is used to amuse; thus, the capitalist society does not separate the two, according to the passage author.
2. This is a Comprehension question, because it asks you to recognize an implicit assumption on which a passage argument is based. The answer is B. In paragraph 3, the author distinguishes the aesthetic approach from the instrumental approach to image-making. The first is matched up with private perception (“Cameras capture and ‘fix’ impressions from that mythical space known as private perception”); and the second linked to the social order (“cameras also arm vision in the service of power – of the state. . . .”). In the next paragraph, this dichotomy is described as a capacity to “subjectify” reality and to “objectify it.”
3. “Cameras are both the antidote and the disease” (paragraph 5) is the only reference to illness in the passage and in this metaphor the camera paradoxically acts as both. But it’s not the capitalist society that separates them, and the analysis doesn’t depend on the separation.
4. “The freedom to consume a variety of images and goods is equated with freedom itself” (paragraph 4); therefore, these two are NOT separated.

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓

C
D

Time Spent:
1 min 54 secs
Difficulty Level:
Expert
Content & Skills:
Skill: CMP
SOC
SSC

Question 81 of 230

Data ID: 631202
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PASSAGE

Passage 4 (Questions 20–26)

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QUESTION + ANSWER CHOICES

Which of the following best exemplifies the author’s assertion that, according to the instrumental approach, cameras can be used to present information that facilitates timely, accurate responses?

- A. Photographs taken to record family history in an album
- B. Photojournalism that produces pictures that tell an amusing story
- C. Photographs used in advertising campaigns to discourage teen-age smoking
- D. Photographs taken by cameras at stop lights to catch traffic offenders

SOLUTION

Solution: The correct answer is D.

1. Photographs to record family history don’t really facilitate “timely, accurate responses;” they preserve the past.
2. Although photojournalism may be timely, an amusing story doesn’t necessarily facilitate a response.
3. This may or may not elicit a response, but there’s nothing to suggest timeliness.
4. **This is a Reasoning Beyond the Text question, because it requires you to apply passage ideas to a novel set of circumstances. The answer is D. The information presented by the traffic camera is a photo of someone in the act of committing an offense, and the response facilitated by such information is a penalty of some sort. Furthermore, this option is close to the claim in the passage that cameras “implement the instrumental view of reality,” an instance of which is “video equipment...used to identify criminal offenders” (paragraph 2).**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓

Time Spent:
33 secs
Difficulty Level:
Easy
Content & Skills:
Skill: RBT
SOC
SSC

Question 82 of 230

Data ID: 631204
Captured: 6/3/2026, 4:40:39 PM

PASSAGE

Passage 4 (Questions 20–26)

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QUESTION + ANSWER CHOICES

Which of the following passage assertions suggests most strongly that the author is critical of capitalist society?

- A. Freedom to consume is equated with freedom itself.
- B. Photographic images make reality problematic.
- C. Social change is expressed by a change in images.
- D. Individual perception is a mythical realm.

SOLUTION

Solution: The correct answer is A.

1. This is a Reasoning Within the Text question because it asks you to assess the way that the author is using a piece of evidence in the passage. A is the correct answer. This statement is made to explain, in a capitalist society, the constant production and consumption of images (“The narrowing of free political choice to free economic consumption [that] requires the constant production and consumption of images” in paragraph 4.) Because the author contends that this increasing supply of images is what leads to the ‘disease’ of a depleted real world (paragraph 5) and our inability to distinguish between images and real objects (paragraph 6), it does suggest that the author is critical of capitalist society. Freedom or liberty, by definition, entails much more than the right to purchase commodities.
2. The term “problematic” can have the connotation of something “debatable,” this can be interpreted as remark that means reality is hard to define or pin down. Such a statement, therefore may not necessarily be a criticism of capitalism. For example, in philosophic inquiries life, love, God, truth, and many other abstractions are considered problematic, or open to inquiry and probing examination.
3. Whether or not this would imply criticism depends on the exact nature of the “social change.” If, for example, it involved increasing equality for disadvantaged segments of the population, its expression in images, helping to implement widespread perception of equality, might earn the passage author’s approval.
4. This observation, which is presented in paragraph 3, pertains to the author’s discussion of the aesthetic and instrumental approaches. Capitalist society isn’t mentioned at all until the next paragraph (4), and the two things are never discussed in relation to each other.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A

✓
B
C
D

Time Spent:
1 min 3 secs
Difficulty Level:
Difficult
Content & Skills:
Skill: RWT
SOC
SSC

Question 83 of 230

Data ID: 631207
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PASSAGE

Passage 4 (Questions 20-26)

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Adapted from S. Sontag, A Sontag Reader. ©1983 Susan Sontag.

QUESTION + ANSWER CHOICES

Suppose an artist created an exhibition consisting entirely of tastefully arranged collages of mug shots and fingerprints. This exemplifies the passage assertion that:

- A. pictures of exploding bombs can be used to advertise a safe.
- B. capitalist societies use photographs to help officials enforce laws.
- C. any photograph, regardless of intention, may be viewed as art.
- D. capitalist societies must provide abundant entertainment.

SOLUTION

Solution: The correct answer is C.

- The scenario in the question does not exemplify this example of a use of photographs.
- This option does not account for the fact that the photographs described in the question are being displayed as art.
- This is a Reasoning Beyond the Text question, because it presents a scenario not discussed in the passage and asks you how it would relate to passage arguments. The answer is C because the scenario in the question talks about items that are intended to identify people as part of the criminal justice system, and these photographs are now being used as part of an art exhibit. Thus, this exemplifies the passage claim that “any photograph, regardless of intention, may be viewed as art.”
- This example does not specifically relate to the ways in which the passage discusses capitalist societies.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
X
B
C
D
Time Spent:
1 min 25 secs

Difficulty Level:
Easy
Content & Skills:
Skill: RBT
SOC
SSC

Question 84 of 230

Data ID: 631208
Captured: 6/3/2026, 4:40:41 PM

PASSAGE

Passage 4 (Questions 20-26)

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The logic of consumption provides the ultimate reason for photographing everything in modern, capitalist society. To consume means to *burn*, to use up—and, therefore, to *require replacement*. As images are made and consumed, the consumers need more and more of them. Images are manufactured at an ever-faster rate. Cameras are both the antidote and the disease. Photographic images add to the natural world the manufactured images that help to bolster a depleted sense of reality. But by so doing, these images further deplete the real world, making it appear drab and obsolete by comparison.

The increasing supply of photographic images, never exhausted by consumerist waste, can overwhelm and threaten to obliterate reality. Because photographic images have come to resemble real things to a greater degree than anyone could have predicted, a distinction between images and real objects, between copies and originals, is less and less plausible. The remedy lies in a new form of conservation—a conservation of manufactured images, as well as natural resources, that would provide ecological balance between the real world and the re-created one.

Adapted from S. Sontag, *A Sivan Sontag Reader*. ©1983 Susan Sontag.

QUESTION + ANSWER CHOICES

Based on passage information, one can infer that implementing the author’s program for conservation (final paragraph) would most likely include suggestions for:

- A. limiting the use of computer images in architectural models.
- B. discouraging the use of photography in magazine advertisements.
- C. increasing police use of cameras to monitor suspected criminals.
- D. decreasing government support of art that might offend the public.

SOLUTION

Solution: The correct answer is B.

1. Architectural models are plans or blueprints for buildings rather than pictures of actual buildings. Therefore, the author would not be likely to single out architectural design as an area in which to implement a conservation program.
2. This is a Reasoning Beyond the Text question, because it asks you to apply passage information to a situation outside of the passage. B is the answer, because it suggests “discouraging the use of photography,” and not only that, discouraging its use in advertising, which, unlike architectural models, is emphasized in the passage. That said, it’s not the use of photographic images per se that is the problem, according to the passage, but the production of new images; if advertising used the same images, if it didn’t constantly replace images (paragraph 5), the problem described by the author would be lessened.
3. This option describes an increase in images and that would only increase the problem, as the author defines it.
4. Decreasing the support of art doesn’t necessarily decrease the production of art. Moreover, the issue in the passage argument pertains to the problem of too many images, not to those with potentially offensive content.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
X
B
C

D
Time Spent:
1 min 1 sec
Difficulty Level:
Difficult
Content & Skills:
Skill: RBT
SOC
SSC

Question 85 of 230

Data ID: 631210
Captured: 6/3/2026, 4:40:41 PM

PASSAGE

Passage 4 (Questions 20–26)

Photography is a medium with endless possibilities: Anything can be said and any purpose served by it. Objects are discrete, but photographic images join them. By means of a photograph, an exploding bomb becomes an advertisement for a safe. Two premises underlie the presumption in the United States that everything is material for the camera: There is nothing that should not be seen; there is nothing that should not be recorded.

Seen through the acute eye of the camera, any object acquires beauty or appears interesting—the most mundane subject constitutes art. The camera empowers everyone to make artistic judgments about importance, interest, or beauty, to assert “that would make a good picture.” Photographs can also be useful, facilitating estimates, decisions, and predictions. Cameras implement the instrumental view of reality by condensing information that enables quick, informed responses for a variety of purposes: Video equipment can be used to identify criminal offenders; X-rays can help to save lives.

The aesthetic approach and the instrumental approach evoke incompatible feelings about human beings and situations—a contradiction inevitable in any society that divorces the sphere of individual contemplation and expression from the sphere of surveillance and social utility. Picture-taking lends itself brilliantly to both realms. Cameras capture and “fix” impressions from that mythical space known as private perception. Cameras also arm vision in the service of power—of the state, of industry, of science.

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The logic of consumption provides the ultimate reason for photographing everything in modern, capitalist society. To consume means to *burn*, to use up—and, therefore, to *require replacement*. As images are made and consumed, the consumers need more and more of them. Images are manufactured at an ever-faster rate. Cameras are both the antidote and the disease. Photographic images add to the natural world the manufactured images that help to bolster a depleted sense of reality. But by so doing, these images further deplete the real world, making it appear drab and obsolete by comparison.

The increasing supply of photographic images, never exhausted by consumerist waste, can overwhelm and threaten to obliterate reality. Because photographic images have come to resemble real things to a greater degree than anyone could have predicted, a distinction between images and real objects, between copies and originals, is less and less plausible. The remedy lies in a new form of conservation—a conservation of manufactured images, as well as natural resources, that would provide ecological balance between the real world and the re-created one.

Adapted from S. Sontag, *A Sontag Reader*. ©1983 Susan Sontag.

QUESTION + ANSWER CHOICES

Recent technological developments like high-resolution satellite imagery and diagnostic positron emission tomography (PET scans) have refined and extended the camera’s capacity to provide information. Which passage assertion does this information support most strongly?

- A. Cameras can illuminate the private, personal lives of individuals.
- B. ~~Capitalist consumption requires the unlimited production of images.~~
- C. ~~Cameras are a means of appropriating reality and making it obsolete.~~
- D. Photography can be used to both control and benefit society.

SOLUTION

Solution: The correct answer is D.

- The examples in the question provide macro- or micro-level scientific information, which is highly impersonal and quite different from “private life,” so this option is incorrect.
- The information provided in the question doesn’t support this option because satellite images and PET scans are not really relevant to capitalist consumption. Moreover, they are unlikely to be characterized as “unlimited.”
- The information provided in the question does not relate to making reality obsolete or out-of-date.
- This is a Reasoning Beyond the Text question, because it asks you to identify information from outside the passage that works to support passage claims. Satellite images of offenders such as speeding drivers could, for example, be used to both control (by penalizing offenders) and benefit society (by protecting them from offenders); PET scans could be used to benefit society by revealing information about the human anatomy. Option D is the correct answer.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓

Time Spent:
1 min 41 secs
Difficulty Level:
Moderate
Content & Skills:
Skill: RBT
SOC
SSC

Question 86 of 230

Data ID: 631213
Captured: 6/3/2026, 4:40:42 PM

PASSAGE

Passage 5 (Questions 27-31)

Of all the ideas for improving education, few are as simple or attractive as reducing the number of pupils per teacher. However, class-size reduction has one obvious drawback: It costs plenty. It requires more teachers and possibly more classrooms and equipment. These expenses can dwarf the price of alternative schemes, such as testing teachers or increasing pay as a means of attracting better candidates. So before deciding to improve schools by reducing class sizes, it is important to answer certain questions: Do class size reductions actually improve student achievement? If so, under what conditions do they do the most good? And most important, how great is the benefit?

Existing data, such as records kept by the U.S. Department of Education, show that over the past few decades, class sizes have fallen substantially at all grade levels. However, data from national tests show no significant improvement in academic achievement over this period. But these statistics do not prove that class size has no effect on academic achievement. It is difficult to isolate the effects of class size from those of the myriad other factors that may influence student performance. In order to do so, controlled experiments must randomly assign large numbers of students and teachers to classes of different sizes. Over the years, hundreds of such experiments have been conducted. Most found evidence that smaller classes can benefit students, at least in the early grades, but only under certain conditions.

The conventional wisdom about small classes is that they minimize disruptions. They also free teachers to bestow individual attention and to use creative approaches, such as letting students work in small groups. Where discipline is not a problem, then, any achievement gains from reduced class sizes probably derive mainly from the teachers' use of methods that take advantage of smaller classes. But studies have found that educators rarely change their instructional styles to match class size. In fact, even a summer professional development program for teachers did not prompt participants to modify their teaching styles. Moreover, educators generally devote the same overall amount of time to individual instruction in small and large classes. With fewer kids in a class, each child gets a bigger share of that time, but the increase is not nearly enough to account for any significant differences in academic performance.

So what could account for the academic benefits of class-size reductions? One likely explanation is that some teachers, who already use methods well suited to smaller classes, do very well when they are actually given small classes. Their improved performance pulls up the average. This interpretation is consistent with findings that substantial performance gains from small classes occur in the early elementary grades and do not accumulate beyond first or second grade. Kindergarten and first-grade teachers tend to use educational techniques that work best in small classes, such as emphasizing hands-on projects and personal interactions.

Although reducing class sizes does appear to improve student performance in early grades, no one has studied the costs and benefits of class-size reductions in comparison to those of alternative educational reforms, such as raising teacher salaries. Legislators and school administrators need much more solid information on these relative costs and benefits before they can make sensible policy decisions. Let's hope they get it before spending billions of dollars more to reduce class sizes in ways that improve academic achievement inefficiently, or not at all.

Adapted from R. G. Ehrenberg, D. J. Brewer, A. Gamoran, and J. D. Wilms, Does class size matter? ©2001 Scientific American, Inc.

QUESTION • ANSWER CHOICES

The central thesis of the passage is that class-size reductions:

- A. can improve academic achievement under some conditions but may not be the most effective way to do so.
- B. improve academic achievement when teachers use techniques well suited to small classes.
- C. improve academic achievement only in the early elementary grades.
- D. are less efficient than other means of improving academic achievement.

SOLUTION

Solution: The correct answer is A.

1. This is a Social Science passage, which falls under the content category "Education." This Comprehension question tests your understanding of the main argument of the passage. The answer is A, as the passage argues that while there is some evidence that reductions in class size do improve academic performance, these reductions are shown to be effective "only under certain conditions" (paragraph 2). The author argues that because reductions in class size are more expensive than are some other alternatives for improving academic performance, researchers should do more to compare the costs and benefits of reductions in class size with those of other interventions "before they can make sensible policy decisions" (final paragraph).
2. The passage suggests that this is a "likely explanation" for the "academic benefits of class-size reductions," but this is not the central thesis of the passage.
3. The author says that research shows that performance gains from small classes do seem to "occur in the early elementary grades and do not accumulate" beyond these, but this is not the central thesis of the passage.
4. The passage does not make this claim, but suggests that the relative efficiency and effectiveness of different means of improving performance must be studied in much more detail to prevent wasted resources.

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:

2 mins 26 secs
Difficulty Level:
Easy
Content & Skills:
Skill: CMP
EDU
SSC

Question 87 of 230

Data ID: 631214
Captured: 6/3/2026, 4:40:43 PM

PASSAGE

Passage 5 (Questions 27-31)

Of all the ideas for improving education, few are as simple or attractive as reducing the number of pupils per teacher. However, class-size reduction has one obvious drawback: It costs plenty. It requires more teachers and possibly more classrooms and equipment. These expenses can dwarf the price of alternative schemes, such as testing teachers or increasing pay as a means of attracting better candidates. So before deciding to improve schools by reducing class sizes, it is important to answer certain questions: Do class size reductions actually improve student achievement? If so, under what conditions do they do the most good? And most important, how great is the benefit?

Existing data, such as records kept by the U.S. Department of Education, show that over the past few decades, class sizes have fallen substantially at all grade levels. However, data from national tests show no significant improvement in academic achievement over this period. But these statistics do not prove that class size has no effect on academic achievement. It is difficult to isolate the effects of class size from those of the myriad other factors that may influence student performance. In order to do so, controlled experiments must randomly assign large numbers of students and teachers to classes of different sizes. Over the years, hundreds of such experiments have been conducted. Most found evidence that smaller classes can benefit students, at least in the early grades, but only under certain conditions.

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Although reducing class sizes does appear to improve student performance in early grades, no one has studied the costs and benefits of class-size reductions in comparison to those of alternative educational reforms, such as raising teacher salaries. Legislators and school administrators need much more solid information on these relative costs and benefits before they can make sensible policy decisions. Let's hope they get it before spending billions of dollars more to reduce class sizes in ways that improve academic achievement inefficiently, or not at all.

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QUESTION • ANSWER CHOICES

The assertion that reducing class sizes requires more teachers (first paragraph) plays which of the following roles in the passage?

- A.
~~It establishes that smaller classes are the most expensive way to improve education.~~
- B.
It provides an explanation for the substantial costs of class-size reductions.
- C.
~~It is a conclusion supported by the observation that class-size reductions are quite expensive.~~
- D.
It establishes the need for more controlled studies on the cost-effectiveness of smaller classes.

SOLUTION

Solution: The correct answer is **B**.

- The passage does not say that class-size reductions are the “most expensive” way to improve academic performance. It does say that they are expensive and that their cost “can dwarf” the cost of other interventions, but it does not support this option.
- This is a Reasoning Within the Text question, because it asks you to evaluate the role played by a particular passage claim. The answer is **B**, because the passage reference to the greater number of teachers required in order to reduce class size comes right after the point that class-size reductions cost “plenty,” and implicitly explains what makes them cost so much.
- The author does not conclude that reducing class size requires more teachers as a result of observing that reducing class size is expensive; rather, the cost of additional teachers helps to explain the cost of the intervention.
- It is not the number of teachers required to reduce class size that establishes the need for more data on the cost-effectiveness of smaller classes. It is the cost of the interventions, relative to their effectiveness, as discussed in the passage.

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
3 mins 20 secs

Difficulty Level:
Moderate
Content & Skills:
Skill: RWT
EDU
SSC

Question 88 of 230

Data ID: 631217
Captured: 6/3/2026, 4:40:43 PM

PASSAGE

Passage 5 (Questions 27-31)

Of all the ideas for improving education, few are as simple or attractive as reducing the number of pupils per teacher. However, class-size reduction has one obvious drawback: It costs plenty. It requires more teachers and possibly more classrooms and equipment. These expenses can dwarf the price of alternative schemes, such as testing teachers or increasing pay as a means of attracting better candidates. So before deciding to improve schools by reducing class sizes, it is important to answer certain questions: Do class size reductions actually improve student achievement? If so, under what conditions do they do the most good? And most important, how great is the benefit?

Existing data, such as records kept by the U.S. Department of Education, show that over the past few decades, class sizes have fallen substantially at all grade levels. However, data from national tests show no significant improvement in academic achievement over this period. But these statistics do not prove that class size has no effect on academic achievement. It is difficult to isolate the effects of class size from those of the myriad other factors that may influence student performance. In order to do so, controlled experiments must randomly assign large numbers of students and teachers to classes of different sizes. Over the years, hundreds of such experiments have been conducted. Most found evidence that smaller classes can benefit students, at least in the early grades, but only under certain conditions.

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QUESTION • ANSWER CHOICES

The observation that some teachers use techniques that work best in small classes (paragraph 4) is used to support which of the following assertions from the passage?

- A. Substantial performance gains from small classes occur in the early elementary grades and do not accumulate beyond first or second grade.
- B. ~~Educators rarely change their instructional styles to match the size of their classes.~~
- C. ~~Small classes free teachers to bestow individual attention and use creative approaches, such as letting students work in small groups.~~
- D. When class sizes are reduced, the improved performance of teachers who already use methods well suited to smaller classes pulls up the average achievement level.

SOLUTION

Solution: The correct answer is D.

- The fact that early elementary grades are the grades that show evidence of improvement from small class size is a building block in the paragraph's larger argument, and not the point that the assertion quoted in the question is intended to make, so A is incorrect.
- The author does make this claim, but it is not the point supported by the claim that early grade teachers often use techniques that are well suited to small class size.
- This is a claim about the benefits of class size, which is neither supported nor challenged by the point about the teachers who use these techniques.
- This is a Reasoning Within the Text question because it asks you to understand the way that a particular piece of evidence is used in the passage. The assertion cited in the question is that early elementary grade teachers are most likely to use techniques that are most successful in smaller classes. The paragraph in which the assertion appears begins by arguing that teachers who already use techniques "well suited to smaller classes" do especially well when they actually have smaller classes, thus improving the average performance in smaller classes. The rest of the paragraph, including the assertion cited in the question, is meant to explain and support this argument, so the answer is D.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
2 mins 44 secs

Difficulty Level:
Moderate
Content & Skills:
Skill: RWT
EDU
SSC

Question 89 of 230

Data ID: 631220
Captured: 6/3/2026, 4:40:44 PM

PASSAGE

Passage 5 (Questions 27-31)

Of all the ideas for improving education, few are as simple or attractive as reducing the number of pupils per teacher. However, class-size reduction has one obvious drawback: It costs plenty. It requires more teachers and possibly more classrooms and equipment. These expenses can dwarf the price of alternative schemes, such as testing teachers or increasing pay as a means of attracting better candidates. So before deciding to improve schools by reducing class sizes, it is important to answer certain questions: Do class size reductions actually improve student achievement? If so, under what conditions do they do the most good? And most important, how great is the benefit?

Existing data, such as records kept by the U.S. Department of Education, show that over the past few decades, class sizes have fallen substantially at all grade levels. However, data from national tests show no significant improvement in academic achievement over this period. But these statistics do not prove that class size has no effect on academic achievement. It is difficult to isolate the effects of class size from those of the myriad other factors that may influence student performance. In order to do so, controlled experiments must randomly assign large numbers of students and teachers to classes of different sizes. Over the years, hundreds of such experiments have been conducted. Most found evidence that smaller classes can benefit students, at least in the early grades, but only under certain conditions.

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Adapted from R. G. Ehrenberg, D. J. Brewer, A. Gamoran, and J. D. Wilms, Does class size matter? ©2001 Scientific American, Inc.

QUESTION + ANSWER CHOICES

Assume that in a particular state, teachers at all grade levels emphasize hands-on projects and personal interactions with individual students. The passage suggests that in this state, class-size reductions would probably:

- A. be more beneficial than alternative educational reforms.
- B. improve student performance only in the early elementary grades.
- C. not substantially improve student performance on average.
- D. improve student performance at all grade levels.

SOLUTION

Solution: The correct answer is D.

- This is not supported by the passage; class-size reductions would still be costly, and there is not enough evidence in the passage to make this comparative claim, based on the scenario in the question.
- The improvements in the early elementary grades, according to the passage author, are due to the kinds of teaching techniques that teachers in those grades generally employ. So if teachers in all grades employed similar techniques, similar improvements would be expected in all grades, not only in the early grades.
- The passage links class-size-related improvements in student performance to the presence of teachers who use teaching techniques suited to small classes; the passage also says that the performance of these teachers and classes pulls up the average level of performance. So this option is incorrect, because performance improvements would be expected in this situation.
- This is a Reasoning Beyond the Text question because it presents a hypothetical scenario that is not discussed in the passage, then asks you to apply passage information to this scenario. The answer is D, because the passage author contends that improvements in student performance as a result of reductions in class size are found primarily in the early elementary grades, because teachers in those grades emphasize teaching techniques (such as hands-on projects) most suited to small classes. So, if teachers in all grades emphasized these techniques, then you would expect to see improvements in student performance due to class-size reductions at all grade levels, and not just in the early grades.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
X

C
D

Time Spent:
1 min 32 secs
Difficulty Level:
Expert
Content & Skills:
Skill: RBT
EDU
SSC

Question 90 of 230

Data ID: 631223
Captured: 6/3/2026, 4:40:44 PM

PASSAGE

Passage 5 (Questions 27-31)

Of all the ideas for improving education, few are as simple or attractive as reducing the number of pupils per teacher. However, class-size reduction has one obvious drawback: it costs plenty. It requires more teachers and possibly more classrooms and equipment. These expenses can dwarf the price of alternative schemes, such as testing teachers or increasing pay as a means of attracting better candidates. So before deciding to improve schools by reducing class sizes, it is important to answer certain questions: Do class size reductions actually improve student achievement? If so, under what conditions do they do the most good? And most important, how great is the benefit?

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QUESTION + ANSWER CHOICES

Which of the following findings would most *weaken* the author's explanation of the observed effects of class-size reductions?

- A. Class-size reductions in the early grades are effective regardless of whether the teachers use methods well suited to small classes.
- B. Raising teacher salaries produces much greater improvements in student performance than does reducing class sizes at all grade levels.
- C. When teachers emphasize personal interactions with students, academic achievement typically improves regardless of class size.
- D. Academic achievement levels are often low even when teachers use methods well suited to small classes.

SOLUTION

Solution: The correct answer is **A**.

- This is another Reasoning Beyond the Text question, because it asks which situation, listed in the options and not presented in the passage, would have a particular effect (in this case, a weakening effect) on a passage claim. The answer is A, because the author's explanation for improved performance in small classes rests on the fact that "some teachers, who already use methods well suited to smaller classes, do very well when they are actually given small classes." If, however, early grades showed improvements in student performance in smaller classes regardless of teaching techniques, then this would clearly weaken the author's explanation for the effects of class-size reductions.
- This is irrelevant to the author's explanation for the effects of class-size reductions.
- This does not weaken the author's explanation for the observed effects of class-size reductions; this describes an effect that seems to be independent of class size.
- Again, the author explains an "observed effect." This claim does not weaken that explanation. Further, the author discusses improvements in academic achievement based on class size; if achievement starts at a very low level, it may still be relatively low, even after improvement.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
1 min 5 secs

Difficulty Level:
Difficult
Content & Skills:
Skill: RBT
EDU
SSC

Question 91 of 230

Data ID: 631229
Captured: 6/3/2026, 4:40:45 PM

PASSAGE

Passage 6 (Questions 32-38)

As citizens of the *cosmo polis*, the mythical “world city,” professors are expected to owe no allegiance to geographical territory; they’re supposed to belong to the boundless world of books, ideas, and eternal truths, not the infinitely particular world of watersheds, growing seasons, and ecological niches. Most professors get their jobs through national searches, and although they may have geographical preferences, most are living wherever they can find work.

Though the majority of U.S. college students attend institutions in their home states, they are taught by this cosmopolitan class of transient exotics. It would be surprising if the rootlessness of the professorial class did not affect their students who go on to become the best-informed, most-educated portion of the U.S. public. Education certainly ought to broaden horizons, but it can and ought to do more. However we define that more, one thing seems clear: rootless professors are systematically ignorant of a key aspect of integrated life, the life that is, after all, a primary goal of a good liberal arts education. They are ignorant of the values of connectedness to place.

This ignorance gives preference to the voice of the abstract at the expense of the voice of the particular. This can only be done by dismissing out of hand the possibility that because moral questions come to us not in the abstract but as concrete, particular problems, solutions to moral questions will necessarily be particular and specific. If we cannot get rid of the causes of this problem, we can alleviate the symptoms. This may be the better course, for it presents the opportunity to keep what is good in both cosmopolitanism and provincialism.

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Second, professors should include local content in their courses. Not abstract theories about distant peoples, but concrete realizations about observable communities; not airy generalizations that transcend student experience and lie beyond their powers of criticism, but specific conclusions whose skeptical testing they can perform themselves; not social-science hearsay taken on faith, but evidence weighed critically, firsthand: these form the substance of a rooted education. Transforming the world immediately outside the classroom into a laboratory will tend to erase the artificial boundary between the roles of student and citizen, thereby encouraging the latter in the habits of the former.

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Adapted from E. Zenoxy, *The rootless professors*. ©1996 Yale University.

QUESTION + ANSWER CHOICES

Which of the following underlying assumptions about professors is implied by the passage argument?

- A. ~~Most of them are generalists, not specialists.~~
- B. Most of them lack a preference for particular geographic areas.
- C. Most of them care more about ideas than they do about actual places.
- D. Most of them in the U.S. attended college or graduate school outside their home states.

SOLUTION

Solution: The correct answer is C.

- The author talks about the abstract and ideal in contrast to the specific and particular, but this is not the same as identifying the professors as generalists rather than specialists on the basis of the overall scope and content of their professional endeavors.
- The author claims just the opposite: “. . . although they [most professors] may have geographical preferences” (paragraph 1).
- This is a Humanities passage, which belongs to the content category “Philosophy.” This is a Comprehension question, because it asks you to recognize a fundamental assumption of the passage argument. The author says that rootless professors (deemed to be most professors) “are ignorant of the values of connectedness to place” (paragraph 2), that they should include local content rather than “abstract theories about distant peoples” in their courses (paragraph 4), and that “what is needed is . . . educators willing to take root and cultivate a sense of place” (paragraph 6). All of these suggest that professors care more about ideas than they do about actual places. C is correct.**
- The author says that “the majority of U.S. college students attend institutions in their home states” (paragraph 2) and that professors often obtain their jobs (not their educations) outside their home states.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓

D
Time Spent:
3 mins 39 secs
Difficulty Level:
Difficult
Content & Skills:
Skill: CMP
HUM
PHI

Question 92 of 230

Data ID: 631232
Captured: 6/3/2026, 4:40:46 PM

PASSAGE

Passage 6 (Questions 32-38)

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QUESTION + ANSWER CHOICES

To fulfill the author’s view of an exemplary professor, which of the following subject areas would be most useful for economics professors to know outside of their discipline?

- A. Mathematics
- B. Sociology
- C. Political science
- D. Ecology

SOLUTION

Solution: The correct answer is D.

1. Mathematics is largely part of the world of “ideals and scholarship” (paragraph 6). It deals with theorems and abstractions rather than descriptive particulars of place.
2. Sociology often focuses on abstract theories (criticized by the author in paragraph 5) and is likely to be considered, by the author, as a repository of “social-science hearsay” (paragraph 5).
3. Political science, like mathematics, is largely part of the world of “ideals and scholarship” (paragraph 6). It generally deals with theory and abstractions rather than with descriptive particulars of place.
4. This is a Reasoning Beyond the Text question, because it asks you to apply the passage author’s argument to a situation outside of the text. D is the answer, because Ecology is less abstract than mathematics and the other academic disciplines. As the study of environmental factors, ecology deals with at least some of those concrete “particulars of place” that the passage author believes most professors are sorely out of touch with. A focus on environmental relationships would help redeem the professoriate from what the author argues is a “rootless” existence limited to “ideals and scholarship.”

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
X
C
D

Time Spent:
57 secs
Difficulty Level:
Expert
Content & Skills:
Skill: RBT
HUM
PHI

Question 93 of 230

Data ID: 631235
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PASSAGE

Passage 6 (Questions 32-38)

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Adapted from E. Zernoy, *The rootless professors*. ©1996 Yale University.

QUESTION + ANSWER CHOICES

Which of the following points in the passage is supported by a concrete example?

- A.
Most professors find their positions through national searches.
- B.
Hiring committees are prejudiced against hiring local candidates to teach at the same university they had attended.
- C.
the world outside the classroom into a laboratory will blur the line between “student” and “citizen.”
- D.
Most U.S. college students attend institutions in their home states.

SOLUTION

Solution: The correct answer is B.

1. This option is not supported by a concrete example.
2. **This is a Reasoning Within the Text question, because it asks you to assess the nature of the evidence provided in the passage. The answer is B, because the claim that university hiring committees are prejudiced against “local candidates” is supported by an anecdote in paragraph 4, in which a job candidate was told that the department (in his home state) from which he had received his degrees would not hire him permanently because they considered him “too much a native.” This is certainly a concrete example to support the claim in option B.**
3. This option is not supported by a concrete example.
4. This option is not supported by a concrete example.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
33 secs

Difficulty Level:
Easy
Content & Skills:
Skill: RWT
HUM
PHI

Question 94 of 230

Data ID: 631237
Captured: 6/3/2026, 4:40:47 PM

PASSAGE

Passage 6 (Questions 32-38)

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QUESTION + ANSWER CHOICES

The author would most likely favor a college course that:

- A. emphasized hands-on experience in a laboratory over textbook learning.
- B. crossed disciplines and combined creative arts with science.
- C. applied classroom concepts to a community garden project.
- D. required a semester of studying and living in another country.

SOLUTION

Solution: The correct answer is C.

1. This option does not include the local element that the author advocates; the laboratory could be anywhere.
2. The author does not argue for this kind of an interdisciplinary approach.
3. **This is a Reasoning Beyond the Text question, because it asks you to apply passage arguments to unfamiliar scenarios and determine what the passage author would most likely say about them. The answer is C, because the passage is an argument for grounding university education in a more “particular,” place-based ethos. The author contends that professors “should include local content in their courses.” Option C emphasizes participation in a local setting, exemplifies the environmental awareness the author advocates in the last paragraph, and fulfills the author’s prescription of “transforming the world immediately outside the classroom into a laboratory” (paragraph 4).**
4. The author advocates an immersion in a local environment, rather than the more “cosmopolitan” ethos that she believes characterizes academia in general.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C ✓
D
Time Spent:
17 secs

Difficulty Level:
Easy
Content & Skills:
Skill: RBT
HUM
PHI

Question 95 of 230

Data ID: 631238
Captured: 6/3/2026, 4:40:47 PM

PASSAGE

Passage 6 (Questions 32-38)

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QUESTION + ANSWER CHOICES

Based on information in the passage, which of the following would be the most likely reason behind the author’s suggested academic reforms?

- A.
To pay more careful attention to the relationship between humans and the world outside the classroom
- B:
~~To encourage students to attend colleges and universities in their home states~~
- C.
To replace a cosmopolitan university education with one that emphasizes the local
- D.
To give a voice to the particular at the expense of the voice of the abstract

SOLUTION

Solution: The correct answer is A.

1. This is a Comprehension question, because it asks you to recognize the author’s purpose in the passage. The answer is A, because the author says that to do anything less (than the suggested academic reforms) is to do a disservice to the future of human culture “in an age when humanity’s relationship to nature is so in need of careful, farsighted attention” (paragraph 6). Other parts of this paragraph, as well as the preceding one, emphasize the need for attention to the world outside the classroom. The author mentions, for example, “watersheds, growing seasons” and the importance of “transforming the world immediately outside the classroom into a laboratory.”
2. According to paragraph 2, students already attend colleges and universities in their home states.
3. The author does not advocate replacement, but addition: “academics ought to work to acquire a kind of dual citizenship – in the world of ideals and scholarship, yes, but also in the very real world of watersheds. . .” (paragraph 6).
4. This is incorrect for reasons similar to the ones that apply to option C; the author does not want to do away with the “abstract,” or theoretical; rather, he wants to correct what he sees as the over-emphasis on abstraction by drawing new attention to the particular and the local.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D

Time Spent:
1 min 41 secs
Difficulty Level:
Difficult
Content & Skills:
Skill: CMP
HUM
PHI

Question 96 of 230

Data ID: 631241
Captured: 6/3/2026, 4:40:48 PM

PASSAGE

Passage 6 (Questions 32-38)

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QUESTION + ANSWER CHOICES

The passage discussion of local content (paragraph 5) assumes which of the following?

- A. Students lack the ability to criticize “concrete realizations about observable communities.”
- B. ~~Materials that deal with social sciences should not be a part of a rooted education.~~
- C. There is an “artificial boundary between the roles of student and citizen.”
- D. Professors have traveled to distant communities more often than have the students they teach.

SOLUTION

Solution: The correct answer is C.

1. The author does not assume this; rather, she suggests that current curricular models have implicitly assumed this.
2. The author does not rule out social science itself; she objects to the way it has been presented to students, as something to be “taken on faith.”
3. **The answer is C. In paragraph 5, the author advocates a locally-based curriculum that is meant to allow students to do more than study “abstract theories [and] . . . airy generalizations. The author writes that this “rooted education . . . will tend to erase the artificial boundary between the roles of student and citizen.” Because the author seeks to erase this boundary, the author must believe that such a boundary has been erected and enforced to begin with. This is a Comprehension question, because it asks you to understand a passage claim and how it functions as part of the author’s argument.**
4. The passage does say that “the majority of U.S. college students attend institutions in their home states” (paragraph 2), but this does not mean that they may not have traveled extensively. The passage does not support this as an assumption of the author’s.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:

1 min 28 secs
Difficulty Level:
Difficult
Content & Skills:
Skill: CMP
HUM
PHI

Question 97 of 230

Data ID: 631244
Captured: 6/3/2026, 4:40:49 PM

PASSAGE

Passage 6 (Questions 32-38)

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This ignorance gives preference to the voice of the abstract at the expense of the voice of the particular. This can only be done by dismissing out of hand the possibility that because moral questions come to us not in the abstract but as concrete, particular problems, solutions to moral questions will necessarily be particular and specific. If we cannot get rid of the causes of this problem, we can alleviate the symptoms. This may be the better course, for it presents the opportunity to keep what is good in both cosmopolitanism and provincialism.

First, academia has to overcome its prejudice against the local and the provincial, so that its hiring committees do not include non-native status as an implicit qualification for employment. Such prejudice is dramatized in the case of one individual who, having received his degrees from the university in his home state and having worked at that institution as a lecturer on temporary contract, was told by the department chair that he was “too much a native” ever to become a permanent faculty member.

Second, professors should include local content in their courses. Not abstract theories about distant peoples, but concrete realizations about observable communities; not airy generalizations that transcend student experience and lie beyond their powers of criticism, but specific conclusions whose skeptical testing they can perform themselves; not social-science hearsay taken on faith, but evidence weighed critically, firsthand: these form the substance of a rooted education. Transforming the world immediately outside the classroom into a laboratory will tend to erase the artificial boundary between the roles of student and citizen, thereby encouraging the latter in the habits of the former.

Finally, academics ought to work to acquire a kind of dual citizenship—in the world of ideals and scholarship, yes, but also in the very real world of watersheds, growing seasons, migratory pathways, food chains, and dependency webs. What is needed is a class of cosmopolitan educators willing to live where they work and to work where they live, a class of educators willing to take root and cultivate a sense of place. These educators could then exemplify in their teaching and in their lives their own manner of accommodation to the fruitful tension between local and universal, particular and general, concrete and abstract. In an age when humanity’s relationship to nature is so in need of careful, farsighted attention, academics do a disservice to their students, and to the future of human culture on the planet, if they do anything less.

Adapted from E. Zenoxy, *The rootless professors*. ©1996 Yale University.

QUESTION + ANSWER CHOICES

Which of the following, if true, most weakens the author’s argument about the majority of college professors?

- A.
Most would prefer to have a career doing something other than teaching.
- B.
Most did not earn their degrees at the school where they teach.
- C.
Most do not teach in areas concerned with social issues.
- D.
Most teach in the city where they prefer to live.

SOLUTION

Solution: The correct answer is D.

1. This is irrelevant to the author’s argument about professors’ cosmopolitan, rather than grounded, ethos.
2. This supports, rather than challenges, the author’s claims.
3. The author does not restrict her argument to those professors who teach in areas specifically concerned with social issues, so this would be irrelevant to the author’s argument.
4. **This is a Reasoning Beyond the Text question; it asks how information not present in the passage, if true, would affect the passage argument about college professors. D is correct, because the passage author maintains that the majority of college professors are essentially indifferent to place, “expected to owe no allegiance to geographic territory” (first paragraph), and to belong instead to the “boundless world of ideas.” But if most professors taught in the cities where they preferred to live, they would be guided by the kind of grounded, local feeling that the argument believes they repudiate—and this would weaken the argument.**

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
33 secs

Difficulty Level:
Easy
Content & Skills:
Skill: RBT
HUM
PHI

Question 98 of 230

Data ID: 631249
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PASSAGE

Passage 7 (Questions 39-43)

To borrow a metaphor, actors are “weavers of dreams”—artists engaged in the creation of the intangible and, in their own estimation, set apart from other occupational groups. Most often, historians have allowed the dreams that actors weave to take precedence over the process of weaving. They have prioritized the moment of consumption over the moment of production. However, actors’ dealings with their employers, like those of many other laborers, were fraught with tension. Locked into an unequal relationship with the businessmen who controlled the U.S. theater industry, actors at every level of the theatrical hierarchy struggled for workplace equity. In August 1919, that struggle led to a work stoppage that left theaters across the U.S. dark for more than a month. The actors’ strike reveals much about early twentieth-century cultural production, showing how the concentration of economic power and the related centralization evident throughout the economy were transforming the nature of work in U.S. culture industries.

According to historian Stuart Ewen, a defining feature of the culture of consumption was the “obliteration of the factory” as the economic world was divided into an unpleasant sphere where goods were produced and a gratifying sphere where they were consumed. Performance, though, is unusual in that it is a labor process exhibited before and consumed by an audience. The individual actor both produces the commodity and embodies it—the weaver and the dream are inextricably bound together. Under normal circumstances, the system of production in theater masks this duality. In transforming actors into objects of consumption, it diverts attention from the economic realities they face. In the 1919 strike, however, the denizens of the U.S. stage were able to exploit their commodity status.

The roots of the strike lay in the radical restructuring of the U.S. theater industry in the late nineteenth century. At that time, *combination companies*, groups of actors touring the country in productions designed to showcase the talents of a single star, began to displace resident stock companies as the principal source of theatrical entertainment. Theater managers began to group their theaters into circuits, a strategy that strengthened their bargaining position immeasurably because they could book several weeks of a touring company’s business in a single transaction. In the wake of this shift, enterprising businessmen began to set themselves up as booking agents and eventually as producers.

In the reconfigured theatrical economy of the early twentieth century, actors’ labor was highly segmented. Most performers never made it onto the New York stage or toured with a combination company. The minority that did had to negotiate the vagaries of an occupational hierarchy that afforded vast salaries for its few stars but meager rewards to the far greater number who eked out a living at its base. Though the theatrical star system created the potential for rapid upward mobility, an absence of job security and frequent unemployment characterized the working lives of most actors.

The shop-floor experience of performers was largely determined by their position in the theatrical hierarchy. The position of the stars was analogous to that of the skilled craftsmen at the center of David Montgomery’s analysis of machine production in the late nineteenth century. Though these stars did highly repetitive labor, often reprising the same role eight times a week, the control that they exercised over their performances permitted them much functional autonomy. Women from the chorus lines, by contrast, had more in common with assembly-line workers. They were treated as interchangeable components in a complex productive process that stripped them of their individuality.

Adapted from S. P. Holmes, *All the world’s a stage!* The actors’ strike of 1919. ©2005 the Organization of American Historians.

QUESTION + ANSWER CHOICES

Based on the passage, during the 1919 strike, which aspect of acting were actors able to make use of that was NOT available to most other industrial workers?

- A.
Their role as “weavers of dreams”
- B.
Their role as commodities
- C.
Their role in producing an object in the culture industry
- D.
Their role in the “obliteration of the factory”

SOLUTION

Solution: The correct answer is B.

1. The actors’ role as “weaver of dreams” would tend to focus only on the aspect of their profession that is consumed—not on their role as producers.
2. **This is a Humanities passage that falls under the content category Theater. This is a Comprehension question, which tests your understanding of a paraphrase of passage material. The answer is B, because the author writes that, in the 1919 strike, actors (or “the denizens of the U.S. stage”) were “able to exploit their commodity status” (paragraph 2). The author explains that consumer culture functioned to obscure the labor that goes into making a commodity available for purchase or consumption. Actors were different from other industrial workers because their labor is what the audience also consumes, and the author maintains that in the strike of 1919, the actors were able to take advantage of their unusual position as both the producers of theater and as what is consumed by the audience.**
3. This is not what differentiates actors from other workers; other producers in the culture industry might not themselves be what is also consumed.
4. “The obliteration of the factory” is another way of talking about the divide between production and consumption created by the “culture of consumption” In the strike of 1919, Ewen argues, actors resisted that very divide, instead highlighting their position on the production side of the equation.

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D

Time Spent:
2 mins 13 secs
Difficulty Level:
Difficult
Content & Skills:
Skill: CMP
HUM
THE

Question 99 of 230

Data ID: 631254
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PASSAGE

Passage 7 (Questions 39-43)

To borrow a metaphor, actors are “weavers of dreams”—artists engaged in the creation of the intangible and, in their own estimation, set apart from other occupational groups. Most often, historians have allowed the dreams that actors weave to take precedence over the process of weaving. They have prioritized the moment of consumption over the moment of production. However, actors’ dealings with their employers, like those of many other laborers, were fraught with tension. Locked into an unequal relationship with the businessmen who controlled the U.S. theater industry, actors at every level of the theatrical hierarchy struggled for workplace equity. In August 1919, that struggle led to a work stoppage that left theaters across the U.S. dark for more than a month. The actors’ strike reveals much about early twentieth-century cultural production, showing how the concentration of economic power and the related centralization evident throughout the economy were transforming the nature of work in U.S. culture industries.

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The roots of the strike lay in the radical restructuring of the U.S. theater industry in the late nineteenth century. At that time, *combination companies*, groups of actors touring the country in productions designed to showcase the talents of a single star, began to displace resident stock companies as the principal source of theatrical entertainment. Theater managers began to group their theaters into circuits, a strategy that strengthened their bargaining position immeasurably because they could book several weeks of a touring company’s business in a single transaction. In the wake of this shift, enterprising businessmen began to set themselves up as booking agents and eventually as producers.

In the reconfigured theatrical economy of the early twentieth century, actors’ labor was highly segmented. Most performers never made it onto the New York stage or toured with a combination company. The minority that did had to negotiate the vagaries of an occupational hierarchy that afforded vast salaries for its few stars but meager rewards to the far greater number who eked out a living at its base. Though the theatrical star system created the potential for rapid upward mobility, an absence of job security and frequent unemployment characterized the working lives of most actors.

The shop-floor experience of performers was largely determined by their position in the theatrical hierarchy. The position of the stars was analogous to that of the skilled craftsmen at the center of David Montgomery’s analysis of machine production in the late nineteenth century. Though these stars did highly repetitive labor, often reprising the same role eight times a week, the control that they exercised over their performances permitted them much functional autonomy. Women from the chorus lines, by contrast, had more in common with assembly-line workers. They were treated as interchangeable components in a complex productive process that stripped them of their individuality.

Adapted from S. P. Holmes, *All the world’s a stage!* The actors’ strike of 1919. ©2005 the Organization of American Historians.

QUESTION + ANSWER CHOICES

Based on the description of performance in the second paragraph of the passage, which of the following occupations is most analogous to professional acting?

- A.
Weaver
- B.
Doctor
- C.
Sculptor
- D.
Massage therapist

SOLUTION

Solution: The correct answer is D.

1. A weaver produces a weaving, which the consumer can view or purchase without being aware of the nature of the labor that produced it.
2. A doctor does not generally embody his or her labor.
3. Like a weaver, a sculptor produces an object for consumption, which can be viewed (or “consumed,” in the terms of the passage) independently of the sculptor. Unlike an actor or a massage therapist, the sculptor does not embody his or her commodity.
4. This is a Reasoning Beyond the Text question, because it asks you to apply an idea from the passage to a novel scenario not discussed in the passage. The answer is D, because the passage describes performance as “unusual in that it is a labor process both exhibited before and consumed by an audience. The . . . actor both produces the commodity and embodies it.” This is most analogous to a massage therapist, who physically and visibly produces the commodity (the massage) for the consumer.

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
1 min 29 secs

Difficulty Level:
Expert
Content & Skills:
Skill: RBT
HUM
THE

Question 100 of 230

Data ID: 631257
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PASSAGE

Passage 7 (Questions 39-43)

To borrow a metaphor, actors are “weavers of dreams”—artists engaged in the creation of the intangible and, in their own estimation, set apart from other occupational groups. Most often, historians have allowed the dreams that actors weave to take precedence over the process of weaving. They have prioritized the moment of consumption over the moment of production. However, actors’ dealings with their employers, like those of many other laborers, were fraught with tension. Locked into an unequal relationship with the businessmen who controlled the U.S. theater industry, actors at every level of the theatrical hierarchy struggled for workplace equity. In August 1919, that struggle led to a work stoppage that left theaters across the U.S. dark for more than a month. The actors’ strike reveals much about early twentieth-century cultural production, showing how the concentration of economic power and the related centralization evident throughout the economy were transforming the nature of work in U.S. culture industries.

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QUESTION + ANSWER CHOICES

In the passage, the author explains that his focus is different from that of most historians because he:

- A. concentrates on the early twentieth-century theater.
- B. accepts that actors are “weavers of dreams.”
- C. deals with the economic realities of the theater.
- D. emphasizes the unequal relationship between businesspeople and their employees.

SOLUTION

Solution: The correct answer is C.

1. This is not what the author suggests makes his focus different from that of other authors.
2. The author suggests that other historians also accept the “weavers of dreams” metaphor; he puts his focus on a different part of the metaphor than do other authors.
3. **This is a Comprehension question, because it tests your understanding of a central passage theme. The answer is C because the author writes in the first paragraph that “Most often, historians have allowed the dreams that actors weave to take precedence over the process of weaving.” He then goes on to distinguish his own perspective from that of “most” historians, writing, “However, actors’ dealings with their employers, like those of many other laborers, were fraught with tension.” The rest of the paragraph talks about the economic circumstances confronted by actors and puts that in the context of the economic conditions in the culture industry more broadly. The author’s concern with the “economic realities” of actors’ lives is part of what makes his work different from that of other historians, according to the passage.**
4. The author implies that most scholars recognize that these unequal relationships exist in business; what distinguishes the author is that he applies these economic insights to the business of acting.

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D

Time Spent:
2 mins 37 secs
Difficulty Level:
Expert
Content & Skills:
Skill: CMP
HUM
THE

Question 101 of 230

Data ID: 631290
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PASSAGE

Passage 7 (Questions 39-43)

To borrow a metaphor, actors are “weavers of dreams”—artists engaged in the creation of the intangible and, in their own estimation, set apart from other occupational groups. Most often, historians have allowed the dreams that actors weave to take precedence over the process of weaving. They have prioritized the moment of consumption over the moment of production. However, actors’ dealings with their employers, like those of many other laborers, were fraught with tension. Locked into an unequal relationship with the businessmen who controlled the U.S. theater industry, actors at every level of the theatrical hierarchy struggled for workplace equity. In August 1919, that struggle led to a work stoppage that left theaters across the U.S. dark for more than a month. The actors’ strike reveals much about early twentieth-century cultural production, showing how the concentration of economic power and the related centralization evident throughout the economy were transforming the nature of work in U.S. culture industries.

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Adapted from S. P. Holmes, *All the world’s a stage! The actors’ strike of 1919*. ©2005 the Organization of American Historians.

QUESTION + ANSWER CHOICES

Which of the following statements best explains Ewen’s use of the phrase “obliteration of the factory,” as it is used in the passage?

- A. Industrial jobs were declining in number.
- B. Industrial jobs produced consumer goods.
- C. Industrial jobs were hidden from public view.
- D. Industrial jobs were being replaced by new models of production.

SOLUTION

Solution: The correct answer is C.

- The passage does not suggest this.
- This is not what the phrase “obliteration of the factory” implies.
- This is another Comprehension question, as it asks you to understand what a particular phrase means in the context of the passage. The answer is C, because the author explains that Ewen talks about the “obliteration of the factory” to describe the division of the economic world into “an unpleasant sphere where goods were produced and a gratifying sphere where they were consumed” (paragraph 2). This suggests that the “obliteration” is meant to make the production side of the consumer economy invisible, so that the gratification of consumption seems to come without the messy realities of industrial labor. The author also explains that theatrical performance is unusual because the labor side of the equation cannot actually be hidden, as it is “embodied by the performer.” As the author opposes theatrical labor to other kinds of labor, it is clear that he suggests that industrial jobs can be hidden.
- The passage does not suggest this.

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RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
X

Time Spent:
2 mins 19 secs
Difficulty Level:
Expert
Content & Skills:
Skill: CMP
HUM
THE

Question 102 of 230

Data ID: 531296
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PASSAGE

Passage 7 (Questions 39-43)

To borrow a metaphor, actors are “weavers of dreams”—artists engaged in the creation of the intangible and, in their own estimation, set apart from other occupational groups. Most often, historians have allowed the dreams that actors weave to take precedence over the process of weaving. They have prioritized the moment of consumption over the moment of production. However, actors’ dealings with their employers, like those of many other laborers, were fraught with tension. Locked into an unequal relationship with the businessmen who controlled the U.S. theater industry, actors at every level of the theatrical hierarchy struggled for workplace equity. In August 1919, that struggle led to a work stoppage that left theaters across the U.S. dark for more than a month. The actors’ strike reveals much about early twentieth-century cultural production, showing how the concentration of economic power and the related centralization evident throughout the economy were transforming the nature of work in U.S. culture industries.

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Adapted from S. P. Holmes, *All the world’s a stage! The actors’ strike of 1919*. ©2005 the Organization of American Historians.

QUESTION + ANSWER CHOICES

The author claims that actors were locked into an unequal relationship with the businessmen who controlled the U.S. theater in the early twentieth century. Which of the following statements by the author provides the strongest support for this claim?

- A.
“[T]he economic world was divided into an unpleasant sphere where goods were produced and a gratifying sphere where they were consumed.”
- B.
Performance is an unusual commodity “in that it is a labor process exhibited before and consumed by an audience.”
- C.
“In the reconfigured theatrical economy of the early twentieth century, actors’ labor was highly segmented.”
- D.
Combination companies allowed theater managers to “book several weeks of a touring company’s business in a single transaction.”

SOLUTION

Solution: The correct answer is **D**.

1. The quotation in option A does not support the point about the unequal relationships between actors and businessmen.
2. The quotation in option B does not support the point about the unequal relationships between actors and businessmen.
3. The quotation in option C talks about the different opportunities that were open to different actors, but this does not specifically support a point about the power held by the businessmen who controlled the theater.
4. **This is a Reasoning Within the Text question, as it asks you to determine which of the choices provides the strongest support for a passage claim. The answer is D, because this option describes a situation in which theater managers controlled a large number of the opportunities for employment, giving the actors much less leverage to negotiate (for higher salaries, better working conditions, etc). As the author writes, once the theater managers were able to book engagements in a whole group of theaters at once, “this strengthened their bargaining position immeasurably.” Because they were able to control a concentrated number of employment opportunities at once, the actors had less opportunity to look elsewhere for a better deal. Thus, this quotation supports the passage claim that the actors were “locked into an “unequal relationship with . . . businessmen.”**

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer

A
B
C
D
✓
Time Spent:
1 min 32 secs
Difficulty Level:
Expert
Content & Skills:
Skill: RWT
HUM
THE

Question 103 of 230

Data ID: 631273
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PASSAGE

Passage 8 (Questions 44-48)

To overburdened building conservators and curators, the idea of commissioning architectural paint research might at first sight appear to be simply another hurdle imposed by over-zealous bureaucracy. But far from being an imposition, architectural paint research has immense potential. The structural analysis of buildings is accepted as an important investigatory procedure, and architectural paint research offers the opportunity to continue the study of the development and use of a building by its occupants, long after the last structural alteration has been completed. For over three hundred years, paint finishes have been repeatedly applied to buildings using a range of techniques and a relatively restricted range of materials.

Changes in fashions of house paint colors and decorative effects are often directly related to technological developments in painting materials. House painters constantly seek brightly colored pigments, which are both inexpensive and stable, to complement an otherwise dull palette of cheaper natural materials. In the seventeenth century, for example, the artificially produced pigment blue verditer was devised as a cheaper and more workable alternative to the expensive pigments azurite and smalt.

Unlike easel paintings or wall paintings, where the artists' work has been retained and carefully conserved, architectural schemes are routinely obliterated by successive decorations. However, the accumulation of successive paint layers on architectural elements is now regarded as an important archaeological resource, and as such requires careful management and conservation.

Recent research carried out in the interiors of the Little Castle at Bolsover in England illustrates how easily this important evidence can be lost. A sketch design dated to 1618 for paneling in the Pillar Parlor was annotated with details of a walnut graining embellished with black and gold. However, as part of a major restoration program carried out during the 1970s, the architectural decorative surfaces of the Pillar Parlor were "smartened up" and the original finishes were stripped to reveal the bare wood in the mistaken belief that the wood had originally been unpainted and the existing layers were applied in the nineteenth century.

This illustrates the importance of collating and synthesizing all existing evidence, which is the vital first phase of any paint research project. The examination of arbitrarily collected paint samples without obtaining a preliminary analysis of the building phases and viewing all documentary evidence is a meaningless exercise. This is the service commonly offered by many paint analysts whose research is based solely on paint samples delivered to the researcher through the mail. A contemporary description of an interior in a painter's bill or a private letter or even a newspaper often provides extremely useful information which is crucial to the planning of a research strategy. A significant paint layer may be more readily dated by a reference in a family letter which states: "Today my chambers painted a fine blue" than by embarking on sophisticated pigment analysis. This is not to say that the identification of pigments is not a vital tool in architectural paint research, but it should be used to answer specific questions and not seen as the purpose of the whole research exercise.

Furthermore, often the unwary client who commissions architectural paint research is presented with a disorganized report which contains reams of unexplained cross-sections, a scattering of elemental spectra, and inconclusive findings. It is no wonder this new discipline is often treated with severe distrust and suspicion, despite its potential to offer detailed insight into the social as well as the structural and decorative history of a building.

Adapted from H. Hughes, Architectural paint research 'expensive, inconclusive ... why bother?' Context 64, 6/1999 the Institute of Historic Building Conservation.

QUESTION + ANSWER CHOICES

What does the passage suggest about the use of blue paint for a house in the sixteenth century?

- A.
It would have been impossible.
- B.
It may have been a statement of wealth.
- C.
The paint would have faded quickly.
- D.
The original paint may have been titanium-white.

SOLUTION

Solution: The correct answer is B.

1. There is no suggestion that it would have been impossible to paint a house blue in the sixteenth century—it just would have been expensive.
2. **This is a Humanities passage that falls under the content category of Art History. This is a Reasoning Beyond the Text question, as it asks you to use passage information to answer a question about a scenario that is not discussed in the passage. The answer is B, because the passage refers to a development in the seventeenth century, when a particular kind of blue pigment was developed "as a cheaper and more workable alternative to the expensive pigments azurite and smalt" (paragraph 2). Thus, you can infer that, in the sixteenth century, it was still expensive to produce blue paint; in these circumstances, using blue paint might well have been a sign of wealth.**
3. The passage says that brightly colored pigments are "stable"; it does not suggest that the sixteenth century paints, containing more expensive pigments, were likely to be less stable (or more likely to fade).
4. The passage does not support this suggestion.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D

Time Spent:
35 secs
Difficulty Level:
Difficult
Content & Skills:
Skill: RBT
AHS
HUM

Question 104 of 230

Data ID: 531275
Captured: 6/3/2026, 4:40:53 PM

PASSAGE

Passage 8 (Questions 44-48)

To overburdened building conservators and curators, the idea of commissioning architectural paint research might at first sight appear to be simply another hurdle imposed by over-zealous bureaucracy. But far from being an imposition, architectural paint research has immense potential. The structural analysis of buildings is accepted as an important investigatory procedure, and architectural paint research offers the opportunity to continue the study of the development and use of a building by its occupants, long after the last structural alteration has been completed. For over three hundred years, paint finishes have been repeatedly applied to buildings using a range of techniques and a relatively restricted range of materials.

Changes in fashions of house paint colors and decorative effects are often directly related to technological developments in painting materials. House painters constantly seek brightly colored pigments, which are both inexpensive and stable, to complement an otherwise dull palette of cheaper natural materials. In the seventeenth century, for example, the artificially produced pigment blue verditer was devised as a cheaper and more workable alternative to the expensive pigments azurite and smalt.

Unlike easel paintings or wall paintings, where the artists' work has been retained and carefully conserved, architectural schemes are routinely obliterated by successive decorations. However, the accumulation of successive paint layers on architectural elements is now regarded as an important archaeological resource, and as such requires careful management and conservation.

Recent research carried out in the interiors of the Little Castle at Bolsover in England illustrates how easily this important evidence can be lost. A sketch design dated to 1618 for paneling in the Pillar Parlor was annotated with details of a walnut graining embellished with black and gold. However, as part of a major restoration program carried out during the 1970s, the architectural decorative surfaces of the Pillar Parlor were "smartened up" and the original finishes were stripped to reveal the bare wood in the mistaken belief that the wood had originally been unpainted and the existing layers were applied in the nineteenth century.

This illustrates the importance of collating and synthesizing all existing evidence, which is the vital first phase of any paint research project. The examination of arbitrarily collected paint samples without obtaining a preliminary analysis of the building phases and viewing all documentary evidence is a meaningless exercise. This is the service commonly offered by many paint analysts whose research is based solely on paint samples delivered to the researcher through the mail. A contemporary description of an interior in a painter's bill or a private letter or even a newspaper often provides extremely useful information which is crucial to the planning of a research strategy. A significant paint layer may be more readily dated by a reference in a family letter which states: "*Today my chambers painted a fine blue*" than by embarking on sophisticated pigment analysis. This is not to say that the identification of pigments is not a vital tool in architectural paint research, but it should be used to answer specific questions and not seen as the purpose of the whole research exercise.

Furthermore, often the unwary client who commissions architectural paint research is presented with a disorganized report which contains reams of unexplained cross-sections, a scattering of elemental spectra, and inconclusive findings. It is no wonder this new discipline is often treated with severe distrust and suspicion, despite its potential to offer detailed insight into the social as well as the structural and decorative history of a building.

Adapted from H. Hughes, Architectural paint research 'expensive, inconclusive ... why bother?' Context 64, 6/1999 the Institute of Historic Building Conservation.

QUESTION + ANSWER CHOICES

Which of the following best describes the author's attitude toward the fact that the original paint of an architectural element has typically been painted over not just once but several times in the course of history?

- A.
It makes much of architectural paint research guesswork.
- B.
It is what makes architectural paint research such a robust source of information.
- C.
It indicates a difference in status between easel/wall painters and house painters.
- D.
It is something that education about architectural paint research might prevent in future.

SOLUTION

Solution: The correct answer is B.

1. This is not supported by the passage; the repeated layers offer specific historical evidence and do not make the process "guesswork."
2. **This is a Reasoning Within the Text question, because it asks you to evaluate the author's position about a piece of information presented in the passage. The answer is B, because the author says that "the accumulation of successive paint layers on architectural elements is now regarded as an important archaeological resource" (paragraph 3). The author contends that the repeated application of paint finishes allows researchers to study a progressive history of the use of the "development and use of the building long after the last structural alteration has been completed" (first paragraph).**
3. The passage distinguishes between easel/wall painters, whose works is preserved in its original state, and house painters, whose work is repeatedly painted over, but this distinction is not presented as a matter of "status" and this is not the significance of the point the question asks about.
4. There is no implication that research should "prevent" successive layers of paint from being used; the repeated layers are part of the historical evolution of the building and house paint is not designed to be preserved or conserved as artworks are.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓

C

D

Time Spent:

1 min 27 secs

Difficulty Level:

Moderate

Content & Skills:

Skill: RWT

AHS

HUM

Question 105 of 230

Data ID: 531276
Captured: 6/3/2026, 4:40:53 PM

PASSAGE

Passage 8 (Questions 44-48)

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QUESTION + ANSWER CHOICES

Why does the author quote the letter which states "Today my chambers painted a fine blue" (paragraph 5)?

- A.
To return to a point made earlier in the passage about blue paint
- B.
To show that interior decoration has been as important to occupants as exterior decoration
- C.
To illustrate the value of consulting documentary evidence as part of architectural paint research
- D.
To suggest the danger of using one type of evidence without corroboration by another type of evidence

SOLUTION

Solution: The correct answer is C.

- The letter is unrelated to the earlier point about the blue pigment developed in the seventeenth century.
- This is not the point of the letter.
- This is a Reasoning Within the Text question, as it asks you to determine the function in the passage of a particular piece of evidence. The answer is C, because the author quotes the letter as part of the argument that good research is holistic, drawing on all available sources. The author uses the example of the letter as an example of a situation in which a document may be more useful than "sophisticated pigment analysis" (when one wishes to date "a significant paint layer").
- The author does not suggest that it is "dangerous" not to corroborate finds with another type of evidence, just that doing so leads to more accurate and efficient work.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
X
Time Spent:
31 secs

Difficulty Level:
Moderate
Content & Skills:
Skill: RWT
AHS
HUM

Question 106 of 230

Data ID: 631279
Captured: 6/3/2026, 4:40:54 PM

PASSAGE

Passage 8 (Questions 44-48)

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Adapted from H. Hughes, Architectural paint research 'openive, inconclusive ... why bother?' Context 64, 6/1999 the Institute of Historic Building Conservation.

QUESTION + ANSWER CHOICES

The author's primary purpose is to:

- A. argue for the value of architectural paint research.
- B. critique unqualified practitioners of architectural paint research.
- C. provide advice to those who commission architectural paint research.
- D. present findings of architectural paint research conducted on 17th century buildings.

SOLUTION

Solution: The correct answer is A.

1. This is a Comprehension question, because it tests your understanding of the central argument, or purpose, of the passage. The answer is A, because the passage describes the valuable information that can be gained by doing this kind of research, including insight into the "development and use of a building, long after" its structural work is finished. While the author expresses some concern about the poor work of some architectural paint researchers, this is more of a caveat; the larger point of the passage is to establish the "potential" of the research "to offer detailed insight into the social as well as the structural and decorative history of a building."
2. The author concludes by acknowledging that some people have legitimate concerns about the discipline, as some do not perform or present their research well, but this is not the main point of the passage.
3. The passage concludes by referring to problems that may be experienced by some people who commission this research, but this is not the main point of the passage.
4. The passage refers to some research done in the seventeenth century, but this is not the main point of the passage.

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D

Time Spent:
28 secs
Difficulty Level:
Moderate
Content & Skills:
Skill: CMP
AHS
HUM

Question 107 of 230

Data ID: 531280
Captured: 6/3/2026, 4:40:55 PM

PASSAGE

Passage 8 (Questions 44-48)

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Adapted from H. Hughes, Architectural paint research 'openive, inconclusive ... why bother?' Content 64, 6/1999 the Institute of Historic Building Conservation.

QUESTION + ANSWER CHOICES

Which of the following approaches to research does the author most support?

- A. Experimental
- B. Theoretical
- C. Statistical
- D. Holistic

SOLUTION

Solution: The correct answer is D.

1. The author does not talk about performing experiments.
2. The type of research that the author advocates involves science and collecting documentary evidence; it is not entirely, or primarily, theoretical.
3. The author does not advocate a statistical approach.
4. This is a Comprehension question, as it asks you to characterize the author's position or viewpoint. The answer is D, because the author argues for the incorporation of multiple sources of evidence in order to plan and execute good architectural paint research; the author points to "the importance of collating and synthesizing all existing evidence" (paragraph 5). This supports the choice of D, "holistic."

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
31 secs

Difficulty Level:
Moderate
Content & Skills:
Skill: CMP
AHS
HUM

Question 108 of 230

Data ID: 631290
Captured: 6/3/2026, 4:40:55 PM

PASSAGE

Passage 9 (Questions 49-53)

For the outsider, Voodoo is almost synonymous with Haiti. The result of this identification is culture-as-commodity. The visitor to Haiti finds that the “real voodoo” is kept hidden, while a staged version is offered readily. Yet these various synthetic versions of voodoo are authentic, being but permutations of this very mutable ritual form.

The ritual process of voodoo is based on a polytheistic belief system that identifies a pantheon of *loas*, divine beings who inhabit an intermediate realm between humans and a higher supreme life force. The *ve-ve*, an invocational drawing made on the ground, is located near the altar. The *loas* are invoked through drumming, dancing, and singing; *mambos* (priestesses) and *houngans* (priests) guide the aspirants, the *hounsis*, as they are possessed by the *loas*. Possession, as a central and inalienable “marker” of any voodoo event, is common to all types of voodoo; however, structural aspects and staging of local performances serve to clarify important differences between the more ritual-based versus theatrical forms. Two examples follow.

At a voodoo session in Mariani, near Port-au-Prince, the audience was composed entirely of non-Haitian tourists and elite Haitians. The audience paid admission to watch the action from rows of seats or tables. Drinks were served, and flaming torches illuminated the elegant and rustic cabaret setting. Although wild events—such as animal sacrifice and hounsis leaping into fire—did occur, the action was tightly choreographed, allowing for the unexpected to take place (predictably) within a selective dramatic presentation of those aspects of voodoo that are palatable to the foreign audience. Simultaneously, Mariani voodoo confirms the popular expectation of “scary voodoo” and enforces a distance from the audience that disallows real transformation (inadvertent possession), thus affirming the other side of the outsiders’ voodoo myth: that voodoo is “fake.” The voodoo at Mariani is a dramatic spectacle, with a hidden back stage, costume changes, skillful execution of performances, and vivid use of theatrical effects.

In the remote village of Nansoucri, I witnessed ritual voodoo for Haitians only. The hounsis and musicians in the village center sang, drummed, and danced in the middle of a large crowd. Excitement grew gradually and steadily; there was a quality of controlled development during the entire ceremony. Villagers of all ages were participating, and children were actively “practicing” or imitating possession. It was often difficult to distinguish serious possession from play. The dancers seemed to respond to certain possessions with teasing and laughter, and to ignore others completely. Those nearest a possessed dancer tossed themselves about, and called out to the loa. The hounsis did not mingle with the crowd but stayed close to the drummers, facing them. The audience-dancers were idiosyncratic, dancing individually or with others. The ambience was comfortable and casual, with no emphasis on “staging” or narrative structure. Small groups outside the peristyle clustered around gambling tables, and peddlers sold candy and cigarettes.

The houngan travels to Nansoucri to learn the “real voodoo” and returns to re-create it. But the essential dynamic of voodoo is self re-creation. Knowledge of voodoo is in the head, but voodoo transmits itself through the body—the social body of the community enacting ritual, and the individual body of the trance-dancer. What is made at Mariani is theatre; it is also a form of preservation. It is voodoo and not voodoo.

Adapted from M. Anderson, *Authentic voodoo is synthetic*. ©1982 the Massachusetts Institute of Technology.

QUESTION + ANSWER CHOICES

Which of the following entities are LEAST likely to be seen at a voodoo session?

- A.
Houngans
- B.
Loas
- C.
Hounsis
- D.
Mambos

SOLUTION

Solution: The correct answer is B.

1. Houngans are priests, who would be seen at a voodoo session, as described in paragraph
2. This is a Humanities passage that falls under the content category of Religion and addresses Studies of Diverse Cultures. This is a Comprehension question, because it depends on your understanding of passage information about the meaning of several different terms. The answer is B, because loas are “divine beings who inhabit an intermediate realm between humans and a higher supreme life force” (paragraph 2). They are unlikely to be visible at a voodoo session, although their presence is very important to participants.
3. Hounsis are the aspirants, who would be present and visible at a voodoo session, as described in paragraph 2.
4. Mambos are priestesses, who would be seen at a voodoo session, like the Houngans.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
27 secs
Difficulty Level:
Moderate

Content & Skills:
Skill: CMP
HUM
RGN

Question 109 of 230

Data ID: 531293
Captured: 6/3/2026, 4:40:56 PM

PASSAGE

Passage 9 (Questions 49-53)

For the outsider, Voodoo is almost synonymous with Haiti. The result of this identification is culture-as-commodity. The visitor to Haiti finds that the “real voodoo” is kept hidden, while a staged version is offered readily. Yet these various synthetic versions of voodoo are authentic, being but permutations of this very mutable ritual form.

The ritual process of voodoo is based on a polytheistic belief system that identifies a pantheon of *loas*, divine beings who inhabit an intermediate realm between humans and a higher supreme life force. The *ve-ve*, an invocalational drawing made on the ground, is located near the altar. The *loas* are invoked through drumming, dancing, and singing; *mambos* (priestesses) and *houngans* (priests) guide the aspirants, the *hounsis*, as they are possessed by the *loas*. Possession, as a central and inalienable “marker” of any voodoo event, is common to all types of voodoo; however, structural aspects and staging of local performances serve to clarify important differences between the more ritual-based versus theatrical forms. Two examples follow.

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QUESTION + ANSWER CHOICES

The single change to the observed ritual at Mariani that would most weaken the author’s distinction between Mariani and Nansoucri voodoo would be a situation in which:

- A. all possessions were outwardly apparent.
- B. rituals included elements of danger.
- C. audience members were invited to dance with the hounsis.
- D. one or more audience members became possessed.

SOLUTION

Solution: The correct answer is D.

1. This would not weaken the distinction, as the author describes outwardly apparent possessions at Mariani.
2. This is consistent with the descriptions of the sessions at Mariani, as “animal sacrifice and hounsis leaping into fire” are included in the sessions.
3. This would not weaken the distinction, as it would still preserve the staged, “choreographed” element of the sessions.
4. This is a Reasoning Beyond the Text question, which presents a series of hypothetical scenarios not discussed in the passage and asks you how these would affect passage arguments. The answer is D, because the author’s distinction between the two voodoo forms turns on the fact that the Mariani session is a “theatrical” session, in which the audience essentially watches the action from a “distance . . . that disallows real transformation (inadvertent possession)” (paragraph 3). Thus, if audience members at Mariani did indeed become possessed, the author’s distinction between Mariani voodoo and the more participatory, ritual form of voodoo at Nansoucri would break down.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
1 min 11 secs
Difficulty Level:

Difficult
Content & Skills:
Skill: RBT
HUM
RGN

Question 110 of 230

Data ID: 531302
Captured: 6/3/2026, 4:40:56 PM

PASSAGE

Passage 9 (Questions 49-53)

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The ritual process of voodoo is based on a polytheistic belief system that identifies a pantheon of *loas*, divine beings who inhabit an intermediate realm between humans and a higher supreme life force. The *ve-ve*, an invocational drawing made on the ground, is located near the altar. The *loas* are invoked through drumming, dancing, and singing; *mambos* (priestesses) and *houngans* (priests) guide the aspirants, the *hounsis*, as they are possessed by the *loas*. Possession, as a central and inalienable “marker” of any voodoo event, is common to all types of voodoo; however, structural aspects and staging of local performances serve to clarify important differences between the more ritual-based versus theatrical forms. Two examples follow.

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In the remote village of Nansoucri, I witnessed ritual voodoo for Haitians only. The hounsis and musicians in the village center sang, drummed, and danced in the middle of a large crowd. Excitement grew gradually and steadily; there was a quality of controlled development during the entire ceremony. Villagers of all ages were participating, and children were actively “practicing” or imitating possession. It was often difficult to distinguish serious possession from play. The dancers seemed to respond to certain possessions with teasing and laughter, and to ignore others completely. Those nearest a possessed dancer tossed themselves about, and called out to the loa. The hounsis did not mingle with the crowd but stayed close to the drummers, facing them. The audience-dancers were idiosyncratic, dancing individually or with others. The ambience was comfortable and casual, with no emphasis on “staging” or narrative structure. Small groups outside the peristyle clustered around gambling tables, and peddlers sold candy and cigarettes.

The houngan travels to Nansoucri to learn the “real voodoo” and returns to re-create it. But the essential dynamic of voodoo is self re-creation. Knowledge of voodoo is in the head, but voodoo transmits itself through the body—the social body of the community enacting ritual, and the individual body of the trance-dancer. What is made at Mariani is theatre; it is also a form of preservation. It is voodoo and not voodoo.

Adapted from M. Anderson, *Authentic voodoo is synthetic*. ©1982 the Massachusetts Institute of Technology.

QUESTION + ANSWER CHOICES

The author uses the term “culture as commodity” to characterize which aspect of Haitian voodoo?

- 1. The charging of admission at voodoo sessions
 - 2. The molding of voodoo to fit audience expectations
 - 3. The sale of voodoo trinkets and other artifacts
- A.
I only
- B.
III only
- C.
I and II only
- D.
II and III only

SOLUTION

Solution: The correct answer is C.

1. Option A is incorrect, as I is not the only correct option.

2. There is no support for option III, as the passage makes no reference to the sale of voodoo-related artifacts, so B is incorrect.

3. **This is a Comprehension question, as it tests your understanding of (the meaning of) a passage phrase, as that phrase is used in the context of the passage. The answer is C, because both options I and II are correct, while option III is not. The author explains that the “outsider” to Haiti expects voodoo to be “almost synonymous” with the country itself. According to the author, this powerful “identification” of Haiti with voodoo leads to “culture as commodity”—or the outsiders essentially being able to get what they want from voodoo. In paragraph 3, the author describes a form of voodoo that is staged for outsiders and which the audience pays admission to watch; this supports option I. Also, in paragraph 3, the author contends that the form of voodoo witnessed at Mariani “confirms the popular expectation of ‘scary voodoo’”; this supports option II.**

4. There is no support for option III, as the passage makes no reference to the sale of voodoo-related artifacts, so D is incorrect.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C

✓

D

Time Spent:

24 secs

Difficulty Level:

Difficult

Content & Skills:

Skill: CMP

HUM

RGN

Question 111 of 230

Data ID: 531305
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PASSAGE

Passage 9 (Questions 49-53)

For the outsider, Voodoo is almost synonymous with Haiti. The result of this identification is culture-as-commodity. The visitor to Haiti finds that the “real voodoo” is kept hidden, while a staged version is offered readily. Yet these various synthetic versions of voodoo are authentic, being but permutations of this very mutable ritual form.

The ritual process of voodoo is based on a polytheistic belief system that identifies a pantheon of *loas*, divine beings who inhabit an intermediate realm between humans and a higher supreme life force. The *ve-ve*, an invocalational drawing made on the ground, is located near the altar. The *loas* are invoked through drumming, dancing, and singing; *mambos* (priestesses) and *houngans* (priests) guide the aspirants, the *hounsis*, as they are possessed by the *loas*. Possession, as a central and inalienable “marker” of any voodoo event, is common to all types of voodoo; however, structural aspects and staging of local performances serve to clarify important differences between the more ritual-based versus theatrical forms. Two examples follow.

At a voodoo session in Mariani, near Port-au-Prince, the audience was composed entirely of non-Haitian tourists and elite Haitians. The audience paid admission to watch the action from rows of seats or tables. Drinks were served, and flaming torches illuminated the elegant and rustic cabaret setting. Although wild events—such as animal sacrifice and hounsis leaping into fire—did occur, the action was tightly choreographed, allowing for the unexpected to take place (predictably) within a selective dramatic presentation of those aspects of voodoo that are palatable to the foreign audience. Simultaneously, Mariani voodoo confirms the popular expectation of “scary voodoo” and enforces a distance from the audience that disallows real transformation (inadvertent possession), thus affirming the other side of the outsiders’ voodoo myth: that voodoo is “fake.” The voodoo at Mariani is a dramatic spectacle, with a hidden back stage, costume changes, skillful execution of performances, and vivid use of theatrical effects.

In the remote village of Nansoucri, I witnessed ritual voodoo for Haitians only. The hounsis and musicians in the village center sang, drummed, and danced in the middle of a large crowd. Excitement grew gradually and steadily; there was a quality of controlled development during the entire ceremony. Villagers of all ages were participating, and children were actively “practicing” or imitating possession. It was often difficult to distinguish serious possession from play. The dancers seemed to respond to certain possessions with teasing and laughter, and to ignore others completely. Those nearest a possessed dancer tossed themselves about, and called out to the loa. The hounsis did not mingle with the crowd but stayed close to the drummers, facing them. The audience-dancers were idiosyncratic, dancing individually or with others. The ambience was comfortable and casual, with no emphasis on “staging” or narrative structure. Small groups outside the peristyle clustered around gambling tables, and peddlers sold candy and cigarettes.

The houngan travels to Nansoucri to learn the “real voodoo” and returns to re-create it. But the essential dynamic of voodoo is self re-creation. Knowledge of voodoo is in the head, but voodoo transmits itself through the body—the social body of the community enacting ritual, and the individual body of the trance-dancer. What is made at Mariani is theatre; it is also a form of preservation. It is voodoo and not voodoo.

Adapted from M. Anderson, *Authentic voodoo is synthetic*. ©1982 the Massachusetts Institute of Technology.

QUESTION + ANSWER CHOICES

The passage offers *no* information about mambos’ and houngans’:

- A. genders.
- B. roles.
- C. costumes.
- D. beliefs.

SOLUTION

Solution: The correct answer is C.

- The distinction between mambos and houngans is a gendered distinction, so information about their genders is available.
- The mambos and houngans are described as priests and priestesses, respectively, so information about their roles is available.
- This is a Reasoning Within the Text question, as it asks you to evaluate the kind of information that is available (either implicitly or explicitly) in the passage. C is the correct answer, because passage information is available about the other options, but there is no information about or even reference to the specific costumes worn by the mambos and houngans. There is a reference to “costume changes” at the end of paragraph 3, but there is no way to know who is wearing these costumes, and there is certainly no information about any costumes specifically worn by mambos or houngans.
- The mambos and houngans are described in the context of the passage depiction of a “polytheistic belief system” centered on loas (paragraph 2), so information about their beliefs is available.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
1 min 33 secs
Difficulty Level:

Moderate
Content & Skills:
Skill: RWT
HUM
RGN

Question 112 of 230

Data ID: 631311
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PASSAGE

Passage 9 (Questions 49-53)

For the outsider, Voodoo is almost synonymous with Haiti. The result of this identification is culture-as-commodity. The visitor to Haiti finds that the “real voodoo” is kept hidden, while a staged version is offered readily. Yet these various synthetic versions of voodoo are authentic, being but permutations of this very mutable ritual form.

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Adapted from M. Anderson, *Authentic voodoo is synthetic*. ©1982 the Massachusetts Institute of Technology.

QUESTION + ANSWER CHOICES

- The passage describes an irony of tourist-oriented voodoo, namely that:
- A. mambos and houngans, although they are human, exert some influence on the loas.
 - B. loas, although they are supreme beings, deign to inhabit the bodies of humans.
 - C. audiences who make the most effort to find authentic voodoo are least likely to actually find it.
 - D. audiences are kept at a safe distance, which reinforces their lack of belief in the powers of voodoo.

SOLUTION

- Solution:** The correct answer is D.
1. The passage actually says that the mambos and houngans guide the aspirants, “as they are possessed by the loas.” It does not say that the mambos and houngans necessarily “influence” the divine loas.
 2. This is not an irony described in the passage, implicitly or explicitly.
 3. The passage actually challenges the notion that there is one kind of “authentic” voodoo, arguing instead that “synthetic” versions of voodoo are also authentic, so this is not a claim the passage makes.
 4. **This is a Comprehension question that tests your understanding of passage information and of the presence of irony. The answer is D, because this is the only option that accurately characterizes a passage claim. The tourist-oriented voodoo that the author describes at Mariani does enforce a distance between the audience and the ceremony, which the author suggests makes “inadvertent possession” impossible. Thus, the way that the tourists experience voodoo (designed to meet their expectations) reinforces their own pre-existing expectations about the limited possibilities of voodoo, which is ironic.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
1 min 6 secs

Difficulty Level:
Difficult
Content & Skills:
Skill: CMP
HUM
RGN

Question 113 of 230

Data ID: 531341
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PASSAGE

Passage 1 (Questions 1-6)

About 15% of colorectal cancers (CRCs), or colon cancers, are caused by defects in the mismatch repair system (MMR) that corrects certain DNA replication errors. In some cases, this MMR defect contributes to microsatellite instability (MSI), which is characterized by genetic hypermutability in DNA microsatellites (small repeated sequences that occur throughout the genome).

Researchers studying MSI in CRC cell lines and primary tumors observed that 100% of those analyzed contained deletions of 3 to 8 base pairs in the 17-thymine nucleotide microsatellite (T_{17}) located in Intron 8 of the gene encoding heat shock protein 110 (HSP110). HSP110 can bind structurally similar heat shock proteins and functions to facilitate proper protein folding and to reduce levels of nonfunctional protein aggregates. The larger HSP110 T_{17} deletions cause Exon 9 to be omitted from the final sequence during pre-mRNA processing. This generates a premature stop codon in Exon 10, resulting in the mutant protein HSP110 Δ E9. People with MSI CRC have HSP110 Δ E9 transcripts in cancerous tissue only.

Table 1 shows, with respect to T_{17} -deletion size, the amount of HSP110 Δ E9 mRNA expressed relative to the total amount of HSP110 mRNA expressed in MSI CRC primary tumor cells. The total amount of HSP110 mRNA expressed includes both wild-type HSP110 (HSP110WT) and HSP110 Δ E9 mRNA.

Table 1 Effect of T_{17} -deletion Size on Relative Expression of HSP110 Δ E9 in MSI CRC Tumor Cells

T_{17} -deletion size (bp)	HSP110 Δ E9 mRNA expressed (% of total HSP110 mRNA expressed)
3	0
4	43
5	89
6	92
7	100
8	100

Table 2 summarizes how HSP110 Δ E9 affects HSP110WT function in cells transfected with HSP110WT and/or HSP110 Δ E9.

Table 2 Effects of HSP110 Δ E9 on HSP110WT Function

Endogenous HSP110 form expressed	Heat shock-induced protein aggregation (% of total protein)	Cellular localization of HSP110 form(s)	Apoptotic cells after induced apoptosis (% of total cells)
none	100	na	nd
HSP110WT	64	nucleus, cytosol	25
HSP110 Δ E9	96	cytosol	nd
HSP110WT + HSP110 Δ E9	100	cytosol	50

(Note: na = not applicable; nd = not determined.)

Figure 1 shows the ability of human CRC cells expressing either HSP110 Δ E9 or HSP110WT to produce tumors after injection into mice.

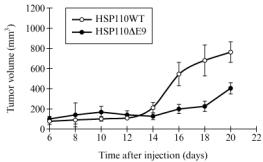


Figure 1 Effect of HSP110 Δ E9 on tumorigenesis

Adapted from C. Dorand et al., Nature Medicine 6:2011 Nature Publishing Group.

QUESTION + ANSWER CHOICES

Based on the data presented in the passage, which statement best describes the HSP110 Δ E9 allele?

- ☒ A. Cancer-promoting and dominant to HSP110WT

- B.
Cancer-promoting and recessive to *HSP110WT*
- **C.**
Cancer-suppressing and dominant to *HSP110WT*
- D.
Cancer-suppressing and recessive to *HSP110WT*

SOLUTION

Solution: The correct answer is C.

1. Based on Figure 1, expression of HSP110ΔE9 results in reduced, not increased, tumorigenicity. Thus, HSP110ΔE9 expression suppresses, it does not promote, cancer.
2. Based on Figure 1, HSP110ΔE9 expression suppresses, it does not promote, cancer. Furthermore, based on Table 2, HSP110ΔE9 expression is dominant, not recessive, to HSP110 because it counteracts HSP110Δ activity in protein aggregation.
3. **Figure 1 shows that HSP110ΔE9 expressing cells have a reduced tumor growth when injected in other mice. Based on Table 2, HSP110ΔE9 counteracts the effects of HSP110 on protein aggregation and stimulates apoptosis, thus HSP110ΔE9 expression is dominant to HSP110.**
4. Based on Figure 1, HSP110ΔE9 expression is cancer-suppressing; however, the effects of HSP110ΔE9 are dominant, and not recessive, to HSP110 because HSP110ΔE9 is able to counteract the effects of HSP110 and stimulate apoptosis.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleic Acid Structure and Function](#); [DNA Replication](#); [Repair of DNA](#).

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer

A

☒ A

B

☐ B

C

☐ C

D

☐ D

Time Spent:

2 mins 51 secs

Difficulty Level:

Expert

Content & Skills:

SIRS4

CC1B

FC1

BIO

Question 114 of 230

Data ID: 531348
Captured: 6/3/2026, 4:40:59 PM

PASSAGE

Passage 1 (Questions 1-6)

About 15% of colorectal cancers (CRCs), or colon cancers, are caused by defects in the mismatch repair system (MMR) that corrects certain DNA replication errors. In some cases, this MMR defect contributes to microsatellite instability (MSI), which is characterized by genetic hypermutability in DNA microsatellites (small repeated sequences that occur throughout the genome).

Researchers studying MSI in CRC cell lines and primary tumors observed that 100% of those analyzed contained deletions of 3 to 8 base pairs in the 17-thymine nucleotide microsatellite (T_{17}) located in Intron 8 of the gene encoding heat shock protein 110 (HSP110). HSP110 can bind structurally similar heat shock proteins and functions to facilitate proper protein folding and to reduce levels of nonfunctional protein aggregates. The larger *HSP110* T_{17} deletions cause Exon 9 to be omitted from the final sequence during pre-mRNA processing. This generates a premature stop codon in Exon 10, resulting in the mutant protein HSP110ΔE9. People with MSI CRC have *HSP110ΔE9* transcripts in cancerous tissue only.

Table 1 shows, with respect to T_{17} -deletion size, the amount of *HSP110ΔE9* mRNA expressed relative to the total amount of *HSP110* mRNA expressed in MSI CRC primary tumor cells. The total amount of *HSP110* mRNA expressed includes both wild-type *HSP110* (HSP110WT) and *HSP110ΔE9* mRNA.

Table 1 Effect of T_{17} -deletion Size on Relative Expression of *HSP110ΔE9* in MSI CRC Tumor Cells

T_{17} -deletion size (bp)	<i>HSP110ΔE9</i> mRNA expressed (% of total <i>HSP110</i> mRNA expressed)
3	0
4	43
5	89
6	92
7	100
8	100

Table 2 summarizes how HSP110ΔE9 affects HSP110WT function in cells transfected with HSP110WT and/or HSP110ΔE9.

Table 2 Effects of HSP110ΔE9 on HSP110WT Function

Endogenous HSP110 form expressed	Heat shock-induced protein aggregation (% of total protein)	Cellular localization of HSP110 form(s)	Apoptotic cells after induced apoptosis (% of total cells)
none	100	na	nd
HSP110WT	64	nucleus, cytosol	25
HSP110ΔE9	96	cytosol	nd
HSP110WT + HSP110ΔE9	100	cytosol	50

(Note: na = not applicable; nd = not determined.)

Figure 1 shows the ability of human CRC cells expressing either HSP110ΔE9 or HSP110WT to produce tumors after injection into mice.

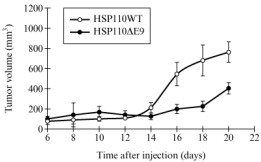


Figure 1 Effect of HSP110ΔE9 on tumorigenesis

Adapted from C. Dorand et al., *Nature Medicine* 6:2011 Nature Publishing Group.

QUESTION + ANSWER CHOICES

Based on the passage, in MSI CRC primary tumor cells, the size of the deletion in the *HSP110* T_{17} microsatellite is inversely correlated with:

- A.
the amount of HSP110ΔE9 protein expressed.

- B.
the number of mature *HSP110WT* transcripts synthesized.
- C.
the frequency of the omission of *HSP110* Exon 9 during splicing.
- D.
the extent of premature translation termination in Exon 10.

SOLUTION

Solution: The correct answer is B.

1. The passage does not report protein levels for HSP110 or HSP110ΔE9.
2. Table 1 shows that as the number of deleted base pairs (bp) increases from 3 to 8, the levels of HSP110ΔE9 increase whereas the levels of HSP110 decrease. This data indicates that there is an inverse relationship between the number of deleted bps and HSP110 levels.
3. The passage does not show a frequency of E9 omissions. It only indicates that the larger omission results in deletion of E9.
4. The passage does not show the extent of premature translation termination, but only the levels of HSP110ΔE9 and HSP110 transcripts (mRNA).

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleic Acid Structure and Function](#), [DNA Replication](#), [Repair of DNA](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer

A

B

✓

C

D

Time Spent:

17 secs

Difficulty Level:

Difficult

Content & Skills:

SIRS4

CC1B

FC1

BIO

Question 115 of 230

Data ID: 831357

Captured: 6/3/2026, 4:41:00 PM

PASSAGE

Passage 1 (Questions 1-6)

About 15% of colorectal cancers (CRCs), or colon cancers, are caused by defects in the mismatch repair system (MMR) that corrects certain DNA replication errors. In some cases, this MMR defect contributes to microsatellite instability (MSI), which is characterized by genetic hypermutability in DNA microsatellites (small repeated sequences that occur throughout the genome).

Researchers studying MSI in CRC cell lines and primary tumors observed that 100% of those analyzed contained deletions of 3 to 8 base pairs in the 17-thymine nucleotide microsatellite (T₁₇) located in Intron 8 of the gene encoding heat shock protein 110 (HSP110). HSP110 can bind structurally similar heat shock proteins and functions to facilitate proper protein folding and to reduce levels of nonfunctional protein aggregates. The larger HSP110 T₁₇ deletions cause Exon 9 to be omitted from the final sequence during pre-mRNA processing. This generates a premature stop codon in Exon 10, resulting in the mutant protein HSP110ΔE9. People with MSI CRC have HSP110ΔE9 transcripts in cancerous tissue only.

Table 1 shows, with respect to T₁₇-deletion size, the amount of HSP110ΔE9 mRNA expressed relative to the total amount of HSP110 mRNA expressed in MSI CRC primary tumor cells. The total amount of HSP110 mRNA expressed includes both wild-type HSP110 (HSP110WT) and HSP110ΔE9 mRNA.

Table 1 Effect of T₁₇-deletion Size on Relative Expression of HSP110ΔE9 in MSI CRC Tumor Cells

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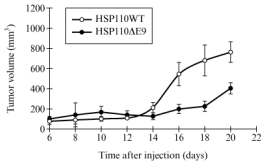


Figure 1 Effect of HSP110ΔE9 on tumorigenesis

Adapted from C. Dorand et al., *Nature Medicine* 6:2011 Nature Publishing Group.

QUESTION + ANSWER CHOICES

A man with a CRC mutation that results in the synthesis of HSP110ΔE9 and a woman that does not carry this mutation in any of her tissues have a child. What is the percent chance that the child will inherit the CRC mutation?

- A.

- 0%
- B.
 - 25%
 - C.
 - 50%
 - D.
 - 100%

SOLUTION

Solution: The correct answer is A.

1. Based on the passage, HSP110ΔE9 is only expressed in somatic cells, not germline. Thus, it cannot be inherited.
2. The passage indicates that HSP110ΔE9 is only expressed in cancerous cells, therefore, it cannot be inherited.
3. As the transcripts are only expressed in cancerous cells, the mutation will not be inherited.
4. The child is not likely to inherit the mutation because, based on the passage, HSPΔE9 is only expressed in cancerous cells.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer

A

✓

B

C

D

Time Spent:

3 secs

Difficulty Level:

Expert

Content & Skills:

SIRS2

CC1C

FC1

BIO

Question 116 of 230

Data ID: 531365
Captured: 6/3/2026, 4:41:00 PM

PASSAGE

Passage 1 (Questions 1-6)

About 15% of colorectal cancers (CRCs), or colon cancers, are caused by defects in the mismatch repair system (MMR) that corrects certain DNA replication errors. In some cases, this MMR defect contributes to microsatellite instability (MSI), which is characterized by genetic hypermutability in DNA microsatellites (small repeated sequences that occur throughout the genome).

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(Note: na = not applicable; nd = not determined.)

Figure 1 shows the ability of human CRC cells expressing either HSP110ΔE9 or HSP110WT to produce tumors after injection into mice.

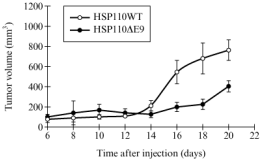


Figure 1 Effect of HSP110ΔE9 on tumorigenesis

Adapted from C. Dorand et al., Nature Medicine 6:2011 Nature Publishing Group.

QUESTION + ANSWER CHOICES

Which nucleotide pairing(s) would be recognized by the MMR system during DNA replication?

1. dTMP and dCMP
2. dGMP and dAMP

3. dAMP and dTMP

- A.
I only
- B.
III only
- C.
I and II only
- D.
I, II, and III

SOLUTION

Solution: The correct answer is C.

- The MMR system recognizes DNA pair mismatches. dTMP-dCMP are a mismatch as well as dGMP and dAMP.
- Option III is not a base mismatch, thus it will not be recognized by the MMR system.
- The passage indicates that MMR system recognizes the pair mismatch. Both options I and II represent mismatches, thus they will be recognized by the MMR system.**
- Option I and II are mismatches and will be recognized by the MMR system. Option III is not a mismatch, and it will not be recognized.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleic Acid Structure and Function: DNA Replication, Repair of DNA](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
6 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC1B
FC1
BIO

Question 117 of 230

Data ID: 531367
Captured: 6/3/2026, 4:41:01 PM

PASSAGE

Passage 1 (Questions 1-6)

About 15% of colorectal cancers (CRCs), or colon cancers, are caused by defects in the mismatch repair system (MMR) that corrects certain DNA replication errors. In some cases, this MMR defect contributes to microsatellite instability (MSI), which is characterized by genetic hypermutability in DNA microsatellites (small repeated sequences that occur throughout the genome).

Researchers studying MSI in CRC cell lines and primary tumors observed that 100% of those analyzed contained deletions of 3 to 8 base pairs in the 17-thymine nucleotide microsatellite (T_{17}) located in Intron 8 of the gene encoding heat shock protein 110 (HSP110). HSP110 can bind structurally similar heat shock proteins and functions to facilitate proper protein folding and to reduce levels of nonfunctional protein aggregates. The larger HSP110 T_{17} deletions cause Exon 9 to be omitted from the final sequence during pre-mRNA processing. This generates a premature stop codon in Exon 10, resulting in the mutant protein HSP110 Δ E9. People with MSI CRC have HSP110 Δ E9 transcripts in cancerous tissue only.

Table 1 shows, with respect to T_{17} -deletion size, the amount of HSP110 Δ E9 mRNA expressed relative to the total amount of HSP110 mRNA expressed in MSI CRC primary tumor cells. The total amount of HSP110 mRNA expressed includes both wild-type HSP110 (HSP110WT) and HSP110 Δ E9 mRNA.

Table 1 Effect of T_{17} -deletion Size on Relative Expression of HSP110 Δ E9 in MSI CRC Tumor Cells

T_{17} -deletion size (bp)	HSP110 Δ E9 mRNA expressed (% of total HSP110 mRNA expressed)
3	0
4	43
5	89
6	92
7	100
8	100

Table 2 summarizes how HSP110 Δ E9 affects HSP110WT function in cells transfected with HSP110WT and/or HSP110 Δ E9.

Table 2 Effects of HSP110 Δ E9 on HSP110WT Function

Endogenous HSP110 form expressed	Heat shock-induced protein aggregation (% of total protein)	Cellular localization of HSP110 form(s)	Apoptotic cells after induced apoptosis (% of total cells)
none	100	na	nd
HSP110WT	64	nucleus, cytosol	25
HSP110 Δ E9	96	cytosol	nd
HSP110WT + HSP110 Δ E9	100	cytosol	50

(Note: na = not applicable; nd = not determined.)

Figure 1 shows the ability of human CRC cells expressing either HSP110 Δ E9 or HSP110WT to produce tumors after injection into mice.

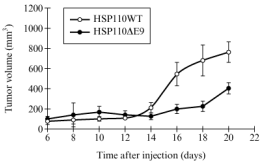


Figure 1 Effect of HSP110 Δ E9 on tumorigenesis

Adapted from C. Dorand et al., *Nature Medicine* 6:2011 Nature Publishing Group.

QUESTION + ANSWER CHOICES

The information in the passage best supports the conclusion that Intron 8 of HSP110 most likely contains which of the following?

- A.
Stop codon

- B. Splice acceptor site
- C. *HSP110* gene promoter
- D. Partial coding sequence of *HSP110*

SOLUTION

Solution: The correct answer is **B**.

1. Based on the passage, intron 8 does not contain a stop codon as translation ends with exon 10.
2. It is **most likely that intron 8 contains a splice acceptor site that allows exon 9 to be omitted upon mutation**.
3. The promoter is usually not located in an intron.
4. The intron does not code for a gene, an exon does.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleic Acid Structure and Function](#); [DNA Replication](#), [Repair of DNA](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
4 secs
Difficulty Level:
Expert
Content & Skills:
SIRS2
CC1B
FC1
BIO

Question 118 of 230

Data ID: 831370
Captured: 6/3/2026, 4:41:01 PM

PASSAGE

Passage 1 (Questions 1-6)

About 15% of colorectal cancers (CRCs), or colon cancers, are caused by defects in the mismatch repair system (MMR) that corrects certain DNA replication errors. In some cases, this MMR defect contributes to microsatellite instability (MSI), which is characterized by genetic hypermutability in DNA microsatellites (small repeated sequences that occur throughout the genome).

Researchers studying MSI in CRC cell lines and primary tumors observed that 100% of those analyzed contained deletions of 3 to 8 base pairs in the 17-thymine nucleotide microsatellite (T_{17}) located in Intron 8 of the gene encoding heat shock protein 110 (*HSP110*). *HSP110* can bind structurally similar heat shock proteins and functions to facilitate proper protein folding and to reduce levels of nonfunctional protein aggregates. The larger *HSP110* T_{17} deletions cause Exon 9 to be omitted from the final sequence during pre-mRNA processing. This generates a premature stop codon in Exon 10, resulting in the mutant protein *HSP110ΔE9*. People with MSI CRC have *HSP110ΔE9* transcripts in cancerous tissue only.

Table 1 shows, with respect to T_{17} -deletion size, the amount of *HSP110ΔE9* mRNA expressed relative to the total amount of *HSP110* mRNA expressed in MSI CRC primary tumor cells. The total amount of *HSP110* mRNA expressed includes both wild-type *HSP110* (*HSP110WT*) and *HSP110ΔE9* mRNA.

Table 1 Effect of T_{17} -deletion Size on Relative Expression of *HSP110ΔE9* in MSI CRC Tumor Cells

T_{17} -deletion size (bp)	<i>HSP110ΔE9</i> mRNA expressed (% of total <i>HSP110</i> mRNA expressed)
3	0
4	43
5	89
6	92
7	100
8	100

Table 2 summarizes how *HSP110ΔE9* affects *HSP110WT* function in cells transfected with *HSP110WT* and/or *HSP110ΔE9*.

Table 2 Effects of *HSP110ΔE9* on *HSP110WT* Function

Endogenous <i>HSP110</i> form expressed	Heat shock-induced protein aggregation (% of total protein)	Cellular localization of <i>HSP110</i> form(s)	Apoptotic cells after induced apoptosis (% of total cells)
none	100	na	nd
<i>HSP110WT</i>	64	nucleus, cytosol	25
<i>HSP110ΔE9</i>	96	cytosol	nd
<i>HSP110WT</i> + <i>HSP110ΔE9</i>	100	cytosol	50

(Note: na = not applicable; nd = not determined.)

Figure 1 shows the ability of human CRC cells expressing either *HSP110ΔE9* or *HSP110WT* to produce tumors after injection into mice.

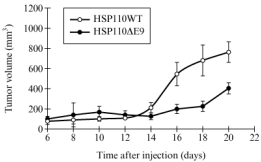


Figure 1 Effect of *HSP110ΔE9* on tumorigenesis

Adapted from C. Dorand et al., *Nature Medicine* 6:2011 Nature Publishing Group.

QUESTION + ANSWER CHOICES

According to the information in the passage, *HSP110* is most likely which type of protein?

- A. Adhesion

- B. Chaperone
- C. Clathrin
- D. Enzyme

SOLUTION

Solution: The correct answer is B.

1. The passage states that HSP110 is involved in protein folding. Adhesion proteins act in binding with other cells or with the extracellular matrix, not protein folding.
2. The passage states that HSP110 is a heat shock protein, which facilitates proper protein folding and inhibits the formation of nonfunctional protein aggregates. Proteins that perform this function are chaperone proteins.
3. Clathrin functions in formation of vesicles for intracellular trafficking, not in protein folding.
4. Based on the passage, HSP110 is involved in protein folding, not enzymatic catalysis of specific biochemical reactions.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
3 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC1A
FC1
BCH

Question 119 of 230

Data ID: 531380
Captured: 6/3/2026, 4:41:02 PM

PASSAGE

Passage 2 (Questions 7-11)

Malnutrition is known to increase susceptibility to colitis (colon inflammation). Colitis can also be caused by pellagra (a condition that may result from tryptophan deficiency) and Hartnup disease (a condition resulting from decreased function of the small intestinal neutral amino acid transporter AT1), the latter of which reduces gut absorption of tryptophan. The plasma membrane-embedded domain of the angiotensin I converting enzyme 2 (ACE2) is necessary for AT1 localization to the luminal plasma membrane of small intestinal epithelial cells. AT1 simultaneously transports tryptophan against its concentration gradient and a sodium ion along its concentration gradient into the cytoplasm of intestinal epithelial cells.

To determine how malnutrition promotes colitis, researchers treated wild-type (WT) mice and mice with loss-of-function mutations in the X-linked *Ace2* gene with dextran sodium sulfate (DSS), which increases colitis susceptibility by damaging intestinal epithelial cells. Mice were then fed one of the following three diets:

1. Standard diet (SD);
2. Protein-free diet (PFD); or
3. Gly–Trp (G–T), a dipeptide not requiring AT1 for intestinal uptake.

Mice were then monitored for weight loss as an indicator of colitis (Table 1).

Table 1 The Effects of *Ace2* Genotype and Diet on Colitis and Circulating Tryptophan

	<i>Ace2</i> genotype	Diet	Percent weight loss 4 days after start of DSS treatment (%)	Tryptophan blood levels (relative to control)
+/y (WT)		SD*	2	1.0
		PFD	22	not determined
		SD + G–T	2	1.4
–/y		SD*	10	0.3
		PFD	20	not determined
		SD + G–T	2	1.0

*Both mice genotypes on SD showed 0% weight loss without DSS.

Adapted from Tatsuo Hashimoto et al., *Nature* ©2012 Macmillan Publishers Limited.

QUESTION + ANSWER CHOICES

Which amino acid, in addition to tryptophan, is most likely to be transported by AT1?

- A. Phenylalanine
- B. Lysine
- C. Arginine
- D. Glutamate

SOLUTION

Solution: The correct answer is A.

1. AT1 is a neutral amino acid transporter. Phenylalanine is a neutral amino acid.
2. Based on the passage AT1 transports neutral amino acids. Lysine is not a neutral amino acid: it is positively charged.
3. Arginine is a positively charged amino acid. The passage indicates that AT1 transports neutral amino acids, thus not arginine.
4. According to the passage AT1 transports neutral amino acids. AT1 cannot transport glutamate because glutamate is negatively charged.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Amino acids and proteins](#); [Enzyme Structure and function](#); and [Enzyme kinetics](#).

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
X
Time Spent:
3 secs
Difficulty Level:

Moderate
Content & Skills:
SIRS1
CC1A
FC1
BCH

Question 120 of 230

Data ID: 631382
Captured: 6/3/2026, 4:41:03 PM

PASSAGE

Passage 2 (Questions 7-11)

Malnutrition is known to increase susceptibility to colitis (colon inflammation). Colitis can also be caused by pellagra (a condition that may result from tryptophan deficiency) and Hartnup disease (a condition resulting from decreased function of the small intestinal neutral amino acid transporter AT1), the latter of which reduces gut absorption of tryptophan. The plasma membrane-embedded domain of the angiotensin I converting enzyme 2 (ACE2) is necessary for AT1 localization to the luminal plasma membrane of small intestinal epithelial cells. AT1 simultaneously transports tryptophan against its concentration gradient and a sodium ion along its concentration gradient into the cytoplasm of intestinal epithelial cells.

To determine how malnutrition promotes colitis, researchers treated wild-type (WT) mice and mice with loss-of-function mutations in the X-linked *Ace2* gene with dextran sodium sulfate (DSS), which increases colitis susceptibility by damaging intestinal epithelial cells. Mice were then fed one of the following three diets:

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2. Protein-free diet (PFD); or
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	SD + G–T	2	1.4
–/y	SD*	10	0.3
	PFD	20	not determined
	SD + G–T	2	1.0

*Both mice genotypes on SD showed 0% weight loss without DSS.

Adapted from Tatsuo Hashimoto et al., *Nature* ©2012 Macmillan Publishers Limited.

QUESTION + ANSWER CHOICES

Based on the information in the passage, a mature AT1 mRNA is most likely to contain a sequence coding for which genetic factor(s)?

- A. Signal sequence
- B. Introns
- C. Promoter
- D. Nuclear localization signal

SOLUTION

Solution: The correct answer is A.

1. Normally, the mature RNA sequence for a protein that is secreted or that will locate to the cell membrane will contain the signal sequence. This portion of the mRNA is located in the 5' region and will signal to the ribosome that translation needs to be continued in the rough endoplasmic reticulum.
2. Introns are absent in mature mRNA.
3. The promoter is not present in mRNA.
4. The nuclear localization signal is a sequence that tags the protein for it to be transported into the nucleus. Based on the passage, AT1 is a protein transporter located on the cell membrane, not inside the nucleus.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
X
Time Spent:
4 secs

Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC2A
FC2
BIO

Question 121 of 230

Data ID: 631385
Captured: 6/3/2026, 4:41:04 PM

PASSAGE

Passage 2 (Questions 7-11)

Malnutrition is known to increase susceptibility to colitis (colon inflammation). Colitis can also be caused by pellagra (a condition that may result from tryptophan deficiency) and Hartnup disease (a condition resulting from decreased function of the small intestinal neutral amino acid transporter AT1), the latter of which reduces gut absorption of tryptophan. The plasma membrane-embedded domain of the angiotensin I converting enzyme 2 (ACE2) is necessary for AT1 localization to the luminal plasma membrane of small intestinal epithelial cells. AT1 simultaneously transports tryptophan against its concentration gradient and a sodium ion along its concentration gradient into the cytoplasm of intestinal epithelial cells.

To determine how malnutrition promotes colitis, researchers treated wild-type (WT) mice and mice with loss-of-function mutations in the X-linked *Ace2* gene with dextran sodium sulfate (DSS), which increases colitis susceptibility by damaging intestinal epithelial cells. Mice were then fed one of the following three diets:

1. Standard diet (SD);
2. Protein-free diet (PFD); or
3. Gly-Trp (G–T), a dipeptide not requiring AT1 for intestinal uptake.

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<i>Ace2</i> genotype	Diet	Percent weight loss 4 days after start of DSS treatment (%)	Tryptophan blood levels (relative to control)
+/y (WT)	SD*	2	1.0
	PFD	22	not determined
	SD + G–T	2	1.4
–/y	SD*	10	0.3
	PFD	20	not determined
	SD + G–T	2	1.0

*Both mice genotypes on SD showed 0% weight loss without DSS.

Adapted from Tatsuo Hashimoto et al., *Nature* ©2012 Macmillan Publishers Limited.

QUESTION + ANSWER CHOICES

Which effect did a protein-free diet have on DSS-treated +/y and –/y mice, respectively?

- A. Both the +/y and the –/y mice gained weight.
- B. The +/y mice lost weight, and the –/y mice gained weight.
- C. The +/y mice gained weight, and the –/y mice lost weight.
- D. Both +/y and the –/y mice lost weight.

SOLUTION

Solution: The correct answer is D.

1. Based on Table 1, both mice genotypes lost weight when fed a protein-free diet.
2. Table 1 summarizes the weight loss at four days after the start of the diet and, in both genotypes, there is a loss of weight.
3. Based on Table 1, WT mice lost 22% of their body weight 4 days after the start of the diet, and the mutant mice lost 20%. Thus, both mice lost weight.
4. Based on Table 1, both mice genotypes lost weight when fed a protein-free diet: WT lost 22% and mutants lost 20%.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Evidence that DNA Is Genetic Material](#), [Mendelian Principles](#), [Analysis Methods](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
4 secs
Difficulty Level:
Easy

Content & Skills:
SIRS4
CC1C
FC1
BIO

Question 122 of 230

Data ID: 531390
Captured: 6/3/2026, 4:41:04 PM

PASSAGE

Passage 2 (Questions 7-11)

Malnutrition is known to increase susceptibility to colitis (colon inflammation). Colitis can also be caused by pellagra (a condition that may result from tryptophan deficiency) and Hartnup disease (a condition resulting from decreased function of the small intestinal neutral amino acid transporter AT1), the latter of which reduces gut absorption of tryptophan. The plasma membrane-embedded domain of the angiotensin I converting enzyme 2 (ACE2) is necessary for AT1 localization to the luminal plasma membrane of small intestinal epithelial cells. AT1 simultaneously transports tryptophan against its concentration gradient and a sodium ion along its concentration gradient into the cytoplasm of intestinal epithelial cells.

To determine how malnutrition promotes colitis, researchers treated wild-type (WT) mice and mice with loss-of-function mutations in the X-linked *Ace2* gene with dextran sodium sulfate (DSS), which increases colitis susceptibility by damaging intestinal epithelial cells. Mice were then fed one of the following three diets:

1. Standard diet (SD);
2. Protein-free diet (PFD); or
3. Gly–Trp (G–T), a dipeptide not requiring AT1 for intestinal uptake.

Mice were then monitored for weight loss as an indicator of colitis (Table 1).

Table 1 The Effects of *Ace2* Genotype and Diet on Colitis and Circulating Tryptophan

<i>Ace2</i> genotype	Diet	Percent weight loss 4 days after start of DSS treatment (%)	Tryptophan blood levels (relative to control)
+/y (WT)	SD*	2	1.0
	PFD	22	not determined
	SD + G–T	2	1.4
–/y	SD*	10	0.3
	PFD	20	not determined
	SD + G–T	2	1.0

*Both mice genotypes on SD showed 0% weight loss without DSS.

Adapted from Tatsuo Hashimoto et al., *Nature* ©2012 Macmillan Publishers Limited.

QUESTION + ANSWER CHOICES

Pellagra also results from a deficiency of nicotinamide, which is synthesized from tryptophan. Nicotinamide nucleotides are neither oxidized nor reduced during which step of cellular respiration?

- A. Glycolysis
- B. Chemiosmosis
- C. Citric acid cycle
- D. Electron transport chain

SOLUTION

Solution: The correct answer is B.

1. During glycolysis NAD⁺ is reduced to form NADH.
2. Chemiosmosis is the only step of cellular respiration where NAD⁺ is neither reduced to form NADH, nor is NADH oxidized to form NAD⁺.
3. During the citric acid cycle, NAD⁺ is reduced to form NADH.
4. During the electron transport chain, NADH is oxidized to form NAD⁺.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Principles of Bioenergetics](#), [Carbohydrates](#), [Glycolysis](#), [Gluconeogenesis](#), and

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
X
B
C
D
Time Spent:
3 secs

Difficulty Level:
Difficult
Content & Skills:
SIRS1
CC1D
FC1
BCH

Question 123 of 230

Data ID: 631394
Captured: 6/3/2026, 4:41:05 PM

PASSAGE

Passage 2 (Questions 7-11)

Malnutrition is known to increase susceptibility to colitis (colon inflammation). Colitis can also be caused by pellagra (a condition that may result from tryptophan deficiency) and Hartnup disease (a condition resulting from decreased function of the small intestinal neutral amino acid transporter AT1), the latter of which reduces gut absorption of tryptophan. The plasma membrane-embedded domain of the angiotensin I converting enzyme 2 (ACE2) is necessary for AT1 localization to the luminal plasma membrane of small intestinal epithelial cells. AT1 simultaneously transports tryptophan against its concentration gradient and a sodium ion along its concentration gradient into the cytoplasm of intestinal epithelial cells.

To determine how malnutrition promotes colitis, researchers treated wild-type (WT) mice and mice with loss-of-function mutations in the X-linked *Ace2* gene with dextran sodium sulfate (DSS), which increases colitis susceptibility by damaging intestinal epithelial cells. Mice were then fed one of the following three diets:

1. Standard diet (SD);
2. Protein-free diet (PFD); or
3. Gly-Trp (G–T), a dipeptide not requiring AT1 for intestinal uptake.

Mice were then monitored for weight loss as an indicator of colitis (Table 1).

Table 1	The Effects of <i>Ace2</i> Genotype and Diet on Colitis and Circulating Tryptophan			
	<i>Ace2</i> genotype	Diet	Percent weight loss 4 days after start of DSS treatment (%)	Tryptophan blood levels (relative to control)
+/y (WT)		SD*	2	1.0
		PFD	22	not determined
		SD + G–T	2	1.4
–/y		SD*	10	0.3
		PFD	20	not determined
		SD + G–T	2	1.0

*Both mice genotypes on SD showed 0% weight loss without DSS.

Adapted from Tatsuo Hashimoto et al., *Nature* ©2012 Macmillan Publishers Limited.

QUESTION + ANSWER CHOICES

If a man with a mutant copy of *Ace2* has a child with a woman that is heterozygous for the mutant *Ace2* allele, what is the probability that the child will be a female and homozygous for the mutant *Ace2* allele?

- A. 0
- B. 0.25
- C. 0.75
- D. 1

SOLUTION

Solution: The correct answer is B.

1. Based on the pattern of inheritance, a daughter will inherit one of the X chromosomes from her father. If the father carries the mutation on his X chromosome, the daughter will surely carry the mutant chromosome. If the mother is heterozygous, then the daughter has more than 0% chances to be homozygous.
2. **Based on the pattern of inheritance, a daughter will inherit one of the X chromosomes from her father. The daughter will inherit the other X chromosome from her mother. If the mother is heterozygous, the daughter will have a 50% chance of being homozygous. As there is a 50% chance of the offspring being a female, based on the rules of probability, there is 25% of probability (50% × 50%) that the child will be a homozygous mutant daughter.**
3. The scenario describes the conditions for the daughter to have a 25% chance of being homozygous for the mutation. There is no scenario possible for the daughter of having a 75% chance of being homozygous for the mutation.
4. For the daughter to have a 100% chance of being a homozygous mutant, the father must carry the mutation and the mother must be homozygous for the mutation. In this case, however, the mother is heterozygous.

Khan Academy Lessons

For more help on this difficulty, concepts, or content associated with this question, see the Khan Academy lessons on [Evidence That DNA Is Passed From Mother to Offspring](#), [Genetics: Mendelian Genetics](#), [Genetics: Methods](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓

C
D
Time Spent:
5 secs
Difficulty Level:
Easy
Content & Skills:
SIRS2
CC1C
FC1
BIO

Question 124 of 230

Data ID: 531401
Captured: 6/3/2026, 4:41:05 PM

QUESTION + ANSWER CHOICES

During the initiation of muscle contraction, myosin binds actin after troponin binds to which ion?

- A. H^+
- B. K^+
- C. Na^+
- D. Ca^{2+}

SOLUTION

Solution: The correct answer is D.

1. H^+ does not play a role in myosin binding to actin during muscle contraction.
2. K^+ ions play a role in the maintenance of membrane potential, not the binding of myosin to actin, during muscle contraction.
3. Na^+ ions are involved in the maintenance of the membrane potential, not in the binding of myosin to actin.
4. **Ca^{2+} ions play a critical role in myosin-actin binding during skeletal muscle contraction. Ca^{2+} binds to troponin and allows tropomyosin to move, freeing the site of interaction between actin and myosin.**

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Excitation/Contraction System](#), [Circulation System](#), [Lymphatic System](#), [Immune System](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
18 secs
Difficulty Level:
Easy
Content & Skills:
SIRS1
CC3B
FC3
BIO

Question 125 of 230

Data ID: 531405
Captured: 6/3/2026, 4:41:06 PM

QUESTION • ANSWER CHOICES

What is the total number of fused rings present in a steroid?

- A.
1
- B.
2
- C.
4
- D.
6

SOLUTION

Solution: The correct answer is **C**.

1. A single ring can be found in a monoterpene, not a steroid.
2. An indole has two fused rings, but it is not a steroid.
3. **A steroid is composed of a 6-6-6-5 fused ring assembly, which is a total of 4 fused rings.**
4. Some natural products have 6 fused rings, but they are not steroids.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
13 secs
Difficulty Level:
Easy
Content & Skills:
SIRS1
CC2A
FC2
OCH

Question 126 of 230

Data ID: 531411
Captured: 6/3/2026, 4:41:07 PM

QUESTION • ANSWER CHOICES

Epilepsy may result in motor seizures due to massive synchronous firing of neurons in a small area of the cerebral cortex (the *epileptic focus*). Excitation spreads from the focus, involving an increasingly larger area of the cortex. A drug for the treatment of epilepsy would be most effective if it caused which of the following changes in the epileptic focus?

- A.
An increase in the neuron-firing threshold
- B.
An increase in extracellular Na⁺ concentration
- C.
~~A decrease in axon-membrane permeability to negative ions~~
- D.
A decrease in the length of the depolarization stage

SOLUTION

Solution: The correct answer is **A**.

1. **An increase in the threshold would make it more challenging for a neuron to fire an action potential. This will reduce the frequency of downstream neuron depolarizations, resulting in relief of the symptoms.**
2. An increase in the concentration of Na⁺ outside the cell will not help reduce the symptoms, because there will still be a flux of ions inside the cell when the firing starts, causing continuous depolarization.
3. Anion influx hyperpolarizes a neuron, reducing likelihood of action potential firing. Therefore, a decrease in axonal-membrane permeability to negative ions will not provide a relief as it will increase action potential firing.
4. A decrease in length of the depolarization stage will allow for a higher frequency of action potential firing. This would worsen the situation instead of improving it.

Khan Academy Lessons

For more help on the skills, concepts, or content associated with this question, see the Khan Academy lessons on [Neuronic System: Structure and Function](#), [Neuronic Cell](#), and [Endocrine System](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
2 mins 12 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC3A
FC3
BIO

Question 127 of 230

Data ID: 531414
Captured: 6/3/2026, 4:41:07 PM

QUESTION • ANSWER CHOICES

When concentrated urine is being produced, in which of the following regions of the kidney will the glomerular filtrate reach its highest concentration?

- A. Proximal convoluted tubule
- B. Distal convoluted tubule
- C. Cortical portion of the collecting duct
- D. Medullary portion of the collecting duct

SOLUTION

Solution: The correct answer is **D**.

- The proximal convoluted tubule is the first tubule where the glomerular filtrate passes through. About 65% of the glomerular filtrate is reabsorbed in the proximal collecting tubule. Thus, this is not the region where the glomerular filtrate is the most concentrated.
- The distal tubule contains relatively dilute glomerular filtrate. Water reabsorption can take place in the collecting duct, which concentrates the filtrate.
- The collecting duct is the last portion of the tubules where reabsorption of water and salts can occur. The glomerular filtrate is, however, slightly less concentrated in the cortical portion of the collecting duct than in the medullary portion.
- The collecting duct is the final structure in which water reabsorption occurs, which concentrates filtrate. The medullary portion of the collecting duct is the last portion of the tubules where reabsorption can occur. In the portion of the tubule that follows, there will be no more reabsorption. Thus, the medullary portion of the collecting duct contains the most concentrated glomerular filtrate that will correspond to the urine.

Khan Academy Lessons

For more help on the skills, concepts, or content associated to this question, see the Khan Academy lessons on [Respiratory System](#), [Circulatory System](#), [Urinary System](#), [Immune System](#).

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
X
C
D
Time Spent:
1 min 27 secs
Difficulty Level:
Expert
Content & Skills:
SIRS1
CC3B
FC3
BIO

Question 128 of 230

Data ID: 531420
Captured: 6/3/2026, 4:41:08 PM

PASSAGE

Passage 3 (Questions 16–19)

The cellular membrane potential is primarily determined by the concentrations of Na^+ and K^+ . The Na^+K^+ ATPase is a membrane protein that maintains the concentrations of these ions within the cell. In addition, the concentration gradient established by the Na^+K^+ ATPase is used as the driving force in the cotransport of other biological molecules. Recent studies have shown that there is a correlation between decreased heart function and decreased Na^+K^+ ATPase activity.

An important question regarding membrane proteins is how the composition of the membrane affects the protein function. Two major factors that are modulated by membrane composition are fluidity and thickness. Membrane composition is changed by both the structure of the component phospholipids and the amount of embedded cholesterol. For instance, equations 1 and 2 can be used to calculate membrane thickness d in angstroms when the membrane is composed of saturated lipids (Equation 1) or monounsaturated lipids (Equation 2), where n_c represents the number of carbon atoms in the alkyl chains.

$$d_{\text{sat}} = 1.75(n_c - 1)$$

Equation 1

$$d_{\text{unsat}} = 1.75(n_c - 2.6)$$

Equation 2

In order to study how membrane composition affects protein function, Na^+K^+ ATPase was reconstituted into artificially engineered lipid bilayers called liposomes. The liposomes used in this study were composed of phosphatidylcholine (PC). Two experiments were conducted.

Experiment 1

Na^+K^+ ATPase was reconstituted with either monounsaturated PC whose acyl chain length is 14 (14:1) or saturated PC of the same chain length (14:0). The temperature dependence of Na^+K^+ ATPase activity was measured in the presence and absence of 40 mol% cholesterol (Figure 1).

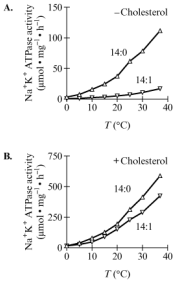


Figure 1 Temperature dependence of reconstituted Na^+K^+ ATPase activity in the absence (A) or presence (B) of cholesterol

Experiment 2

Na^+K^+ ATPase was reconstituted with monounsaturated PC of various acyl chain lengths (n_c), and the Na^+K^+ ATPase activity was measured for each in the presence and absence of 40 mol% cholesterol at 25°C (Figure 2).

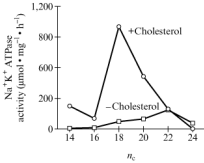


Figure 2 Acyl chain length dependence of Na^+K^+ ATPase activity

Adapted from Cornelius, F. ©2001 American Chemical Society

QUESTION • ANSWER CHOICES

What aspect of Experiment 1 does NOT address whether membrane composition has an effect on Na^+K^+ ATPase activity? The activity of the Na^+K^+ ATPase:

- A. showed less temperature dependence in the 14:1 liposome than the 14:0 liposome.
- B. was highest in the 14:0 liposomes at all temperatures.
- C. increased with temperature in both the 14:1 liposome and the 14:0 liposome.
- D. was greater at all temperatures when cholesterol was present.

SOLUTION

Solution: The correct answer is C.

1. Figure 1 shows that Na⁺K⁺ ATPase activity in 14:1 liposome is indeed less dependent on temperature than in the 14:0 liposome.
2. Data obtained in Experiment 1 show that Na⁺K⁺ ATPase activity in the 14:0 liposome is indeed highest at all temperatures.
3. In kinetic studies, protein activity is expected to increase with temperature, assuming the protein does not thermally denature. Therefore, the researchers did not raise the temperature to study the effects of temperature on protein activity, they instead raised the temperature to study how membrane fluidity plays a role in protein activity.
4. In Experiment 1, researchers analyzed the temperature dependence of Na⁺K⁺ ATPase activity in the presence and absence of cholesterol and showed that that it was greater at all temperatures when cholesterol was present.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
X
Time Spent:
3 secs
Difficulty Level:
Expert
Content & Skills:
SIRS3
CC1A
FC1
BCH

Question 129 of 230

Data ID: 531423
Captured: 6/3/2026, 4:41:08 PM

PASSAGE

Passage 3 (Questions 16–19)

The cellular membrane potential is primarily determined by the concentrations of Na⁺ and K⁺. The Na⁺K⁺ ATPase is a membrane protein that maintains the concentrations of these ions within the cell. In addition, the concentration gradient established by the Na⁺K⁺ ATPase is used as the driving force in the cotransport of other biological molecules. Recent studies have shown that there is a correlation between decreased heart function and decreased Na⁺K⁺ ATPase activity.

An important question regarding membrane proteins is how the composition of the membrane affects the protein function. Two major factors that are modulated by membrane composition are fluidity and thickness. Membrane composition is changed by both the structure of the component phospholipids and the amount of embedded cholesterol. For instance, equations 1 and 2 can be used to calculate membrane thickness *d* in angstroms when the membrane is composed of saturated lipids (Equation 1) or monounsaturated lipids (Equation 2), where *n_c* represents the number of carbon atoms in the alkyl chains.

$$d_{\text{sat}} = 1.75(n_c - 1)$$

Equation 1

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Equation 2

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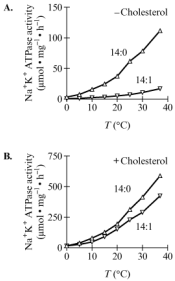


Figure 1 Temperature dependence of reconstituted Na⁺K⁺ ATPase activity in the absence (A) or presence (B) of cholesterol

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Na⁺K⁺ ATPase was reconstituted with monounsaturated PC of various acyl chain lengths (*n_c*), and the Na⁺K⁺ ATPase activity was measured for each in the presence and absence of 40 mol% cholesterol at 25°C (Figure 2).

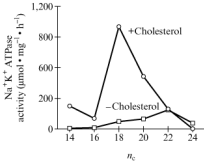


Figure 2 Acyl chain length dependence of Na⁺K⁺ ATPase activity

Adapted from Cornelius, F. ©2001 American Chemical Society

QUESTION + ANSWER CHOICES

Which approach does NOT measure the activity of the Na⁺K⁺ ATPase?

- A. Measuring the rate of ATP hydrolysis
- **B.** Measuring the free energy of the ion transport
- C. Measuring the rate of ADP production
- D. Measuring the change in ion concentration within the liposome

SOLUTION

Solution: The correct answer is **B**.

1. The activity of the Na⁺K⁺ ATPase is directly related to the rate of ATP hydrolysis.
2. **Free energy does not correlate with enzyme activity because it is NOT a measure of reaction rate/activity.**
3. The rate of ADP production does correlate with ATPase activity because ADP is a product of Na⁺K⁺ ATPase catalysis.
4. Na⁺K⁺ ATPase catalyzes ion transport through the liposome, therefore a change in ion concentration correlates with Na⁺K⁺ ATPase activity.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
33 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS3
CC1D
FC1
GCH

Question 130 of 230

Data ID: 531426
Captured: 6/3/2026, 4:41:09 PM

PASSAGE

Passage 3 (Questions 16–19)

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An important question regarding membrane proteins is how the composition of the membrane affects the protein function. Two major factors that are modulated by membrane composition are fluidity and thickness. Membrane composition is changed by both the structure of the component phospholipids and the amount of embedded cholesterol. For instance, equations 1 and 2 can be used to calculate membrane thickness *d* in angstroms when the membrane is composed of saturated lipids (Equation 1) or monounsaturated lipids (Equation 2), where *n_c* represents the number of carbon atoms in the alkyl chains.

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Equation 1

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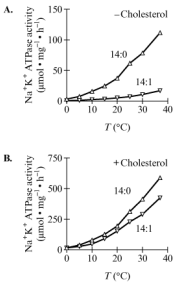


Figure 1 Temperature dependence of reconstituted Na⁺K⁺ ATPase activity in the absence (A) or presence (B) of cholesterol

Experiment 2

Na⁺K⁺ ATPase was reconstituted with monounsaturated PC of various acyl chain lengths (*n_c*), and the Na⁺K⁺ ATPase activity was measured for each in the presence and absence of 40 mol% cholesterol at 25°C (Figure 2).

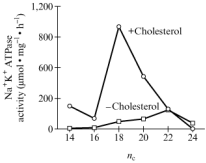


Figure 2 Acyl chain length dependence of Na⁺K⁺ ATPase activity

Adapted from Cornelius, F. ©2001 American Chemical Society

QUESTION + ANSWER CHOICES

What is the function of the Na⁺K⁺ ATPase during a neuronal action potential?

- A. Stimulation of the action potential
- B. Depolarization of the membrane
- C. Hyperpolarization of the membrane
- D. Restoration of the resting potential

SOLUTION

Solution: The correct answer is D.

1. The action potential is stimulated by the release of a neurotransmitter that binds to ligand gated channels, not the Na^+/K^+ ATPase.
2. Membrane depolarization is due to the opening of Na^+ channels, a process that is independent from the Na^+/K^+ ATPase.
3. Hyperpolarization of the cell membrane occurs when K^+ channels open and Na^+ channels close. This process is independent of the Na^+/K^+ ATPase.
4. **The restoration and maintenance of the resting potential relies on Na^+/K^+ ATPase. This is achieved by moving three Na^+ out of the cell for every two K^+ ions that are brought into the cell.**

Khan Academy Lessons

For more help on this skills, concepts, or content associated with this question, see the Khan Academy lessons on [Membrane Structure](#), [Structure and Function: Membrane Cell](#), and [Endocrine System](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer

- A
B
C
D
✓

Time Spent:
7 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS:1
CC3A
FC3
BIO

Question 131 of 230

Data ID: 531429
Captured: 6/3/2026, 4:41:10 PM

PASSAGE

Passage 3 (Questions 16–19)

The cellular membrane potential is primarily determined by the concentrations of Na^+ and K^+ . The Na^+/K^+ ATPase is a membrane protein that maintains the concentrations of these ions within the cell. In addition, the concentration gradient established by the Na^+/K^+ ATPase is used as the driving force in the cotransport of other biological molecules. Recent studies have shown that there is a correlation between decreased heart function and decreased Na^+/K^+ ATPase activity.

An important question regarding membrane proteins is how the composition of the membrane affects the protein function. Two major factors that are modulated by membrane composition are fluidity and thickness. Membrane composition is changed by both the structure of the component phospholipids and the amount of embedded cholesterol. For instance, equations 1 and 2 can be used to calculate membrane thickness d in angstroms when the membrane is composed of saturated lipids (Equation 1) or monounsaturated lipids (Equation 2), where n_c represents the number of carbon atoms in the alkyl chains.

$$d_{\text{sat}} = 1.75(n_c - 1)$$

Equation 1

$$d_{\text{unsat}} = 1.75(n_c - 2.6)$$

Equation 2

In order to study how membrane composition affects protein function, Na^+/K^+ ATPase was reconstituted into artificially engineered lipid bilayers called liposomes. The liposomes used in this study were composed of phosphatidylcholine (PC). Two experiments were conducted.

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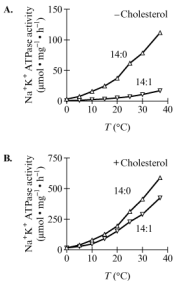


Figure 1 Temperature dependence of reconstituted Na^+/K^+ ATPase activity in the absence (A) or presence (B) of cholesterol

Experiment 2

Na^+/K^+ ATPase was reconstituted with monounsaturated PC of various acyl chain lengths (n_c), and the Na^+/K^+ ATPase activity was measured for each in the presence and absence of 40 mol% cholesterol at 25°C (Figure 2).

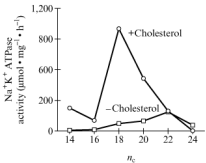


Figure 2 Acyl chain length dependence of Na^+/K^+ ATPase activity

Adapted from Cornelius, F. ©2001 American Chemical Society

QUESTION + ANSWER CHOICES

In which direction does the Na^+/K^+ ATPase transport ions across the cell membrane upon ATP hydrolysis?

- A.
Na⁺ is transported out of the cell; K⁺ is transported into the cell.
- B.
Na⁺ is transported into the cell; K⁺ is transported out of the cell.
- C.
Both Na⁺ and K⁺ are transported into the cell.
- D.
Both Na⁺ and K⁺ are transported out of the cell.

SOLUTION

Solution: The correct answer is A.

1. Upon ATP hydrolysis, three Na⁺ are transported outside the cell and two K⁺ are transported inside the cell against their concentration gradient.
2. Na⁺ and K⁺ are transported against their concentration gradient, thus Na⁺ is transported outside the cell and K⁺ is transported inside the cell, not the opposite.
3. Na⁺ and K⁺ are transported against their concentration gradient, thus Na⁺ is transported outside the cell and K⁺ is transported inside the cell.
4. If both Na⁺ and K⁺ were transported out of the cell, K⁺ would not be transported against its concentration gradient. Instead, both Na⁺ and K⁺ are transported against their concentration gradient. Thus Na⁺ is transported outside the cell and K⁺ is transported inside the cell.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
5 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC2A
FC2
BIO

Question 132 of 230

Data ID: 531435
Captured: 6/3/2026, 4:41:10 PM

PASSAGE

Passage 4 (Questions 20-23)

Diffuse large B-cell lymphoma (DLBCL) is the most common subtype of malignant lymphoma. The most common feature of the molecular pathogenesis of DLBCL is the constant activation of the NF- κ B pathway. The NF- κ B proteins p65 and cRel are transcription factors that can be continuously activated when mutations occur to proteins upstream in the signaling pathway (Figure 1). Their activation leads to increased cell proliferation and suppression of apoptosis. Activation occurs when the inhibitory I κ B α protein is degraded. This frees the nuclear localization sequence (NLS) on p65. The NLS is composed of four amino acids (KRKR) and interacts with an importin on the nuclear envelope.

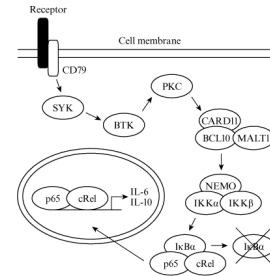


Figure 1 The major proteins in the NF- κ B signaling pathway (Note: The I κ B α is crossed-out to show it is degraded in response to the upstream signal.)

Drugs that target the variant signaling proteins may represent an effective therapeutic tool in the treatment of DLBCL. One such drug is a selective protein kinase C (PKC) inhibitor called Sotrastaurin (STN, Figure 2).

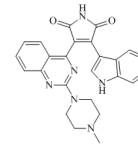


Figure 2 The structure of STN

STN was tested in four different DLBCL cell lines. Each cell line had a different mutation in the NF- κ B pathway that resulted in the overproduction of interleukins 6 and 10 (IL-6 and IL-10, respectively). The cell lines HBL1, TMD8, and Ly10 all had mutations in the CD79 protein, and the cell line Ly3 had a mutation in the CARD11 protein. These mutations resulted in continuous activation of these proteins regardless of upstream signals. The effectiveness of STN was determined by the concentration dependent drop in IL-6 mRNA levels (Figure 3). It was also found that STN treatment caused TMD8 and HBL1 cells to undergo cell-cycle arrest that was characterized by a decrease in the number of cells with 4N DNA content.

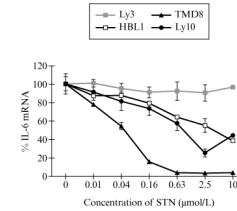


Figure 3 The STN concentration dependent expression of IL-6 mRNA in the DLBCL cell lines discussed in the passage

Adapted from T. Naylor et al. ©2011 American Association for Cancer Research.

QUESTION + ANSWER CHOICES

Upon activation, p65 and cRel control the level of IL-6 mRNA by:

- A. binding RNA.
- **B. binding DNA.**
- C. replicating RNA.
- D. replicating DNA.

SOLUTION

Solution: The correct answer is **B**.

1. Transcription factors regulate the expression of other genes by binding to DNA, not RNA.
2. **p65 and cRel are transcription factors and regulate the expression of other genes by binding to the promoter or the enhancer of the gene located on the DNA.**
3. It is DNA, and not RNA, that interacts with the transcription factors.
4. Transcription factors regulate the expression of specific genes, they do not replicate the entire genome.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleic Acid Structure and Function](#), [DNA Replication](#), [Repair of DNA](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
B
✓
C
D
Time Spent:
3 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS1
CC1B
FC1
BIO

Question 133 of 230

Data ID: 831438
Captured: 6/3/2026, 4:41:11 PM

PASSAGE

Passage 4 (Questions 20-23)

Diffuse large B-cell lymphoma (DLBCL) is the most common subtype of malignant lymphoma. The most common feature of the molecular pathogenesis of DLBCL is the constant activation of the NF- κ B pathway. The NF- κ B proteins p65 and cRel are transcription factors that can be continuously activated when mutations occur to proteins upstream in the signaling pathway (Figure 1). Their activation leads to increased cell proliferation and suppression of apoptosis. Activation occurs when the inhibitory I κ B α protein is degraded. This frees the nuclear localization sequence (NLS) on p65. The NLS is composed of four amino acids (KRKR) and interacts with an importin on the nuclear envelope.

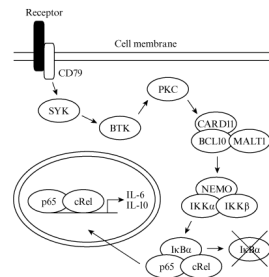


Figure 1 The major proteins in the NF- κ B signaling pathway (Note: The I κ B α is crossed-out to show it is degraded in response to the upstream signal.)

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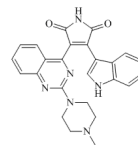


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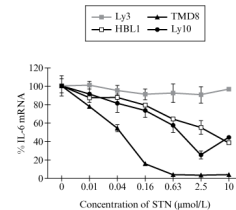


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Adapted from T. Naylor et al. ©2011 American Association for Cancer Research.

QUESTION + ANSWER CHOICES

Considering the structure of STN, what is the most likely mechanism for its entry into the cell?

- A. Active transport
- B. Receptor mediated endocytosis
- C. Diffusion directly through the membrane
- D. Passage through an ion channel

SOLUTION

Solution: The correct answer is C.

1. The structure of STN shows that it is a hydrophobic molecule. These type of molecules can pass through the membrane by simple diffusion. In addition the transport is not carried against a concentration gradient which would require an active transport.
2. The structure of STN shows that it is a hydrophobic molecule. These types of molecules can pass through the membrane by simple diffusion and therefore do not need a transporter.
3. **The structure of STN shows that it is a hydrophobic molecule. These types of molecules pass through the membrane by simple diffusion.**
4. The structure of STN shows that it is a hydrophobic molecule, not an ion.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
7 secs
Difficulty Level:
Expert
Content & Skills:
SIRS2
CC2A
FC2
BCH

Question 134 of 230

Data ID: 531441
Captured: 6/3/2026, 4:41:11 PM

PASSAGE

Passage 4 (Questions 20-23)

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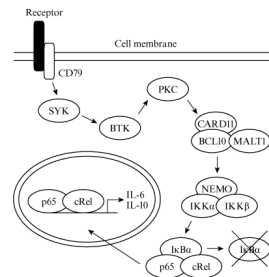


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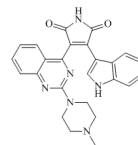


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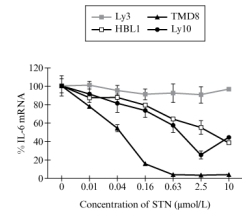


Figure 3 The STN concentration dependent expression of IL-6 mRNA in the DLBCL cell lines discussed in the passage

Adapted from T. Naylor et al. ©2011 American Association for Cancer Research.

QUESTION + ANSWER CHOICES

Assuming no mutations to the signaling pathway described in the passage, what event directly activates CARD11?

- A. Proteolytic cleavage
- B. GTP binding
- C. Membrane association
- D. Phosphorylation

SOLUTION

Solution: The correct answer is D.

1. The passage implies that phosphorylation by PKC and not proteolytic cleavage activates CARD11.
2. The passage implies that phosphorylation, and not GTP binding, activates CARD11.
3. The passage implies that phosphorylation, and not membrane association, activates CARD11.
4. **Based on the passage, the protein that activates CARD11 is a protein kinase. This implies that phosphorylation is the activating event.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
7 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS1
CC1A
FC1
BCH

Question 135 of 230

Data ID: 531442
Captured: 6/3/2026, 4:41:12 PM

PASSAGE

Passage 4 (Questions 20-23)

Diffuse large B-cell lymphoma (DLBCL) is the most common subtype of malignant lymphoma. The most common feature of the molecular pathogenesis of DLBCL is the constant activation of the NF- κ B pathway. The NF- κ B proteins p65 and cRel are transcription factors that can be continuously activated when mutations occur to proteins upstream in the signaling pathway (Figure 1). Their activation leads to increased cell proliferation and suppression of apoptosis. Activation occurs when the inhibitory I κ B α protein is degraded. This frees the nuclear localization sequence (NLS) on p65. The NLS is composed of four amino acids (KRKR) and interacts with an importin on the nuclear envelope.

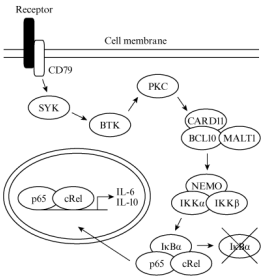


Figure 1 The major proteins in the NF- κ B signaling pathway (Note: The I κ B α is crossed-out to show it is degraded in response to the upstream signal.)

Drugs that target the variant signaling proteins may represent an effective therapeutic tool in the treatment of DLBCL. One such drug is a selective protein kinase C (PKC) inhibitor called Sotrastaurin (STN, Figure 2).

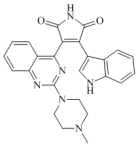


Figure 2 The structure of STN

STN was tested in four different DLBCL cell lines. Each cell line had a different mutation in the NF- κ B pathway that resulted in the overproduction of interleukins 6 and 10 (IL-6 and IL-10, respectively). The cell lines HBL1, TMD8, and Ly10 all had mutations in the CD79 protein, and the cell line Ly3 had a mutation in the CARD11 protein. These mutations resulted in continuous activation of these proteins regardless of upstream signals. The effectiveness of STN was determined by the concentration dependent drop in IL-6 mRNA levels (Figure 3). It was also found that STN treatment caused TMD8 and HBL1 cells to undergo cell-cycle arrest that was characterized by a decrease in the number of cells with 4N DNA content.

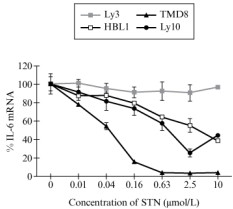


Figure 3 The STN concentration dependent expression of IL-6 mRNA in the DLBCL cell lines discussed in the passage

Adapted from T. Naylor et al. ©2011 American Association for Cancer Research.

QUESTION + ANSWER CHOICES

According to the passage, what is most likely correct about the genes that are expressed as a result of the constant activation of the NF- κ B pathway?

- ~~A.~~
They cause disruption of the mitochondria.
- **B.**
They contain a p65/cRel binding site in their promoter region.
- C.
They have accumulated mutations that alter function.
- D.
~~They bind to STN in the cytoplasm.~~

SOLUTION

Solution: The correct answer is **B**.

1. The passage does not describe the role of NF- κ B in mitochondria.
2. **Based on the passage, activation of the NF- κ B pathway results in translocation of the transcription factor subunits p65/cRel in the nucleus, where they regulate the expression of downstream target genes. Expression regulation is possible because the downstream target genes contain in their promoter or enhancer, DNA regions that are able to interact with p65 and cRel.**
3. Genes downstream of NF- κ B signaling have not accumulated mutations; these genes have only been activated.
4. STN is a protein kinase inhibitor, and it inhibits IL-6 expression by acting upstream of the NF- κ B pathway; there is no binding to STN in the cytoplasm.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleic Acid Structure and Function](#); [DNA Replication](#), [Repair of DNA](#).

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
Time Spent:
8 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC1B
FC1
BIO

Question 136 of 230

Data ID: 531451
Captured: 6/3/2026, 4:41:13 PM

PASSAGE

Passage 5 (Questions 24–27)

P-glycoprotein (P-gp) is a plasma membrane ABC transporter that has broad substrate specificity for hydrophobic compounds. P-gp is responsible for multidrug resistance (MDR) in tumor cells by preventing drugs from accumulating to cytotoxic levels. Studies show that P-gp overproduction in the membrane of tumor cells produces an MDR phenotype.

It has been shown that P-gp exhibits ATPase activity and is localized in cholesterol-rich domains within the membrane. To determine whether a functional link exists between P-gp and cholesterol, researchers conducted two experiments.

Experiment 1

Methyl- β -cyclodextrin (MBCD) is a cyclic oligosaccharide that binds and removes cholesterol from cell membranes. MBCD was added to cells that either overexpressed P-gp (MDR) or were P-gp free (control). The ATPase activity was measured at 37°C, and the reactions were supplemented with 1 mM ATP (Figure 1).

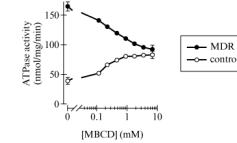


Figure 1 The MBCD-dependent ATPase activity of MDR and control cells

Experiment 2

The ATPase activity of P-gp was measured in the presence of increasing concentrations of cholesterol in MDR cells that were never treated with MBCD (native MDR) and MDR cells that were treated with MBCD for 1 hour at 37°C.

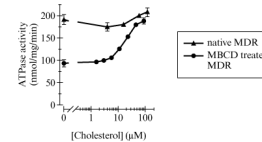


Figure 2 The cholesterol-dependent ATPase activity of native MDR and MDR cells pre-treated with MBCD for 1 hour at 37°C

Adapted from A. Garrigue et al., *PNAS* ©2002 the Authors.

QUESTION + ANSWER CHOICES

Which mechanism best describes how P-gp facilitates drug resistance?

- A.
~~P-gp binds to antitumor drugs in the presence of ATP and degrades the drugs.~~
- **B.**
P-gp serves as a pump and uses active transport to move antitumor drugs outside the cell.
- C.
~~P-gp prevents the entry of anti-tumor drugs into the cell.~~
- D.
P-gp causes increased membrane permeability, which causes antitumor drugs to exit the cell.

SOLUTION

Solution: The correct answer is **B**.

1. The passage notes that P-gp is an ABC transporter protein, not a binding protein.
2. **The passage notes that P-gp is an ABC transporter protein with ATPase activity. These transporters use ATP to actively transport molecules such as drugs out of the cell.**
3. The passage notes that P-gp is an ABC transporter protein. Transporter proteins facilitate, rather than prevent, transfer of molecules across the membrane bilayer.
4. The passage notes that P-gp is an ABC transporter protein. Transporter proteins are not involved in increased membrane permeability, they function to carry molecules across membranes.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
7 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC2A
FC2
BCH

Question 137 of 230

Data ID: 531454
Captured: 6/3/2026, 4:41:13 PM

PASSAGE

Passage 5 (Questions 24-27)

P-glycoprotein (P-gp) is a plasma membrane ABC transporter that has broad substrate specificity for hydrophobic compounds. P-gp is responsible for multidrug resistance (MDR) in tumor cells by preventing drugs from accumulating to cytotoxic levels. Studies show that P-gp overproduction in the membrane of tumor cells produces an MDR phenotype.

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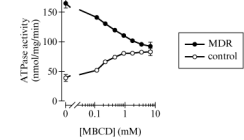


Figure 1 The MBCD-dependent ATPase activity of MDR and control cells

Experiment 2

The ATPase activity of P-gp was measured in the presence of increasing concentrations of cholesterol in MDR cells that were never treated with MBCD (native MDR) and MDR cells that were treated with MBCD for 1 hour at 37°C.

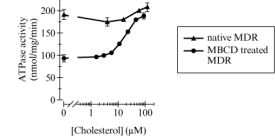


Figure 2 The cholesterol-dependent ATPase activity of native MDR and MDR cells pre-treated with MBCD for 1 hour at 37°C

Adapted from A. Garrigue et al., *PNAS* ©2002 the Authors.

QUESTION + ANSWER CHOICES

The formation of which reaction products is increased in MBCD-treated MDR cells after exposure to 100 μ M cholesterol?

- A. AMP + ADP
- B. ADP + P_i**
- C. ATP + P_i
- D. ADP + ATP

SOLUTION

Solution: The correct answer is B.

- Data in Figure 2 show that exposure of cholesterol-depleted MDR cells to 100 μ M cholesterol results in increased ATPase activity (ATP hydrolysis). ATP hydrolysis produces ADP and inorganic phosphate, not AMP.
- Figure 2 shows that exposure of cholesterol-depleted MDR cells to 100 μ M cholesterol results in complete restoration of ATPase activity (ATP hydrolysis). ATP hydrolysis produces ADP and inorganic phosphate.
- Data in Figure 2 show that exposure of cholesterol-depleted MDR cells to 100 μ M cholesterol results in increased ATP hydrolysis. ATP is the reagent in the hydrolysis, not a product.
- Exposure of cholesterol-depleted MDR cells to 100 μ M cholesterol results in increased ATP hydrolysis. While ADP is a product of the hydrolysis, ATP is the reagent in the hydrolysis, not a product.

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Principles of Bioenergetics](#), [Carbohydrates](#), [Glycolysis](#), [Gluconeogenesis](#), and

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
4 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC1D
FC1
BCH

Question 138 of 230

Data ID: 531457
Captured: 6/3/2026, 4:41:14 PM

PASSAGE

Passage 5 (Questions 24-27)

P-glycoprotein (P-gp) is a plasma membrane ABC transporter that has broad substrate specificity for hydrophobic compounds. P-gp is responsible for multidrug resistance (MDR) in tumor cells by preventing drugs from accumulating to cytotoxic levels. Studies show that P-gp overproduction in the membrane of tumor cells produces an MDR phenotype.

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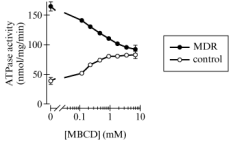


Figure 1 The MBCD-dependent ATPase activity of MDR and control cells

Experiment 2

The ATPase activity of P-gp was measured in the presence of increasing concentrations of cholesterol in MDR cells that were never treated with MBCD (native MDR) and MDR cells that were treated with MBCD for 1 hour at 37°C.

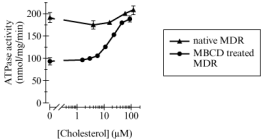


Figure 2 The cholesterol-dependent ATPase activity of native MDR and MDR cells pre-treated with MBCD for 1 hour at 37°C

Adapted from A. Garrigue et al., *PNAS* ©2002 the Authors.

QUESTION + ANSWER CHOICES

What is the most likely location of P-gp within the plasma membrane?

- A. Associated with lipids on the cytoplasmic side only
- B. Associated with lipids on the extracellular side only
- C. Peripheral to the plasma membrane
- D. Within a lipid raft

SOLUTION

Solution: The correct answer is D.

- Based on the passage, P-gp is localized in cholesterol-rich domains within the membrane which are known as lipid rafts. These structures are not associated with lipids on the cytoplasmic side of the membrane, but are part of the plasma membrane.
- Based on the passage, P-gp is localized in cholesterol-rich domains within the membrane which are known as lipid rafts. These structures are not associated with lipids on the extracellular side of the membrane, but are part of the plasma membrane.
- The passage specifically states that P-gp is localized in cholesterol-rich domains within the membrane, not peripheral to the plasma membrane. Lipid rafts are cholesterol-rich domains within the membrane.
- The passage states the P-gp is found in cholesterol rich domains, pointing to lipid rafts.

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
5 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS1
CC2A
FC2
BCH

Question 139 of 230

Data ID: 531459
Captured: 6/3/2026, 4:41:14 PM

PASSAGE

Passage 5 (Questions 24–27)

P-glycoprotein (P-gp) is a plasma membrane ABC transporter that has broad substrate specificity for hydrophobic compounds. P-gp is responsible for multidrug resistance (MDR) in tumor cells by preventing drugs from accumulating to cytotoxic levels. Studies show that P-gp overproduction in the membrane of tumor cells produces an MDR phenotype.

It has been shown that P-gp exhibits ATPase activity and is localized in cholesterol-rich domains within the membrane. To determine whether a functional link exists between P-gp and cholesterol, researchers conducted two experiments.

Experiment 1

Methyl- β -cyclodextrin (MBCD) is a cyclic oligosaccharide that binds and removes cholesterol from cell membranes. MBCD was added to cells that either overexpressed P-gp (MDR) or were P-gp free (control). The ATPase activity was measured at 37°C, and the reactions were supplemented with 1 mM ATP (Figure 1).

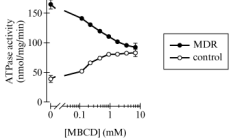


Figure 1 The MBCD-dependent ATPase activity of MDR and control cells

Experiment 2

The ATPase activity of P-gp was measured in the presence of increasing concentrations of cholesterol in MDR cells that were never treated with MBCD (native MDR) and MDR cells that were treated with MBCD for 1 hour at 37°C.

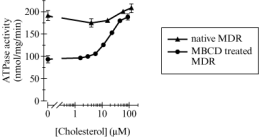


Figure 2 The cholesterol-dependent ATPase activity of native MDR and MDR cells pre-treated with MBCD for 1 hour at 37°C

Adapted from A. Garrigue et al., *PNAS* ©2002 the Authors.

QUESTION + ANSWER CHOICES

MBCD is composed of which type of molecule?

- A. Amino acids
- B. Fatty acids
- C. Nucleotides
- D. Carbohydrates

SOLUTION

Solution: The correct answer is D.

- Oligosaccharides are composed of carbohydrates, not amino acids.
- Oligosaccharides are composed of carbohydrates, not fatty acids.
- Oligosaccharides are composed of carbohydrates, not nucleotides.
- The passage notes that MBCD is a cyclic oligosaccharide, indicating that it is composed of carbohydrate molecules.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Principles of Bioenergetics](#); [Carbohydrates](#); [Glycolysis](#); [Gluconeogenesis](#), and

RESULT / TIMING / TAGS

Correct

Correct Answer Your Answer

A

B

C

D

✓

Time Spent:

3 secs

Difficulty Level:

Moderate

Content & Skills:

SIRS1

CC1D

FC1

BCH

Question 140 of 230

Data ID: 531494
Captured: 6/3/2026, 4:41:15 PM

QUESTION • ANSWER CHOICES

Colchicine is a drug that prevents the formation of microtubules. Which of the following mitotic processes would NOT occur after exposure to this drug?

- A. Formation of new cell plate
- B. Replication of DNA during interphase
- C. Breakdown of the nucleolus during prophase
- D. Movement of the chromosomes toward opposite poles of the cell during anaphase

SOLUTION

Solution: The correct answer is D.

- Microtubules do not play a role in the formation of the cell plate.
- DNA replication is not regulated by microtubules.
- Nucleolus breakdown during prophase is not regulated by microtubules.
- Microtubules bind to chromosomes at the level of the kinetochore and regulate their migration toward the opposite poles of the cell during anaphase.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct

Correct Answer Your Answer

A

B

C

D

✓

Time Spent:

30 secs

Difficulty Level:

Easy

Content & Skills:

SIRS2

CC2C

FC2

BIO

QUESTION + ANSWER CHOICES

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Content & Skills:

CC3B

QUESTION + ANSWER CHOICES

Content & Skills:

CC3B

Question 143 of 230

Data ID: 531475
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QUESTION + ANSWER CHOICES

Enzymes alter the rate of chemical reactions by all of the following methods EXCEPT:

- A. co-localizing substrates.
- B. altering local pH.
- C. altering substrate shape.
- D. altering substrate primary structure.

SOLUTION

Solution: The correct answer is **D**.

1. The enzyme needs to co-localize with the substrates to be able to modify the rate of chemical reactions.
2. It is one of the enzyme functions to change the pH of the environment so that the reaction can occur.
3. The enzyme will alter the shape of the substrate to modify the rate of chemical reactions.
4. **Altering the primary structure of the substrate is not a way for the enzyme to modify the rate of chemical reaction.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer

A
B
~~C~~
C
D

Time Spent:
58 secs
Difficulty Level:
Expert
Content & Skills:
SIRS1
CC1A
FC1
BIO

Question 144 of 230

Data ID: 531478
Captured: 6/3/2026, 4:41:18 PM

PASSAGE

Passage 6 (Questions 32-35)

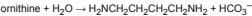
The amino acids from which most proteins are made are called the common amino acids. Most share the general formula shown in Figure 1.



Figure 1

Amino acids can be classified on the basis of the chemical properties of their R groups. Some have hydrophobic, nonpolar R groups (e.g., valine: R = $-\text{CH}(\text{CH}_3)_2$); some have polar, uncharged R groups and are hydrophilic (e.g., serine: R = $-\text{CH}_2\text{OH}$); and others have polar, hydrophilic R groups that are acidic (e.g., aspartic acid: R = $-\text{CH}_2\text{COOH}$) or basic (e.g., lysine: R = $-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$).

Ornithine (R = $-\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$) is an amino acid that is found in cells, but not incorporated into proteins. In mammals and many other organisms, it can be converted to diaminobutane via the ornithine decarboxylase reaction, an early event in cell division. (See Equation 1.)



Equation 1

In the laboratory, the rate of this reaction is measured by an enzyme assay that uses ornithine in which the carboxylic acid carbon is radioactively labeled. The reaction is terminated by acidification of the reaction mixture, which converts the HCO_3^- ion to CO_2 and H_2O .

Thin-layer chromatography is an effective way of separating amino acids. Table 1 below shows R_f values for four amino acids using various solvents. "Orn" represents ornithine. "X," "Y," and "Z" represent three other amino acids.

Table 1

Amino acid	Solvent 1	Solvent 2	Solvent 3
Orn	0.04	0.14	0.24
X	0.15	0.47	0.74
Y	0.06	0.19	0.17
Z	0.03	0.11	0.26

QUESTION + ANSWER CHOICES

Which of the following statements gives the most fundamental reason why ornithine is unlikely to be found in proteins synthesized in vivo?

- A. There is no codon for it in the standard genetic code.
- B. It cannot form a peptide bond.
- C. ~~It is not available in the diet.~~
- D. ~~It has a net positive charge in aqueous solution.~~

SOLUTION

Solution: The correct answer is **A**.

1. **Proteins are generated during translation. In this process, mRNA triplets are read by ribosomes, and tRNA molecules provide the amino acids. Each amino acid is matched to a tRNA molecule carrying a specific anti-codon. A specific amino acid will be added only if it is recognized by its specific codon. Based on the triplet code, no codon recognizes ornithine, thus ornithine cannot integrate proteins primary structure.**
2. As ornithine is an amino acid, it is capable of forming peptide bonds.
3. As the passage indicates, ornithine is most likely provided by the diet.
4. The charge of the amino acid does not prevent it from being incorporated in the protein primary sequence.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleic Acid Structure and Function](#), [DNA Replication](#), [Repair of DNA](#).

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
X
C
D
Time Spent:
12 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC1B
FC1
BIO

Question 145 of 230

Data ID: 531479
Captured: 6/3/2026, 4:41:18 PM

PASSAGE

Passage 6 (Questions 32-35)

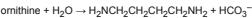
The amino acids from which most proteins are made are called the common amino acids. Most share the general formula shown in Figure 1.



Figure 1

Amino acids can be classified on the basis of the chemical properties of their R groups. Some have hydrophobic, nonpolar R groups (e.g., valine: $R = -CH(CH_3)_2$); some have polar, uncharged R groups and are hydrophilic (e.g., serine: $R = -CH_2OH$); and others have polar, hydrophilic R groups that are acidic (e.g., aspartic acid: $R = -CH_2COOH$) or basic (e.g., lysine: $R = -CH_2CH_2CH_2CH_2NH_2$).

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Equation 1

In the laboratory, the rate of this reaction is measured by an enzyme assay that uses ornithine in which the carboxylic acid carbon is radioactively labeled. The reaction is terminated by acidification of the reaction mixture, which converts the HCO_3^- ion to CO_2 and H_2O .

Thin-layer chromatography is an effective way of separating amino acids. Table 1 below shows R_f values for four amino acids using various solvents. "Orn" represents ornithine. "X," "Y," and "Z" represent three other amino acids.

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QUESTION + ANSWER CHOICES

Which of the following terms best describes the role of ornithine decarboxylase in the reaction shown in Equation 1?

- A. Catalyst
- B. Cofactor
- C. Substrate
- D. Activator

SOLUTION

Solution: The correct answer is A.

- The passage indicates that ornithine is converted to diaminobutane via ornithine decarboxylase reaction, inferring that ornithine decarboxylase is an enzyme. Enzymes are classified as catalysts in biochemical reactions.
- The passage infers that ornithine decarboxylase is an enzyme. Enzymes are classified as catalysts, rather than cofactors in biochemical reactions.
- As described in the passage, ornithine decarboxylase converts ornithine to diaminobutane. This infers that ornithine decarboxylase is an enzyme rather than a substrate.
- The information in the passage indicates that ornithine decarboxylase is involved in a biochemical reaction, inferring that it is an enzyme, rather than an activator.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer

A

✓

B

C

D

Time Spent:

3 secs

Difficulty Level:

Difficult

Content & Skills:

SIRS1

CC1A

FC1

BCH

Question 146 of 230

Data ID: 631480
Captured: 6/3/2026, 4:41:19 PM

PASSAGE

Passage 6 (Questions 32-35)

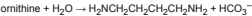
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Z	0.03	0.11	0.26

QUESTION + ANSWER CHOICES

The ornithine decarboxylase reaction has been studied extensively by biomedical researchers. The most likely reason for the interest of these researchers is that the reaction is:

- A. a good source of ornithine.
- B. an early event in cell division.
- C. unique to mammals.
- D. one in which a chiral reactant is converted to an achiral product.

SOLUTION

Solution: The correct answer is B.

1. The ornithine decarboxylase reaction is an event that will result in ornithine degradation rather than being a source of ornithine.
2. The passage states that, in mammals and other organisms, ornithine decarboxylase is active during the early events of cell division.
3. The passage indicates that ornithine decarboxylase reaction takes place in mammals and other organisms, thus it is not unique to mammals.
4. Although a carboxylase can change a chiral substrate into an achiral product, it is not the most likely reason that the researchers are interested in this enzyme.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleic Acid Structure and Function](#), [DNA Replication](#), [Repair of DNA](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer

A

B

✓

C

D

Time Spent:

3 secs

Difficulty Level:

Moderate

Content & Skills:

SIRS3

CC1B

FC1

BIO

Question 147 of 230

Data ID: 531481
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PASSAGE

Passage 6 (Questions 32-35)

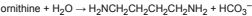
The amino acids from which most proteins are made are called the common amino acids. Most share the general formula shown in Figure 1.



Figure 1

Amino acids can be classified on the basis of the chemical properties of their R groups. Some have hydrophobic, nonpolar R groups (e.g., valine: R = –CH(CH₃)₂); some have polar, uncharged R groups and are hydrophilic (e.g., serine: R = –CH₂OH); and others have polar, hydrophilic R groups that are acidic (e.g., aspartic acid: R = –CH₂COOH) or basic (e.g., lysine: R = –CH₂CH₂CH₂CH₂NH₂).

Ornithine (R = –CH₂CH₂CH₂NH₂) is an amino acid that is found in cells, but not incorporated into proteins. In mammals and many other organisms, it can be converted to diaminobutane via the ornithine decarboxylase reaction, an early event in cell division. (See Equation 1.)



Equation 1

In the laboratory, the rate of this reaction is measured by an enzyme assay that uses ornithine in which the carboxylic acid carbon is radioactively labeled. The reaction is terminated by acidification of the reaction mixture, which converts the HCO₃[–] ion to CO₂ and H₂O.

Thin-layer chromatography is an effective way of separating amino acids. Table 1 below shows R_f values for four amino acids using various solvents. “Orn” represents ornithine. “X,” “Y,” and “Z” represent three other amino acids.

Table 1

Amino acid	Solvent 1	Solvent 2	Solvent 3
Orn	0.04	0.14	0.24
X	0.15	0.47	0.74
Y	0.06	0.19	0.17
Z	0.03	0.11	0.26

QUESTION + ANSWER CHOICES

The statement that the ornithine decarboxylase assay is highly specific means that it:

- A. requires radioactive ornithine of high specific activity.
- B. generates diaminobutane of high specific activity.
- C. can distinguish ornithine decarboxylase activity from the many other enzymatic reactions in a cell.
- D. can measure the small amount of ornithine present in a cell.

SOLUTION

Solution: The correct answer is C.

- Enzyme assays are typically very specific and only measures the activity of a single chemical reaction catalyzed by a single enzyme. Ornithine is an amino acid and does not possess an enzymatic specific activity.
- Diaminobutane is a four carbon diamine and does not possess specific activity.
- Enzymes such as ornithine decarboxylase are highly specific both in the reactions that they catalyze and in their choice of substrates. Therefore, ornithine decarboxylase assay is very specific and only measures the activity of a single chemical reaction catalyzed by ornithine decarboxylase.**
- Enzyme specificity relates to the type of substrate for the reaction being catalyzed and not the quantity of substrate present in the cell.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct

Correct Answer Your Answer

A
B
C
✓
D

Time Spent:
10 secs

Difficulty Level:
Moderate

Content & Skills:

SIRS1

CC1A

FC1

BCH

Question 148 of 230

Data ID: 531503

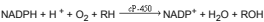
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PASSAGE

Passage 7 (Questions 36-39)

Many lipid-soluble toxins must be metabolized before they can be excreted from an organism. In mammalian systems, this metabolic process is often catalyzed by a type of membrane-bound heme protein known as cytochrome P-450 (cP-450). Many forms (isoenzymes) of cP-450 have been isolated, all of which contain iron with 4 coordinate bonds to nitrogen atoms in the heme group and 1 coordinate bond to a cysteine residue in the protein portion of the enzyme.

The general reaction by which cP-450 detoxifies many compounds is shown in Reaction 1 where RH is a lipid-soluble toxin.



Reaction 1

cP-450 is inducible. That is, when an organism is challenged by a new toxin or by increased concentrations of a toxin, the organism can increase its concentration of cP-450 and thus metabolize and excrete the toxin more effectively.

However, cP-450 does not metabolize all toxins at the same rate. For example, although cP-450 usually metabolizes barbiturates, alcohol acts as a competitive inhibitor of barbiturate metabolism.

The role of cP-450 in the metabolism of toxins may be exemplified by the following case in which a chronic alcoholic died of a barbiturate overdose.

While sober, the alcoholic was having trouble falling asleep and, therefore, took the recommended dosage of sleeping pills (barbiturates). When no drowsiness was felt, the alcoholic consumed more sleeping pills, then drank an alcoholic beverage. Severe respiratory depression and death soon followed caused by an overdose of barbiturates. An autopsy revealed that if alcohol had not been present, the barbiturates would have been metabolized rapidly enough to be eliminated by the body, and death would not have occurred.

QUESTION + ANSWER CHOICES

In the example of the alcoholic in the passage, the presence of alcohol caused death because the alcohol:

- A.
denatured the cP-450.
- **B.**
inhibited the cP-450.
- C.
decreased the concentration of cP-450.
- D.
increased the concentration of cP-450.

SOLUTION

Solution: The correct answer is **B**.

1. The passage indicates that alcohol is an inhibitor of cP-450, not a molecule able to degrade it.
2. **Based on the passage alcohol inhibits cP-450, thus preventing it from metabolizing barbiturates.**
3. According to the passage alcohol does not decrease cP-450 concentration.
4. If the concentration of cP-450 was increased, the barbiturate would have been eliminated faster, avoiding the death of the individual.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer

A
B
✓
C
D

Time Spent:

6 secs

Difficulty Level:

Moderate

Content & Skills:

SIRS2

CC1A

FC1

BIO

Question 149 of 230

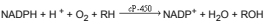
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PASSAGE

Passage 7 (Questions 36-39)

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QUESTION + ANSWER CHOICES

Ethanol may be metabolized to acetic acid, then condensed with a coenzyme to form acetyl coenzyme A. Acetyl coenzyme A may then participate in:

- A. the Krebs (citric acid) cycle.
- B. glycolysis.
- C. electron transport.
- D. oxidative phosphorylation.

SOLUTION

Solution: The correct answer is A.

- Acetyl-CoA is the molecule that enters the Krebs cycle.
- The molecule that enters glycolysis is glucose, not Acetyl-CoA.
- Acetyl-CoA enters the Krebs cycle, not the electron transport chain.
- Acetyl-CoA enters the Krebs cycle, not oxidative phosphorylation. Oxidative phosphorylation is the process that follows the Krebs cycle.

Khan Academy Lessons

For more help on the skills mentioned or content assessed by this question, see the Khan Academy lessons on Glycolysis, Gluconeogenesis, and the Pentose Phosphate Pathway; Citric Acid

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
6 secs
Difficulty Level:
Easy
Content & Skills:
SIRS1
CC1D
FC1
BIO

Question 150 of 230

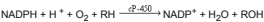
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PASSAGE

Passage 7 (Questions 36-39)

Many lipid-soluble toxins must be metabolized before they can be excreted from an organism. In mammalian systems, this metabolic process is often catalyzed by a type of membrane-bound heme protein known as cytochrome P-450 (cP-450). Many forms (isoenzymes) of cP-450 have been isolated, all of which contain iron with 4 coordinate bonds to nitrogen atoms in the heme group and 1 coordinate bond to a cysteine residue in the protein portion of the enzyme.

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Reaction 1

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However, cP-450 does not metabolize all toxins at the same rate. For example, although cP-450 usually metabolizes barbiturates, alcohol acts as a competitive inhibitor of barbiturate metabolism.

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QUESTION + ANSWER CHOICES

The lung cells of heavy smokers would be expected to have greatly increased concentrations of cP-450 and:

- A. DNA sequences that code for cP-450.
- B. mRNA sequences that code for cP-450.
- C. rRNA that process cP-450.
- D. tRNA that are specific for cysteine.

SOLUTION

Solution: The correct answer is B.

- cP-450 DNA is present in the genome, and its levels do not change.
- The levels of cP-450 mRNA should be elevated in smokers as they are almost directly related to the levels of cP-450 protein produced to get rid of the toxins.
- The levels of ribosomal RNA (rRNA) should not change in the epithelial cells of smokers.
- tRNA levels specific for cysteine are not related to the production of cP-450 protein.

Khan Academy Lessons

For more help on the skills mentioned or content assessed by this question, see the Khan Academy lessons on Nucleic Acid Structure and Function; DNA Replication; Repair of DNA

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
18 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC1B
FC1
BIO

Question 151 of 230

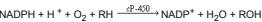
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PASSAGE

Passage 7 (Questions 36-39)

Many lipid-soluble toxins must be metabolized before they can be excreted from an organism. In mammalian systems, this metabolic process is often catalyzed by a type of membrane-bound heme protein known as cytochrome P-450 (cP-450). Many forms (isoenzymes) of cP-450 have been isolated, all of which contain iron with 4 coordinate bonds to nitrogen atoms in the heme group and 1 coordinate bond to a cysteine residue in the protein portion of the enzyme.

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Reaction 1

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QUESTION + ANSWER CHOICES

No drowsiness was initially felt by the alcoholic because the previous abuse of alcohol had:

- A. denatured the cP-450.
- B. inhibited the cP-450.**
- C. reduced the cP-450.
- D. induced the cP-450.

SOLUTION

Solution: The correct answer is D.

1. If cP-450 was denatured, the individuals would have felt drowsiness as the barbiturate would have had an immediate effect.
2. Inhibition of cP-450 would have resulted in drowsiness felt by the individual, not the opposite.
3. The passage indicates that cP-450 expression is inducible. Thus previous alcohol abuse will induce, rather than reduce, cP-450.
4. **The passage indicates that cP-450 is involved in barbiturate elimination and that its expression can be induced by higher levels of toxin, alcohol in this case. This suggests that higher levels of alcohol result in a rise of cP-450 levels and faster barbiturate elimination. As a consequence, higher doses of barbiturate will be needed to obtain drowsiness.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
x
C
D
Time Spent:
8 secs
Difficulty Level:
Expert
Content & Skills:
SIRS2
CC1A
FC1

BIO

Question 152 of 230

Data ID: 531519
Captured: 6/3/2026, 4:41:23 PM

PASSAGE

Passage 8 (Questions 40-43)

Endothelial cells (ECs) define the lumen of capillaries and are often associated with elongated, contractile cells called *pericytes* (Figure 1).

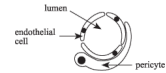


Figure 1

The following experiments were performed to test the effect of pericytes on the growth rate of ECs. In both experiments, ECs were cocultured with either pericytes, smooth-muscle cells, or fibroblasts. All cells were obtained from mammalian cell cultures.

Experiment 1: Coculture with Contact

Equal numbers of endothelial cells were placed in three separate containers (labeled A–C). Pericytes, smooth-muscle cells, and fibroblasts were growth-arrested—that is, treated so that they would not divide but other metabolic processes would function normally. The growth-arrested cells were mixed directly with the ECs in Containers A–C. Three matched control containers received only ECs. The number of cells in each container was counted electronically over 14 days; results are shown in Figure 2.

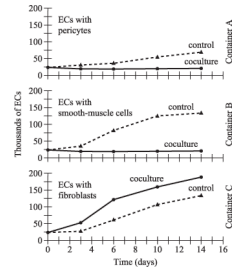


Figure 2

Experiment 2: Coculture without Contact

Equal numbers of ECs were added to two containers (D and E). Growth-arrested pericytes and smooth-muscle cells were then placed in Containers D and E, respectively; however, these cells were separated from the ECs by a semipermeable membrane. Matched control containers received only ECs. The number of cells was again counted for 14 days; results are shown in Figure 3.

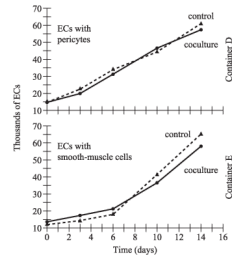


Figure 3

Figures 2 and 3 adapted from Alicia Oridge and Patricia A. D'Amore, "Inhibition of Capillary Endothelial Cell Growth by Pericytes and Smooth Muscle Cells." ©1987 by The Rockefeller University Press.

The pericytes used in these experiments were probably in which phase of the cell cycle?

- A. Telophase
- B. Metaphase
- C. Anaphase
- D. Interphase

SOLUTION

Solution: The correct answer is D.

1. Telophase is one of the last phases of mitosis. Cells that are in telophase are dividing.
2. Metaphase is the phase of mitosis during which the chromosomes are located in the middle of the cell plate. Cells that are in mitosis are dividing.
3. Anaphase is the phase of mitosis where sister chromatids are separated. Cells that are in anaphase are dividing.
4. The passage indicates that pericytes and other supportive cells were growth arrested, thus they were not dividing. Cells that are in interphase are not actively dividing.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
5 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC2C
FC2
BIO

Data ID: 831524
Captured: 6/3/2026, 4:41:23 PM

Passage 8 (Questions 40-43)

A schematic diagram of a blood vessel cross-section. The central cavity is labeled 'lumen'. The inner lining is labeled 'endothelial cell'. A cell on the outer wall is labeled 'pericyte'.

The following experiments were performed to test the effect of pericytes on the growth rate of ECs. In both experiments, ECs were cocultured with either pericytes, smooth-muscle cells, or fibroblasts. All cells were obtained from mammalian cell cultures.

Equal numbers of endothelial cells were placed in three separate containers (labeled A-C). Pericytes, smooth-muscle cells, and fibroblasts were *growth-arrested*—that is, treated so that they would not divide but other metabolic processes would function normally. The growth-arrested cells were mixed directly with the ECs in Containers A-C. Three matched control containers received only ECs. The number of cells in each container was counted electronically over 14 days; results are shown in Figure 2.



Equal numbers of ECs were added to two containers (D and E). Growth-arrested pericytes and smooth-muscle cells were then placed in Containers D and E, respectively; however, these cells were separated from the ECs by a semipermeable membrane. Matched control containers received only ECs. The number of cells was again counted for 14 days; results are shown in Figure 3.



QUESTION + ANSWER CHOICES

- A.
No; the DNA in circulating erythrocytes is needed to help transport O_2 through the capillaries.
- B.
No; circulating erythrocytes do not contain DNA.
- C.
Yes; DNA is responsible for cell division in most cells.
- D.
Yes; circulating erythrocytes carry DNA nutrients through the capillaries.

Solution: The correct answer is **B**.

1. Mature erythrocytes do not contain DNA. It is proteins, not DNA, that help transport oxygen.
2. **Mature erythrocytes are enucleated cells that do not contain DNA.**
3. It is the levels of specific proteins, like cyclins and MPF, that regulate cell division.
4. Erythrocytes do not carry DNA, nor DNA nutrients, through the capillaries because erythrocytes are enucleated cells.

For more help on the skills, concepts, or content associated with this question, see the Khan Academy lessons on [Daerinstani Suetam](#), [Circulastani Suetam](#), [Imunastani Suetam](#), [Immuna Suetam](#).

Incorrect
Correct Answer Your Answer
A
B
C
X
D
Time Spent:
11 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC3B
FC3
BIO

Question 154 of 230

Data ID: 531526
Captured: 6/3/2026, 4:41:24 PM

PASSAGE

Passage 8 (Questions 40-43)

Endothelial cells (ECs) define the lumen of capillaries and are often associated with elongated, contractile cells called pericytes (Figure 1).

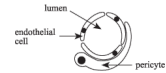


Figure 1

The following experiments were performed to test the effect of pericytes on the growth rate of ECs. In both experiments, ECs were cocultured with either pericytes, smooth-muscle cells, or fibroblasts. All cells were obtained from mammalian cell cultures.

Experiment 1: Coculture with Contact

Equal numbers of endothelial cells were placed in three separate containers (labeled A–C). Pericytes, smooth-muscle cells, and fibroblasts were growth-arrested—that is, treated so that they would not divide but other metabolic processes would function normally. The growth-arrested cells were mixed directly with the ECs in Containers A–C. Three matched control containers received only ECs. The number of cells in each container was counted electronically over 14 days; results are shown in Figure 2.

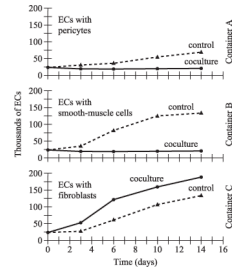


Figure 2

Experiment 2: Coculture without Contact

Equal numbers of ECs were added to two containers (D and E). Growth-arrested pericytes and smooth-muscle cells were then placed in Containers D and E, respectively; however, these cells were separated from the ECs by a semipermeable membrane. Matched control containers received only ECs. The number of cells was again counted for 14 days; results are shown in Figure 3.

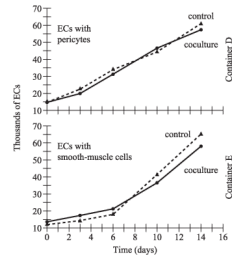


Figure 3

Figures 2 and 3 adapted from Alicia Orledge and Patricia A. D'Amore, "Inhibition of Capillary Endothelial Cell Growth by Pericytes and Smooth Muscle Cells." ©1987 by The Rockefeller University Press.

QUESTION + ANSWER CHOICES

Pericytes were growth-arrested in Experiment 1 so that the:

- A. pericyte growth would not interfere with the measurement of EC growth.
- B. pericytes would not inhibit EC growth.
- C. metabolic wastes of the pericytes would not interfere with the measurement of EC growth.
- D. ECs would not cause the pericytes to grow.

SOLUTION

Solution: The correct answer is A.

1. The passage indicates that endothelial cells and supportive cells were mixed together and the cell count was done electronically. This method will not allow identification of the type of dividing cell. Thus, the best way to ensure only dividing endothelial cells are counted would be to prevent other cells, in this case pericytes, from dividing.
2. Although it might be possible that the presence of too many pericytes negatively affects the growth of endothelial cells, this is not the reason why pericyte growth was arrested.
3. It is quite unlikely that the metabolic waste might interfere with endothelial cell growth. It is most likely that the presence of more pericytes might favor endothelial cell growth.
4. The purpose of the experiment is to evaluate whether the pericytes support endothelial cell growth, and not the opposite.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Respiratory System](#); [Circulatory System](#); [Lymphatic System](#); [Immune System](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
4 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS3
CC3B
FC3
BIO

Question 155 of 230

Data ID: 531528
Captured: 6/3/2026, 4:41:24 PM

PASSAGE

Passage 8 (Questions 40-43)

Endothelial cells (ECs) define the lumen of capillaries and are often associated with elongated, contractile cells called pericytes (Figure 1).

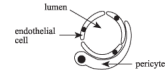


Figure 1

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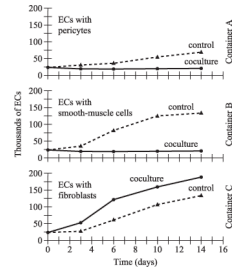


Figure 2

Experiment 2: Coculture without Contact

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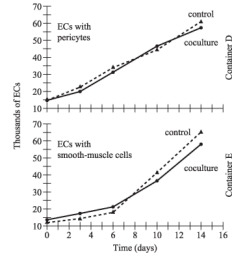


Figure 3

Figures 2 and 3 adapted from Alicia Orledge and Patricia A. D'Amore, "Inhibition of Capillary Endothelial Cell Growth by Pericytes and Smooth Muscle Cells." ©1987 by The Rockefeller University Press.

Based on Figure 1 and the passage, which of the following cells are most important in the exchange of O₂ between the blood and the surrounding tissues?

- A. Pericytes
- B. Endothelial cells**
- C. Smooth-muscle cells
- D. Fibroblasts

SOLUTION

Solution: The correct answer is B.

- Pericytes are located between the endothelial cells and the matrix and they do not have direct contact with circulating blood.
- Endothelial cells are the cells that are in direct contact with blood and the surrounding matrix so these are the cells that play the most important role in gas exchange.**
- Smooth-muscle cells are not in direct contact with blood, thus they are least likely to play a critical role in gas exchange between blood and the surrounding tissues.
- Fibroblasts are supportive cells that are not in direct contact with the circulating blood, thus they are not likely to play a role in gas exchange between blood and the surrounding tissues.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Respiratory System](#); [Circulatory System](#); [Lymphatic System](#); [Immune System](#).

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
X
D
Time Spent:
3 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS4
CC3B
FC3
BIO

Question 156 of 230

Data ID: 531530
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QUESTION • ANSWER CHOICES

The process of culturing bacteria often involves inoculation of cells on a noncellular, agar-based medium. Such a methodology would NOT result in growth of animal viruses because animal viruses:

- **A.** are obligate parasites.
- B. lack DNA.
- C. assimilate carbon.
- D. require essential vitamin supplements for growth.

SOLUTION

Solution: The correct answer is **A**.

1. **Animal viruses can only infect animal cells, not bacteria. Thus, animal viruses cannot grow in agar plates with bacteria.**
2. Some animal viruses contain DNA. The inability to culture animal viruses on agar plates is not due to the structure of their genome, but rather to their selective capability of infection.
3. Carbon assimilation is not the reason why viruses cannot be grown in bacteria seeded agar plates, but rather because animal viruses cannot infect bacteria.
4. It is the animal cells infected by the animal viruses that will require essential vitamin supplements for growth, and not the viruses per se. Animal viruses will infect only animal cells, thus, they cannot be grown in bacteria seeded agar plates.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Cell Theory, Classification, and Structure of Prokaryotic Cells](#), [Growth and](#)

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
1 min 30 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC2B
FC2
BIO

Question 157 of 230

Data ID: 531531
Captured: 6/3/2026, 4:41:26 PM

QUESTION • ANSWER CHOICES

In general, telomeres are NOT important to bacterial cells because most bacterial chromosomes:

- A. do not replicate.
- **B.** are circular.
- C. replicate quickly and efficiently.
- D. are composed of single-stranded DNA.

SOLUTION

Solution: The correct answer is **B**.

1. Bacterial chromosomes replicate and their DNA copies are distributed to their daughter cells like in eukaryotic cells.
2. **The term telomere refers to a distal end of a chromosome. Telomeres contain non-coding repeats. As opposed to eukaryotic cells, bacterial DNA is constituted by one single circular molecule, thus it does not contain telomeres.**
3. Lack of telomeres is not due to the speed of replication.
4. Bacterial DNA is double stranded like the eukaryotic DNA.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleic Acid Structure and Function](#), [DNA Replication](#), [Repair of DNA](#),

RESULT / TIMING / TAGS

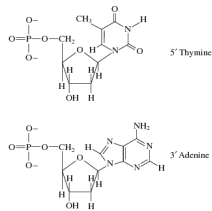
Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
20 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC1B
FC1
BIO

Question 158 of 230

Data ID: 831532
Captured: 6/3/2026, 4:41:26 PM

QUESTION • ANSWER CHOICES

Which of the following best describes the bond that would form between the following two nucleotides if they were located adjacent to each other as shown in a single strand of DNA?



- A.
A bond between the phosphate of the thymine and the phosphate of the adenine
- B.
A bond between an oxygen in the thymine base and a nitrogen in the adenine base
- C.
A bond between the phosphate of the thymine and the sugar of the adenine
- D.
A bond between the phosphate of the adenine and the sugar of the thymine

SOLUTION

Solution: The correct answer is **D**.

1. Nucleotides are linked to one another by phosphodiester bonds between a hydroxyl group of the sugar base of the nucleotide at the 5' (thymine) end and the phosphate group of the adjacent nucleotide at the 3' end (adenine). In this option, thymine is placed 5' to adenine.
2. The oxygens and nitrogens of nucleotides are not involved in bond formation between adjacent nucleotides.
3. This is the opposite of the bond that would form between a thymine at 5' end and an adjacent adenine at 3' end.
4. **Nucleotides are linked to one another by phosphodiester bonds between the sugar base of one nucleotide (thymine) and the phosphate group of the adjacent nucleotide (adenine) in a way that the 5' end bears a phosphate, and the 3' end a hydroxyl group.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
1 min 31 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS1
CC1B
FC1
BCH

Question 159 of 230

Data ID: 831534
Captured: 6/3/2026, 4:41:27 PM

QUESTION • ANSWER CHOICES

During an action potential, the movement of sodium ions into a neuron causes the neuronal membrane to do which of the following?

- A.
Inactivate
- B.
Hyperpolarize
- C.
Repolarize
- D.
Depolarize

SOLUTION

Solution: The correct answer is **D**.

1. The flux of sodium into the cytoplasm of the cell results in membrane depolarization. This is required for activation, not inactivation, of neuronal action potentials.
2. Hyperpolarization occurs when the membrane potential becomes more electronegative. On the contrary, the flux of sodium into the cytoplasm of the cell would result in a reduction of the membrane potential and thus depolarization instead of hyperpolarization.
3. Repolarization occurs after the membrane has been depolarized and is due to the opening of potassium channels. Sodium channels are typically inactivated during this period.
4. **As sodium enters the cytoplasm of the cell, the membrane potential will increase and the cell will depolarize.**

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Neuronic System: Structure and Function](#), [Neuronic Cell](#), and [Endocrine System](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
13 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC3A
FC3
BIO

Question 160 of 230

Data ID: 531547
Captured: 6/3/2026, 4:41:27 PM

PASSAGE

Passage 9 (Questions 48-51)

A 52-year-old female patient shows symptoms of generalized weakness, low blood pressure (80/50 mmHg), weight loss, nausea, and an extreme craving for salt. Her adrenal cortex is found to be atrophied and to contain both lymphocytes and adrenal autoantibodies. A diagnosis of primary adrenal gland insufficiency, or *Addison's disease*, is made.

Addison's disease occurs when cells of the adrenal cortex are destroyed, leaving the gland unable to secrete either glucocorticoids or mineralocorticoids. A major function of cortisol, the body's primary glucocorticoid, is to stimulate gluconeogenesis (formation of glucose from noncarbohydrate sources) in the liver by activating DNA transcription to produce liver enzymes and by mobilizing amino acids from muscle tissue. Aldosterone, the primary mineralocorticoid, maintains ionic balance by causing conservation of Na⁺ and excretion of K⁺.

Addison's disease may arise either as a result of tuberculosis or as an autoimmune disease. Adrenal autoantibodies are not present when tuberculosis is the cause.

To treat her Addison's disease, this patient must take both glucocorticoids and mineralocorticoids for the remainder of her life. With this medication, her life span is expected to be normal.

QUESTION • ANSWER CHOICES

The aldosterone deficiency associated with Addison's disease will cause a decrease in the serum levels of all of the following ions EXCEPT:

- A. Na⁺ ions.
- B. Cl⁻ ions.
- C. K⁺ ions.
- D. HCO₃⁻ ions.

SOLUTION

Solution: The correct answer is C.

- The passage indicates that aldosterone is responsible for sodium ion conservation in the body, thus in the absence of aldosterone, the levels of sodium ions inside the body will decrease.
- Chloride ions movements follow the movements of sodium ions. Thus, if the levels of sodium ions decline, the levels of chloride ions will decline as well.
- The passage indicates that aldosterone is responsible for the elimination of potassium ions from the body. Thus, in the absence of aldosterone, the levels of potassium will increase.**
- The passage indicates a role for corticosteroids in the regulation of sodium and potassium. The passage does not provide any information on the role of the adrenal gland on HCO₃⁻ levels, which most likely stay constant.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Respiratory System](#), [Circulatory System](#), [Urinary System](#), [Immune System](#).

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
X
Time Spent:
11 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC3B
FC3
BIO

Question 161 of 230

Data ID: 531551
Captured: 6/3/2026, 4:41:28 PM

PASSAGE

Passage 9 (Questions 48-51)

A 52-year-old female patient shows symptoms of generalized weakness, low blood pressure (80/50 mmHg), weight loss, nausea, and an extreme craving for salt. Her adrenal cortex is found to be atrophied and to contain both lymphocytes and adrenal autoantibodies. A diagnosis of primary adrenal gland insufficiency, or *Addison's disease*, is made.

Addison's disease occurs when cells of the adrenal cortex are destroyed, leaving the gland unable to secrete either glucocorticoids or mineralocorticoids. A major function of cortisol, the body's primary glucocorticoid, is to stimulate gluconeogenesis (formation of glucose from noncarbohydrate sources) in the liver by activating DNA transcription to produce liver enzymes and by mobilizing amino acids from muscle tissue. Aldosterone, the primary mineralocorticoid, maintains ionic balance by causing conservation of Na⁺ and excretion of K⁺.

Addison's disease may arise either as a result of tuberculosis or as an autoimmune disease. Adrenal autoantibodies are not present when tuberculosis is the cause.

To treat her Addison's disease, this patient must take both glucocorticoids and mineralocorticoids for the remainder of her life. With this medication, her life span is expected to be normal.

QUESTION • ANSWER CHOICES

Normally, a hypothalamic factor stimulates the release of adrenocorticotrophic hormone (ACTH) from the pituitary gland. In a patient with Addison's disease, the secretion of the hypothalamic factor will:

- A. be lower than normal.
- B. be higher than normal.
- C. be unchanged.
- D. increase before disease onset and decrease thereafter.

SOLUTION

Solution: The correct answer is B.

- Negative feedback is the mechanism that regulates the release of corticotropin releasing hormone (CRH) from the hypothalamus. In the absence or in the presence of low levels of circulating cortisol, like in Addison's disease, CRH secretion is not inhibited, thus CRH levels will be higher, not lower, than normal.
- Release of ACTH from the pituitary is regulated by negative feedback. In normal conditions, high levels of circulating glucose and other stressors activate the production of corticotropin-releasing hormone (CRH) from the hypothalamus. CRH will stimulate the pituitary gland to release ACTH which will trigger cortisol release from the adrenal cortex. Finally, the presence of high levels of circulating cortisol will inhibit CRH secretion (negative feedback) thus closing the loop. In Addison's disease, the circulating levels of ACTH will be higher than normal because the factor that triggers the inhibition of CRH production, high cortisol levels, is absent or low.**
- Corticotropin releasing hormone (CRH) levels will be higher than normal, and not unchanged, in individuals suffering from Addison's disease because the levels of circulating cortisol, that normally inhibits CRH secretion, are low.
- High levels of circulating cortisol are the stimulus to reduce corticotropin releasing hormone (CRH) secretion from the hypothalamus. In Addison's disease, the levels of cortisol are low, thus the inhibitory stimulus is absent, triggering continuous secretion and high levels of CRH. Thus, the levels of CRH are most likely low before the disease onset and high thereafter, not the opposite.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Muscular System: Structure and Function](#), [Nervous System: Structure and Function](#), [Endocrine System](#).

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
X
Time Spent:
4 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC3A
FC3
BIO

Question 162 of 230

Data ID: 531565
Captured: 6/3/2026, 4:41:29 PM

PASSAGE

Passage 9 (Questions 48-51)

A 52-year-old female patient shows symptoms of generalized weakness, low blood pressure (80/50 mmHg), weight loss, nausea, and an extreme craving for salt. Her adrenal cortex is found to be atrophied and to contain both lymphocytes and adrenal autoantibodies. A diagnosis of primary adrenal gland insufficiency, or *Addison's disease*, is made.

Addison's disease occurs when cells of the adrenal cortex are destroyed, leaving the gland unable to secrete either glucocorticoids or mineralocorticoids. A major function of cortisol, the body's primary glucocorticoid, is to stimulate gluconeogenesis (formation of glucose from noncarbohydrate sources) in the liver by activating DNA transcription to produce liver enzymes and by mobilizing amino acids from muscle tissue. Aldosterone, the primary mineralocorticoid, maintains ionic balance by causing conservation of Na⁺ and excretion of K⁺.

Addison's disease may arise either as a result of tuberculosis or as an autoimmune disease. Adrenal autoantibodies are not present when tuberculosis is the cause.

To treat her Addison's disease, this patient must take both glucocorticoids and mineralocorticoids for the remainder of her life. With this medication, her life span is expected to be normal.

QUESTION • ANSWER CHOICES

If a patient with Addison's disease is given too high a replacement dose of glucocorticoids, the effect over time will be an increase in:

- A. muscle mass.
- B. muscle weakness.
- C. white blood cell count.
- D. heart rate.

SOLUTION

Solution: The correct answer is **B**.

- One of the effects of glucocorticoids is to increase protein degradation in various tissues, including muscles. Thus, high levels of circulating glucocorticoids will most likely reduce, rather than increase, muscle mass.
- High levels of circulating glucocorticoids will increase protein degradation in various tissues, muscles included. A direct consequence of protein degradation in muscles is muscle weakness.**
- Glucocorticoids suppress immune function, which is mediated by white blood cells. Therefore, the activity of white blood cells will decrease, not increase, in response to excess glucocorticoid exposure.
- Heart rate will be decreased in Addison's affected individuals treated with high doses of glucocorticoids, but these individuals will suffer from hypertension due to increased systemic vascular resistance, increased extracellular volume, and increased cardiac contractility.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Respiratory System](#), [Circulatory System](#), [Urinary System](#), [Immune System](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
4 secs
Difficulty Level:
Expert
Content & Skills:
SIRS2
CC3B
FC3
BIO

Question 163 of 230

Data ID: 531568
Captured: 6/3/2026, 4:41:29 PM

PASSAGE

Passage 9 (Questions 48-51)

A 52-year-old female patient shows symptoms of generalized weakness, low blood pressure (80/50 mmHg), weight loss, nausea, and an extreme craving for salt. Her adrenal cortex is found to be atrophied and to contain both lymphocytes and adrenal autoantibodies. A diagnosis of primary adrenal gland insufficiency, or *Addison's disease*, is made.

Addison's disease occurs when cells of the adrenal cortex are destroyed, leaving the gland unable to secrete either glucocorticoids or mineralocorticoids. A major function of cortisol, the body's primary glucocorticoid, is to stimulate gluconeogenesis (formation of glucose from noncarbohydrate sources) in the liver by activating DNA transcription to produce liver enzymes and by mobilizing amino acids from muscle tissue. Aldosterone, the primary mineralocorticoid, maintains ionic balance by causing conservation of Na⁺ and excretion of K⁺.

Addison's disease may arise either as a result of tuberculosis or as an autoimmune disease. Adrenal autoantibodies are not present when tuberculosis is the cause.

To treat her Addison's disease, this patient must take both glucocorticoids and mineralocorticoids for the remainder of her life. With this medication, her life span is expected to be normal.

QUESTION • ANSWER CHOICES

The most rapid rate of gluconeogenesis will most likely occur in the body when:

- A. blood glucose levels are high.
- B. cortisol release is inhibited.
- C. the body's stores of carbohydrates are low.
- D. the body's stores of proteins are low.

SOLUTION

Solution: The correct answer is **C**.

- Gluconeogenesis becomes active when the body needs more glucose to generate energy. Therefore, gluconeogenesis will most likely occur when blood glucose levels are low, not high.
- Gluconeogenesis becomes active when the body needs energy, so high cortisol levels increase gluconeogenesis. Therefore, gluconeogenesis is most likely downregulated when cortisol release is inhibited.
- Gluconeogenesis is the pathway for the synthesis of glucose (a carbohydrate) from other metabolic compounds such as lipids and amino acids. Therefore, this pathway is activated when the body's stores of carbohydrates are low.**
- Gluconeogenesis occurs when blood glucose levels, rather than protein storage, is low.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Principles of Bioenergetics](#), [Carbohydrates](#), [Glycolysis](#), [Gluconeogenesis](#), and

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
18 secs
Difficulty Level:
Easy
Content & Skills:
SIRS1
CC1D
FC1
BCH

Question 164 of 230

Data ID: 531585
Captured: 6/3/2026, 4:41:30 PM

PASSAGE

Passage 10 (Questions 52-56)

A 40-year-old patient developed a constant thirst and a frequent need to urinate. The attending physician suspected the patient had diabetes because in addition to these symptoms, the patient was overweight and had a family history of diabetes, which both are risk factors for the disease.

Diabetes mellitus is a condition in which the body cannot produce or is unable to effectively respond to insulin, a peptide hormone made in pancreatic β cells. Normally, insulin promotes facilitated diffusion of glucose into cells from the bloodstream by inducing the translocation of glucose transporter proteins to the plasma membrane. It also helps regulate the metabolism of fatty and amino acids.

In diabetes, the ability of many types of cells to take up glucose is compromised, leading to elevated blood glucose levels. Many tissues then metabolize fatty acids as an alternative energy source, which can lead to *metabolic acidosis*, a condition of low blood pH. Diabetes can also cause protein degradation, resulting in the release of excess amino acids that can be used as substrates for gluconeogenesis, which further increases blood glucose levels. There are two main types of diabetes. Many cases of Type 1 diabetes, in which the body is unable to produce insulin, are caused by an autoimmune response directed against pancreatic β cells. While the body often produces normal-to-elevated amounts of insulin in Type 2 diabetes, its ability to respond to insulin is compromised.

QUESTION • ANSWER CHOICES

Glucose transporter proteins in the liver do not require the presence of insulin to facilitate the uptake of glucose. However, insulin does stimulate the first step in the glycolytic pathway within the liver. Therefore, in liver cells, insulin most likely:

- A. hinders glucose uptake by increasing the cellular concentration of glucose.
- B. aids glucose uptake by decreasing the cellular concentration of glucose.
- C. hinders glucose uptake by using the ATP needed by the glucose transporter proteins.
- D. aids glucose uptake by providing the ATP needed by the glucose transporter proteins.

SOLUTION

Solution: The correct answer is **B**.

1. Based on the passage, insulin is involved in the first step of the glycolytic pathway. Therefore, it promotes, rather than hinders, glucose uptake.
2. **The question states that insulin stimulates the first step in the glycolytic pathway in the liver. This will result in a decrease of the cellular concentration of glucose. To compensate for the low cellular glucose concentration, glucose uptake is increased.**
3. The question states that insulin stimulates the first step in the glycolytic pathway in the liver and depletion of intracellular glucose levels. Therefore, insulin will promote, rather than hinder, glucose uptake. This process occurs because of the decrease in the cellular concentration of glucose, rather than using the ATP needed by the glucose transporter proteins.
4. Based on the question, insulin stimulates the first step in the glycolytic pathway within the liver. This infers that insulin promotes glucose uptake via depletion of intracellular glucose, rather than providing the ATP needed by the glucose transporter proteins.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Principles of Bioenergetics](#); [Carbohydrates](#); [Glycolysis](#); [Gluconeogenesis](#); [and](#)

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
14 secs
Difficulty Level:
Expert
Content & Skills:
SIRS2
CC1D
FC1
BCH

Question 165 of 230

Data ID: 531588
Captured: 6/3/2026, 4:41:30 PM

PASSAGE

Passage 10 (Questions 52-56)

A 40-year-old patient developed a constant thirst and a frequent need to urinate. The attending physician suspected the patient had diabetes because in addition to these symptoms, the patient was overweight and had a family history of diabetes, which both are risk factors for the disease.

Diabetes mellitus is a condition in which the body cannot produce or is unable to effectively respond to insulin, a peptide hormone made in pancreatic β cells. Normally, insulin promotes facilitated diffusion of glucose into cells from the bloodstream by inducing the translocation of glucose transporter proteins to the plasma membrane. It also helps regulate the metabolism of fatty and amino acids.

In diabetes, the ability of many types of cells to take up glucose is compromised, leading to elevated blood glucose levels. Many tissues then metabolize fatty acids as an alternative energy source, which can lead to *metabolic acidosis*, a condition of low blood pH. Diabetes can also cause protein degradation, resulting in the release of excess amino acids that can be used as substrates for gluconeogenesis, which further increases blood glucose levels. There are two main types of diabetes. Many cases of Type 1 diabetes, in which the body is unable to produce insulin, are caused by an autoimmune response directed against pancreatic β cells. While the body often produces normal-to-elevated amounts of insulin in Type 2 diabetes, its ability to respond to insulin is compromised.

QUESTION • ANSWER CHOICES

Exercise promotes the insulin-independent uptake of glucose in working skeletal muscles. Given this, regular exercise would most likely reduce blood glucose levels in patients with which type(s) of diabetes?

- A. Type 1 only
- B. Type 2 only
- C. Both Type 1 and Type 2
- D. Neither Type 1 nor Type 2

SOLUTION

Solution: The correct answer is **C**.

1. As working skeletal muscles are insulin independent, exercise will reduce the levels of insulin in both types of diabetes, not just Type 1.
2. The item indicates that working skeletal muscles use glucose independently from the availability of insulin. Thus, exercise will reduce circulating glucose levels in both types of diabetes, not just Type 2.
3. **The item indicates that working skeletal muscles use glucose in an insulin-independent manner. Thus, exercise will be able to reduce the levels of glucose in both types of diabetes.**
4. Working skeletal muscle can use glucose independently from insulin availability, thus the levels of circulating glucose will be reduced in both types of diabetes during exercise.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Respiratory System](#); [Circulatory System](#); [Lymphatic System](#); [Immune System](#);

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
6 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC3B
FC3
BIO

Question 166 of 230

Data ID: 531592
Captured: 6/3/2026, 4:41:31 PM

PASSAGE

Passage 10 (Questions 52-56)

A 40-year-old patient developed a constant thirst and a frequent need to urinate. The attending physician suspected the patient had diabetes because in addition to these symptoms, the patient was overweight and had a family history of diabetes, which both are risk factors for the disease.

Diabetes mellitus is a condition in which the body cannot produce or is unable to effectively respond to insulin, a peptide hormone made in pancreatic β cells. Normally, insulin promotes facilitated diffusion of glucose into cells from the bloodstream by inducing the translocation of glucose transporter proteins to the plasma membrane. It also helps regulate the metabolism of fatty and amino acids.

In diabetes, the ability of many types of cells to take up glucose is compromised, leading to elevated blood glucose levels. Many tissues then metabolize fatty acids as an alternative energy source, which can lead to *metabolic acidosis*, a condition of low blood pH. Diabetes can also cause protein degradation, resulting in the release of excess amino acids that can be used as substrates for gluconeogenesis, which further increases blood glucose levels. There are two main types of diabetes. Many cases of Type 1 diabetes, in which the body is unable to produce insulin, are caused by an autoimmune response directed against pancreatic β cells. While the body often produces normal-to-elevated amounts of insulin in Type 2 diabetes, its ability to respond to insulin is compromised.

QUESTION • ANSWER CHOICES

During the production of insulin, the translated polypeptide is cleaved into the mature form and secreted from the cell. The cleavage most likely takes place in which of the following locations?

- A. Nucleus
- B. Ribosomes
- C. Endomembrane system
- D. Cytoplasm

SOLUTION

Solution: The correct answer is C.

1. Insulin is not a transcription factor and its polypeptide is most likely not present in the nucleus.
2. Insulin polypeptide is translated, not cleaved, in the ribosomes.
3. **The endomembrane system is the portion of the cells that is in charge of modifying proteins that will be secreted. Thus, it is most likely that insulin cleavage will occur in the endomembrane system.**
4. Insulin cleavage is most likely to occur in the endomembrane system, rather than the cytoplasm, because insulin is a secreted protein.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Protein Movement](#), [Membranes](#), [Bound Membranes](#), and [Protein Secretion](#).

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
X
Time Spent:
33 secs
Difficulty Level:
Expert
Content & Skills:
SIRS1
CC2A
FC2
BIO

Question 167 of 230

Data ID: 531596
Captured: 6/3/2026, 4:41:32 PM

PASSAGE

Passage 10 (Questions 52-56)

A 40-year-old patient developed a constant thirst and a frequent need to urinate. The attending physician suspected the patient had diabetes because in addition to these symptoms, the patient was overweight and had a family history of diabetes, which both are risk factors for the disease.

Diabetes mellitus is a condition in which the body cannot produce or is unable to effectively respond to insulin, a peptide hormone made in pancreatic β cells. Normally, insulin promotes facilitated diffusion of glucose into cells from the bloodstream by inducing the translocation of glucose transporter proteins to the plasma membrane. It also helps regulate the metabolism of fatty and amino acids.

In diabetes, the ability of many types of cells to take up glucose is compromised, leading to elevated blood glucose levels. Many tissues then metabolize fatty acids as an alternative energy source, which can lead to *metabolic acidosis*, a condition of low blood pH. Diabetes can also cause protein degradation, resulting in the release of excess amino acids that can be used as substrates for gluconeogenesis, which further increases blood glucose levels. There are two main types of diabetes. Many cases of Type 1 diabetes, in which the body is unable to produce insulin, are caused by an autoimmune response directed against pancreatic β cells. While the body often produces normal-to-elevated amounts of insulin in Type 2 diabetes, its ability to respond to insulin is compromised.

QUESTION • ANSWER CHOICES

Despite the effects of diabetes, the brains of diabetic patients still receive adequate nourishment. This is most likely because the brain uses:

- A. less glucose than do other body tissues.
- B. insulin-independent transporters for the uptake of glucose.
- C. fatty acids for energy instead of glucose.
- D. insulin-dependent transporters for the uptake of glucose.

SOLUTION

Solution: The correct answer is B.

1. The brain is the main consumer of glucose in the body.
2. **The brain cells contain glucose transporters that are insulin independent.**
3. Although under certain specific conditions the brain can use fuels other than glucose, under physiological conditions glucose is the main fuel for brain cells. Since the brains of diabetic patients receive an adequate amount of glucose, the brain must use an insulin-independent mechanism for glucose uptake
4. If the brain cells had insulin-dependent transporters, they will not be functioning correctly in individuals affected by diabetes.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Muscle System: Structure and Function](#), [Muscle Cells](#), and [Endocrine System](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
11 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC3A
FC3
BIO

Question 168 of 230

Data ID: 831602
Captured: 6/3/2026, 4:41:32 PM

PASSAGE

Passage 10 (Questions 52-56)

A 40-year-old patient developed a constant thirst and a frequent need to urinate. The attending physician suspected the patient had diabetes because in addition to these symptoms, the patient was overweight and had a family history of diabetes, which both are risk factors for the disease.

Diabetes mellitus is a condition in which the body cannot produce or is unable to effectively respond to insulin, a peptide hormone made in pancreatic β cells. Normally, insulin promotes facilitated diffusion of glucose into cells from the bloodstream by inducing the translocation of glucose transporter proteins to the plasma membrane. It also helps regulate the metabolism of fatty and amino acids.

In diabetes, the ability of many types of cells to take up glucose is compromised, leading to elevated blood glucose levels. Many tissues then metabolize fatty acids as an alternative energy source, which can lead to *metabolic acidosis*, a condition of low blood pH. Diabetes can also cause protein degradation, resulting in the release of excess amino acids that can be used as substrates for gluconeogenesis, which further increases blood glucose levels. There are two main types of diabetes. Many cases of Type 1 diabetes, in which the body is unable to produce insulin, are caused by an autoimmune response directed against pancreatic β cells. While the body often produces normal-to-elevated amounts of insulin in Type 2 diabetes, its ability to respond to insulin is compromised.

QUESTION + ANSWER CHOICES

Based on the passage, which of the following is LEAST likely to be a symptom of diabetes mellitus?

- A. Loss of appetite
- B. Sweet-tasting urine
- C. Unexplained weight loss
- D. Feelings of fatigue

SOLUTION

Solution: The correct answer is A.

- Based on the passage, diabetes affected individuals will use proteins and lipids as a source of glucose, thus increasing, rather than decreasing, their appetite.
- High levels of glucose are a symptom of diabetes. This is due to the fact that kidneys can reabsorb glucose up to 320 mg/min (tubular maximum). The levels of glucose in the filtrate that are in excess will pass into urine, leading to a sweet taste.
- Unexplained weight loss might occur as the body uses proteins and lipids as a source of energy.
- Feeling of fatigue might occur as the body uses amino acids and lipids as a source of energy.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Chlorophyll](#), [Glucose transport](#), and the [Diabetes Mellitus Pathway](#).

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
X
D
Time Spent:
4 secs
Difficulty Level:
Expert
Content & Skills:
SIRS2
CC1D
FC1
BIO

Question 169 of 230

Data ID: 831609
Captured: 6/3/2026, 4:41:33 PM

QUESTION + ANSWER CHOICES

A certain bacterium was cultured for several generations in medium containing ^{15}N , transferred to medium containing ^{14}N , and allowed to complete two rounds of cell division. Given that the bacterium's genome mass is 5.4 fg when grown in ^{14}N media and 5.5 fg when grown in ^{15}N medium, individual bacteria with which of the following genome masses would most likely be isolated from this culture?

- A. 5.4 fg only
- B. 5.4 fg and 5.45 fg
- C. 5.4 fg and 5.5 fg
- D. 5.45 fg only

SOLUTION

Solution: The correct answer is B.

- This would be the genome mass if the bacterium was cultured only in ^{14}N .
- DNA replication occurs in a semiconservative way and, after two rounds of bacterial growth, there will be bacteria containing only ^{14}N and bacteria containing the same amount of ^{15}N and ^{14}N . Thus, the genome mass will be 5.4 fg for the bacteria that only contain ^{14}N , and 5.45 fg for the bacteria that contain equal levels of ^{14}N and ^{15}N .
- This would be the genome mass if the bacteria were cultured only with ^{15}N or ^{14}N separately. The two rounds of division with ^{14}N will generate bacteria that contain ^{14}N and ^{15}N , as well as bacteria that only contain ^{14}N .
- Based on the semiconservative method of DNA replication, after two rounds of cell division some bacteria will only contain ^{14}N (genome mass 5.4 fg) and some others will contain a mix of ^{14}N and ^{15}N (genome mass 5.45 fg), thus the genome mass of the entire culture cannot be only 5.45 fg.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Cell Theory](#), [Classification and Structure of Prokaryotic Cells](#), [Growth and](#)

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
X
Time Spent:
1 min 14 secs
Difficulty Level:
Expert
Content & Skills:
SIRS2
CC2B
FC2
BIO

Question 170 of 230

Data ID: 531618
Captured: 6/3/2026, 4:41:33 PM

QUESTION • ANSWER CHOICES

Assume that *K* and *M* are two unlinked genes that affect hearing. The dominant *K* allele is necessary for hearing, and the dominant *M* allele causes deafness regardless of the other genes present. Given this, what fraction of the offspring of parents with the genotypes *KkMm* and *Kkmm* will most likely be deaf?

- A.
1/4
- B.
3/8
- C.
1/2
- D.
5/8

SOLUTION

Solution: The correct answer is **D**.

1. This would be correct if both individuals were heterozygous for *K* and homozygous recessive for *M*.
2. There are no possible genotype combination that will result in this ratio.
3. This would be correct if one parent was *Kkmm* and the other *kkmm*, or if one parent was *KkMm* and the other was *Kkmm*.
4. **Based on the possible allele combination, out of 16 possible combinations, there will be ten individuals that will be hearing impaired, carrying either one allele *M* or the two *kk*. Thus 10/16 = 5/8 individuals.**

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Evidence that DNA is Genetic Material: Mendelian Inheritance](#), [Analyze Methods](#).

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
X
D
Time Spent:
1 min 41 secs
Difficulty Level:
Expert
Content & Skills:
SIRS2
CC1C
FC1
BIO

Question 171 of 230

Data ID: 531618
Captured: 6/3/2026, 4:41:34 PM

QUESTION • ANSWER CHOICES

An RNA molecule has 1500 bases. What is the maximum number of amino acids it can encode?

- A.
500
- B.
1000
- C.
1500
- D.
4500

SOLUTION

Solution: The correct answer is **A**.

1. **Each amino acid is coded by a triplet of nucleobases. Thus, $\frac{1500}{3} = 500$ amino acids.**
2. This is the number of amino acids coded by a sequence of 3000 bases, not 1500 bases.
3. This is the number of amino acids coded by a sequence of 4500 bases, not 1500 bases.
4. This is the number of amino acids coded by a sequence of 13500 bases, not 1500 bases.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleic Acid Structure and Function: DNA Replication](#), [Repair of DNA](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
1 sec
Difficulty Level:
Easy
Content & Skills:
SIRS1
CC1B
FC1
BIO

Question 172 of 230

Data ID: 531643
Captured: 6/3/2026, 4:41:35 PM

PASSAGE

Passage 1 (Questions 1-4)

Categorical bias refers to people's tendency to judge the physical distance between objects from the same category as being smaller compared to the distance between objects from different categories. Seven-, nine-, and eleven-year-old children and a group of adults were recruited to participate in a study of categorical bias. In the training phase, each participant was presented with 20 objects in a box and was asked to study the location of these objects. The training objects were selected from four categories: vehicles, animals, food, and clothing. There were two different types of training conditions. The participants in the simultaneous condition were presented with all the training objects all at once and were asked to study their locations simultaneously. The participants in the serial condition were presented with the training objects one at a time.

Testing began immediately after the training. The objects were removed from the box, and the participants were asked to position the objects in the empty box exactly as they were arranged in the training phase.

The researchers were interested in whether the participants showed categorical bias, which was measured by central placement error. To assess this, the researchers first calculated the central point between objects of the same category based on their original locations during training. The central placement error was determined based on the errors participants made in placing objects from the same category nearer to the central point than their original locations. The results of the experiment are shown in Figure 1.

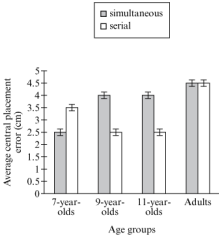


Figure 1 Average central placement error by age and type of presentation

Adapted from K. M. Recker and J. M. Plumert ©2008 Taylor & Francis Group.

QUESTION + ANSWER CHOICES

Compared to the simultaneous condition, the serial condition of the experiment would be more likely to cause:

- A. a primacy effect.
- B. a state dependency effect.
- C. a misinformation effect.
- D. a dual-coding effect.

SOLUTION

Solution: The correct answer is A.

- In the serial condition, the objects are presented to the participants one at a time, which might result in better memory for events at the beginning of the series (a primacy effect).
- A state dependency effect refers to the benefit of a match between the conditions (in particular, the subject's mental and emotional state) under which information is encoded into memory and the conditions present when the subject attempts to retrieve that information.
- Misinformation effects refer to memory errors (usually errors of commission), in which some information introduced and encoded after the target information is retrieved along with some portions of the target information. In such cases, the subject usually has trouble identifying which retrieved information had been originally encoded and which information was introduced subsequently, a situation known as source confusion.
- Dual-coding refers to the retrieval advantage of verbal items that are imageable. These items are encoded with both a verbal code and a visual code, which benefits their retrievability.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A

✓
B
C
D
Time Spent:
2 mins 45 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC6B
FC6
PSY

Question 173 of 230

Data ID: 531651
Captured: 6/3/2026, 4:41:35 PM

PASSAGE

Passage 1 (Questions 1-4)

Categorical bias refers to people's tendency to judge the physical distance between objects from the same category as being smaller compared to the distance between objects from different categories. Seven-, nine-, and eleven-year-old children and a group of adults were recruited to participate in a study of categorical bias. In the training phase, each participant was presented with 20 objects in a box and was asked to study the location of these objects. The training objects were selected from four categories: vehicles, animals, food, and clothing. There were two different types of training conditions. The participants in the simultaneous condition were presented with all the training objects all at once and were asked to study their locations simultaneously. The participants in the serial condition were presented with the training objects one at a time.

Testing began immediately after the training. The objects were removed from the box, and the participants were asked to position the objects in the empty box exactly as they were arranged in the training phase.

The researchers were interested in whether the participants showed categorical bias, which was measured by central placement error. To assess this, the researchers first calculated the central point between objects of the same category based on their original locations during training. The central placement error was determined based on the errors participants made in placing objects from the same category nearer to the central point than their original locations. The results of the experiment are shown in Figure 1.

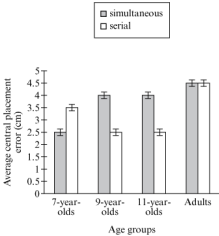


Figure 1 Average central placement error by age and type of presentation

Adapted from K. M. Ricker and J. M. Plumert ©2008 Taylor & Francis Group.

QUESTION + ANSWER CHOICES

The researchers change the procedure such that instead of placing the objects in a box, the participants have to recall all the objects that they have seen during training. According to the spreading of activation theory, which type of memory error is most likely?

- A. Making source monitoring errors regarding the location of the training objects
- B. Poorer memory for the training objects seen at the later points in the experiment**
- C. Selective forgetting of the training objects that were placed in the center of the box
- D. Recalling objects that were not presented but are from the same category as the training objects

SOLUTION

Solution: The correct answer is D.

- Source monitoring refers to a subject's ability to retrieve the details of the situation extant when memory items were encoded. More specifically, when some additional information is introduced after the original encoding, this post-event information can be mistakenly included in recall of the original event, leading to a misinformation effect.
- This option is incorrect for two reasons. First, poorer memory for items seen later is unlikely, because it would entail the absence of a recency effect (i.e., items at the end of a list are generally recalled better than items presented in the middle). Second, spreading activation would not predict such an outcome.
- There is no evidence to suggest the validity of such selective forgetting based on specific spatial locations.
- Spreading activation suggests that, when the representation of a concept is activated in memory, the activation spreads to concepts that are semantically or associatively related to it. Thus, people often retrieve unrepresented members of a category when tested on their memory for a series of presented concepts from that category.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A

B
C
D
✓
Time Spent:
55 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC6B
FC6
PSY

Question 174 of 230

Data ID: 531662
Captured: 6/3/2026, 4:41:36 PM

PASSAGE

Passage 1 (Questions 1-4)

Categorical bias refers to people's tendency to judge the physical distance between objects from the same category as being smaller compared to the distance between objects from different categories. Seven-, nine-, and eleven-year-old children and a group of adults were recruited to participate in a study of categorical bias. In the training phase, each participant was presented with 20 objects in a box and was asked to study the location of these objects. The training objects were selected from four categories: vehicles, animals, food, and clothing. There were two different types of training conditions. The participants in the simultaneous condition were presented with all the training objects all at once and were asked to study their locations simultaneously. The participants in the serial condition were presented with the training objects one at a time.

Testing began immediately after the training. The objects were removed from the box, and the participants were asked to position the objects in the empty box exactly as they were arranged in the training phase.

The researchers were interested in whether the participants showed categorical bias, which was measured by central placement error. To assess this, the researchers first calculated the central point between objects of the same category based on their original locations during training. The central placement error was determined based on the errors participants made in placing objects from the same category nearer to the central point than their original locations. The results of the experiment are shown in Figure 1.

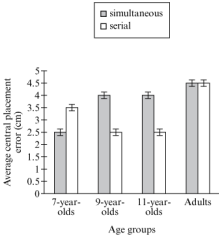


Figure 1 Average central placement error by age and type of presentation

Adapted from K. M. Ricker and J. M. Plumert ©2008 Taylor & Francis Group.

QUESTION + ANSWER CHOICES

The findings described in the passage suggest that memory for locations:

- A. is comparable to adults by nine years of age.
- B. is influenced by categorical knowledge.
- C. becomes more accurate with age.
- D. relies solely on recall of distances.

SOLUTION

Solution: The correct answer is B.

- The nine-year-olds in fact performed differently from adults.
- The results showed that categorical bias was observed in adults.
- The fact that categorical bias was present in adults, hence spatial memory does not necessarily become more accurate with age.
- Because world knowledge about categories did not affect people's judgments of distance, the findings do not support the conclusion that spatial memory relies solely on the recall of distances.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C

D
Time Spent:
36 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC6B
FC6
PSY

Question 175 of 230

Data ID: 531667
Captured: 6/3/2026, 4:41:36 PM

PASSAGE

Passage 1 (Questions 1-4)

Categorical bias refers to people's tendency to judge the physical distance between objects from the same category as being smaller compared to the distance between objects from different categories. Seven-, nine-, and eleven-year-old children and a group of adults were recruited to participate in a study of categorical bias. In the training phase, each participant was presented with 20 objects in a box and was asked to study the location of these objects. The training objects were selected from four categories: vehicles, animals, food, and clothing. There were two different types of training conditions. The participants in the simultaneous condition were presented with all the training objects all at once and were asked to study their locations simultaneously. The participants in the serial condition were presented with the training objects one at a time.

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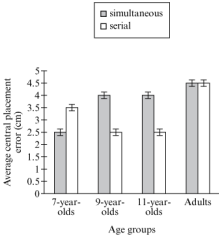


Figure 1 Average central placement error by age and type of presentation

Adapted from K. M. Recker and J. M. Plumert ©2008 Taylor & Francis Group.

QUESTION + ANSWER CHOICES

The children from the experiment in the passage participate in a separate study using Piaget's water conservation task. They are shown two identical beakers, containing equal amounts of water. The water from one of the containers is poured into a thinner and taller beaker. Which prediction is most likely to be confirmed?

- A. All three age groups will state that the water in the taller beaker is greater in quantity.
- B. The majority of the 11-year-olds will state that the amount of water in the taller beaker is the same as in the original beaker.
- C. The majority of the 9- and 11-year-olds will state that the amount of water in the taller beaker is greater in quantity.
- D. The majority of the 7-year-olds will state that the amount of water in the taller beaker is the same as in the original beaker.

SOLUTION

Solution: The correct answer is B.

- Most of the 7-year-olds and many 9-year-olds will fail at tests of conservation.
- According to Piaget's theory of cognitive development, mastery of conservation is the criterion for entering the concrete operations stage, which starts at approximately 7 years and ends around 11 to 12 years. The vast majority of the 11-year-olds will have entered the concrete operational stage. Thus, they will be able to judge that pouring water into the taller beaker does not change its quantity.
- The concrete operations stage starts at approximately 7 years and ends around 11 to 12 years. Solving conservation problems is the criterion for entering this stage. Most of the 9-year-olds will have developed the ability to solve conservation problems.
- The concrete operations stage starts at approximately 7 years and ends around 11 to 12 years. Solving conservation problems is the criterion for entering this stage. Although some 7-year-old children already may have developed the ability to solve conservation problems, many will not.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer

A
B ✓
C
D
Time Spent:
45 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC6B
FC6
PSY

Question 176 of 230

Data ID: 531681
Captured: 6/3/2026, 4:41:37 PM

PASSAGE

Passage 2 (Questions 5-10)

Panic attacks co-occur with various other psychological disorders. Panic disorder (PD) is diagnosed when a patient is preoccupied with the possibility of panic attacks and experiences behavioral changes as a result of this preoccupation.

The following scenarios demonstrate two different patients' responses to the question, "Do you worry about the panic attacks that you experience?"

Patient 1: No. They are somewhat unpleasant, but they are the least of my concerns. I am more worried about my daughter and our future. I don't get any sleep most nights. I just lie awake feeling nervous about what would happen to my daughter if something happened to me and my husband.

Patient 2: I am terrified of having a panic attack at a meeting. I dread the thought of others noticing how nervous I am. I'm worried that others will think I am weird. I don't go to meetings anymore.

Studies have identified temperament as a risk factor for PD. Specifically, patients who are diagnosed with PD tend to score high on neuroticism. Patients with PD also tend to have high interoceptive awareness (a heightened ability to detect bodily sensations of arousal, such as increases in heartbeat or blood pressure).

According to some researchers, interoceptive awareness contributes to maintenance factors for PD. Specifically, patients with high interoceptive awareness are likely to associate mild changes in bodily sensations (such as slightly heightened heartbeat) with the panic they experienced during panic attacks; this leads to conditioned fear in response to changes in bodily sensations (Maintenance Factor 1). Another maintenance factor for PD is the misappraisal of bodily sensations (Maintenance Factor 2), such as interpreting bodily sensations as signs of imminent death or loss of control. Some researchers suggest that such misappraisals are largely responsible for the excessive panic and fear that PD patients experience during a panic attack.

QUESTION + ANSWER CHOICES

Maintenance Factor 1 is most closely related to which approach to psychological disorders?

- ☐ A. Psychodynamic
- ☒ B. Behaviorist
- ☐ C. Trait
- ☐ D. Humanistic

SOLUTION

Solution: The correct answer is **B**.

- Maintenance Factor 1 is centered on classical conditioning, which is completely unrelated to psychodynamic theory.
- Maintenance Factor 1 focuses on the role of classical conditioning, which is closely related to the behaviorist approach.**
- Trait theory refers to an approach to personality. Maintenance Factor 1 is concerned with classical conditioning, which is completely unrelated to the trait approach.
- Maintenance Factor 1 is concerned with classical conditioning, which is completely unrelated to the humanistic approach.

Khan Academy Lessons

For more help on the skills *concepts* or content associated to this question, see the Khan Academy lessons on [Behavioral Basis of Behavior: Human Development: Behavior and Cognition](#).

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
☒ B
C
D
Time Spent:
2 mins 26 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS1
CC7A
FC7
PSY

Question 177 of 230

Data ID: 531686
Captured: 6/3/2026, 4:41:38 PM

PASSAGE

Passage 2 (Questions 5-10)

Panic attacks co-occur with various other psychological disorders. Panic disorder (PD) is diagnosed when a patient is preoccupied with the possibility of panic attacks and experiences behavioral changes as a result of this preoccupation.

The following scenarios demonstrate two different patients' responses to the question, "Do you worry about the panic attacks that you experience?"

Patient 1: No. They are somewhat unpleasant, but they are the least of my concerns. I am more worried about my daughter and our future. I don't get any sleep most nights. I just lie awake feeling nervous about what would happen to my daughter if something happened to me and my husband.

Patient 2: I am terrified of having a panic attack at a meeting. I dread the thought of others noticing how nervous I am. I'm worried that others will think I am weird. I don't go to meetings anymore.

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QUESTION + ANSWER CHOICES

According to Maintenance Factor 1, mild changes in bodily sensations act as:

- ☐ A. unconditioned stimuli.
- ☒ B. conditioned stimuli.
- ☐ C. unconditioned responses.
- ☐ D. conditioned responses.

SOLUTION

Solution: The correct answer is **B**.

- Mild changes in bodily sensations become conditioned stimuli (CS), NOT unconditioned stimuli. These stimuli become associated with excessive panic. The excessive panic is the unconditioned stimulus (US).
- According to Maintenance Factor 1, mild changes in bodily sensations become cues that are associated with excessive panic. Those mild changes in bodily sensations become conditioned stimuli, which come to elicit the conditioned fear response.**
- Mild changes in bodily sensations become conditioned stimuli, which become associated with excessive panic, which is the unconditioned stimulus. The excessive panic is NOT the unconditioned response.
- Mild changes in bodily sensations become conditioned stimuli, which elicit the conditioned fear response. But stimuli are not responses.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
☒ B
C
D
Time Spent:
56 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS1
CC7C
FC7
PSY

Question 178 of 230

Data ID: 531701
Captured: 6/2/2026, 4:41:38 PM

PASSAGE

Passage 2 (Questions 5-10)

Panic attacks co-occur with various other psychological disorders. Panic disorder (PD) is diagnosed when a patient is preoccupied with the possibility of panic attacks and experiences behavioral changes as a result of this preoccupation.

The following scenarios demonstrate two different patients' responses to the question, "Do you worry about the panic attacks that you experience?"

Patient 1: No. They are somewhat unpleasant, but they are the least of my concerns. I am more worried about my daughter and our future. I don't get any sleep most nights. I just lie awake feeling nervous about what would happen to my daughter if something happened to me and my husband.

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QUESTION + ANSWER CHOICES

For Patient 2, panic attacks act as:

- A. discriminatory stimuli.
- B. signaling stimuli.
- C. positive punishers.
- D. negative punishers.

SOLUTION

Solution: The correct answer is C.

- Discriminatory stimuli allow an organism to tell whether an appetitive stimulus or an aversive stimulus is forthcoming in an operant conditioning situation.
- Signaling stimuli are neutral stimuli that may potentially become conditioned stimuli.
- Positive punishment refers to the decrease in the frequency of a response resulting from the introduction of an aversive stimulus following that response. The patient describes the panic attacks as highly aversive and mentions that he no longer goes to meetings for fear of a panic attack. Because the frequency of the patient's attendance at meetings has decreased as the result of the panic attacks, the panic attacks have acted as positive punishers.**
- Negative punishment refers to the decrease in the frequency of a response resulting from the removal of an appetitive stimulus following that response. The patient describes the panic attacks as highly aversive. In addition, they are introduced, rather than removed. Thus, the panic attacks do not match either of the criteria for negative punishers.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Discriminative and Punishing Stimuli](#), [Associative Learning](#), [Operant Conditioning](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
58 secs
Difficulty Level:
Expert
Content & Skills:
SIRS1
CC7C
FC7
PSY

Question 179 of 230

Data ID: 531711
Captured: 6/2/2026, 4:41:39 PM

PASSAGE

Passage 2 (Questions 5-10)

Panic attacks co-occur with various other psychological disorders. Panic disorder (PD) is diagnosed when a patient is preoccupied with the possibility of panic attacks and experiences behavioral changes as a result of this preoccupation.

The following scenarios demonstrate two different patients' responses to the question, "Do you worry about the panic attacks that you experience?"

Patient 1: No. They are somewhat unpleasant, but they are the least of my concerns. I am more worried about my daughter and our future. I don't get any sleep most nights. I just lie awake feeling nervous about what would happen to my daughter if something happened to me and my husband.

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QUESTION + ANSWER CHOICES

Interoceptive awareness involves sensitivity to increases in the activity of the:

- A. reticular activation system.
- B. autonomic nervous system.
- C. limbic system.
- D. somatic nervous system.

SOLUTION

Solution: The correct answer is B.

- The reticular formation (RF), deep in the brainstem, is concerned with functions involving arousal, particularly the sleep-wake cycle, and attention. Changes in the RF's activity level are not described as "interoceptive."
- Interoceptive awareness is described as increased sensitivity to internal bodily sensations, such as heartbeat and blood pressure. These functions are regulated by the autonomic nervous system.**
- The limbic system consists of brain structures, such as the thalamus, hypothalamus, and hippocampus. Changes in the activity levels of these structures are generally not sensed and, therefore, we are not aware of such changes. In addition, such changes would not be considered "interoceptive."
- The somatic nervous system is the division of the peripheral nervous system that controls sensory and motor functions of effectors that enable the organism to deal with its external environment. It does not function in the ability to sense interoceptive changes.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Behavioral Basis of Behavior: Human Development: Behavior and Cognition](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
1 min 0 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC7A
FC7

Question 180 of 230

Data ID: 531715
Captured: 6/3/2026, 4:41:40 PM

PASSAGE

Passage 2 (Questions 5-10)

Panic attacks co-occur with various other psychological disorders. Panic disorder (PD) is diagnosed when a patient is preoccupied with the possibility of panic attacks and experiences behavioral changes as a result of this preoccupation.

The following scenarios demonstrate two different patients' responses to the question, "Do you worry about the panic attacks that you experience?"

Patient 1: No. They are somewhat unpleasant, but they are the least of my concerns. I am more worried about my daughter and our future. I don't get any sleep most nights. I just lie awake feeling nervous about what would happen to my daughter if something happened to me and my husband.

Patient 2: I am terrified of having a panic attack at a meeting. I dread the thought of others noticing how nervous I am. I'm worried that others will think I am weird. I don't go to meetings anymore.

Studies have identified temperament as a risk factor for PD. Specifically, patients who are diagnosed with PD tend to score high on neuroticism. Patients with PD also tend to have high interoceptive awareness (a heightened ability to detect bodily sensations of arousal, such as increases in heartbeat or blood pressure).

According to some researchers, interoceptive awareness contributes to maintenance factors for PD. Specifically, patients with high interoceptive awareness are likely to associate mild changes in bodily sensations (such as slightly heightened heartbeat) with the panic they experienced during panic attacks; this leads to conditioned fear in response to changes in bodily sensations (Maintenance Factor 1). Another maintenance factor for PD is the misappraisal of bodily sensations (Maintenance Factor 2), such as interpreting bodily sensations as signs of imminent death or loss of control. Some researchers suggest that such misappraisals are largely responsible for the excessive panic and fear that PD patients experience during a panic attack.

QUESTION + ANSWER CHOICES

- Which two processes best summarize maintenance factors 1 and 2, respectively?
- A. Operant conditioning and top-down processes
 - B. Classical conditioning and affective processes
 - C. Operant conditioning and physiological processes
 - D. Classical conditioning and cognitive processes

SOLUTION

- Solution:** The correct answer is **D**.
- Maintenance Factor 1 focuses on classical (not operant) conditioning. Maintenance Factor 2 focuses on the cognitive appraisal of bodily sensations, not on any cognitive control of those sensations.
 - Although it is true that Maintenance Factor 1 focuses on classical conditioning of a fear response to bodily sensations, Maintenance Factor 2 did not involve affective (i.e., emotional) processing.
 - Maintenance Factor 1 focuses on classical conditioning, not operant conditioning. Maintenance Factor 2 focuses on cognitive appraisal of bodily sensations, not the physiological processes behind those sensations.
 - Maintenance Factor 1 focuses on classical conditioning of a fear response to bodily sensations, whereas Maintenance Factor 2 focuses on cognitive appraisal (interpretation) of bodily sensations.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
1 min 41 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS1
CC6B
FC6

Question 181 of 230

Data ID: 531727
Captured: 6/3/2026, 4:41:40 PM

PASSAGE

Passage 2 (Questions 5-10)

Panic attacks co-occur with various other psychological disorders. Panic disorder (PD) is diagnosed when a patient is preoccupied with the possibility of panic attacks and experiences behavioral changes as a result of this preoccupation.

The following scenarios demonstrate two different patients' responses to the question, "Do you worry about the panic attacks that you experience?"

Patient 1: No. They are somewhat unpleasant, but they are the least of my concerns. I am more worried about my daughter and our future. I don't get any sleep most nights. I just lie awake feeling nervous about what would happen to my daughter if something happened to me and my husband.

Patient 2: I am terrified of having a panic attack at a meeting. I dread the thought of others noticing how nervous I am. I'm worried that others will think I am weird. I don't go to meetings anymore.

Studies have identified temperament as a risk factor for PD. Specifically, patients who are diagnosed with PD tend to score high on neuroticism. Patients with PD also tend to have high interoceptive awareness (a heightened ability to detect bodily sensations of arousal, such as increases in heartbeat or blood pressure).

According to some researchers, interoceptive awareness contributes to maintenance factors for PD. Specifically, patients with high interoceptive awareness are likely to associate mild changes in bodily sensations (such as slightly heightened heartbeat) with the panic they experienced during panic attacks; this leads to conditioned fear in response to changes in bodily sensations (Maintenance Factor 1). Another maintenance factor for PD is the misappraisal of bodily sensations (Maintenance Factor 2), such as interpreting bodily sensations as signs of imminent death or loss of control. Some researchers suggest that such misappraisals are largely responsible for the excessive panic and fear that PD patients experience during a panic attack.

QUESTION + ANSWER CHOICES

- Data on the role of Maintenance Factor 1 on PD is LEAST likely to come from which type of research?
- A. Correlational studies
 - B. Case studies
 - C. Longitudinal studies
 - D. Experimental studies

SOLUTION

Solution: The correct answer is **D**.

1. Most of the data on the course of a disorder like PD are going to come from correlational research.

2. Case studies of individuals, and groups of individuals, with PD are entirely possible.

3. The developmental course of a disorder like PD can appropriately be studied in longitudinal studies.

4. It would be extremely difficult to systematically manipulate participants' physiological states and their sensitivity to changes in those states. In addition, random assignment to various conditions involving maintenance of PD would pose insurmountable ethical restrictions. For these reasons, it is very unlikely that experimental research could be designed to study the role of Maintenance Factor 1 on PD.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
1 min 45 secs
Difficulty Level:
Expert
Content & Skills:
SIRS3
CC7C
FC7
PSY

Question 182 of 230

Data ID: 531736
Captured: 6/3/2026, 4:41:41 PM

QUESTION • ANSWER CHOICES

Harlequin ichthyosis, a rare genetic disorder, causes the skin to become thick and scaly. Flaking skin behind the eyelids of individuals with this condition is most likely to damage which structure of the eye?

- A. Choroid
- **B. Cornea**
- C. Lens
- D. Retina

SOLUTION

Solution: The correct answer is **B**.

1. The choroid does not form the outermost layer of the eyeball, but instead is located deep to the sclera. Therefore, the choroid is unlikely to be significantly damaged by flaking skin off the back of the eyelids.

2. **The cornea forms the outermost layer of the front of the eye, in which it directly contacts the back of the eyelids. Due to this direct interaction, the cornea is highly susceptible to damage caused by flaking skin from the back of the eyelids.**

3. While the lens is positioned toward the front of the eye, it is located deep when compared to the cornea. Therefore, compared to the lens, the cornea will be more likely to be damaged by flaking skin from the back of the eyelids.

4. The retina forms the innermost layer along the posterior portion of the eye. Therefore, the retina is unlikely to be significantly damaged by flaking skin off the back of the eyelids.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
51 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC6A
FC6
BIO

Question 183 of 230

Data ID: 531741
Captured: 6/3/2026, 4:41:41 PM

QUESTION • ANSWER CHOICES

Sound-induced vibrations depolarize hair cells of the cochlea by opening ion channels that are gated in what way?

- A. Chemically
- **B. Mechanically**
- C. Electrically
- D. Synaptically

SOLUTION

Solution: The correct answer is **B**.

1. Chemically gated ion channels require the binding of a molecule to the ion channel, causing it to open. This is observed during olfaction and gustation, but not during the activation of hair cells, which occurs upon the physical perturbation of the hair cell membrane.

2. **Sound wave-induced vibration of hair cells generates tension within the cell membrane that directly activate ion channels responsible for auditory signaling. This type of ion channel is a mechanically-gated channel, such that upon its activation, it generates a depolarizing current within the cell.**

3. Electrically gated ion channels activate upon a change in membrane potential. Activity of electrically gated ion channels is associated with action potential propagation, but not vibration of the hair cell membrane.

4. While some ion channels are localized to the synapse, synaptic gating is not a mechanism by which ion channels can be activated.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
1 min 10 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC6A
FC6
BIO

Question 184 of 230

Data ID: 531744
Captured: 6/3/2026, 4:41:42 PM

QUESTION + ANSWER CHOICES

Max attends a party and does not make eye contact with, or approach, his acquaintance, Sam. According to the actor–observer bias, which graph best represents how Max and Sam view this behavior?

- A.

Judgment	not feeling well	socially awkward
Max's self-judgment	High	Low
Sam's judgment of Max	Low	High
- B.

Judgment	not feeling well	socially awkward
Max's self-judgment	Low	High
Sam's judgment of Max	High	Low
- C.

Judgment	not feeling well	socially awkward
Max's self-judgment	Low	High
Sam's judgment of Max	Low	High
- D.

Judgment	not feeling well	socially awkward
Max's self-judgment	High	Low
Sam's judgment of Max	High	Low

SOLUTION

Solution: The correct answer is A.

- According to the actor-observer bias, actors will attribute their own bad behavior to situational factors (not feeling well), whereas observers will attribute actors' behavior to dispositional factors (social awkwardness).
- This figure shows the opposite pattern of attributions for the observer's and actor's behavior than the pattern described as the actor-observer bias.
- This figure shows no difference in the attributions of the observer's and actor's behaviors. Both are shown as dispositional attributions.
- This figure inaccurately shows a situational attribution for both the observer's and the actor's behavior.

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
1 min 18 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS4
CC8B
FC8
PSY

Question 185 of 230

Data ID: 531751
Captured: 6/3/2026, 4:41:43 PM

QUESTION + ANSWER CHOICES

In a study, each trial involves administering a drop of lemon juice to the participant's tongue and measuring the participant's level of salivation. As more trials are conducted, the researcher finds that the magnitude of salivation declines. After a certain point, the researcher switches to administering lime juice. This researcher is most likely studying which process?

- A. Sensory perception
- B. Habituation and dishabituation
- C. Stimulus generalization in classical conditioning
- D. Conditioned responses in classical conditioning

SOLUTION

Solution: The correct answer is B.

1. The study of sensory perception usually involves either a psychophysics paradigm (involving thresholds) or a signal detection theory paradigm (involving signal-to-noise ratio and response criterion), which were not described in the stem.
2. The study involves reduced responding to a repeating stimulus, which is best described as habituation. The researcher then changes the stimulus, which will likely lead to dishabituation.
3. No presentation of a conditioned stimulus (CS) was described in the stem.
4. Since no CS was presented along with the Unconditioned CS (citrus juice), the stem does not describe a classical conditioning paradigm.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Habituation and Dishabituation](#), [Associative Learning](#), [Observational Learning](#)

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
2 mins 41 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS1
CC7C
FC7
PSY

Question 186 of 230

Data ID: 531822
Captured: 6/3/2026, 4:41:43 PM

PASSAGE

Passage 3 (Questions 15-18)

Research has indicated that stereotypes associated with a social group can affect the performance of its members. To counteract this effect, researchers have been trying to understand the mechanism behind it. In the United States, women are stereotyped as having lower math ability. Asian Americans, on the other hand, are stereotyped as having high math ability. Study 1 was designed to investigate the math performance of Asian American women, who have two conflicting stereotypes associated with their identity. The participants were randomly assigned to one of three groups. In the gender identity condition, the participants were asked to indicate their gender and answered personal questions related to their gender identity. In the ethnic identity condition, they indicated their ethnicity and answered personal questions related to their ethnic identity. In the control condition, the participants were not asked to indicate their gender or ethnicity, but answered general questions about their background. After answering the questions, all three groups worked on a math test. Findings from this study are represented in Figure 1.

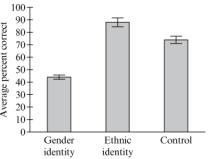


Figure 1 Math performance as a function of experimental condition

Stereotypes associated with social groups are not consistent across cultures. In Canada, women are stereotyped to have lower math ability, but Asian Canadians are not stereotyped to have higher math ability. Study 2 attempted to test whether the difference in stereotypes would affect the math performance of Asian Canadian women. The same procedure that was used for Study 1 was used in Study 2, using Asian Canadian women as participants.

Adapted from M. Shih, T. L. Pittinsky, and N. Ambady, ©1999 American Psychological Society.

QUESTION + ANSWER CHOICES

Which explanation of the results of Study 1 presents the most appropriate application of the optimal arousal theory? Reminding participants of:

- A. their membership in a stereotyped group leads to optimal arousal, which results in varying levels of performance depending on ability.
- B. their membership in a stereotyped group leads to fluctuating arousal and results in poor performance.
- C. a negative stereotype associated with their identity increases arousal beyond what is optimal and leads to poor performance.
- D. a positive stereotype associated with them eliminates arousal completely, which results in optimal performance.

SOLUTION

Solution: The correct answer is C.

1. The theory does not predict that arousal interacts with any factor other than the difficulty (or complexity) of the task. In particular, arousal and ability are not posited to interact.
2. The theory does not say anything about fluctuating arousal levels.
3. Optimal arousal theory states that optimal performance requires optimal arousal and that arousal levels that are too high or too low will impede performance. This option implies that the negative stereotype will cause poor performance by increasing arousal levels beyond the optimal level.
4. The theory does not state that the suggestion of a positive stereotype will eliminate, or even reduce, arousal levels.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Biological Bases of Behavior](#), [Hormone Development](#), [Behavior and Cognition](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
2 mins 21 secs
Difficulty Level:
Difficult

Content & Skills:
SIRS2
CC7A
FC7
PSY

Question 187 of 230

Data ID: 831828
Captured: 6/3/2026, 4:41:44 PM

PASSAGE

Passage 3 (Questions 15-18)

Research has indicated that stereotypes associated with a social group can affect the performance of its members. To counteract this effect, researchers have been trying to understand the mechanism behind it. In the United States, women are stereotyped as having lower math ability. Asian Americans, on the other hand, are stereotyped as having high math ability. Study 1 was designed to investigate the math performance of Asian American women, who have two conflicting stereotypes associated with their identity. The participants were randomly assigned to one of three groups. In the gender identity condition, the participants were asked to indicate their gender and answered personal questions related to their gender identity. In the ethnic identity condition, they indicated their ethnicity and answered personal questions related to their ethnic identity. In the control condition, the participants were not asked to indicate their gender or ethnicity, but answered general questions about their background. After answering the questions, all three groups worked on a math test. Findings from this study are represented in Figure 1.

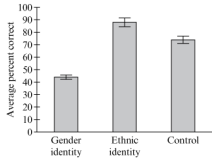


Figure 1 Math performance as a function of experimental condition

Stereotypes associated with social groups are not consistent across cultures. In Canada, women are stereotyped to have lower math ability, but Asian Canadians are not stereotyped to have higher math ability. Study 2 attempted to test whether the difference in stereotypes would affect the math performance of Asian Canadian women. The same procedure that was used for Study 1 was used in Study 2, using Asian Canadian women as participants.

Adapted from M. Shih, T. L. Pittinsky, and N. Ambady, ©1999 American Psychological Society.

QUESTION + ANSWER CHOICES

If the participants in Study 1 differed in the degree to which their ethnic identity was a central part of their self-concept, which outcome would be most likely? Participants whose ethnic identity was a central part of their self-concepts would:

- A. be more affected by the gender stereotypes regarding their math performance.
- B. not be affected by the negative stereotype regarding women's math ability.
- C. be more affected by the positive stereotype regarding Asian Americans' math ability.
- D. perform the same regardless of which condition they were placed in.

SOLUTION

Solution: The correct answer is C.

1. There is no reason that someone whose ethnic identity is central to their self-concept would be more affected by gender stereotypes or affected by the conditions of the experiment.
2. There is no reason to suppose that someone whose ethnic identity is central to their self-concept would be unaffected by gender stereotypes.
3. **People whose ethnic identity is a central part of their self-concept would be more prone to the effects of stereotype threat regarding their ethnic identity.**
4. There is no reason that someone whose ethnic identity is central to their self-concept would not be affected by the conditions of the experiment.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
3 mins 4 secs
Difficulty Level:
Moderate

Content & Skills:
SIRS2
CC8A
FC8
PSY

Question 188 of 230

Data ID: 831833
Captured: 6/3/2026, 4:41:45 PM

PASSAGE

Passage 3 (Questions 15-18)

Research has indicated that stereotypes associated with a social group can affect the performance of its members. To counteract this effect, researchers have been trying to understand the mechanism behind it. In the United States, women are stereotyped as having lower math ability. Asian Americans, on the other hand, are stereotyped as having high math ability. Study 1 was designed to investigate the math performance of Asian American women, who have two conflicting stereotypes associated with their identity. The participants were randomly assigned to one of three groups. In the gender identity condition, the participants were asked to indicate their gender and answered personal questions related to their gender identity. In the ethnic identity condition, they indicated their ethnicity and answered personal questions related to their ethnic identity. In the control condition, the participants were not asked to indicate their gender or ethnicity, but answered general questions about their background. After answering the questions, all three groups worked on a math test. Findings from this study are represented in Figure 1.

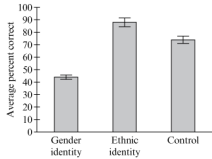


Figure 1 Math performance as a function of experimental condition

Stereotypes associated with social groups are not consistent across cultures. In Canada, women are stereotyped to have lower math ability, but Asian Canadians are not stereotyped to have higher math ability. Study 2 attempted to test whether the difference in stereotypes would affect the math performance of Asian Canadian women. The same procedure that was used for Study 1 was used in Study 2, using Asian Canadian women as participants.

Adapted from M. Shih, T. L. Pittinsky, and N. Ambady, ©1999 American Psychological Society.

QUESTION + ANSWER CHOICES

Which measure would be most useful if the researchers were interested in the degree of sympathetic arousal experienced in the different conditions of Study 1?

- A. A measure of electrical conductivity of the skin
- B. A CT scan of the hindbrain
- C. A PET scan of the parietal cortex
- D. A measure of melatonin levels

SOLUTION

Solution: The correct answer is A.

- Increases in the electrical conductivity of the skin is a physiological indication of increased sympathetic arousal, which is associated with anxiety.
- Brain imaging is very unlikely to provide an indication of the level of sympathetic arousal and the hindbrain is not involved in anxiety.
- Brain imaging of the cerebral lobes will not provide an indication of sympathetic arousal.
- Melatonin is a hormone (involved in the sleep-wake cycle) secreted by the pineal gland. Levels of this hormone will not provide any indication of the level of sympathetic arousal.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
42 secs
Difficulty Level:
Difficult
Content & Skills:

Question 189 of 230

Data ID: 531838
Captured: 6/3/2026, 4:41:45 PM

PASSAGE

Passage 3 (Questions 15-18)

Research has indicated that stereotypes associated with a social group can affect the performance of its members. To counteract this effect, researchers have been trying to understand the mechanism behind it. In the United States, women are stereotyped as having lower math ability. Asian Americans, on the other hand, are stereotyped as having high math ability. Study 1 was designed to investigate the math performance of Asian American women, who have two conflicting stereotypes associated with their identity. The participants were randomly assigned to one of three groups. In the gender identity condition, the participants were asked to indicate their gender and answered personal questions related to their gender identity. In the ethnic identity condition, they indicated their ethnicity and answered personal questions related to their ethnic identity. In the control condition, the participants were not asked to indicate their gender or ethnicity, but answered general questions about their background. After answering the questions, all three groups worked on a math test. Findings from this study are represented in Figure 1.

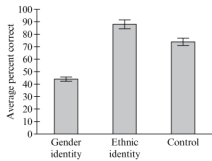


Figure 1 Math performance as a function of experimental condition

Stereotypes associated with social groups are not consistent across cultures. In Canada, women are stereotyped to have lower math ability, but Asian Canadians are not stereotyped to have higher math ability. Study 2 attempted to test whether the difference in stereotypes would affect the math performance of Asian Canadian women. The same procedure that was used for Study 1 was used in Study 2, using Asian Canadian women as participants.

Adapted from M. Shih, T. L. Pittinsky, and N. Ambady, ©1999 American Psychological Society.

QUESTION + ANSWER CHOICES

Which conclusion is supported by Figure 1?

- A. Priming stereotypes always has a negative impact on performance.
- B. Priming a negative stereotype interferes with performance.
- C. Priming a positive stereotype does not affect performance.
- D. Priming stereotypes does not affect performance as much as one's self-concept.

SOLUTION

Solution: The correct answer is B.

1. Priming of positive stereotypes facilitates performance. The figure shows that participants' performance improved after they were reminded of their ethnic identity.
2. The figure indicates that participants performed worse after being reminded of their gender identity. This reminder may serve to prime the negative stereotype about performance in the math task.
3. The figure indicates that participants' performance improved after they were reminded of their ethnic identity. This reminder may serve to prime the positive stereotype about performance in the math task.
4. The figure indicates that subjects asked to describe their gender identity performed worse, and those who described their ethnic identity performed better, compared to the controls who did not describe either their gender or ethnic identity. Thus, being reminded of either a positive or negative stereotype about math performance affected subjects' performance.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B ✓
C
D
Time Spent:
57 secs

Difficulty Level:
Easy
Content & Skills:
SIRS4
CC8B
FC8
PSY

Question 190 of 230

Data ID: 531867
Captured: 6/3/2026, 4:41:46 PM

PASSAGE

Passage 4 (Questions 19–22)

Studies have shown that the availability of social support, before and after exposure to a stressor, is associated with fewer stress responses. Based on these findings, a research group hypothesized that when exposed to stressors, people will be more inclined to engage in prosocial behaviors (such as sharing and trusting) as a coping mechanism. To test this hypothesis, college-aged male participants were recruited. All the participants were given an anxiety assessment, and five participants who reported unusually high levels of anxiety were dropped. The remaining participants were randomly assigned to either the stress group or the control group. In the first phase of the experiment, the stress group was asked to study a set of slides and give a public presentation, a condition that is considered a social stressor. The control group was asked to study the same set of slides as the stress group but did not have to give a public presentation.

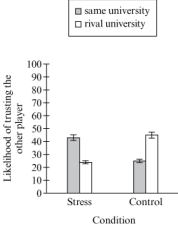
In the second phase of the experiment, the participants were told that they would be playing a “trust game” with another player. According to the trust game, each participant could choose between two monetary rewards: walking away with \$30 or choosing to share \$120 with another player. Sharing required trusting the other player because the other player would decide how the \$120 would be divided. If the other player was fair, he could choose to divide the money into equal halves; in which case, the participant would get \$60. If the other player wasn’t fair, he could keep the \$120, and the participant would get \$0. The results showed that the participants who had been in the stress condition were more likely to trust the other player than those who had been in the control condition. The researchers concluded that their findings supported the hypothesis that stress promotes prosocial behavior. However, the experiment has been criticized for not accounting for the possibility that stress might lead to an increase in risk-taking rather than an increase in prosocial behavior.

Adapted from B. von Dawans, U. Fischbacher, C. Kirschbaum, E. Fehr, & M. Heinrichs. ©2012 the Authors.

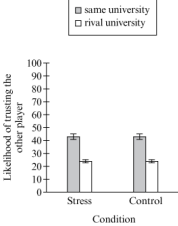
QUESTION + ANSWER CHOICES

The experiment was replicated with the addition of another variable: when playing the trust game, the participants were told either that the other player was a student at their university or was a student at a rival university. Based on the findings in the passage and research on in-group and out-group dynamics, which outcome is most likely?

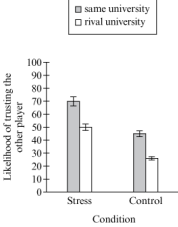
- A.



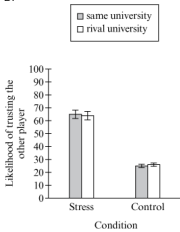
- B.



- C.



• D.



SOLUTION

Solution: The correct answer is C.

1. This figure shows the reverse of what has been observed in research on group dynamics, namely that people are more likely to trust in-group members than out-group members. In addition, this figure shows no effect of stress level, whereas the study described in the passage suggested that stress caused participants to be more trusting of the other player.
2. This figure shows no effect of stress level. The study described in the passage suggested that stress caused participants to be more trusting of the other player.
3. **Research on group dynamics has shown that people are more likely to trust in-group members than out-group members. The study described in the passage suggested that stress caused participants to be more trusting of the other player.**
4. This figure shows no difference between the levels of trust in members of an in-group and members of an out-group. Studies of group dynamics show that people are more likely to trust in-group members than out-group members.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
2 mins 25 secs
Difficulty Level:
Easy
Content & Skills:
SIRS4
CC8A
FC8
PSY

Question 191 of 230

Data ID: 631872
Captured: 6/3/2026, 4:41:46 PM

PASSAGE

Passage 4 (Questions 19–22)

Studies have shown that the availability of social support, before and after exposure to a stressor, is associated with fewer stress responses. Based on these findings, a research group hypothesized that when exposed to stressors, people will be more inclined to engage in prosocial behaviors (such as sharing and trusting) as a coping mechanism. To test this hypothesis, college-aged male participants were recruited. All the participants were given an anxiety assessment, and five participants who reported unusually high levels of anxiety were dropped. The remaining participants were randomly assigned to either the stress group or the control group. In the first phase of the experiment, the stress group was asked to study a set of slides and give a public presentation, a condition that is considered a social stressor. The control group was asked to study the same set of slides as the stress group but did not have to give a public presentation.

In the second phase of the experiment, the participants were told that they would be playing a “trust game” with another player. According to the trust game, each participant could choose between two monetary rewards: walking away with \$30 or choosing to share \$120 with another player. Sharing required trusting the other player because the other player would decide how the \$120 would be divided. If the other player was fair, he could choose to divide the money into equal halves; in which case, the participant would get \$60. If the other player wasn’t fair, he could keep the \$120, and the participant would get \$0. The results showed that the participants who had been in the stress condition were more likely to trust the other player than those who had been in the control condition. The researchers concluded that their findings supported the hypothesis that stress promotes prosocial behavior. However, the experiment has been criticized for not accounting for the possibility that stress might lead to an increase in risk-taking rather than an increase in prosocial behavior.

Adapted from B. von Dawans, U. Fischbacher, C. Kirschbaum, E. Fehr, & M. Heinrichs. ©2012 the Authors.

QUESTION + ANSWER CHOICES

The researchers collected data on participants’ heart rates and electrical skin conductance at different stages of the experiment. Which pattern would verify that the independent variable had the intended effect? Heart rate and skin conductance are higher:

- A.
for all the participants before they participate in the first phase of the experiment compared to after.
- B.
for the participants who chose to share \$120 compared to those who chose to walk away with \$30.
- **C.**
for the participants in the stress group than for the participants in the control group.
- D.
for all the participants before they participate in the second phase of the experiment compared to after.

SOLUTION

Solution: The correct answer is C.

1. The independent variable was whether participants were exposed to a stressor (i.e., being told that they would give a public presentation). Elevated heart rate and skin conductance are part of the stress response. Therefore, participants in the stress condition (but not those in the control condition) were expected to show higher levels of these dependent measures. Therefore, measuring heart rate and skin conductance before and after participation for all the participants may not verify the effect of independent variable.
2. Whether participants in either condition would display greater levels of trust was the research question. The independent variable was whether participants were exposed to a stressor (i.e., being told that they would give a public presentation). Elevated heart rate and skin conductance are part of the stress response, which is associated with prosocial behavior.
3. **The independent variable was whether participants were exposed to a stressor (i.e., being told that they would give a public presentation). Elevated heart rate and skin conductance are part of the stress response. Therefore, participants in the stress condition, but not those in the control condition, were expected to show higher levels of these dependent measures.**
4. All participants (in both the stress and control conditions) participated in the second phase of the experiment. The independent variable was whether participants were exposed to a stressor (i.e., being told that they would give a public presentation). Elevated heart rate and skin conductance are part of the stress response. Therefore, participants in the stress condition, but not those in the control condition, are expected to show higher levels of these dependent measures.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
33 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS3
CC8C
FC6

Question 192 of 230

Data ID: 531877
Captured: 6/3/2026, 4:41:47 PM

PASSAGE

Passage 4 (Questions 19–22)

Studies have shown that the availability of social support, before and after exposure to a stressor, is associated with fewer stress responses. Based on these findings, a research group hypothesized that when exposed to stressors, people will be more inclined to engage in prosocial behaviors (such as sharing and trusting) as a coping mechanism. To test this hypothesis, college-aged male participants were recruited. All the participants were given an anxiety assessment, and five participants who reported unusually high levels of anxiety were dropped. The remaining participants were randomly assigned to either the stress group or the control group. In the first phase of the experiment, the stress group was asked to study a set of slides and give a public presentation, a condition that is considered a social stressor. The control group was asked to study the same set of slides as the stress group but did not have to give a public presentation.

In the second phase of the experiment, the participants were told that they would be playing a “trust game” with another player. According to the trust game, each participant could choose between two monetary rewards: walking away with \$30 or choosing to share \$120 with another player. Sharing required trusting the other player because the other player would decide how the \$120 would be divided. If the other player was fair, he could choose to divide the money into equal halves; in which case, the participant would get \$60. If the other player wasn’t fair, he could keep the \$120, and the participant would get \$0. The results showed that the participants who had been in the stress condition were more likely to trust the other player than those who had been in the control condition. The researchers concluded that their findings supported the hypothesis that stress promotes prosocial behavior. However, the experiment has been criticized for not accounting for the possibility that stress might lead to an increase in risk-taking rather than an increase in prosocial behavior.

Adapted from B. von Dawans, U. Fischbacher, C. Kirschbaum, E. Fehr, & M. Heinrichs. ©2012 the Authors.

QUESTION + ANSWER CHOICES

Which statement provides the LEAST likely explanation for why the researchers dropped the highly anxious participants from their sample?

- A. Ethical committees do not allow research with participants who may have psychological disorders.
- B. If all the anxious participants get assigned to the stress condition, this may pose a confounding variable.
- C. Highly anxious participants are not representative of the population and may reduce the generalizability of the findings.
- D. If all the anxious participants get assigned to the control condition, this may lead to a false rejection of the experimental hypothesis.

SOLUTION

Solution: The correct answer is A.

1. **Ethical standards of research with human participants allow research to be conducted with anxious participants.**
2. One of the reasons for excluding some volunteers from participation is to prevent the possibility that more of the outliers may be assigned to one of the conditions (i.e., either the stress or control group). Such confounding may bring about errors in inference.
3. The research question was whether exposure to stressors causes people in general to engage in prosocial behaviors. Persons who are already at unusually high levels of stress may respond differently, or perhaps not at all (ceiling effect) to the stressor manipulation used in the study. Thus, their responses may limit the generalizability of the research findings.
4. The main reason for screening the volunteers and excluding the extremely anxious ones from participation was to prevent the possibility that more of those outliers would be assigned to one of the conditions (i.e., either the stress or control group). Such confounding could cause errors in interpreting the results.

Khan Academy Lessons

For more help on the skills, concepts, or content covered by this question, see the Khan Academy Lessons on [Biological Bases of Behavior](#), [Human Development](#), [Behavior](#), and [Cognition](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
51 secs
Difficulty Level:
Expert
Content & Skills:
SIRS3
CC7A
FC7
PSY

Question 193 of 230

Data ID: 531892
Captured: 6/3/2026, 4:41:48 PM

PASSAGE

Passage 4 (Questions 19–22)

Studies have shown that the availability of social support, before and after exposure to a stressor, is associated with fewer stress responses. Based on these findings, a research group hypothesized that when exposed to stressors, people will be more inclined to engage in prosocial behaviors (such as sharing and trusting) as a coping mechanism. To test this hypothesis, college-aged male participants were recruited. All the participants were given an anxiety assessment, and five participants who reported unusually high levels of anxiety were dropped. The remaining participants were randomly assigned to either the stress group or the control group. In the first phase of the experiment, the stress group was asked to study a set of slides and give a public presentation, a condition that is considered a social stressor. The control group was asked to study the same set of slides as the stress group but did not have to give a public presentation.

In the second phase of the experiment, the participants were told that they would be playing a “trust game” with another player. According to the trust game, each participant could choose between two monetary rewards: walking away with \$30 or choosing to share \$120 with another player. Sharing required trusting the other player because the other player would decide how the \$120 would be divided. If the other player was fair, he could choose to divide the money into equal halves; in which case, the participant would get \$60. If the other player wasn’t fair, he could keep the \$120, and the participant would get \$0. The results showed that the participants who had been in the stress condition were more likely to trust the other player than those who had been in the control condition. The researchers concluded that their findings supported the hypothesis that stress promotes prosocial behavior. However, the experiment has been criticized for not accounting for the possibility that stress might lead to an increase in risk-taking rather than an increase in prosocial behavior.

Adapted from B. von Dawans, U. Fischbacher, C. Kirschbaum, E. Fehr, & M. Heinrichs. ©2012 the Authors.

QUESTION + ANSWER CHOICES

A researcher replicates the experiment with the addition of a physical stressor to the first phase of the experiment. According to Selye’s general adaptation syndrome, this change is:

- **A.** necessary, because humans respond differently to different types of stressors.
- **B.** necessary, because physical stressors cause avoidance–avoidance conflict, whereas social stressors cause approach–approach conflict.
- **C.** unnecessary, because the human stress response is not specific to the type of stressor.
- **D.** unnecessary, because both physical stressors and social stressors cause avoidance–avoidance conflict.

SOLUTION

Solution: The correct answer is C.

1. According to Selye’s General Adaptation Syndrome (GAS), an organism’s stress response always follows a similar course, regardless of the exact nature of the stressor.
2. Selye’s General Adaptation Syndrome (GAS) contains no provisions about the exact nature of the stressor. In particular, no mention of whether any specific stressor causes an approach–approach conflict or an avoidance–avoidance conflict.
3. **According to Selye’s General Adaptation Syndrome (GAS), an organism’s stress response always follows a similar course, regardless of the exact nature of the stressor.**
4. Selye’s General Adaptation Syndrome (GAS) makes no mention of whether any specific stressor causes an avoidance–avoidance conflict. The nature of the stressor is irrelevant.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
x
B
C
D
Time Spent:
1 min 3 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC6C
FC6
PSY

Question 194 of 230

Data ID: 531898
Captured: 6/3/2026, 4:41:48 PM

PASSAGE

Passage 5 (Questions 23–26)

Synchronized behaviors refer to coordinated actions that are matched in time. Study 1 demonstrated that these behaviors can affect affiliation ratings. The participants’ task was to tap their fingers to a metronome in two different testing conditions: with an experimenter who tapped her finger to the same metronome (synchronous condition) or with an experimenter who tapped her finger to another metronome (asynchronous condition). Afterwards, the participants were asked to rate the degree of affiliation they felt toward the experimenter. The results showed that the participants felt more affiliated with the experimenter under the synchronous condition.

Study 2 was designed to test the hypothesis that synchronous actions make participants more prone to comply when prompted to engage in aggression. Participants were assigned to either the synchronous or asynchronous condition in groups of three. Synchrony was manipulated similarly to Study 1, by having the participants in each group finger tap to the same metronome or to different metronomes. After the finger-tapping task, each group was told that a similar experiment was going to be conducted with either music or noise in the background and that their group would pick what the participants in the other group would be exposed to. They were asked to make a selection from a list composed of ten songs and an option labeled “noise blast” that was described as “90 seconds of static played at an extremely high volume. Can be aggravating.” In the persuasion trials, one of the participants was a research confederate posing as a participant. As the group was deciding, the confederate said, “Oooh, I think we should choose the noise blast.” In the no persuasion trials, there was no confederate in the group. The results supported the researchers’ hypothesis.

Adapted from S. S. Wittemouth ©2011 Elsevier, M. J. Howe and J. L. Risen ©2009 Social Cognition.

QUESTION + ANSWER CHOICES

Which conclusion is best supported by the outcome of Study 2?

- **A.** Synchronous activity increases group polarization.
- **B.** Synchronous activity increases conformity.
- **C.** ~~Asynchronous activity increases cognitive dissonance.~~
- **D.** ~~Asynchronous activity increases social loafing.~~

SOLUTION

Solution: The correct answer is B.

1. The results of Study 2 were unrelated to the phenomenon of group polarization.
2. **The results of Study 2 showed that synchronous activity made participants in the persuasion trials more likely to conform to the confederate’s suggestion to expose the other group to the “noise blast.”**
3. No results from Study 2 were reported that were related to cognitive dissonance.
4. The passage did not mention any results from Study 2 that were related to the phenomenon of social loafing.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
4 mins 16 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CCTB
FC7
PSY

Question 195 of 230

Data ID: 531907
Captured: 6/3/2026, 4:41:49 PM

PASSAGE

Passage 5 (Questions 23–26)

Synchronized behaviors refer to coordinated actions that are matched in time. Study 1 demonstrated that these behaviors can affect affiliation ratings. The participants' task was to tap their fingers to a metronome in two different testing conditions: with an experimenter who tapped her finger to the same metronome (synchronous condition) or with an experimenter who tapped her finger to another metronome (asynchronous condition). Afterwards, the participants were asked to rate the degree of affiliation they felt toward the experimenter. The results showed that the participants felt more affiliated with the experimenter under the synchronous condition.

Study 2 was designed to test the hypothesis that synchronous actions make participants more prone to comply when prompted to engage in aggression. Participants were assigned to either the synchronous or asynchronous condition in groups of three. Synchrony was manipulated similarly to Study 1, by having the participants in each group finger tap to the same metronome or to different metronomes. After the finger-tapping task, each group was told that a similar experiment was going to be conducted with either music or noise in the background and that their group would pick what the participants in the other group would be exposed to. They were asked to make a selection from a list composed of ten songs and an option labeled "noise blast" that was described as "90 seconds of static played at an extremely high volume. Can be aggravating." In the persuasion trials, one of the participants was a research confederate posing as a participant. As the group was deciding, the confederate said, "Oooh, I think we should choose the noise blast." In the no persuasion trials, there was no confederate in the group. The results supported the researchers' hypothesis.

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QUESTION + ANSWER CHOICES

Damage to which nervous system structure is most likely to cause problems in the participants' ability to perform the synchronous behaviors in studies 1 and 2?

- A. Hippocampus
- B. Hypothalamus
- C. Cochlea
- D. Cerebellum

SOLUTION

Solution: The correct answer is D.

- The primary functions of the hippocampus involve memory.
- The hypothalamus is primarily concerned with functions like the processing of sympathetic nervous system inputs and control of the endocrine system (i.e., through its connection to the pituitary gland).
- The cochlea, a structure in the inner ear, is primarily concerned with the reception of auditory sensory inputs.
- The coordinated finger tapping in the synchronous behavior condition is a motor task. Therefore, it most likely requires the proper functioning of the cerebellum.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
X
D
Time Spent:
1 min 40 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS1
CC7A
FC7
PSY

Question 196 of 230

Data ID: 531910
Captured: 6/3/2026, 4:41:50 PM

PASSAGE

Passage 5 (Questions 23–26)

Synchronized behaviors refer to coordinated actions that are matched in time. Study 1 demonstrated that these behaviors can affect affiliation ratings. The participants' task was to tap their fingers to a metronome in two different testing conditions: with an experimenter who tapped her finger to the same metronome (synchronous condition) or with an experimenter who tapped her finger to another metronome (asynchronous condition). Afterwards, the participants were asked to rate the degree of affiliation they felt toward the experimenter. The results showed that the participants felt more affiliated with the experimenter under the synchronous condition.

Study 2 was designed to test the hypothesis that synchronous actions make participants more prone to comply when prompted to engage in aggression. Participants were assigned to either the synchronous or asynchronous condition in groups of three. Synchrony was manipulated similarly to Study 1, by having the participants in each group finger tap to the same metronome or to different metronomes. After the finger-tapping task, each group was told that a similar experiment was going to be conducted with either music or noise in the background and that their group would pick what the participants in the other group would be exposed to. They were asked to make a selection from a list composed of ten songs and an option labeled "noise blast" that was described as "90 seconds of static played at an extremely high volume. Can be aggravating." In the persuasion trials, one of the participants was a research confederate posing as a participant. As the group was deciding, the confederate said, "Oooh, I think we should choose the noise blast." In the no persuasion trials, there was no confederate in the group. The results supported the researchers' hypothesis.

Adapted from S. S. Wittemouth ©2011 Elsevier, M. J. Howe and J. L. Rosen ©2009 Social Cognition.

QUESTION + ANSWER CHOICES

A researcher suggests that the effect of the testing condition on the affiliation ratings is caused by increased sympathetic nervous system activity due to performing a repetitive behavior. Is this explanation likely to be supported?

- A. Yes, because studies show that increased sympathetic nervous system arousal can affect how people feel about others.
- B. Yes, because the design used in Study 1 allows researchers to establish the effect of sympathetic nervous system arousal on affiliation ratings.
- C. No, because performing repetitive behaviors is more likely to increase parasympathetic nervous system activity.
- D. No, because both of the conditions involve repetitive behaviors, but participants' ratings differ under the two conditions.

SOLUTION

Solution: The correct answer is D.

- Although it is entirely plausible that increased sympathetic activity affects people's feelings towards others, the repetitive activities were performed by participants in both the synchronous and asynchronous conditions.
- The study's design did not allow any inferences about the potential effects of increases in sympathetic arousal.
- The study's design did not allow any inferences about the potential effects of increases in parasympathetic arousal. Also, it is unlikely that the repetitive behaviors would affect responses of the parasympathetic nervous system.
- Participants in both the synchronous and asynchronous conditions engaged in repetitive activities. Therefore, if differences in the affiliation ratings had been caused by sympathetic nervous system arousal, only the synchronous condition would have been affected.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Disinhibited Social Behavior](#), [Human Development: Behavior and Cognition](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
2 mins 57 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC7A
FC7
PSY

Question 197 of 230

Data ID: 531914
Captured: 6/3/2026, 4:41:50 PM

PASSAGE

Passage 5 (Questions 23–26)

Synchronized behaviors refer to coordinated actions that are matched in time. Study 1 demonstrated that these behaviors can affect affiliation ratings. The participants' task was to tap their fingers to a metronome in two different testing conditions: with an experimenter who tapped her finger to the same metronome (synchronous condition) or with an experimenter who tapped her finger to another metronome (asynchronous condition). Afterwards, the participants were asked to rate the degree of affiliation they felt toward the experimenter. The results showed that the participants felt more affiliated with the experimenter under the synchronous condition.

Study 2 was designed to test the hypothesis that synchronous actions make participants more prone to comply when prompted to engage in aggression. Participants were assigned to either the synchronous or asynchronous condition in groups of three. Synchrony was manipulated similarly to Study 1, by having the participants in each group finger tap to the same metronome or to different metronomes. After the finger-tapping task, each group was told that a similar experiment was going to be conducted with either music or noise in the background and that their group would pick what the participants in the other group would be exposed to. They were asked to make a selection from a list composed of ten songs and an option labeled "noise blast" that was described as "90 seconds of static played at an extremely high volume. Can be aggravating." In the persuasion trials, one of the participants was a research confederate posing as a participant. As the group was deciding, the confederate said, "Oooh, I think we should choose the noise blast." In the no persuasion trials, there was no confederate in the group. The results supported the researchers' hypothesis.

Adapted from S. S. Wittemouth ©2011 Elsevier, M. J. Howe and J. L. Risen ©2009 Social Cognition.

QUESTION + ANSWER CHOICES

Based on the findings in the passage, engaging in synchronous behaviors is most likely to lead to an increase in:

- A. egocentrism.
- B. observational learning.
- C. obedience.
- D. social facilitation.

SOLUTION

Solution: The correct answer is **C**.

- Neither study involved the measurement of any response that would indicate whether participants were behaving in an egocentric manner.
- There was no modeling of behavior in either study. Therefore, no inference about observational learning is possible.
- Based on the finding (Study 2) that engaging in repetitive synchronous behaviors can increase compliance, it can be inferred that it may also increase levels of obedience.**
- There was no manipulation of group size in either study, which would be needed to make any inference about social facilitation.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
1 min 11 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC7B
FC7
PSY

Question 198 of 230

Data ID: 531931
Captured: 6/3/2026, 4:41:51 PM

QUESTION + ANSWER CHOICES

Which type of design is LEAST appropriate for research on residential segregation?

- A. Ethnographic methods
- B. Experimental methods
- C. Quantitative methods
- D. Survey methods

SOLUTION

Solution: The correct answer is **B**.

- Ethnography typically involves the systematic observation of people in their cultural settings. Sometimes ethnographic methods also include in-depth interviews as well, to better understand participants' culture. The study of residential segregation can be appropriately conducted using ethnographic methods.
- Experimental methods are generally not appropriate for the study of residential segregation. Experiments require random assignment to different groups (usually an intervention group and a control group). Except under specific circumstances, it would be unethical and impractical to randomly assign participants to different groups in which residential segregation varies.**
- Quantitative methods simply refer to methods that utilize numerical data. Residential segregation can be appropriately studied using quantitative methods.
- Survey methods involve the use of a questionnaire in which respondents provide answers to various questions. Residential segregation can be appropriately studied using surveys.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
25 secs
Difficulty Level:
Easy
Content & Skills:
SIRS3
CC10A
FC10
SOC

Question 199 of 230

Data ID: 531944
Captured: 6/3/2026, 4:41:51 PM

QUESTION • ANSWER CHOICES

Participants in a weight-loss program agree to have their body mass index (BMI) measured to track their progress in the program. Among a sample of 72 participants, the mean BMI is 30 and the median BMI is 25. Which statement provides an accurate description of the sample?

- ☐ A. More participants had a BMI over 25 than a BMI under 25.
- ☐ B. The majority of participants had a BMI between 25 and 30.
- ☒ C. Half of the participants had a BMI over 25 and half had a BMI under 25.
- ☐ D. ~~More outliers among the participants had a BMI under 25 than a BMI over 30.~~

SOLUTION

Solution: The correct answer is **C**.

1. The median splits the sample distribution of values in half. Thus, there would not be more participants over (or under) the median BMI of 25.
2. With the information given in the question, it cannot be determined whether a majority of participants had a BMI between 25 and 30 (to make such a determination, one would need to have some information on the variance of the sample distribution).
3. **The median represents the middle of the distribution, splitting the sample distribution of values in half. Thus, a median of 25 can be accurately described as when half of the sample has a BMI over 25 and the other half of the sample has a BMI under 25.**
4. The number of outliers cannot be determined based on the information provided in the question alone. Given that the mean is greater than the median, however, it is more likely that there would be more outliers over the mean or median values, as opposed to under them.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
X
B
C
D
Time Spent:
6 mins 17 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS4
CC9A
FC9
SOC

Question 200 of 230

Data ID: 531951
Captured: 6/3/2026, 4:41:52 PM

QUESTION • ANSWER CHOICES

Often utilized when studying communicable diseases, which type of analysis maps the series of relationships among a set of individuals?

- ☐ A. Social support analysis
- ☒ B. Social network analysis
- ☐ C. Social stratification analysis
- ☐ D. Social reproduction analysis

SOLUTION

Solution: The correct answer is **B**.

1. Social support typically refers to the tangible and intangible forms of assistance that individuals receive from their social ties. Social support describes the assistance that individuals receive, rather than the series of relationships among a set of individuals.
2. **Used in some epidemiological studies, social network analysis is the method described in the question. Network analysis involves mapping the social relationships that exist among a set of individuals.**
3. Social stratification refers to the hierarchy of social positions in a society (often according to either social class or social status). Stratification does not describe the relationship among individuals but rather the relationship among social positions in a society.
4. Social reproduction refers to the perpetuation of inequality through social institutions. Because of the focus on institutions, social reproduction would not apply to relationships among a set of individuals.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
40 secs
Difficulty Level:
Easy
Content & Skills:
SIRS3
CC8C
FC8
SOC

Question 201 of 230

Data ID: 531957
Captured: 6/3/2026, 4:41:53 PM

QUESTION • ANSWER CHOICES

Which effect is LEAST likely to occur with the process of gentrification?

- A. Development of affordable housing
- B. Increased neighborhood stratification
- C. Displacement of lower-income residents
- D. Expanded tax base for local government

SOLUTION

Solution: The correct answer is **A**.

1. **Gentrification describes the process in which relatively affluent individuals move into a neighborhood that recently consisted of residents with moderate to low income. With the arrival of more affluent residents, housing demand increases and often leads to a decrease in affordable housing available to lower income residents. In contrast, the other options are more likely to occur with the process of gentrification.**
2. Given the increase in more affluent residents that occurs from gentrification, there would likely be an increase in neighborhood stratification (meaning a larger difference in the social class or status positions across the residents in the neighborhood).
3. Due to the influx of more affluent residents, and the increased demand for housing that then raises prices, gentrification often leads to the displacement of lower-income residents who can no longer afford to live in the neighborhood.
4. The increase in more affluent residents that occurs with gentrification is likely to increase the tax base for the local government (given the increase in real estate prices, which would yield increased tax revenue).

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
17 secs
Difficulty Level:
Easy
Content & Skills:
SIRS2
CC9B
FC9
SOC

Question 202 of 230

Data ID: 531973
Captured: 6/3/2026, 4:41:53 PM

PASSAGE

Passage 6 (Questions 31-34)

A study was designed to examine the role of peer pressure on risky behaviors across different age groups. The participants were post-pubescent adolescents, young adults, and adults. The majority of the participants were White. The measures were a decision-making questionnaire (DMQ) and a computer game (CG). The DMQ presented the participants with hypothetical dilemmas that involved risky decisions and asked them what they would do if they were faced with these dilemmas. In the CG, the player watched an animated car move toward a stoplight that was yellow. The player was instructed to press a button to stop the car before the light turned red. The objective of the game was for the car to advance as much as possible without running a red light. For both the DMQ and the CG, higher scores indicated riskier behavior.

The participants completed the measures under one of two testing conditions: alone or in a group of three. For the DMQ, the participants in the group condition were encouraged to discuss the scenarios with each other. In the CG, the participants in the group condition took turns playing. The participants who were not playing at a given time were told to offer the player advice. The young adults and the adults came to the experiments in groups of three friends. The adolescents were assigned to groups of three by the experimenters. All of the groups were composed of same-sex participants.

The results showed a negative correlation between age and scores on the DMQ. The effect of testing condition on risky behavior scores from the CG is shown in Figure 1.

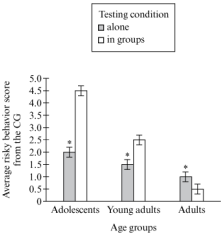


Figure 1 Risky behavior by age group and testing condition
(Note: An asterisk (*) indicates that the difference between the testing conditions was statistically significant.)

Adapted from M. Gardner and L. Steinberg. ©2005 the American Psychological Association.

QUESTION • ANSWER CHOICES

On average, adults are found to be more risk averse than adolescents. Based on group polarization, how would testing condition affect adults' scores on the DMQ? Adults would have:

- A. lower scores on the DMQ in a group than alone.
- B. higher scores on the DMQ in a group than alone.
- C. lower scores on the DMQ in small groups but higher scores on the DMQ in large groups.
- D. higher scores on the DMQ in small groups but lower scores on the DMQ in large groups.

SOLUTION

Solution: The correct answer is **A**.

1. **After interacting with one another, a group of risk averse individuals will most likely become more risk averse. The DMQ (Decision Making Questionnaire) scores will most likely be lower after a group discussion, indicating a higher level of risk aversion. This prediction is based on the phenomenon of group polarization, in which people's attitudes become more extreme after they discuss the attitude object with like-minded individuals. Thus, their scores will most likely indicate a greater risk aversion than participants who were tested alone.**
2. After interacting with one another, a group of risk averse individuals will most likely become more risk averse. Their DMQ (Decision Making Questionnaire) scores will most likely be lower than the DMQ scores of the participants who are tested individually.
3. Group size is expected to influence the results. Thus, the effect of group polarization for adult participants will be greater (i.e., lower DMQ scores, indicating greater risk aversion) in the group condition.
4. Group size is expected to influence the results. There will still be evidence of the group polarization effect (i.e., low DMQ scores, indicating greater risk aversion) in adults with any group size. However, participants were studied either alone or with two friends, and no social influence is expected in the alone condition.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer

A

✓

B

C

D

Time Spent:

4 mins 11 secs

Difficulty Level:

Difficult

Content & Skills:

SIRS2

CC7B

FC7

PSY

Question 203 of 230

Data ID: 631980

Captured: 6/3/2026, 4:41:54 PM

PASSAGE

Passage 6 (Questions 31-34)

A study was designed to examine the role of peer pressure on risky behaviors across different age groups. The participants were post-pubescent adolescents, young adults, and adults. The majority of the participants were White. The measures were a decision-making questionnaire (DMQ) and a computer game (CG). The DMQ presented the participants with hypothetical dilemmas that involved risky decisions and asked them what they would do if they were faced with these dilemmas. In the CG, the player watched an animated car move toward a stoplight that was yellow. The player was instructed to press a button to stop the car before the light turned red. The objective of the game was for the car to advance as much as possible without running a red light. For both the DMQ and the CG, higher scores indicated riskier behavior.

The participants completed the measures under one of two testing conditions: alone or in a group of three. For the DMQ, the participants in the group condition were encouraged to discuss the scenarios with each other. In the CG, the participants in the group condition took turns playing. The participants who were not playing at a given time were told to offer the player advice. The young adults and the adults came to the experiments in groups of three friends. The adolescents were assigned to groups of three by the experimenters. All of the groups were composed of same-sex participants.

The results showed a negative correlation between age and scores on the DMQ. The effect of testing condition on risky behavior scores from the CG is shown in Figure 1.

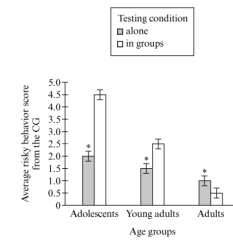


Figure 1 Risky behavior by age group and testing condition
(Note: An asterisk (*) indicates that the difference between the testing conditions was statistically significant.)

Adapted from M. Gardner and L. Steinberg. ©2005 the American Psychological Association.

QUESTION • ANSWER CHOICES

Which statement best applies the behaviorist theory to explain the association between age and DMQ scores? Compared to younger participants, older participants:

- A.
have had more experiences in which they have been punished for risky behaviors.
- B.
are less prone to the effects of peer pressure.
- C.
have a more mature prefrontal cortex that inhibits impulsive behavior.
- D.
have a more developed superego that inhibits the impulses of the id.

SOLUTION

Solution: The correct answer is A.

1. **Older participants most likely have had greater levels of experience with their risky decisions leading to aversive consequences. Thus, their risky behavior has been punished more often than that of younger participants.**
2. Although the statement is accurate, peer pressure, a form of normative social influence, is not a concept used in behaviorist theory.
3. Although the statement is accurate, behaviorist theory does not explain behavioral variations in terms of neural mechanisms.
4. Superego and id are constructs used in psychoanalytic theory, which is antithetical to behaviorist theory.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct

Correct Answer Your Answer

A

✓

B

C

D
Time Spent:
56 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC7C
FC7
PSY

Question 204 of 230

Data ID: 531985
Captured: 6/3/2026, 4:41:54 PM

PASSAGE

Passage 6 (Questions 31-34)

A study was designed to examine the role of peer pressure on risky behaviors across different age groups. The participants were post-pubescent adolescents, young adults, and adults. The majority of the participants were White. The measures were a decision-making questionnaire (DMQ) and a computer game (CG). The DMQ presented the participants with hypothetical dilemmas that involved risky decisions and asked them what they would do if they were faced with these dilemmas. In the CG, the player watched an animated car move toward a stoplight that was yellow. The player was instructed to press a button to stop the car before the light turned red. The objective of the game was for the car to advance as much as possible without running a red light. For both the DMQ and the CG, higher scores indicated riskier behavior.

The participants completed the measures under one of two testing conditions: alone or in a group of three. For the DMQ, the participants in the group condition were encouraged to discuss the scenarios with each other. In the CG, the participants in the group condition took turns playing. The participants who were not playing at a given time were told to offer the player advice. The young adults and the adults came to the experiments in groups of three friends. The adolescents were assigned to groups of three by the experimenters. All of the groups were composed of same-sex participants.

The results showed a negative correlation between age and scores on the DMQ. The effect of testing condition on risky behavior scores from the CG is shown in Figure 1.

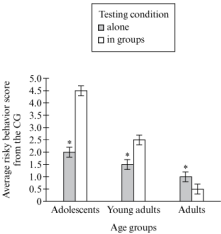


Figure 1 Risky behavior by age group and testing condition
(Note: An asterisk (*) indicates that the difference between the testing conditions was statistically significant.)

Adapted from M. Gardner and L. Steinberg. ©2005 the American Psychological Association.

QUESTION + ANSWER CHOICES

Which finding would support the validity of the DMQ measure?

- A. A positive correlation between the DMQ scores and the CG scores
- B. A negative correlation between the DMQ scores and the CG scores
- C. A positive correlation between participants' scores on two different administrations of the DMQ
- D. A negative correlation between participants' scores on two different administrations of the DMQ

SOLUTION

Solution: The correct answer is A.

- Both the DMQ and the CG were designed to assess risky behavior. Lower scores indicated greater risk aversion on both measures. Therefore, a positive correlation between the DMQ scores and the CG scores can be interpreted as supporting their validity (i.e., that both measure what they are intended to measure).
- Both the DMQ and the CG were designed to assess the same tendency toward engaging in risky behavior. Thus, a negative correlation between these measures would be a contraindication of the validity of either or both of them.
- The consistency of scores on the DMQ would indicate reliability, but not the validity of the assessment.
- Although it would not really tell us anything about the validity of the assessment, such inconsistency of scores on the DMQ would indicate that this assessment lacked reliability, which is normally regarded as a necessary condition for validity.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A

✓
B
C
D

Time Spent:

3 mins 41 secs

Difficulty Level:

Difficult

Content & Skills:

SIRS4
CC7B
FC7
PSY

Question 205 of 230

Data ID: 531990
Captured: 6/3/2026, 4:41:55 PM

PASSAGE

Passage 6 (Questions 31-34)

A study was designed to examine the role of peer pressure on risky behaviors across different age groups. The participants were post-pubescent adolescents, young adults, and adults. The majority of the participants were White. The measures were a decision-making questionnaire (DMQ) and a computer game (CG). The DMQ presented the participants with hypothetical dilemmas that involved risky decisions and asked them what they would do if they were faced with these dilemmas. In the CG, the player watched an animated car move toward a stoplight that was yellow. The player was instructed to press a button to stop the car before the light turned red. The objective of the game was for the car to advance as much as possible without running a red light. For both the DMQ and the CG, higher scores indicated riskier behavior.

The participants completed the measures under one of two testing conditions: alone or in a group of three. For the DMQ, the participants in the group condition were encouraged to discuss the scenarios with each other. In the CG, the participants in the group condition took turns playing. The participants who were not playing at a given time were told to offer the player advice. The young adults and the adults came to the experiments in groups of three friends. The adolescents were assigned to groups of three by the experimenters. All of the groups were composed of same-sex participants.

The results showed a negative correlation between age and scores on the DMQ. The effect of testing condition on risky behavior scores from the CG is shown in Figure 1.

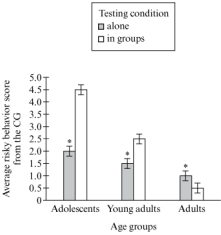


Figure 1 Risky behavior by age group and testing condition
(Note: An asterisk (*) indicates that the difference between the testing conditions was statistically significant.)

Adapted from M. Gardner and L. Steinberg. ©2005 the American Psychological Association.

QUESTION + ANSWER CHOICES

According to Erikson's theory of psychosocial development, which group of participants are most likely dealing with the challenges of generativity versus stagnation?

1. Adolescents

2. Young adults

3. Adults
- A.
I only

• **B.**
III only

• C.
I and II only

• D.
II and III only

SOLUTION

Solution: The correct answer is **B**.

1. Adolescents have not yet reached the age at which the major crisis is that of generativity versus stagnation.

2. **In Erikson's theory, the major crisis facing adults is that of generativity versus stagnation.**

3. Neither adolescents nor young adults have reached the stage at which the major life crisis is that of generativity versus stagnation.

4. In Erikson's theory, the major crisis facing adults is that of generativity versus stagnation. Young adults have not yet reached the age at which this is the major life crisis.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A

B
✓
C
D

Time Spent:
1 min 2 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC8A
FC9
PSY

Question 206 of 230

Data ID: 632000
Captured: 6/3/2026, 4:41:56 PM

PASSAGE

Passage 7 (Questions 35-38)

Studies have found that being bilingual is associated with better self-rated physical and mental health among immigrants in the United States. Research has also found that, on average, assimilation is associated with worse health. Thus, the processes behind both bilingualism and assimilation raise the question of how social integration influences immigrant health.

Using data from the National Latino and Asian American Analysis Project, researchers designed a study to investigate the relationship between language and self-rated health. The measures of language included English and native language proficiency, as well as frequency of English and native language use. The immigrant groups included in the sample were Asian and Latino immigrants.

Based on a national area probability sample, the researchers oversampled geographical areas that have a moderate-to-high density of Latinos and Asian Americans. Bilingual interviewers asked the respondents to rate their overall physical health status and overall mental health status on a 5-point scale (with 1 being poor health and 5 being excellent health). The researchers' statistical models considered multiple variables that may affect the relationship between language and health. These control variables included socioeconomic status (SES), assimilation, family and social support, stress and discrimination, and healthcare access and health behaviors.

Results showed strong associations between bilingualism and both physical and mental self-rated health for Latino and Asian immigrants. These associations are partially explained by SES and family and social support, but the researchers did not find evidence that assimilation, stress and discrimination, or healthcare access and health behaviors affected the language–health relationship. Based on their findings, the researchers suggested that future studies should consider how the unique cultural, linguistic, and cognitive skills of bilingual immigrants might shape their physical and mental health statuses.

Adapted from A. Schachter, R. T. Kimbro, and B. K. Gorman, "Language Proficiency and Health Status: Are Bilingual Immigrants Healthier?" ©2012 American Sociological Association.

QUESTION + ANSWER CHOICES

Which statement does NOT identify an aspect of the concept of assimilation?

- A.
Assimilation is the influence that cultural changes have on an individual's health.
- B:
~~Assimilation is the process of cultural adaptation that results from geographic mobility.~~
- C:
~~Assimilation occurs when individuals adopt the cultural norms of a dominant culture.~~
- D.
Assimilation occurs when individuals relinquish the cultural norms of their childhood.

SOLUTION

Solution: The correct answer is A.

1. **Assimilation refers to the process in which new members of a society adopt the norms and values of the dominant culture. Each of the other options provides an element of assimilation (cultural adaptation, adopting new norms, and relinquishing old norms). However, this option does NOT describe an aspect of assimilation itself, but instead speculates about a possible consequence of assimilation.**
2. Assimilation is often used in the context of migration (movement both within a nation and across nations), describing the process of cultural adaptation that occurs as a result of moving to a different location.
3. A central tenet of assimilation is that individuals will adopt the norms and values of a new culture.
4. Assimilation usually presumes that individuals will not retain the norms that they previously held when they were younger. The assimilation perspective has been criticized for this assumption of relinquishing norms.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
4 mins 3 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC9A
FC9
SOC

Question 207 of 230

Data ID: 632003
Captured: 6/3/2026, 4:41:56 PM

PASSAGE

Passage 7 (Questions 35-38)

Studies have found that being bilingual is associated with better self-rated physical and mental health among immigrants in the United States. Research has also found that, on average, assimilation is associated with worse health. Thus, the processes behind both bilingualism and assimilation raise the question of how social integration influences immigrant health.

Using data from the National Latino and Asian American Analysis Project, researchers designed a study to investigate the relationship between language and self-rated health. The measures of language included English and native language proficiency, as well as frequency of English and native language use. The immigrant groups included in the sample were Asian and Latino immigrants.

Based on a national area probability sample, the researchers oversampled geographical areas that have a moderate-to-high density of Latinos and Asian Americans. Bilingual interviewers asked the respondents to rate their overall physical health status and overall mental health status on a 5-point scale (with 1 being poor health and 5 being excellent health). The researchers’ statistical models considered multiple variables that may affect the relationship between language and health. These control variables included socioeconomic status (SES), assimilation, family and social support, stress and discrimination, and healthcare access and health behaviors.

Results showed strong associations between bilingualism and both physical and mental self-rated health for Latino and Asian immigrants. These associations are partially explained by SES and family and social support, but the researchers did not find evidence that assimilation, stress and discrimination, or healthcare access and health behaviors affected the language–health relationship. Based on their findings, the researchers suggested that future studies should consider how the unique cultural, linguistic, and cognitive skills of bilingual immigrants might shape their physical and mental health statuses.

Adapted from A. Schachter, R. T. Kimbro, and B. K. Gorman, “Language Proficiency and Health Status: Are Bilingual Immigrants Healthier?” ©2012 American Sociological Association.

QUESTION + ANSWER CHOICES

A researcher suggests that the benefits of bilingualism are related to the idea that the structure of language affects the perceptions of its speakers. This researcher is referring to which concept?

- A. Weber’s Law
- B. The nativist hypothesis
- C. Schacter–Singer theory
- D. Linguistic relativity

SOLUTION

Solution: The correct answer is D.

1. Weber’s Law is the name given to the robust psychophysical observation that the just noticeable difference (JND) between two physical stimuli is a constant proportion (i.e., Weber’s constant) of the original (or standard) stimulus. It is unrelated to the relationship between language and cognition.
2. The nativist hypothesis is concerned with the acquisition of a child’s first language, and not with the relationship between language and cognition.
3. The Schachter-Singer two-factor theory is concerned with explaining the relation between emotions and cognitive awareness of one’s emotional state. It is unrelated to language and cognition.
4. **Linguistic relativity, which represents the weak version of the Whorf-Sapir hypothesis, is the only term related to the idea referenced in the question.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
26 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC6B
FC6
PSY

Question 208 of 230

Data ID: 632007
Captured: 6/3/2026, 4:41:57 PM

PASSAGE

Passage 7 (Questions 35-38)

Studies have found that being bilingual is associated with better self-rated physical and mental health among immigrants in the United States. Research has also found that, on average, assimilation is associated with worse health. Thus, the processes behind both bilingualism and assimilation raise the question of how social integration influences immigrant health.

Using data from the National Latino and Asian American Analysis Project, researchers designed a study to investigate the relationship between language and self-rated health. The measures of language included English and native language proficiency, as well as frequency of English and native language use. The immigrant groups included in the sample were Asian and Latino immigrants.

Based on a national area probability sample, the researchers oversampled geographical areas that have a moderate-to-high density of Latinos and Asian Americans. Bilingual interviewers asked the respondents to rate their overall physical health status and overall mental health status on a 5-point scale (with 1 being poor health and 5 being excellent health). The researchers’ statistical models considered multiple variables that may affect the relationship between language and health. These control variables included socioeconomic status (SES), assimilation, family and social support, stress and discrimination, and healthcare access and health behaviors.

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Adapted from A. Schachter, R. T. Kimbro, and B. K. Gorman, “Language Proficiency and Health Status: Are Bilingual Immigrants Healthier?” ©2012 American Sociological Association.

QUESTION + ANSWER CHOICES

The study found that Mexican immigrants reported worse physical and mental health statuses than the other Latino immigrant groups. Based on other results reported in the passage, which statement provides the best explanation of this finding?

- A. Other Latino immigrants have higher average SES than Mexican immigrants.
- B. **Mexican immigrants have more extensive social support networks than other Latino immigrants.**
- C. Mexican immigrants report greater levels of assimilation than other Latino immigrants.
- D. Other Latino immigrants report higher stress levels than Mexican immigrants.

SOLUTION

Solution: The correct answer is A.

1. **The results in the passage suggest that the relationship between language and health is partly explained by socioeconomic status (SES), along with family and social support. If the average SES of other Latino immigrant groups is higher than that of Mexican immigrants, then average differences in SES across racial/ethnic groups (or sub-groups) could help explain the observed differences in health. Thus, SES could be an explanatory factor for the reported health status differentials across Latino immigrant groups.**
2. Social support is generally associated with better health, and thus it would not be likely to explain the difference in health status referenced in the question.
3. The passage reports that the researchers did not find evidence that assimilation affected the health-language relationship. It can be inferred from this result that assimilation alone, without any other factors, is unlikely to explain the reported difference in health status among Latino immigrant groups.
4. Although high stress levels are generally associated with worse health status, this option does not follow the finding in the question. To provide a possible explanation, the groups would have to be reversed in this option (with Mexican immigrants reporting higher stress levels than the other two groups).

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
1 min 37 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC9B
FC9
SOC

Question 209 of 230

Data ID: 632020
Captured: 6/3/2026, 4:41:57 PM

PASSAGE

Passage 7 (Questions 35-38)

Studies have found that being bilingual is associated with better self-rated physical and mental health among immigrants in the United States. Research has also found that, on average, assimilation is associated with worse health. Thus, the processes behind both bilingualism and assimilation raise the question of how social integration influences immigrant health.

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Adapted from A. Schachter, R. T. Kimbro, and B. K. Gorman, "Language Proficiency and Health Status: Are Bilingual Immigrants Healthier?" ©2012 American Sociological Association.

QUESTION + ANSWER CHOICES

Sociologists describe the health benefits of social support as resulting from all of the following EXCEPT:

- A. friends and family members that help patients adhere to medical treatment.
- B. friends and family members that help patients to reduce harmful behaviors.
- C. social relationships that help individuals to cope with stressful events.
- D. social activities that help individuals to avoid loneliness and boredom.

SOLUTION

Solution: The correct answer is D.

1. This option refers to help with treatment adherence from family and friends, which would reflect a health benefit of social support.
2. Some social ties can help reduce harmful behaviors, which would reflect a health benefit of social support.
3. Social relationships that help an individual cope with stress would indicate a benefit of social support.
4. Social support refers to social network ties (friends, family, and other relationships) that provide an individual with various types of assistance, some of which can improve health or reduce harms. All of the response options except for this one address potential health benefits of social support. This option refers to social activities rather than social relationships (or social ties) and thus does not identify social support.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Elements of Social Interaction: Self-association and Interaction with Others](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
47 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS1
CC8C
FC8
SOC

Question 210 of 230

Data ID: 632027
Captured: 6/3/2026, 4:41:58 PM

PASSAGE

Passage 8 (Questions 39-44)

Researchers examined whether adding gaming elements to a computerized working memory task for children with attention deficit hyperactivity disorder (ADHD) was beneficial. Fifty-one children between 7 and 12 years of age with ADHD were randomly assigned to a computer task with either a non-gaming or gaming format.

In the non-gaming condition, squares on a four-by-four grid lit up in random order on a screen. Participants were asked to replicate the pattern by clicking their computer mouse on the squares in the same sequence. Trial difficulty was adjusted depending on participants' performance.

In the gaming condition, animated figures were added. Points were earned by responding correctly, which strengthened participants' efforts and weakened animated opponents. Points were lost by responding incorrectly, strengthening the opponents' power. Trials of varying difficulty were presented randomly to participants. The training session for each condition lasted 35 minutes. After 15 minutes, participants were given a choice to continue playing the game or read a magazine.

Data were collected once weekly for three weeks. Participants' working memory capacity for visual–spatial information was assessed with a block-tapping task. A researcher tapped 3-dimensional blocks in a specific sequence that participants were asked to replicate.

Motivation was assessed in three ways. First, the computer tracked how many grid sequences the participants attempted to solve. Second, absence time was recorded when each participant was not interacting with the mouse. Third, participants answered an exit questionnaire asking how much they liked the task and whether they would like to have it at home.

Results indicated that participants in the gaming condition spent more time playing the game, attempted to replicate more sequences, and made fewer errors. These participants also earned higher scores on the block-tapping task compared to children in the non-gaming condition. The same participants were also more likely to report liking the game and wanting a version to play at home.

Adapted from P. J. M. Prins et al., ©2011 Mary Ann Liebert, Inc.

QUESTION + ANSWER CHOICES

If adults performed the block tapping task as a measure of working memory capacity, they would most likely replicate sequences of what length correctly?

- A. 9 plus or minus 1
- B. 7 plus or minus 2
- C. 10 plus or minus 1
- D. 16 plus or minus 2

SOLUTION

Solution: The correct answer is B.

1. The capacity of STM was believed to be lower, approximately between 5 and 9 items (plus or minus 2).
2. Based on early research into the capacity of the short-term memory store, the belief was that STM capacity lies between 5 and 9 items. George Miller (1956) named this idea the "Magical Number 7 + 2".
3. The range in this option is too high. Based on early research into the capacity of the short-term memory store, the belief was that STM capacity lies between 5 and 9 items.
4. Based on early research into the capacity of the short-term memory store, the belief was that STM capacity lies between 5 and 9 items. The range in this option is much too high.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
X
D
Time Spent:
1 min 56 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS1
CC6B
FC6
PSY

Question 211 of 230

Data ID: 632032
Captured: 6/3/2026, 4:41:59 PM

PASSAGE

Passage 8 (Questions 39-44)

Researchers examined whether adding gaming elements to a computerized working memory task for children with attention deficit hyperactivity disorder (ADHD) was beneficial. Fifty-one children between 7 and 12 years of age with ADHD were randomly assigned to a computer task with either a non-gaming or gaming format.

In the non-gaming condition, squares on a four-by-four grid lit up in random order on a screen. Participants were asked to replicate the pattern by clicking their computer mouse on the squares in the same sequence. Trial difficulty was adjusted depending on participants' performance.

In the gaming condition, animated figures were added. Points were earned by responding correctly, which strengthened participants' efforts and weakened animated opponents. Points were lost by responding incorrectly, strengthening the opponents' power. Trials of varying difficulty were presented randomly to participants. The training session for each condition lasted 35 minutes. After 15 minutes, participants were given a choice to continue playing the game or read a magazine.

Data were collected once weekly for three weeks. Participants' working memory capacity for visual-spatial information was assessed with a block-tapping task. A researcher tapped 3-dimensional blocks in a specific sequence that participants were asked to replicate.

Motivation was assessed in three ways. First, the computer tracked how many grid sequences the participants attempted to solve. Second, absence time was recorded when each participant was not interacting with the mouse. Third, participants answered an exit questionnaire asking how much they liked the task and whether they would like to have it at home.

Results indicated that participants in the gaming condition spent more time playing the game, attempted to replicate more sequences, and made fewer errors. These participants also earned higher scores on the block-tapping task compared to children in the non-gaming condition. The same participants were also more likely to report liking the game and wanting a version to play at home.

Adapted from P. J. M. Prins et al., ©2011 Mary Ann Liebert, Inc.

QUESTION • ANSWER CHOICES

Based on the study, which assessment represents a subjective measure of motivation?

- A. The number of grid sequences participants attempted to solve on the computer task
- B. The participants' decision to continue playing the game or stop to read a magazine
- C. Participants' answers on the exit questionnaire
- D. Participants' time spent not playing the game using the computer mouse

SOLUTION

Solution: The correct answer is C.

- This was an objective measure, which more directly measured motivation.
- The duration of time playing the game was an objective measure, which more directly measured motivation.
- Answers participants gave on their exit questionnaires were the only subjective measures obtained in the study. They provided an indirect measure of each participant's motivation.**
- This duration was another objective measure, which was a direct measure of motivation.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Biological Basis of Behavior: Human Development: Behavior and Cognition](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
38 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC7A
FC7
PSY

Question 212 of 230

Data ID: 632040
Captured: 6/3/2026, 4:41:59 PM

PASSAGE

Passage 8 (Questions 39-44)

Researchers examined whether adding gaming elements to a computerized working memory task for children with attention deficit hyperactivity disorder (ADHD) was beneficial. Fifty-one children between 7 and 12 years of age with ADHD were randomly assigned to a computer task with either a non-gaming or gaming format.

In the non-gaming condition, squares on a four-by-four grid lit up in random order on a screen. Participants were asked to replicate the pattern by clicking their computer mouse on the squares in the same sequence. Trial difficulty was adjusted depending on participants' performance.

In the gaming condition, animated figures were added. Points were earned by responding correctly, which strengthened participants' efforts and weakened animated opponents. Points were lost by responding incorrectly, strengthening the opponents' power. Trials of varying difficulty were presented randomly to participants. The training session for each condition lasted 35 minutes. After 15 minutes, participants were given a choice to continue playing the game or read a magazine.

Data were collected once weekly for three weeks. Participants' working memory capacity for visual-spatial information was assessed with a block-tapping task. A researcher tapped 3-dimensional blocks in a specific sequence that participants were asked to replicate.

Motivation was assessed in three ways. First, the computer tracked how many grid sequences the participants attempted to solve. Second, absence time was recorded when each participant was not interacting with the mouse. Third, participants answered an exit questionnaire asking how much they liked the task and whether they would like to have it at home.

Results indicated that participants in the gaming condition spent more time playing the game, attempted to replicate more sequences, and made fewer errors. These participants also earned higher scores on the block-tapping task compared to children in the non-gaming condition. The same participants were also more likely to report liking the game and wanting a version to play at home.

Adapted from P. J. M. Prins et al., ©2011 Mary Ann Liebert, Inc.

QUESTION • ANSWER CHOICES

Based on the study, which extrinsic factor most likely accounts for the motivational differences between groups?

- A.** Rewards and losses tied to memory performance for the gaming group
- B. The presence of animated characters in the visual display for the gaming group
- C. The opportunity to stop playing the game for the gaming and non-gaming groups
- D. The opportunity to state they disliked the game for the gaming and non-gaming groups

SOLUTION

Solution: The correct answer is A.

- In the gaming condition, each correct response strengthened the participant and weakened the opponent, and each incorrect response weakened the participant and strengthened the opponent.**
- The presence or absence of the animated figures in the gaming condition was not made contingent on a participant's responses. Thus, it was not an extrinsic motivational factor.
- The decision to stop playing represents an intrinsic, rather than an extrinsic, factor.
- This opportunity represents an intrinsic factor, rather than an extrinsic factor, related to their motivation.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Biological Basis of Behavior: Human Development: Behavior and Cognition](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
3 mins 44 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC7A
FC7
PSY

Question 213 of 230

Data ID: 632046
Captured: 6/3/2026, 4:42:00 PM

PASSAGE

Passage 8 (Questions 39-44)

Researchers examined whether adding gaming elements to a computerized working memory task for children with attention deficit hyperactivity disorder (ADHD) was beneficial. Fifty-one children between 7 and 12 years of age with ADHD were randomly assigned to a computer task with either a non-gaming or gaming format.

In the non-gaming condition, squares on a four-by-four grid lit up in random order on a screen. Participants were asked to replicate the pattern by clicking their computer mouse on the squares in the same sequence. Trial difficulty was adjusted depending on participants' performance.

In the gaming condition, animated figures were added. Points were earned by responding correctly, which strengthened participants' efforts and weakened animated opponents. Points were lost by responding incorrectly, strengthening the opponents' power. Trials of varying difficulty were presented randomly to participants. The training session for each condition lasted 35 minutes. After 15 minutes, participants were given a choice to continue playing the game or read a magazine.

Data were collected once weekly for three weeks. Participants' working memory capacity for visual-spatial information was assessed with a block-tapping task. A researcher tapped 3-dimensional blocks in a specific sequence that participants were asked to replicate.

Motivation was assessed in three ways. First, the computer tracked how many grid sequences the participants attempted to solve. Second, absence time was recorded when each participant was not interacting with the mouse. Third, participants answered an exit questionnaire asking how much they liked the task and whether they would like to have it at home.

Results indicated that participants in the gaming condition spent more time playing the game, attempted to replicate more sequences, and made fewer errors. These participants also earned higher scores on the block-tapping task compared to children in the non-gaming condition. The same participants were also more likely to report liking the game and wanting a version to play at home.

Adapted from P. J. M. Prins et al., ©2011 Mary Ann Liebert, Inc.

QUESTION • ANSWER CHOICES

Suppose squares 1, 2, 3, and 4, moving from left to right at the top of the grid light up and some participants think "top row lights up first," to remember the sequence. Which process is most likely being used to enhance working memory capacity?

- A. Divided attention
- B. Priming
- C. Automatic processing
- D. Chunking

SOLUTION

Solution: The correct answer is **D**.

1. The participants are engaging in selective attention, rather than divided attention.
2. No evidence of priming has been presented.
3. The selective attention described in the stem requires some amount of effort, which rules out the possibility that the sequence has been processed automatically.
4. **The participants are engaging in chunking, in that a sequence of lights is combined into a single unit, which is easier to recognize.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
1 min 7 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC6B
FC6
PSY

Question 214 of 230

Data ID: 632051
Captured: 6/3/2026, 4:42:00 PM

PASSAGE

Passage 8 (Questions 39-44)

Researchers examined whether adding gaming elements to a computerized working memory task for children with attention deficit hyperactivity disorder (ADHD) was beneficial. Fifty-one children between 7 and 12 years of age with ADHD were randomly assigned to a computer task with either a non-gaming or gaming format.

In the non-gaming condition, squares on a four-by-four grid lit up in random order on a screen. Participants were asked to replicate the pattern by clicking their computer mouse on the squares in the same sequence. Trial difficulty was adjusted depending on participants' performance.

In the gaming condition, animated figures were added. Points were earned by responding correctly, which strengthened participants' efforts and weakened animated opponents. Points were lost by responding incorrectly, strengthening the opponents' power. Trials of varying difficulty were presented randomly to participants. The training session for each condition lasted 35 minutes. After 15 minutes, participants were given a choice to continue playing the game or read a magazine.

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Adapted from P. J. M. Prins et al., ©2011 Mary Ann Liebert, Inc.

QUESTION • ANSWER CHOICES

Which type of memory would most likely be activated first during the computer task, before working memory is activated?

- A. Implicit memory
- B. Sensory memory**
- C. Short-term memory
- D. Episodic memory

SOLUTION

Solution: The correct answer is **B**.

1. Implicit memory is assessed using indirect measures, none of which involves working memory.
2. **In the Atkinson-Shiffrin multistore memory model, the initial registration of the information is briefly processed in a form of sensory memory, which is extremely brief in duration.**
3. Prior to the creation of Baddeley's working memory (WM) model, the Atkinson-Shiffrin multistore memory model included a short-term store, which came to be termed short-term memory (STM).
4. Episodic memory refers to long-term memory representations of events experienced by the participant. In this study, participants were not required (and made no attempt) to retain any information about the game.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
25 secs
Difficulty Level:
Expert
Content & Skills:
SIRS1
CC6B
FC6

Question 215 of 230

Data ID: 632090
Captured: 6/3/2026, 4:42:01 PM

PASSAGE

Passage 8 (Questions 39-44)

Researchers examined whether adding gaming elements to a computerized working memory task for children with attention deficit hyperactivity disorder (ADHD) was beneficial. Fifty-one children between 7 and 12 years of age with ADHD were randomly assigned to a computer task with either a non-gaming or gaming format.

In the non-gaming condition, squares on a four-by-four grid lit up in random order on a screen. Participants were asked to replicate the pattern by clicking their computer mouse on the squares in the same sequence. Trial difficulty was adjusted depending on participants' performance.

In the gaming condition, animated figures were added. Points were earned by responding correctly, which strengthened participants' efforts and weakened animated opponents. Points were lost by responding incorrectly, strengthening the opponents' power. Trials of varying difficulty were presented randomly to participants. The training session for each condition lasted 35 minutes. After 15 minutes, participants were given a choice to continue playing the game or read a magazine.

Data were collected once weekly for three weeks. Participants' working memory capacity for visual-spatial information was assessed with a block-tapping task. A researcher tapped 3-dimensional blocks in a specific sequence that participants were asked to replicate.

Motivation was assessed in three ways. First, the computer tracked how many grid sequences the participants attempted to solve. Second, absence time was recorded when each participant was not interacting with the mouse. Third, participants answered an exit questionnaire asking how much they liked the task and whether they would like to have it at home.

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Adapted from P. J. M. Prins et al., ©2011 Mary Ann Liebert, Inc.

QUESTION + ANSWER CHOICES

Based on the findings in the study, which conclusion about ADHD is most likely?

- A. Immediate performance feedback reduces impulsive responding in children with ADHD.
- B. Exposure to a gaming element increases impulsive responding in children with ADHD.
- C. Gaming elements on working memory tasks place high demands on divided attention and inhibit focus.
- D. Low levels of intrinsic motivation in children with ADHD are minimally enhanced with external reinforcers.

SOLUTION

Solution: The correct answer is A.

- Children in the gaming condition received immediate feedback after making each of their responses.
- Children in the gaming condition attempted more sequences and made fewer errors. Therefore, they were less impulsive (not more) than children in the non-gaming condition.
- There were no demands of divided attention in the game.
- External reinforcers are relevant to extrinsic (not intrinsic) motivation.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
X
C
D
Time Spent:
2 mins 57 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS4
CC6B
FC6
PSY

Question 216 of 230

Data ID: 632071
Captured: 6/3/2026, 4:42:02 PM

QUESTION • ANSWER CHOICES

A researcher wants to gain insight into how a social movement organization presents its beliefs to the general public in an effort to attract new members. Which methodological approach would be most appropriate for this study?

- A. Observing meetings about the organization's marketing budget
- B. Conducting a content analysis of the organization's website**
- C. Surveying organization members about their beliefs
- D. Randomly sampling the general population about social movement organizations

SOLUTION

Solution: The correct answer is **B**.

- Internal meetings about the organization's budget would not specifically indicate how the organization presents its beliefs to the general public.
- Conducting content analysis of online information would provide data on how the social movement organization frames issues to the broader public. Content analysis is a sociological method that is used to make inferences about a group's communication. As applied to studying a website, content analysis could help determine which beliefs the organization emphasizes to the general public.**
- Although this option could provide some insights into the organization's beliefs, it would more precisely yield data on the organization's members and their specific beliefs. It would not be the best approach for understanding how the organization presents its beliefs to the general public for attracting new members.
- A random sample of the general public would only provide data on how the public perceives the organizations, but it would not provide data on how the social movement organization presents its beliefs to the general public (for attracting new members).

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
1 min 11 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS3
CC9B
FC9
SOC

Question 217 of 230

Data ID: 632077
Captured: 6/3/2026, 4:42:02 PM

QUESTION • ANSWER CHOICES

An immigrant teen stops participating in the ethnic customs of his family and instead identifies with the dominant culture of the new country by dressing, speaking, and acting like teens from that culture. In this scenario, the teens of the new country's dominant culture become which type of group for the teen?

- A. Assimilated group
- B. Reference group**
- C. Majority group
- D. Peer group

SOLUTION

Solution: The correct answer is **B**.

- An assimilated group would refer to a group that adopts the norms and values of a new culture. However, the question asks for the type of group that is represented by the dominant culture.
- Individuals often emulate the attitudes and behaviors of groups that they would like to join. Sociologists refer to these groups as reference groups, which are important for self-evaluation and identity formation. The scenario in the question suggests that the teen constructs an identity, through modifying behaviors (dressing, speaking, acting), to match that of teens from the dominant culture. In the context of the specific information provided in the question, the dominant culture is closest to a reference group for the teen.**
- Although the dominant culture may reflect that of a majority group, the question does not provide information on majority and minority group dynamics in order to make a determination that the dominant culture is also the same as a majority group.
- A peer group is made up of individuals in a similar social position or social role. Because the immigrant teen is trying to emulate teens from the new country's culture, it is better described as a reference group (group used for evaluation) rather than a peer group.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
41 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC8A
FC8
SOC

Question 218 of 230

Data ID: 832081
Captured: 6/3/2026, 4:42:03 PM

QUESTION • ANSWER CHOICES

Researchers hypothesize that experiencing a certain scent while viewing paintings of flowers will influence the viewers' ratings of attractiveness of the paintings. Researchers dispensed a floral scent in one room with paintings and no scent in another room with paintings. Which statement is accurate regarding the study?

- A. The study's design is an observational study since viewers were observing paintings and noting their ratings.
- **B.** The independent variable is the presence or absence of scent in the rooms.
- C. The dependent variable is the amount of time viewers spent observing paintings.
- D. Informing viewers ahead of time whether a scent was absent or present in the room is important for accurate ratings.

SOLUTION

Solution: The correct answer is **B**.

1. The presence versus absence of a scent in the room was systematically manipulated. The systematic manipulation of an independent variable is the defining feature of an experiment, not the observation of paintings and their ratings.

2. **The presence versus absence of a scent in the room was systematically manipulated. The independent variable is the variable that is systematically manipulated in an experiment.**

3. No mention was made of observation times. The dependent variable was the viewers' attractiveness ratings of the paintings.

4. This conclusion can neither be verified nor refuted, because there was no mention of providing viewers with this information ahead of time.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
51 secs
Difficulty Level:
Easy
Content & Skills:
SIRS3
CC6A
FC6
PSY

Question 219 of 230

Data ID: 832087
Captured: 6/3/2026, 4:42:03 PM

QUESTION • ANSWER CHOICES

After initially learning to ride a bike, riding a bike becomes easy for an individual. Each time the individual rides a bike thereafter, what type of memory is being used?

- A. Semantic
- B. Episodic
- C. Explicit
- **D.** Procedural

SOLUTION

Solution: The correct answer is **D**.

1. Semantic memory refers to memory for general world knowledge and other meaningful information, including the meanings of words and phrases.

2. Episodic memory refers to long-term memory representations of events experienced by the participant.

3. Explicit memory performance refers to information retrieval as demonstrated by a direct memory task, such as a free recall, serial recall, yes-no recognition, or forced-choice recognition task.

4. **Procedural memory, which is a variety of implicit memory, refers to memory for performing particular actions (i.e., what many people call "muscle memory").**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
21 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC6B
FC6
PSY

Question 220 of 230

Data ID: 632096
Captured: 6/3/2026, 4:42:04 PM

PASSAGE

Passage 9 (Questions 49-52)

Sociodemographic variables are some of the most significant determinants of health outcomes at both the macro- and micro-levels. Despite substantial steps toward racial equality in the United States, inequalities in health outcomes persist across racial categories (see Table 1).

Table 1 Health-Related Factors in the U.S. by Race

Health-Related Factors	Race	
	White	Black
Private Insurance (%)	68	50
Life Expectancy*		
Males	75.7	69.8
Females	80.8	76.5
Mortality Rates		
AIDS**	2.1	18.7
Cancer**	179.9	217.4
Heart Disease**	197	257.7
Infant***	5.7	13.3

*Age in years; **Age-adjusted death rate per 100,000 people; ***Deaths per 1000 infants

Research by medical sociologists indicates that health disparities are due to a number of structural factors including socioeconomic inequalities. In particular, lower educational attainment and marginal economic security, both more highly associated with racial and ethnic minorities in the United States, lead to segregated living conditions. Studies have shown that racial and ethnic minorities are more likely to live in environments characterized by high crime, neighborhood disorder, and lack of institutional support. These environments create immediate negative effects in terms of psychological distress, toxic exposure, risky behavior, and hypertension. The negative effects of segregated neighborhoods contribute to the growing social distance between residents of lower-socioeconomic neighborhoods and those living in higher-socioeconomic environments.

At the micro-level, the quality of the physician–patient interaction is one of the strongest predictors of healthcare quality and treatment outcomes. The amount of social distance between patients and physicians impacts healthcare quality and treatment adherence in important ways. The social distance hypothesis suggests that patients who have more in common with their physicians in terms of class, cultural background, and social status receive more attention from physicians. Primarily, this has been attributed to differences in communication styles and cultural competencies. For example, Black patients report having less input in health decisions and lower levels of physician trust if their race differs from that of their physician. Race concordant physician–patient relationships result in higher levels of patient trust and physician attention. In order to mediate this dynamic, specially trained cultural liaisons who work with both patients and physicians have been introduced into many clinics.

QUESTION + ANSWER CHOICES

Because cultural liaisons work closely with physicians and are advocates for patients, which phenomenon are cultural liaisons most likely to experience?

- A. Network redundancy because of the amount of people with whom they come into contact
- B. Homogeneity because they will see the same patients over and over
- C. Role strain because of their collegiality with physicians and their responsibility to patients
- D. Role conflict because of their status as a non-medical professional in a clinic setting

SOLUTION

Solution: The correct answer is C.

- Network redundancy refers to repetition of ties within a social network. The cultural liaisons would not be likely to experience such repetition from their work with patients and physicians. Instead, they would likely have a variety of professional ties.
- Homogeneity generally refers to similarity in demographic characteristics of social ties in a network (such as based on gender identity, race/ethnicity, or social class). Not only would cultural liaisons be unlikely to have homogeneous professional networks, but this option also does not accurately describe homogeneity.
- Role strain is the tension that results from competing demands within the context of a single social role. Individuals in the social role of cultural liaison will likely have to manage different expectations when working with physicians compared to when they are advocating for patients. As a result, cultural liaisons are likely to experience role strain from their occupational role.
- Role conflict refers to the tension that can exist between social expectations of two or more social roles that a single individual holds. In the context of the question, there is only single social role (the occupational role of cultural liaison).

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
1 min 55 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC8C
FC8
SOC

Question 221 of 230

Data ID: 532113
Captured: 6/3/2026, 4:42:05 PM

PASSAGE

Passage 9 (Questions 49-52)

Sociodemographic variables are some of the most significant determinants of health outcomes at both the macro- and micro-levels. Despite substantial steps toward racial equality in the United States, inequalities in health outcomes persist across racial categories (see Table 1).

Table 1 Health-Related Factors in the U.S. by Race

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Heart Disease**	197	257.7
Infant***	5.7	13.3

*Age in years; **Age-adjusted death rate per 100,000 people; ***Deaths per 1000 infants

Research by medical sociologists indicates that health disparities are due to a number of structural factors including socioeconomic inequalities. In particular, lower educational attainment and marginal economic security, both more highly associated with racial and ethnic minorities in the United States, lead to segregated living conditions. Studies have shown that racial and ethnic minorities are more likely to live in environments characterized by high crime, neighborhood disorder, and lack of institutional support. These environments create immediate negative effects in terms of psychological distress, toxic exposure, risky behavior, and hypertension. The negative effects of segregated neighborhoods contribute to the growing social distance between residents of lower-socioeconomic neighborhoods and those living in higher-socioeconomic environments.

At the micro-level, the quality of the physician–patient interaction is one of the strongest predictors of healthcare quality and treatment outcomes. The amount of social distance between patients and physicians impacts healthcare quality and treatment adherence in important ways. The social distance hypothesis suggests that patients who have more in common with their physicians in terms of class, cultural background, and social status receive more attention from physicians. Primarily, this has been attributed to differences in communication styles and cultural competencies. For example, Black patients report having less input in health decisions and lower levels of physician trust if their race differs from that of their physician. Race concordant physician–patient relationships result in higher levels of patient trust and physician attention. In order to mediate this dynamic, specially trained cultural liaisons who work with both patients and physicians have been introduced into many clinics.

QUESTION + ANSWER CHOICES

Research findings indicate higher levels of patient trust and physician attention in race-concordant physician–patient relationships because:

- A. both the physician and patient belong to the same in-group.
- B. the physician and patient share similar levels of status.
- C. of racial hostility present in race-discordant relationships.
- D. of the lack of role strain experienced by the physician.

SOLUTION

Solution: The correct answer is A.

- Members of in-groups tend to share common backgrounds and similar social group identities, which generally result in high levels of trust among in-group members. Racial/ethnic identity can be an important characteristic of an in-group, resting in a shared culture, language, or community. Thus, the higher levels of patient trust and physician attention in race-concordant physician-patient relationships can be explained by in-group affinity.
- Even if race-concordant, the patient and physician could have different levels of social status based on characteristics such as income or education.
- Although there could be hostility in some race-discordant relationships, this option does not provide an explanation for why there would be higher levels of patient trust and physician attention in race-concordant relationships.
- Role strain is the tension that can result from competing demands of a single social role. The physician may or may not still experience role strain in a race concordant physician-patient relationship. Given this uncertainty, role strain does not provide the best explanation of the finding reported in the question.

Khan Academy Lessons

Correct
Correct Answer Your Answer
A ✓
B
C
D
Time Spent:
1 min 18 secs
Difficulty Level:
Easy
Content & Skills:
SIRS2
CC8C
FC8
SOC

Question 222 of 230

Data ID: 532125
Captured: 6/3/2026, 4:42:05 PM

PASSAGE

Passage 9 (Questions 49-52)

Sociodemographic variables are some of the most significant determinants of health outcomes at both the macro- and micro-levels. Despite substantial steps toward racial equality in the United States, inequalities in health outcomes persist across racial categories (see Table 1).

Table 1 Health-Related Factors in the U.S. by Race

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Heart Disease**	197	257.7
Infant***	5.7	13.3

*Age in years; **Age-adjusted death rate per 100,000 people; ***Deaths per 1000 infants

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QUESTION + ANSWER CHOICES

What is the advantage of including a specially trained cultural liaison as opposed to training physicians in cultural competency?

- A. Communicating across racial and cultural lines is difficult and requires a trained specialist to do it effectively.
- B. Physicians are inherently resistant to cultural training because of their medical biases.
- C. Adding a cultural liaison creates a triadic group, a more stable arrangement than a dyadic group.
- D. Larger groups facilitate the transfer of information more effectively than smaller groups.

SOLUTION

Solution: The correct answer is C.

- Although a trained specialist may have specific skills that others would not, it does not mean that physicians with cultural competency training could not be effective at communicating across racial and cultural lines. Thus, this option fails to provide a clear advantage.
- This option is overgeneralizing and thus inaccurate. Physicians are not necessarily resistant to cultural training because of assumed medical biases.
- From the perspective of basic group dynamics in sociology, larger groups are generally considered more stable but less intimate, whereas smaller groups are usually considered less stable but more intimate. Dyads, two-person groups such as the physician–patient group, are unstable because either party can break the single social tie. The triad, three person groups such as the physician–patient–cultural liaison group, is considered relatively more stable because of the additional social tie. As a result, the triadic group provides the advantage of added stability in group dynamics.
- Larger groups do not necessarily transfer information more effectively than smaller groups, and can actually lead to less effective information transfer at times. The advantage of larger group size is related to group stability rather than to the transfer of information.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
1 min 6 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC8C
FC8
SOC

Question 223 of 230

Data ID: 832135
Captured: 6/3/2026, 4:42:06 PM

PASSAGE

Passage 9 (Questions 49-52)

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QUESTION + ANSWER CHOICES

Neighborhood-level socioeconomic inequalities are most likely to affect physician–patient interactions through which phenomenon?

- A. Status hierarchies rely on socioeconomic differences above other master status characteristics.
- B. Physical boundaries create social boundaries and closed networks which develop their own cultures.
- C. Socioeconomic inequalities create role conflict in physicians who are trained to treat all patients equally.
- D. The physician–patient relationship is not subject to the same external forces that govern other relationships.

SOLUTION

Solution: The correct answer is B.

- Status hierarchies do not necessarily rely on socioeconomic differences, as status can refer to general social status or to other types of social categories (such as gender identity or race/ethnicity). Master status specifically identifies a status of an individual that tends to supersede all other statuses that the individual has. This option is incorrect because it inaccurately focuses on socioeconomic differences as the most significant factor in status hierarchies.
- Neighborhoods that are segregated by socioeconomic status create increased social distance and different experiences, such that people from different social backgrounds can have trouble relating to one another. The correct answer succinctly explains this by referring to the closed networks and social boundaries that can exist across neighborhoods. In turn, these closed networks and social boundaries will impact physician-patient interactions when the physician and patient have different experiences from living in distinct, and often very socially separated, neighborhoods.
- Role conflict occurs when a person experiences stress or tension from having to manage the social responsibilities of two or more social roles. The option does not accurately describe role conflict because it does not describe different social roles. Instead the option refers to a disjuncture between social stratification (socioeconomic inequalities) and a professional value (to treat all patients equally).
- Although the physician-patient relationship has some unique dynamics, it is not totally exempt from the external forces that affect nearly all social relationships.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
X
B
C
D
Time Spent:
2 mins 3 secs
Difficulty Level:
Expert
Content & Skills:
SIRS2
CC10A
FC10
SOC

Question 224 of 230

Data ID: 532156
Captured: 6/3/2026, 4:42:07 PM

PASSAGE

Passage 10 (Questions 53-56)

If gender and race are considered in decision-making processes in the workplace, it can lead to discrimination and inequality. This inequality includes hiring and firing practices, amount of pay, and opportunities for advancement. More than 60,000 complaints citing employer bias and racial or gender discrimination were filed with the Ohio Civil Rights Committee (OCRC) from 1988 to 2003. Researchers used these data to study the dynamics of workplace discrimination. For Sample 1, they compiled 15,175 claims that were found to be justified with supporting evidence by the OCRC. The researchers divided justifiable claims into five categories: exclusion (not being hired), expulsion (getting fired), promotion, demotion, and harassment on the job (overt antagonism, neglect, or differential treatment). These data are shown in Table 1.

Table 1 Discrimination Claims by Race, Gender, and Category of Discrimination

Category of discrimination	Type of claim	
	Race n = 9013	Gender n = 6162
Exclusion	5%	4%
Expulsion	56%	55%
Promotion	7%	4%
Demotion	2%	3%
Harassment	30%	34%

For Sample 2, the researchers used a random subsample of 740 cases for an in-depth analysis of the claims to examine how inequality in the workplace is created and maintained. They concluded that gatekeeping actors, such as workplace supervisors, are driven by preset cultural views and stereotypes. The gatekeeping actors operate within an organizational structure where procedures and rules allow them the flexibility to act on stereotypes. Thus, they recreate status hierarchies within work organizations and engage in discriminatory practices that harm women and minorities.

Adapted from V. Rosignio, L. Garcia, and D. Bibbitt-Zaher. "Social closure and processes of race/sex discrimination." ©2007 Sage Publications.

QUESTION + ANSWER CHOICES

The research in the passage is best described as a:

- A. qualitative study.
- B. quantitative study.
- C. mixed-methods study.
- D. case study.

SOLUTION

Solution: The correct answer is C.

1. A qualitative study involves research methods using non-numerical data, which does not accurately describe the research in the passage.
2. A quantitative study involves research methods collecting numerical data, which does not provide the best description of the study in the passage.
3. The study in the passage is best described as taking a mixed-methods approach. The research design employs both a quantitative component (the statistical analysis of the total claims) and a qualitative component (the in-depth analysis of a subset of cases). Bringing together quantitative methods and qualitative methods is referred to as mixed methods in the social sciences.
4. A case study focuses on the examination of a single case, and the research in the passage involves more than a single case.

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
2 mins 0 secs
Difficulty Level:
Expert
Content & Skills:

SIRS3
CC9B
FC9
SOC

Question 225 of 230

Data ID: 532158
Captured: 6/3/2026, 4:42:07 PM

PASSAGE

Passage 10 (Questions 53-56)

If gender and race are considered in decision-making processes in the workplace, it can lead to discrimination and inequality. This inequality includes hiring and firing practices, amount of pay, and opportunities for advancement. More than 60,000 complaints citing employer bias and racial or gender discrimination were filed with the Ohio Civil Rights Committee (OCRC) from 1988 to 2003. Researchers used these data to study the dynamics of workplace discrimination. For Sample 1, they compiled 15,175 claims that were found to be justified with supporting evidence by the OCRC. The researchers divided justifiable claims into five categories: exclusion (not being hired), expulsion (getting fired), promotion, demotion, and harassment on the job (overt antagonism, neglect, or differential treatment). These data are shown in Table 1.

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QUESTION + ANSWER CHOICES

Which interpretation of the research findings is consistent with the conflict theory perspective? Status hierarchies facilitate:

- A. efficiency in the organization, whereas discrimination is a byproduct of these structures.
- B. the preservation of structural power while being maintained by practices of discrimination.
- C. effective social interactions, whereas discrimination is a subjective quality of those interactions.
- D. the necessary operation of the workplace while removing ineffective members by practices of discrimination.

SOLUTION

Solution: The correct answer is B.

- This option is more in line with functionalist theory, which assumes that social phenomena have specific functions that work within a systematic whole. A functionalist interpretation of status hierarchies would assume that they exist to serve a function like organizational efficiency while at the same time recognizing that discrimination can occur as a possible dysfunction of a social system (as a "byproduct").
- Conflict theory emphasizes competition between social groups over the allocation of resources. The theory assumes that power and authority are unequally distributed across a society, and that groups attempt to maintain their advantages. This option captures the basic elements of conflict theory, whereas the other options are incorrect because they are more aligned with other sociological theories.
- This option is more consistent with symbolic interactionism, which focuses on social interactions and, in particular, the subjective meaning that emerges from social interactions.
- This option is also closest to functionalist theory because of the assumption that hierarchies and discrimination are social phenomena with a purpose for the effective operation of the workplace. It provides a different type of explanation than option A, but is still closest to functionalist theory.

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
27 secs

Difficulty Level:
Moderate
Content & Skills:
SIRS2
CC9A
FC9
SOC

Question 226 of 230

Data ID: 532194
Captured: 6/3/2026, 4:42:08 PM

PASSAGE

Passage 10 (Questions 53-56)

If gender and race are considered in decision-making processes in the workplace, it can lead to discrimination and inequality. This inequality includes hiring and firing practices, amount of pay, and opportunities for advancement. More than 60,000 complaints citing employer bias and racial or gender discrimination were filed with the Ohio Civil Rights Committee (OCRC) from 1988 to 2003. Researchers used these data to study the dynamics of workplace discrimination. For Sample 1, they compiled 15,175 claims that were found to be justified with supporting evidence by the OCRC. The researchers divided justifiable claims into five categories: exclusion (not being hired), expulsion (getting fired), promotion, demotion, and harassment on the job (overt antagonism, neglect, or differential treatment). These data are shown in Table 1.

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QUESTION + ANSWER CHOICES

If this study were replicated, which hypothesis would most likely be supported?

- A. Regular employees with stereotypical views would exclude minority group members, regardless of organizational culture.
- B. Workplace supervisors with stereotypical views would exclude minority group members, regardless of organizational culture.
- C. Organizational culture is the primary determinant of whether minority populations are excluded from a workplace.
- D. The interplay between a supervisor's stereotypical views and an organization that fosters those stereotypes would be more detrimental to minority group members.

SOLUTION

Solution: The correct answer is D.

- This option is not consistent with the study because it only refers to regular employees whereas the study focused on gatekeeping actors (such as workplace supervisors).
- This option is not consistent with the study because it only includes workplace supervisors but without reference to organizational culture, which was part of the study in the passage.
- The study in the passage suggests that organizational culture alone is not the primary determinant, without reference to gatekeeping actors (such as workplace supervisors).
- The study in the passage calls attention to the interplay of individual attitudes and overall organizational culture. Given the results of the study, it is expected that minority group members would be more likely to experience unfair treatment when individual bias is allowed to exist in an organization. This option correctly addresses the combination of individual factors (supervisor's stereotypical views) and organizational culture (fostering stereotypes) that is discussed with the results of the study in the passage.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Elements of Social Interaction](#); [Self-presentation and Interacting with Others](#); and [Discrimination](#).

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:

2 mins 35 secs
Difficulty Level:
Difficult
Content & Skills:
SIRS2
CC8C
FC8
SOC

Question 227 of 230

Data ID: 532171
Captured: 6/3/2026, 4:42:09 PM

PASSAGE

Passage 10 (Questions 53-56)

If gender and race are considered in decision-making processes in the workplace, it can lead to discrimination and inequality. This inequality includes hiring and firing practices, amount of pay, and opportunities for advancement. More than 60,000 complaints citing employer bias and racial or gender discrimination were filed with the Ohio Civil Rights Committee (OCRC) from 1988 to 2003. Researchers used these data to study the dynamics of workplace discrimination. For Sample 1, they compiled 15,175 claims that were found to be justified with supporting evidence by the OCRC. The researchers divided justifiable claims into five categories: exclusion (not being hired), expulsion (getting fired), promotion, demotion, and harassment on the job (overt antagonism, neglect, or differential treatment). These data are shown in Table 1.

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QUESTION + ANSWER CHOICES

Based on the passage, which conclusion is best supported?

- A. Gender discrimination is more likely to occur than racial discrimination.
- B. Race and gender discrimination manifest in similar ways.
- C. The incidence of race and gender discrimination is equivalent.
- D. Minority women are especially subject to discrimination.

SOLUTION

Solution: The correct answer is B.

- The study does not show that gender discrimination is more likely than racial discrimination, as the data in Table 1 is mostly similar across race discrimination and gender discrimination claims.
- As shown in the table in the passage, both the race discrimination claims and the gender discrimination claims were very similar in their distribution across the researchers' categories (exclusion, expulsion, promotion, demotion, and harassment). Thus, the conclusion in this option is supported by the passage.
- The passage only provides information on the incidence of race and gender discrimination claims together, but not on each type of discrimination separately. Table 1 displays the distribution of claims across the 5 categories, but does not show the incidence when comparing race and gender discrimination.
- This statement cannot be concluded from the passage, as there is no information on the combination of race and gender when it comes to discrimination claims.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
1 min 10 secs
Difficulty Level:
Moderate

Content & Skills:
SIRS2
CC10A
FC10
SOC

Question 228 of 230

Data ID: 532172
Captured: 6/3/2026, 4:42:09 PM

QUESTION • ANSWER CHOICES

A student is investigating the bystander effect and receives permission to conduct the study in the school cafeteria. What is a potential dependent variable that can be measured by the student?

- A. The time of day that the experiment is carried out
- B. The number of people present in the cafeteria
- C. The amount of time a student takes to assist another student in distress
- D. The temperature in the cafeteria

SOLUTION

Solution: The correct answer is **C**.

1. Although time of day may be used as an independent variable in a study of the bystander effect, it would not be used as a dependent measure in such a study.
2. Although the number of persons (i.e., group size) may be used as an independent variable in a study of the bystander effect, it would not be used as a dependent measure in such a study.
3. **This option is the only choice that represents helping behavior, which is the primary topic within which the bystander effect is studied.**
4. The room temperature does not involve any human behavior.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
24 secs
Difficulty Level:
Easy
Content & Skills:
SIRS3
CC7B
FC7
PSY

Question 229 of 230

Data ID: 532179
Captured: 6/3/2026, 4:42:10 PM

QUESTION • ANSWER CHOICES

Which research methodology involves the extended, systematic observation of a complete social environment?

- A. Comparative methods
- B. Ethnographic methods
- C. Experimental methods
- D. Survey methods

SOLUTION

Solution: The correct answer is **B**.

1. Comparative methods involve comparing an organization, geographic region, government, or other entity. The question does not refer to comparison, making this option incorrect.
2. **Ethnographic methods are accurately described in the question as involving systematic observation of a complete social environment. Systematic observation tends to be one of the key features of ethnography.**
3. Experimental methods involve the manipulation of an independent variable and random assignment to different groups (usually to an intervention group or a control group). The question does not describe any of these specific elements of an experiment.
4. Survey methods involve respondents providing responses to a questionnaire. The data collected in a survey are those responses, rather than the systematic observations that this question asks about.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
19 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS3
CC9A
FC9
SOC

Question 230 of 230

Data ID: 532196
Captured: 6/3/2026, 4:42:10 PM

QUESTION • ANSWER CHOICES

Which term refers to closed status positions that hinder social mobility?

- A.
Class systems
- B.
Caste systems
- C.
Economic systems
- D.
Financial systems

SOLUTION

Solution: The correct answer is **B**.

1. Class systems have status positions, but are generally not considered to be closed. Opportunities for social mobility can also vary significantly in class systems.
2. **Caste systems are generally closed stratification systems that do not provide many opportunities for social mobility. Thus, this option answers the question.**
3. Economic systems would refer to how the economy of a society is structured, but not to the characteristics of a society's status positions or level of social mobility.
4. Financial systems would refer to how a society finances its economy and organizations, but not to the society's status positions or possibilities for social mobility.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
0 secs
Difficulty Level:
Moderate
Content & Skills:
SIRS1
CC10A
FC10
SOC